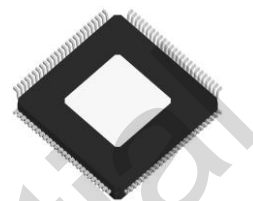


Features

- AEC-Q100 qualified
- Engineered for ISO26262 compliant system
- 1 pre boost regulator with 2.4 A output current
- 1 pre buck regulator with 2.3 A output current
- 1 linear 5 V regulator with 1.6 A output current
- 3 independent self-protection 5 V tracking regulator with 150 mA output.
- 1 input voltage pin for monitor external tracking.
- Coordinated soft start-up of all regulators
- 4 channels LS injector LS drivers
- 2 channels LS drivers for O2H load with current sense
- 2 channels LS camshaft or solenoid drivers
- 5 channels LS relay drivers
- 2 channels LS LED drivers
- 3 channels LS/HS drivers with low battery function for smart start
- 1 channel LS main relay driver (MRD) with internal diode for reverse battery protection
- 5 channels pre-drivers for external FET drivers. Pre-driver 1&3 configurable for O2H load with external Rshunt-on the source of Ext.N-channel MOS
- 6 channels pre-drivers for internal or external igniter drivers
- 1 K-Line ISO9141/LIN 2.1 compliant
- Integrated charge-pump
- VRS-interface
- Watchdog
- Wake-up pin
- Temperature sensor and monitoring
- Stop-counter with wake up
- Dual bandgap reference & oscillator
- Micro-second-channel MSC for differential single ended mode
- SEO function

- CAN-FD with wake up by CAN function
- Package ETQFP exposed pad



Description

The HE9788N is an integrated circuit designed for automotive engine management system.

HE9788N is a device realized in BCD proprietary technology, able to provide the full set of power supplies and signal preprocessing peripherals needed to control a 4 cylinders internal combustion engine.



Part Number	Package	Marking	Packing
HE9788Nxxxx	ETQFP100	HE9788N	1000/Tape&Reel

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks!

*The full part number will be confirmed after the official version of the product.

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1 Block diagram and pin description

1.1 Block diagram

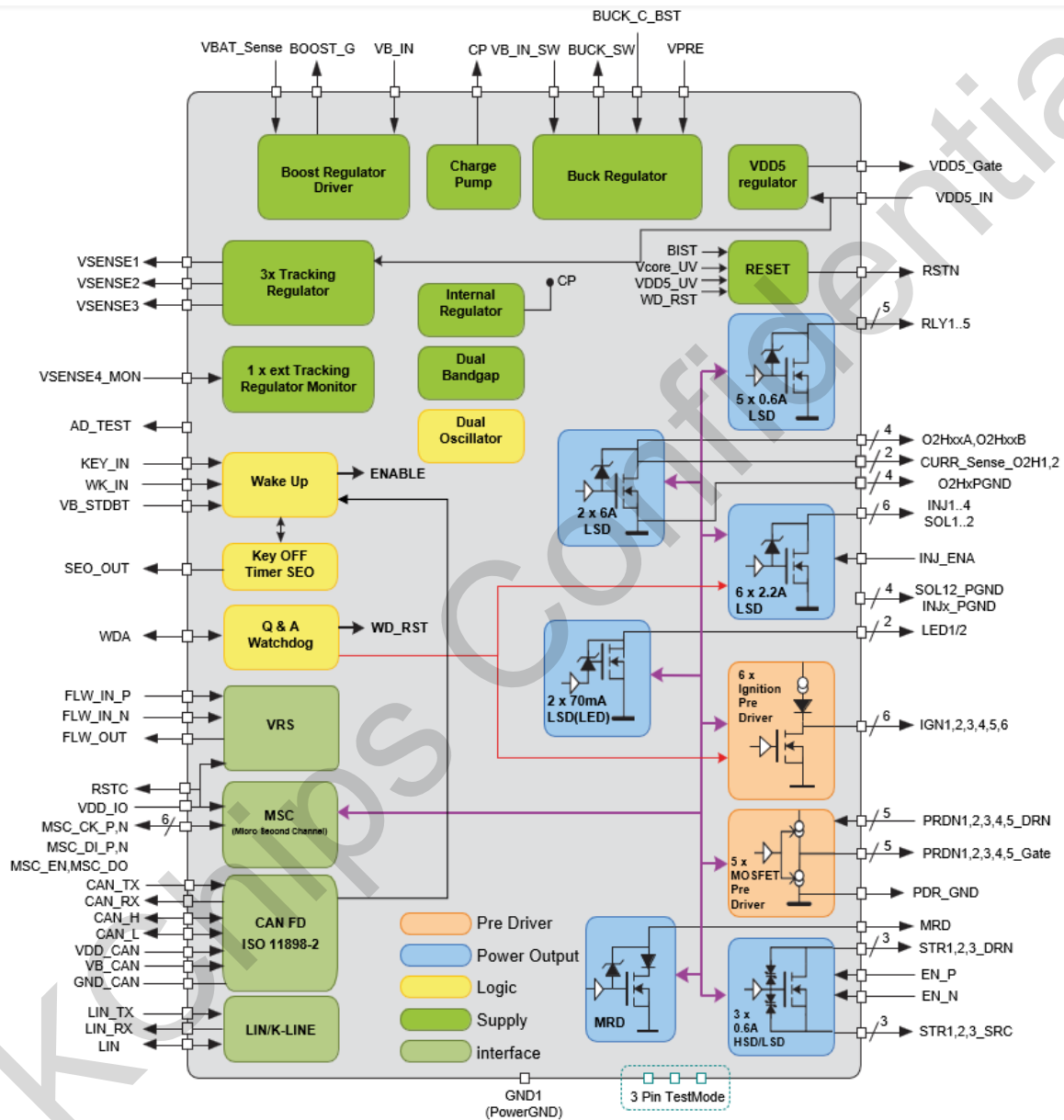


Figure 1-1 Block diagram

1.2 Pin description

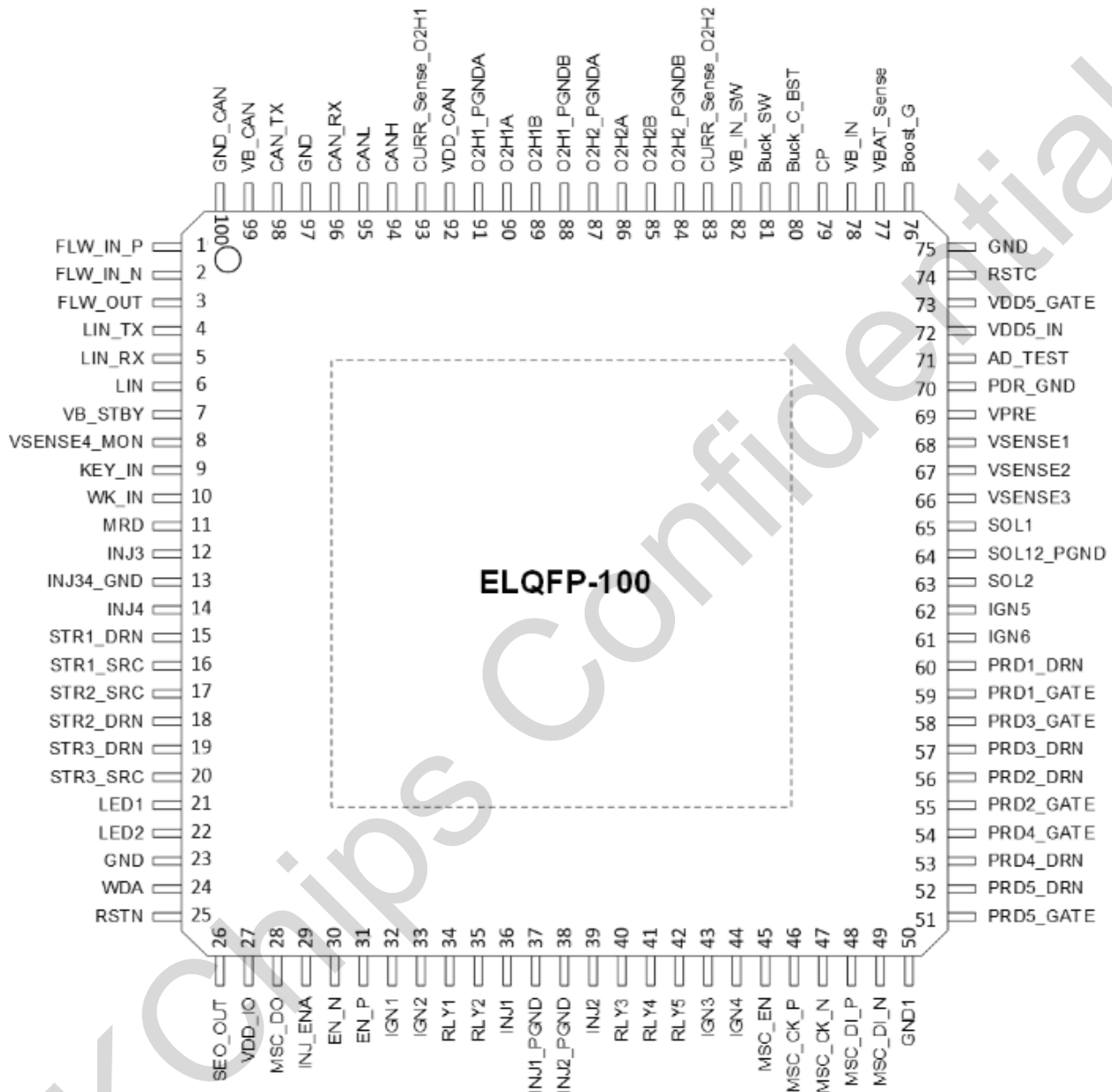


Figure 1-2 Pin connection diagram (top view)

Table 1-1 Pin function

Pin	Pin name	Function	Type	Note
1	FLW_IN_P	Flying wheel inputs voltage positive	Power In	-
2	FLW_IN_N	Flying wheel inputs voltage negative	Power In	-
3	FLW_OUT	Flying wheel output voltage	Digital Out	-
4	LIN_TX	LIN data input	Digital In	-

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Block diagram and pin description

5	LIN_RX	LIN data output	Digital Out	-
6	LIN	LIN driver output	Power In/Out	-
7	VB_STBY	Standby function supply voltage	Supply	-
8	VSENSE4_MON	tracking sensor supply Monitor	Analog In	-
9	KEY_IN	Key signal input	Power In	-
10	WK_IN	wake up signal input	Power In	-
11	MRD	Main relay drivers output voltage	Power Out	-
12	INJ3	Injector Driver output voltage	Power Out	-
13	INJ34_PGND	Ground	GND	-
14	INJ4	Injector Driver output voltage	Power Out	-
15	STR1_DRN	Configurable high/low-side drain voltage	Power Out	-
16	STR1_SRC	Configurable high/low-side source voltage	Power Out	-
17	STR2_SRC	Configurable high/low-side source voltage	Power Out	-
18	STR2_DRN	Configurable high/low-side drain voltage	Power Out	-
19	STR3_DRN	Configurable high/low-side drain voltage	Power Out	-
20	STR3_SRC	Configurable high/low-side source voltage	Power Out	-
21	LED1	LED Driver output voltage	Power Out	-
22	LED2	LED Driver output voltage	Power Out	-
23	GND	Ground	GND	-
24	WDA	Watchdog output	Digital In/Out	-
25	RSTN	Reset output for VDD5	Digital Out	-
26	SEO_OUT	SEO output pin	Digital Out	-
27	VDD_IO	Dedicated supply for FLW_OUT, MSC_DO, LIN_RX, CAN_RX	Supply	-
28	MSC_DO	MSC digital I/O voltage	Digital Out	-
29	INJ_ENA	Enable pin for injector driver	Digital In	-
30	EN_N	enable signals	Digital In	-
31	EN_P	enable signals	Digital In	-
32	IGN1	Ignition pre-driver output voltage	Power Out	-
33	IGN2	Ignition pre-driver output voltage	Power Out	-
34	RLY1	Relay Driver output voltage	Power Out	-
35	RLY2	Relay Driver output voltage	Power Out	-
36	INJ1	Injector Driver output voltage	Power Out	-
37	INJ1_PGND	Ground	GND	-
38	INJ2_PGND	Ground	GND	-
39	INJ2	Injector Driver output voltage	Power Out	-
40	RLY3	Relay Driver output voltage	Power Out	-
41	RLY4	Relay Driver output voltage	Power Out	-
42	RLY5	Relay Driver output voltage	Power Out	-
43	IGN3	Ignition pre-driver output voltage	Power Out	-
44	IGN4	Ignition pre-driver output voltage	Power Out	-
45	MSC_EN	MSC digital I/O voltage	Digital In	-
46	MSC_CK_P	MSC digital I/O voltage	Digital In	-
47	MSC_CK_N	MSC digital I/O voltage	Digital In	-
48	MSC_DI_P	MSC digital I/O voltage	Digital In	-
49	MSC_DI_N	MSC digital I/O voltage	Digital In	-
50	GND1	Power ground	GND	Shorted to GND by package
51	PRD5_GATE	General purpose pre-drivers gate voltage	Power Out	-
52	PRD5_DRN	General purpose pre-drivers feedback voltage	Power In	-
53	PRD4_DRN	General purpose pre-drivers feedback voltage	Power In	-
54	PRD4_GATE	General purpose pre-drivers gate voltage	Power Out	-
55	PRD2_GATE	General purpose pre-drivers gate voltage	Power Out	-
56	PRD2_DRN	General purpose pre-drivers feedback voltage	Power In	-
57	PRD3_DRN	General purpose pre-drivers feedback voltage	Power In	-
58	PRD3_GATE	General purpose pre-drivers gate voltage	Power Out	-
59	PRD1_GATE	General purpose pre-drivers gate voltage	Power Out	-
60	PRD1_DRN	General purpose pre-drivers feedback voltage	Power In	-
61	IGN6	Ignition pre-driver output voltage	Power Out	-

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Block diagram and pin description

62	IGN5	Ignition pre-driver output voltage	Power Out	-
63	SOL2	Valve driver output voltage	Power Out	-
64	SOL12_PGND	Ground	GND	-
65	SOL1	Valve driver output voltage	Power Out	-
66	VSENSE3	5V tracking sensor supply output voltage	Power Out	-
67	VSENSE2	5V tracking sensor supply output voltage	Supply Out	-
68	VSENSE1	5V tracking sensor supply output voltage	Supply Out	-
69	VPRE	Buck output voltage	Supply Out	-
70	PDR_GND	Ground pin for predriver supply feedback input	GND	-
71	AD_TEST	AD test pin	Analog Out	-
72	VDD5_IN	5V regulator feedback voltage	Power In	-
73	VDD5_GATE	5V linear regulator pre-driver output	Power Out	-
74	RST _{case}	Reset output for VDDIO supply	Digital Out	-
75	GND	Ground	GND	-
76	Boost_G	Boost LS gate voltage	Power Out	-
77	VBAT_Sense	Battery supply voltage	Supply In	-
78	VB_IN	Battery supply voltage	Supply In	-
79	CP	Charge pump	Power Out	-
80	Buck_C_BST	Bootstrap capacitor pin	Power Out	-
81	Buck_SW	Buck switching pin	Power Out	-
82	VB_IN_SW	BUCK supply voltage	Power In	-
83	CURR_Sense_O2H2	O2heater drivers output current value	Analog Out	-
84	O2H2_PGND	Ground	GND	-
85	O2H2B	O2H2B Driver output voltage	Power Out	-
86	O2H2A	O2H2A Driver output voltage	Power Out	-
87	O2H2_PGND	Ground	GND	-
88	O2H1_PGND	Ground	GND	-
89	O2H1B	O2H1B Driver output voltage	Power Out	-
90	O2H1A	O2H1A Driver output voltage	Power Out	-
91	O2H1_PGND	Ground	GND	-
92	VDD_CAN	CAN Supply 5 V	Supply In	-
93	CURR_Sense_O2H1	O2heater drivers output current value	Analog Out	-
94	CANH	CANH output	Power In/Out	-
95	CANL	CANL output	Power In/Out	-
96	CAN_RX	CAN RX data output	Digital In/Out	-
97	GND	Ground	GND	-
98	CAN_TX	CAN TX data input	Digital In	-
99	VB_CAN	CAN enable signal	Supply In	-
100	GND_CAN	Ground CAN	GND	-

2 Absolute maximum ratings

The component withstands all the following stimuli without any damage or latch-up.

Exceeding any of these values or sustaining it for extended periods may lead to characteristics degradation or component damage.

All voltages are related to analog ground

T_j -40 to 175 °C unless otherwise specified.

Table 2-1 Absolute maximum ratings

Pin name	Parameter	Referred to	Constr.	Min	Max	Unit	Pin(1) Direction	Number
VB_IN	Chip supply voltage	GND1	-	-0.3	+40	V	S	P_2.1
			10 ms (ISOpulse1) VB_IN pin shorted with VB_IN_SW	-2	-	V	S	P_2.2
VB_IN_SW	Buck supply voltage	GND1	-	-0.3	+40	V	S	P_2.3
			10 ms (ISOpulse1)	-2	-	V	S	P_2.4
VBAT_Sense	Battery supply voltage sense	GND1	-	-0.3	+40	V	S	P_2.5
VBAT_Sense	Input current	GND1	Protect from Reverse battery with 1kΩ resistor	-16	-	mA	S	P_2.6
VBAT_Sense	Input current	GND1	Protect from ISOPULSE 1 with 1kΩ resistor	-150	-	mA	S	P_2.7
VB_STBY	Standby function supply voltage for EOT	GND1	-	-16	+40	V	S	P_2.8
Boost_G	Boost LS gate voltage	GND1	-	-0.3	+20	V	O	P_2.9
VPRE	Buck output voltage	GND1	-	-0.3	+20	V	S	P_2.10
Buck_SW	Buck switching pin	GND1	-	-0.3(2)	+40	V	S	P_2.11
Buck_C_BST	Bootstrap capacitor pin	GND1	-	-0.3	+45	V	I	P_2.12
VDD5_GATE	5 V linear regulator predriver output	GND1	-	-0.3	+20	V	O	P_2.13
VDD5_IN	5 V regulator feedback voltage	GND1	-	-0.3	+20	V	S	P_2.14
VDD_IO	Dedicated 5 V voltage for FLW_OUT, MSC_DO,	GND1	-	-0.3	+20	V	S	P_2.15

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Absolute maximum ratings

	TxDC, RxDC							
VSENSE1, VSENSE2, VSENSE3	5 V tracking sensor supply output voltage	GND1	-	-2	+40	V	S	P_2.16
VSENSE4_MON	External tracking sensor supply Monitor	GND1	-	-2	+40	V	I	P_2.17
CP	Charge pump	GND1	-	-0.3	VB_IN+ 5V	V	S	P_2.18
GND1	Power ground	-	-	-0.3	+0.3	V	O	P_2.19
GND_CAN	Power ground	GND1	-	-0.3	+0.3	V	O	P_2.20
WDA	Digital output (open drain) / Digital input	GND1	-	-0.3	+20	V	I/O	P_2.21
SEO_OUT	Digital output	GND1	-	-0.3	+20	V	O	P_2.22
RSTN, RSTC	Digital output / open drain	GND1	-	-0.3	+20	V	I/O	P_2.23
AD_TEST	Digital output voltage	GND1	-	-0.3	+20	V	O	P_2.24
INJ_ENA,	Digital input / Digital output	GND1	-	-0.3	+20	V	I/O	P_2.25
KEY_IN, WK_IN	Input voltage	GND1	-	-0.3	+40	V	I	P_2.26
KEY_IN, WK_IN	Input current	GND1	Protect from Reverse battery with 1kΩ resistor	-16	-	mA	I	P_2.27
KEY_IN, WK_IN	Input current	GND1	Protect from ISOPULSE 1 with 1kΩ resistor	-150	-	mA	I	P_2.28
MSC_EN, MSC_CK_P, MSC_CK_N, MSC_DI_P, MSC_DI_N, MSC_DO	MSC digital I/O voltage	GND1	-	-0.3	+20	V	I/O	P_2.29
FLW_IN_P, FLW_IN_N	Flying wheel inputs voltage	GND1	With 20 mA (Max) inputs current	-0.3	Vclamp	V	I	P_2.30
FLW_IN_P, FLW_IN_N	Flying wheel inputs current	-	-	-20	+20	mA	I	P_2.31
FLW_OUT	Flying wheel output voltage (open drain)	GND1	-	-0.3	+20	V	O	P_2.32
MRD	Main relay drivers output voltage	GND1	-	-16	Vclamp	V	O	P_2.33
INJ1 INJ2 INJ3 INJ4 O2H1A O2H1B O2H2A O2H2B SOL1 SOL2 RLY1 RLY2 RLY3	Low-side drivers output voltage	GND1	-	-1	Vclamp	V	O	P_2.34

RLY4 RLY5 LED1 LED2								
STR1_DRN STR2_DRN STR3_DRN	Configurable high/low-side drain voltage	GND1	-	-1	Vclamp	V	O	P_2.35
STR1_SRC STR2_SRC STR3_SRC	Configurable high/low-side source voltage	GND1	-	Vclamp*	+40	V	O	P_2.36
IGN1 IGN2 IGN3 IGN4 IGN5 IGN6	Ignition predriver Output voltage	GND1	Negative voltage with limited current at 20 mA, protected by external resistor	-1	+40	V	O	P_2.37
CURR_Sense_O2H1 CURR_Sense_O2H2	O2heater drivers output current value	GND1	-	-0.3	+20	V	O	P_2.38
INJ1_PGND INJ2_PGND INJ34_PGND O2H1_PGND O2H1_PGND O2H2_PGND O2H2_PGND SOL12_PGND PDR_GND	Low-side drivers GND	GND1	-	-0.3	+0.3	V	O	P_2.39
PRD1_GATE PRD2_GATE PRD3_GATE PRD4_GATE PRD5_GATE	General purpose pre-drivers gate voltage	GND1	-	-0.3	+20	V	O	P_2.40
PRD1_DRN PRD2_DRN PRD3_DRN PRD4_DRN PRD5_DRN	General purpose pre-drivers feedback voltage	GND1	-	-0.3	+60	V	I	P_2.41
EN_N EN_P	Digital input (Cranking drivers enable signals)	GND1	-	-0.3	+20	V	I	P_2.42
LIN_TX	LIN data input	GND1	-	-0.3	+20	V	I	P_2.43
LIN_RX	LIN data output	GND1	-	-0.3	+20	V	O	P_2.44
LIN	LIN driver output	GND1	T = -40 °C	-18	+40	V	O	P_2.45
LIN	LIN driver output	GND1	T = -27 °C, 175 °C	-27	+40	V	O	P_2.46
CAN_TX	CAN data input	GND1	-	-0.3	+20	V	I	P_2.47
CAN_RX	CAN data output	GND1	-	-0.3	+20	V	O	P_2.48
CANH	CAN driver output	GND1	-	-27	+40	V	O	P_2.49
CANL	CAN driver output	GND1	-	-27	+40	V	O	P_2.50
VDD_CAN	CAN Supply 5V	GND1	-	-0.3	+20	V	S	P_2.51

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Absolute maximum ratings

VB_CAN	CAN Supply Battery	GND1	-	-16	+40	V	S	P_2.52
VCANH-VCANL	Differential CAN-bus voltage	-	-	-5	10	V	-	P_2.53

1. S: supply pin;

I: input pin;

O: output pin;

I/O: input/output pin

2. -2 V allowed during transients.

2.1 Latch-up trials

Latch-up tests performed according to JEDEC 78 class 2 Level

2.2 ESD

Table 2-2 ESD

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
ESD HBM Global	HBM global pins	ESD according to Human Body Model (HBM), Q100-002 for global pins; (100pF/1.5kΩ)	-4	-	4	kV	Global	P_2.2.1
ESD HBM	HBM local pins	ESD according to Human Body Model (HBM), Q100-002 for all pins; (100pF/1.5kΩ)	-2	-	2	kV	ALL	P_2.2.2
ESD HBM Global CAN	CAN_H, CAN_L pins	ESD according to Human Body Model (HBM), Q100-002 for global pins; (100pF/1.5kΩ) (1)	-5	-	5	kV	Global	P_2.2.3
ESD HBM Global CAN	CAN_H, CAN_L pins	Direct ESD Test according to IEC 61000-4-2 (150 pF, 330Ω) and Hardware Requirements for LIN, CAN and Flexray Interfaces in Automotive Applications"	-6	-	6	kV	Global	P_2.2.4
ESD CDM Corner	CDM corner Pins	ESD according to Charged Device Model (CDM), Q100-011 Corner pins(Except for pin 1)	-750	-	750	V	Corner	P_2.2.5
ESD CDM	CDM all pins	ESD according to Charged Device Model (CDM), Q100-011 All pins	-500	-	500	V	ALL	P_2.2.6

1. HBM with all not zapped pins grounded.;

2. ESD CDM Corner pin1 ranges from +-500V;

3. HBM: Pin 73/94/15~20/77/81 to GND;

Table 2-3 Pin classify

Pin classify	
Global pin (ECU connector)	VB_IN VBAT_Sense VB_STBY VSENSE1/2 VESNSE3/4_MON KEY_IN WK_IN MRD INJ1/2/3/4 O2H1/2 SOL1/2 RLY1/2/3/4/5 LED1/2 STR1/2/3_DRN STR1/2/3_SRC IGN1/2/3/4/5/6 LIN PRD1/2/3/4/5_DRN
Direction pin (power and I/O)	VDD_IO WDA SOE_OUT RSTN INJ_ENA MSC_EN MSC_CK_P MSC_CK_N MSC_DI_P MSC_DI_N MSC_DO FLW_OUT CURR_Sense_O2H1/2 PRD1/2/3/4/5_GATE EN_P EN_N LIN_TX LIN_RX VDD5_IN CP EN_N EN_P
Local pins	VB_IN_SW Boost_G VPRE BUCK_SW Buck_C_BST VDD5_GATE FLW_IN_P FLW_IN_N
GND pins	GND1 INJ1/2/3/4_PGND O2H1/2_PGND_A O2H1/2_PGND_B GND

2.3 Temperature ranges and thermal data

Table 2-4 Temperature ranges and thermal data

Pin	Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Number
-	Ta	Operating temperature	-	-40	-	125	°C	P_2.3.1
-	Tj	Junction temperature	-	-40	-	175	°C	P_2.3.2
-	Tstr	Storage temperature	-	-50	-	175	°C	P_2.3.3
-	R _{THj-c} ⁽¹⁾	Thermal resistance junction to case	-	-	1	-	°C/W	P_2.3.4
-	R _{THj-a} ⁽¹⁾	Thermal Resistance junction to ambient	supposing a mounting on board 2s2p + heatsink vias (refer to application note for PCB parameters)	-	10	-	°C/W	P_2.3.5

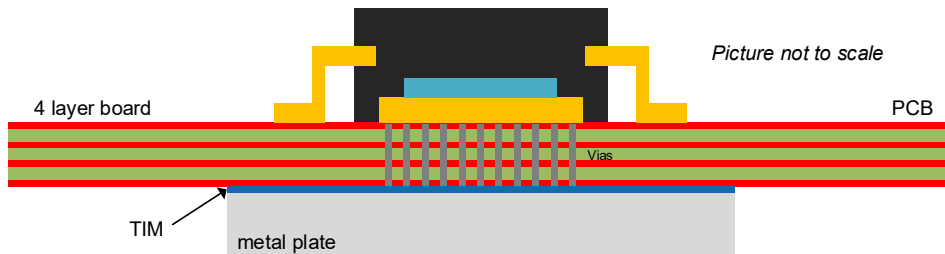
1. With 2s2p PCB thermally enhanced, cold plate as per std Jedec best practice guidelines (JESD51), assuming Pdiss = 5 W dissipated statically and homogeneously, thermal vias under package exposed pad assuming filling ratio 50% and 35 μm vias diameter (157 vias estimation); refer to Figure 2-1 Thermal simulation set-up.

Note:

All parameters are tested in the temperature range Tj -40 - 150°C; device functionality at high temperature is guaranteed by bench validation, electrical parameters are guaranteed by correlation with ATE tests at reduced temperature and adjusted limits (if needed). Temperature sensor present, please refer to [Section 3.5](#).

Boundary conditions

- Thermal simulations carried on using typical boundaries for ECU applications:
- 4 layer board with thermal vias
- metal plate in contact with board
- Thermal interface material (TIM) between PCB and metal plate
- $T_{AMB} = 135^{\circ}\text{C}$ / natural convection



With these boundaries and considering uniform power on overall chip surface, package thermal resistance is:
 $\text{Theat}_{JA} = 10^{\circ}\text{C/W}$

Figure 2-1 Thermal simulation set-up

2.4 Functional range

Table 2-5 Functional range

Symbol	Parameter	Note	Min	Typ	Max	Unit	Pin	Number
-	Cranking condition	u-chip in cranking condition, reduced operation range, main relay and delayed off power stages keep the status if delay-off function is triggered. All other driver/regulators /CAN/LIN are in off condition. See description in the Reset Matrix Table, Section 12.10 .	4.25	-	4.8	V	VB_IN	P_2.4.1
-	Low drop condition	u-chip in low drop operation condition, all functions are guaranteed all parameters are guaranteed unless specified. Vpre and VDD5 are in low drop condition	4.8	-	7	V	VB_IN	P_2.4.2
-	Normal condition	u-chip in normal operation condition. All functions and parameters are guaranteed, considering the description range of electrical characteristic table and parameters test set up	7	-	19	V	VB_IN	P_2.4.3
-	Jump start condition	u-chip in jump start condition, all functions are guaranteed. Parameters are guaranteed unless specified.	19	-	28	V	VB_IN	P_2.4.4
-	Load dump condition	u-chip in load dump condition. U-chip is no damage. All drivers/regulators/CAN/LIN are in off condition. See description in the Reset Matrix Table, Section 3.3 and Section 12.10 .	28	-	40	V	VB_IN	P_2.4.5

3 Operation behavior

3.1 Power up/down

3.1.1 Power up/down state diagram chart

The power up/down sequences are shown as state diagram in the following figure:

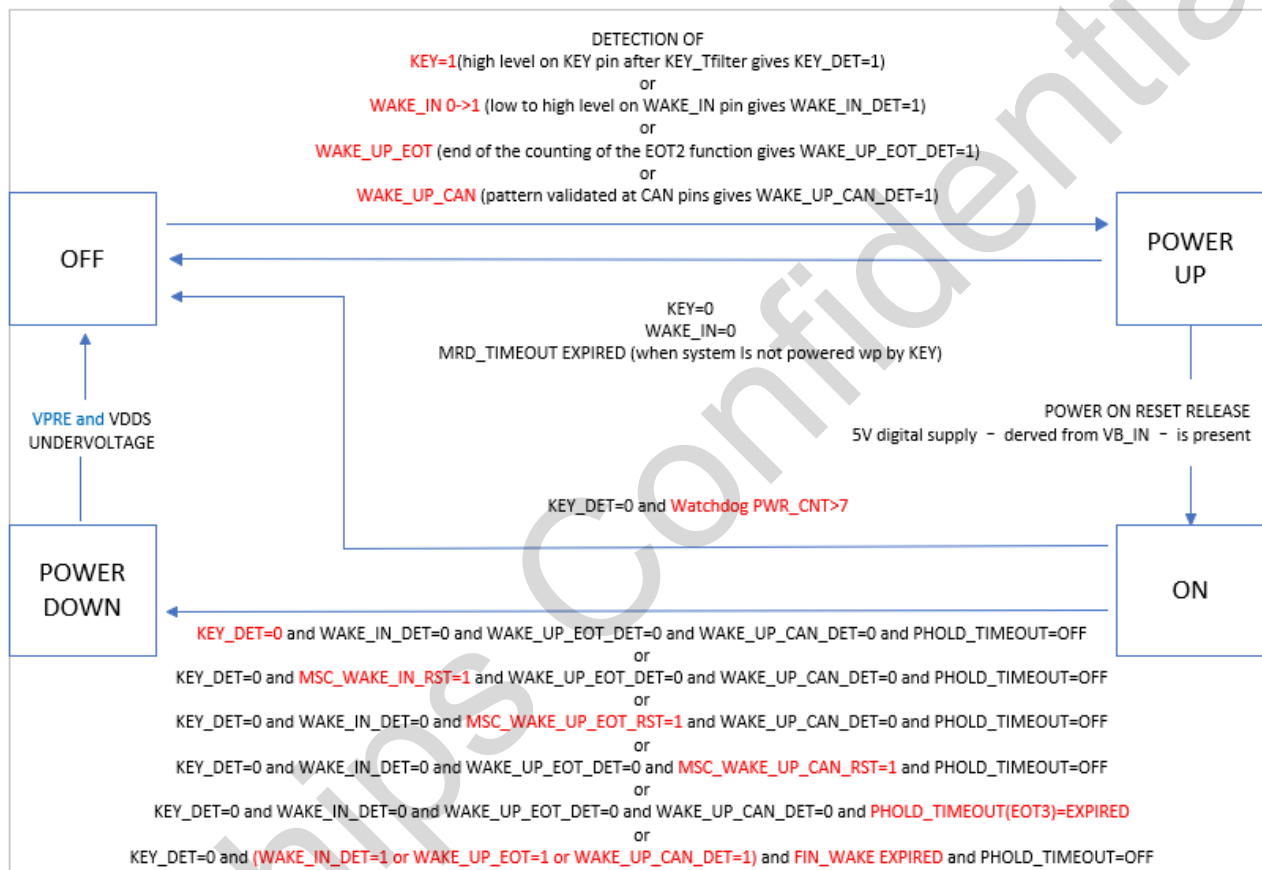


Figure 3-1 Power up state diagram

3.1.2 Power up sources and actions summary

Table 3-1 Power up sources and actions summary

KEY_IN	WK_IN	EOT	CAN_FD	VB_IN	MRD	Actions
active	-	-	-	permanent	on	MRD on by KEY_IN pin
-	active	-	-	permanent	off	MRD used as normal low-side driver (is controlled by MSC)
-	-	active	-	permanent	off	MRD used as normal low-side driver (is controlled by MSC)
-	-	-	active	permanent	off	MRD used as normal low-side driver (is controlled by MSC)

3.1.3 Power up from KEY_IN

The chip is powered up if KEY_IN signal stays asserted for Key_Tfilter. There is a MSC bit (*KEY_IN_PIN_FLT* in read command 15 frame 2) to indicate the status of KEY_IN pin signal after filter time.

3.1.4 Power down from KEY_IN when PHOLD = 0

The chip is powered down if KEY_IN signal stays de-asserted for Key_Tfilter if no other power up source is active.

Condition:

$4.8 \leq V_{B_IN} \leq 18 \text{ V}$, T_j -40 to 175 °C unless otherwise specified.

Table 3-2 Key pins electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
V_{IH}	Logic input high voltage threshold	-	3.6	-	4.1	V	KEY_IN	P_3.1.4.1
V_{IL1}	Logic input low voltage threshold Main logic in POR not released	-	3.1	-	3.6	V	KEY_IN	P_3.1.4.2
V_{IL2}	Logic input low voltage threshold Main logic POR released	-	1.7	-	2.3	V	KEY_IN	P_3.1.4.3
V_{IHYS}	Logic input hysteresis	-	0.1	-	0.8	V	KEY_IN	P_3.1.4.4
Key_Tfilter	Filter time for switching on/off	Guaranteed by test scan(1)	7.5	16	24	ms	KEY_IN	P_3.1.4.5
I_{input}	Input current of KEY_IN	KEY_IN $\leq 5 \text{ V}$	-	-	340	μA	KEY_IN	P_3.1.4.6
R_{PD}	Pull-down resistor	KEY_IN = 1 V	170	300	500	k Ω	KEY_IN	P_3.1.4.7

1. Filter time for switching on, is active until KEY logic is out of POR, Filter time for switching off, is active until main logic out of POR.

3.1.5 Power up from WK_IN

The chip is powered up if WK_IN signal has a rising edge and stays asserted for WK_Tfilter.

The u-chip will not power up until further rising edge on WK_IN pin. This detection of a WK_IN condition flag is named *WK_IN_DET* (detection of a WK_IN condition) and is clear by VDD5/VDDIO under voltage or by the MSC command *WK_IN_RST*. In these cases, if no other power-on source is active, a power down sequence is started. When U chip is already in on state (the logic is out of power on reset), the rising edge on pin WK_IN only sets a flag, with no impact on power supply.

The Main Relay Driver is not switched on by *WAKE_IN_DET* rising edge.

There is a MSC bit (*WAKE_IN_PIN_FLT* in read command 15 frame 2) to indicate the status of WK_IN pin signal after filter time.

A negative glitch on WK_IN can produce unwanted power up and also prevent the correct

power down.

To cover the major cases the falling edge filtering is done both in the VB_STBY domain and in the VB_IN domain. Depending on which supply is present the behavior is described in the following cases:

1. If VB_STBY is present the falling edge filtering is applied also to power up so a negative glitch on WK_IN will not power up the device.
2. If VB_IN is present (after the device is already powered up) the falling edge filtering on WK_IN avoids that a glitch (happening in PHOLD after the MSC has cleared the WK_IN detection) is recognized as valid detection and the device is kept on.

3.1.6 Power down from WK_IN when PHOLD=0

The chip is powered down if the MSC clears, by the MSC command *WK_IN_RST*, the detection of WK_IN positive edge if no other power up source is active. The power down is actuated even if WK_IN is still high. The power down is not activated by WK_IN going low or WK_IN low level. The detection reset by MSC command, *WK_IN_RST* (CONFIG_REG12[7]), is active at level, so from when MSC writes them at 1 they keep on clearing the detections. If there is further wake detection signal, the detection signal will be cleared, until the MSC writes 0 in the bits.

Condition:

$4.8 \leq V_{B_IN} \leq 18 \text{ V}$, $T_j -40$ to $175 \text{ }^\circ\text{C}$ unless otherwise specified.

Table 3-3 WK pins electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
V_{IH}	Logic input high voltage threshold	-	3.6	-	4.1	V	WK_IN	P_3.1.6.1
V_{IL}	Logic input low voltage threshold	-	3.1	-	3.6	V	WK_IN	P_3.1.6.2
V_{IHYS}	Logic input hysteresis	-	0.1	-	0.8	V	WK_IN	P_3.1.6.3
I_{input}	Input current of WK_IN	$W_{K_IN} \leq 5 \text{ V}$	-	-	340	μA	WK_IN	P_3.1.6.4
WK_Tfilter	Filter time for switching on/off	Guaranteed by test scan	1	2	3	ms	WK_IN	P_3.1.6.5
RPD	Pull-down resistor	$W_{K_IN}=1\text{V}$	170	300	500	k Ω	WK_IN	P_3.1.6.6
Tbattery_det	Main logic POR timeout (used to recognize	-	210	-	450	μs	-	P_3.1.6.7

	permanent battery condition)							
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3.1.7 Power up from WAKE_UP_TIMER

The chip is powered up when an end counting is reached inside the wake_up_timer block.

The end and enable counting are set by MSC before the chip is powered down.

When U chip is already in on state (the logic is out of power on reset), the wake_up_timer end counting only sets a flag, with no impact on power supply. This flag is named *WAKE_UP_EOT_DET* (detection of a wake_up end of time condition) and is clear by VDD5/VDDIO under voltage or by the MSC command *WAKE_UP_EOT_RST*. If no other power on source is active a power down sequence is started.

The Main Relay Driver is not switched on by *WAKE_UP_EOT* if the device is powered in permanent battery.

The MSC sets the programmable time on *Wake up timer_SET Registers (CONFIG-REG15-0 to CONFIG_REG15-2)*. The counting can be started/stopped by MSC command - *CONFIG_REG1[6:7] = WAKE_UP_TIMER_START_STOP[0:1] (CONFIG_REG1 D6;D7)* - or by switching the pin *KEY_IN* according to a selection set by MSC bit - *CONFIG_REG20[2] = WAKE_UP_TIMER_EN_SEL*.

The counter counts time on the *WAKE_UP_TIMER_CNT* Register.

WAKE_UP_TIMER_CNT Register and *Wake up timer_SET* Register are being compared, and when they match, *WAKE_UP_EOT_DET* signal will be set to high level to trigger a power up sequence.

Condition:

$4.8\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, $3.5\text{ V} \leq \text{VB_STBY} \leq 18\text{ V}$ T_j -40 to 175 °C unless otherwise specified;

$\text{VB_STBY} > 3.5\text{ V}$, EOT can keep the counter value (allow to stop counting),

$5\text{ V} < \text{VB_STBY} < 18\text{ V}$, EOT full function, full parameter.

Table 3-4 VB_STBY pins electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VVB_STBY_Th	VB_STBY voltage range that EOT counter keep data or operating correctly	-	3.5	-	-	V	VB_STBY	P_3.1.7.1
IVB_STBY_EOT_disable	VB_STBY current with EOT 32 kHz Clock device off	$3.5 < \text{VB_STBY} < 18\text{ V}$ $\text{VB_CAN} < 0.7\text{ V}$	-	-	30	μA	VB_STBY	P_3.1.7.2
IVB_STBY_EOT_enable	VB_STBY current with EOT 32 kHz	$3.5 < \text{VB_STBY} < 18\text{ V}$ $\text{VB_CAN} < 0.7\text{ V}$ $-40\text{ °C} < T_{\text{amb}} <$	-	-	50	μA	VB_STBY	P_3.1.7.3

	Clock device off	80 °C						
IVB_STBY_EOT_enable	VB_STBY current with EOT 32 kHz Clock device off	$3.5 < \text{VB_STBY} < 18 \text{ V}$ $\text{VB_CAN} < 0.7 \text{ V}$ $80^\circ\text{C} < T_{\text{amb}} < 125^\circ\text{C}$	-	-	70	μA	VB_STBY	P_3.1.7.4
EOT_clock_frequency	Freq EOT 32 kHz Clock internal components	$6 \text{ V} < \text{VB_STBY} < 18 \text{ V}$ $-40^\circ\text{C} < T_{\text{amb}} < 80^\circ\text{C}$	-6%	1	+6%	Hz	VB_STBY	P_3.1.7.5
EOT_clock_frequency	Freq EOT 32 kHz Clock internal components	$6 \text{ V} < \text{VB_STBY} < 18 \text{ V}$ $80^\circ\text{C} < T_{\text{amb}} < 125^\circ\text{C}$	-7%	1	+7%	Hz	VB_STBY	P_3.1.7.6
VB_STBY_UV	VB_STBY under voltage threshold the 32 kHz clock is active but counter is stopped during crank.	$\text{VB_STBY} > 3.1 \text{ V}$	3.5	-	5	V	VB_STBY	P_3.1.7.7

When VB_STBY_UV happens, it stops the counter.

The counter will resume when vb_stby_uv goes low (no undervoltage).

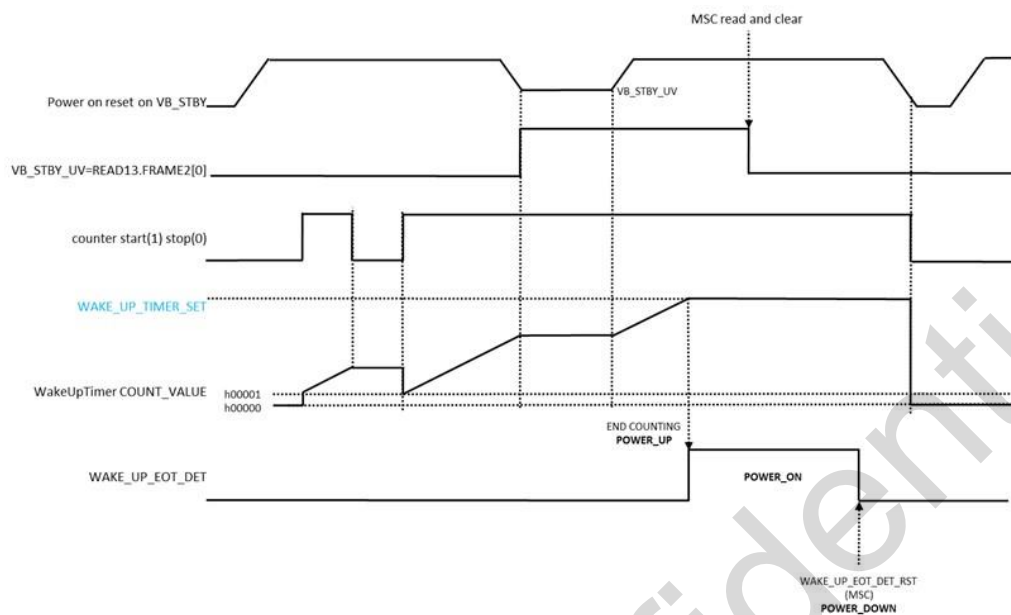


Figure 3-2 Timing for Wake Up Timer - driver by MSC start/stop commands

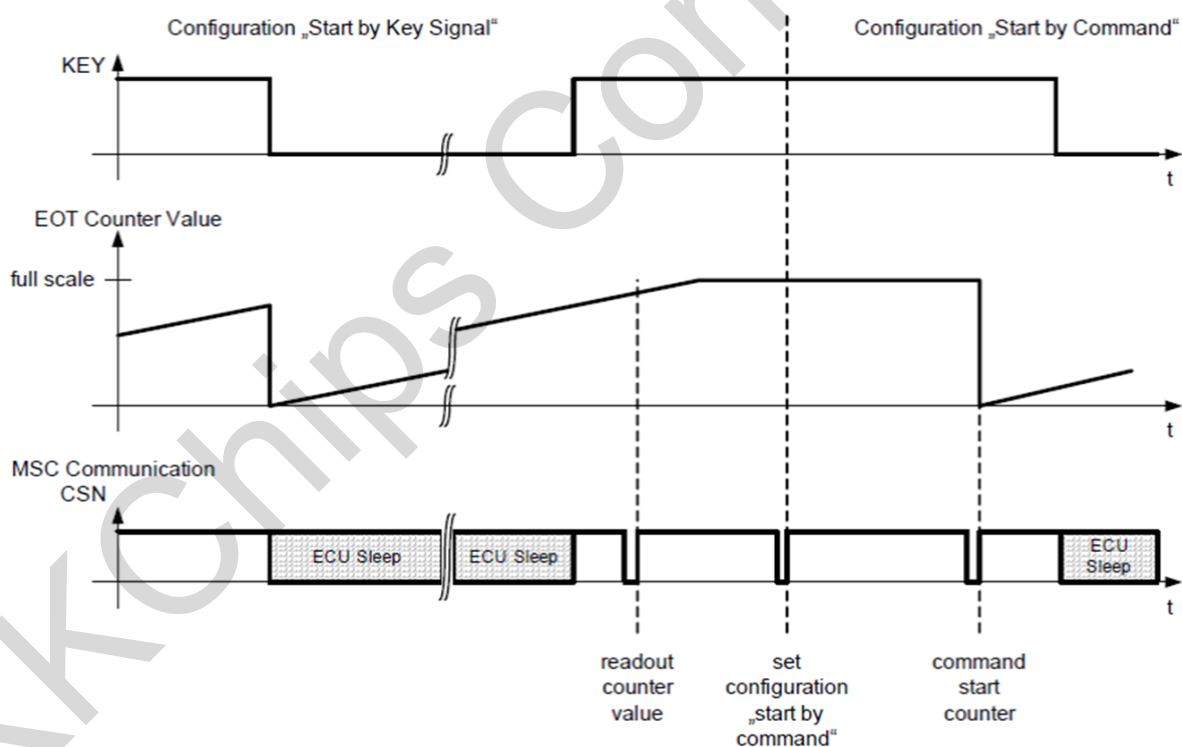


Figure 3-3 Timing for Wake Up Timer - activated (started) by KEY negative edge

Reset condition

VB_STBY under voltage stops (not reset) EOT counter during cranking. The counter is reset in case of power on reset generated on VB_STBY supply.

Count up

Wake up Timer shall count up the timer from 1 while the MSC *Wake Up Timer Start* command is given. When *Wake Up Timer Stop* command is given, the Wake Up Timer shall stop counting and the value will clear. While the counter is counting the *EOT_STATUS* will be 1. If it's not counting *EOT_STATUS* will be 0.

Wake Up Timer start/stop commands

Wake Up Timer counts up when msc start command in the MSC *WAKE_UP_TIMER_START_STOP* register is given. When the msc stop command in the MSC *WAKE_UP_TIMER_START_STOP* register is given the timer shall stop counting up but the *WAK_UP_TIMER_CNT* register will clear. The bit can be set or reset via MSC.

WAKE_UP_TIMER_CNT Register

WAKE_UP_TIMER_CNT Register consists of 24 bits it's Read Only. The range of *WAKE_UP_TIMER_CNT* is from 0 (h'000000) to 16777215 (h'FFFFFF). *WAKE_UP_TIMER_CNT* Register is set to 0 (h'000000) when power on reset of *VB_STBY* occurs. *WAKE_UP_TIMER_CNT* Register is set to all "0" via MSC when power on reset period or *VB_STBY* is open.

Wake up timer_SET time register

The *Wake up timer_SET* register consists of 24 bits. And the register is Read and Write register.

The range of *Wake up timer_SET* is from 0 (h'000000) to 16777215 (h'FFFFFF). *Wake Up Timer_SET* Register is set to 0(h'000000) when power on reset of *VB_STBY* occurs.

EOT function 1

EOT Function 1 - count "stand" time: After the start of the counter, the counter starts to count up till the value that has to be set before starting it using *Wake_up_timer_SET*. If ECU switches on before the counter reaches the wake up timer_set value, the μ C can read the "stand" time via MSC. If the counter reach the wake up timer_set value before ECU switch on, the count keeps the wake up timer_set value.

This function is selected with *MSC CONFIG_REG6[6]=EOT_MODE=0*. The EOT function 1 does not set *WAKE_UP_EOT_DET* at 1. *EOT_MODE* bit can be correctly read only when wake up timer logic is working, else it is read 0.

VB_STBY must be provided to correctly read the *EOT_MODE* bit. If *VB_STBY* is removed the register content will be lost.

EOT function 2

EOT Function 2 - wake-up via *WAKE_UP_TIMER_SET*. Set *WAKE_UP_TIMER_SET* (*CONFIG-REG15-0 / CONFIG-REG15-1/ CONFIG-REG15-2*) to a certain value, then start the counter before shutting down the ECU. The counter counts up until the *WAKE_UP_TIMER_SET* is reached. The reaching of *WAKE_UP_TIMER_SET* activates the wake signal and will start a wake-up sequence. This function is selected with *MSC CONFIG_REG6[6]=EOT_MODE=1*. *EOT_MODE* bit can be correctly read only when wut logic is working, else it is read 0.

VB_STBY supply must be provided to correctly read the *EOT_MODE* bit.

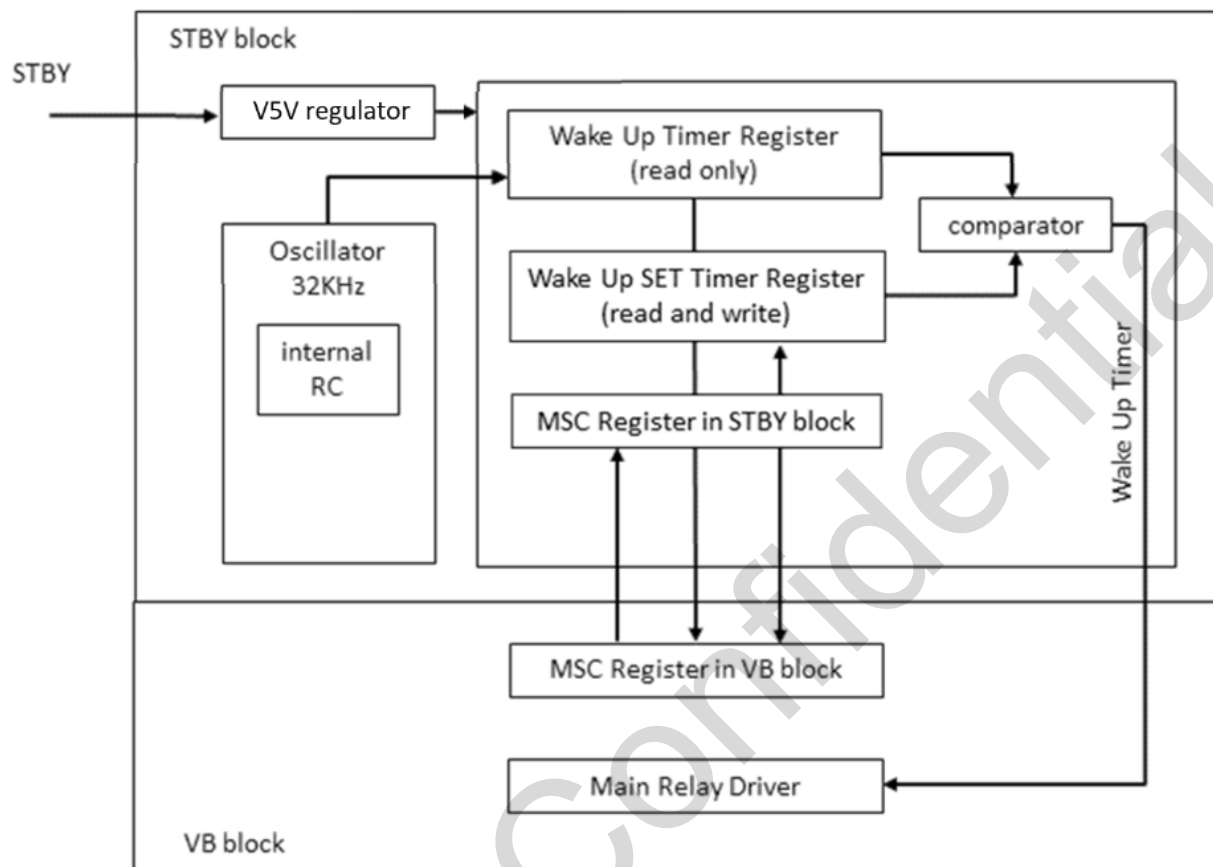


Figure 3-4 Wake up timer block diagram

Table 3-5 Wake up timer resolution

Range	resolution
0 - 16777215	1 second

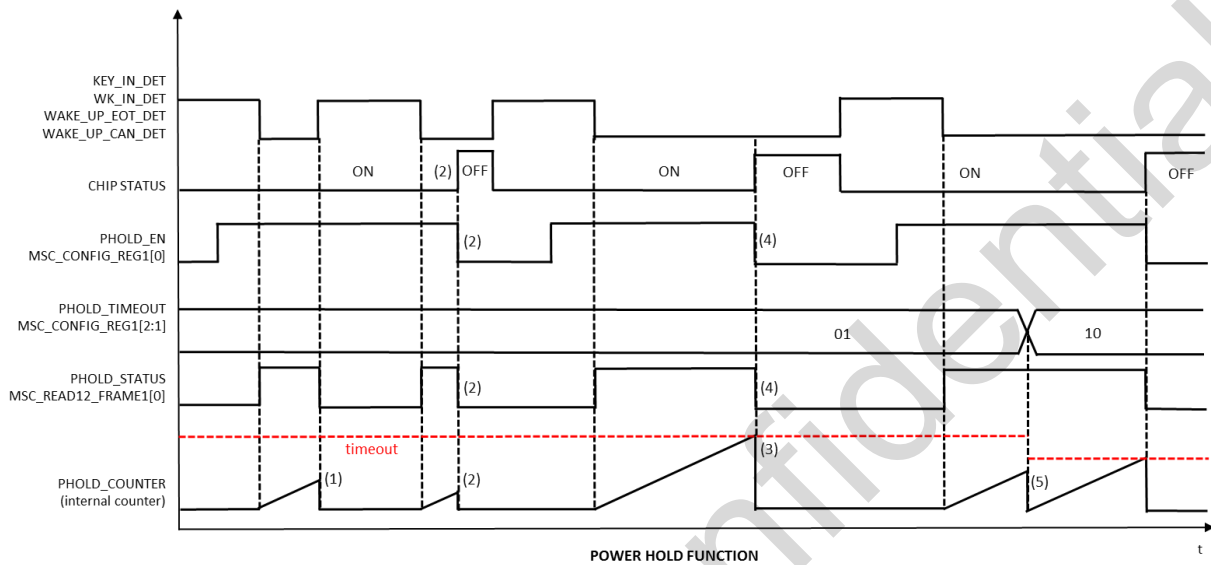
WAKE_UP_TIMER_CNT Register can be read via MSC. **Wake up timer_SET** Register can be read and write via MSC.

3.1.8 Power on with PHOLD_EN (EOT function 3)

The chip is kept switched on by an MSC command (*PHOLD_EN CONFIG_REG1[0]*) even though one of the power down sources has been asserted (*KEY_IN*, *WK_IN*, *WAKE_UP_TIMER*, *WAKE_UP_CAN*). The chip is kept on in power hold mode for a time programmed (default is infinity) by MSC (*PHOLD_TIME[0:1] CONFIG_REG1 (D1;D2)*). This is also EOT function 3. The power hold mode is enabled by MSC and is cleared by the conditions reported in the reset matrix.

After setting the *PHOLD_EN=CONFIG_REG1[0]=1* the PHOLD and the PHOLD timer are enabled when the *KEY_DET. WK IN DET. WAKE UP EOT DET. WAKE UP CAN DET*

detections are not active. The $PHOLD_TIMER[14:0]=READ12.FRAME[2:1]$ starts counting when the KEY_DET , WK_IN_DET , $WAKE_UP_EOT_DET$, $WAKE_UP_CAN_DET$ detections are cleared. When the $PHOLD_TIME=CONFIG_REG1[2:1]$ is configured as 00 (default) the timeout is disabled and the $PHOLD_TIMER[14:0]=READ12.FRAME[2:1]$ is read always at 0.



- (1) The timeout counter is cleared on the detection of at least one of the power up sources.
- (2) The timeout counter is cleared if MSC disables the PHOLD function, MRD is switched off, a power down sequence is started.
- (3) The counter has reached the timeout, MRD is switched off, a power down sequence is started.
- (4) The PHOLD_EN and the PHOLD_STATUS bits are clear along with the power down sequence.
- (5) A new MSC setting for the timeout makes the timeout counter clear and restart.

Figure 3-5 Power hold diagram

3.1.9 Power down from WAKE_UP_TIMER when PHOLD_EN=0

The chip is powered down if the MSC clears, by the MSC command $WAKE_UP_EOT_RST$, $WAKE_UP_EOT_DET$ (The counter counts up until the $WAKE_UP_TIMER_SET$ is reached) if no other power up source is active and $PHOLD_EN = 0$. The detection reset by MSC command, $WAKE_UP_EOT_RST=CONFIG_REG12[6]$, is active at level not at change, so from when MSC writes them at 1 they keep on clearing the detections. If there is further wake detection signal, the detection signal will be cleared, until the MSC writes 0 in the bits.

3.1.10 Power down from PHOLD

The chip is powered down after the power hold timer programmed by MSC has expired or if the PHOLD function is disabled by MSC. The PHOLD timer can be programmed as timeout disabled so that the chip stays powered up.

3.1.11 Power up from CAN

The chip can be powered up from CAN according to what reported in the Wake Up (From CAN) chapter. To allow the power up from CAN the pins VB_CAN , VB_STBY must be supplied.

When U chip is already in on state (the logic is out of power on reset), the validation of a valid CAN pattern only sets a flag, with no impact on power supply. This flag is named

WAKE_UP_CAN_DET (detection of a valid can pattern) and is cleared by VDD5/VDDIO under voltage or by the MSC command *WAKE_UP_CAN_RST*. In this case if no other power on source is active a power down sequence is started.

The Main Relay Driver is not switched on by *WAKE_UP_CAN* if the device is powered in permanent battery.

3.1.12 Power down from *WAKE_UP_CAN* when *PHOLD_EN*=0

The chip is powered down if the MSC clears, by the MSC command *WAKE_UP_CAN_RST*, *WAKE_UP_CAN_DET* (the detection of *WAKE_UP_CAN* high) if no other power up source is active and *PHOLD_EN* = 0. The detection reset by MSC command, *WAKE_UP_CAN_RST*=*CONFIG_REG12*[5], is active at level, so from when MSC writes them at 1 they keep on clearing the detections. If there is further wake detection signal, the detection signal will be cleared, until the MSC writes 0 in the bits.

3.1.13 Finish wake function

There is a Finish wake function (auto clear mechanism) of *WK_IN_DET*, *WAKE_UP_EOT_DET* and *WAKE_UP_CAN_DET*. The purpose is to ensure the system could power down from wake up state after the impulse timer timeout (typ. 1s).

If the software does not stop the impulse timer before time out, the *WK_IN_DET*, *WAKE_UP_EOT_DET* and *WAKE_UP_CAN_DET* will be cleared as long as time out. If the software stops the impulse timer (*FIN_WAKE* = 1) before time out, the *WK_IN_DET*, *WAKE_UP_EOT_DET* and *WAKE_UP_CAN_DET* will not be cleared.

The impulse timer is not affected by watchdog(*ERR_CNT*>4) and EOT. The Finish wake function is not disabled by *RSTN* pulled low externally and *MSC_SW_RESET* active. The Finish wake function can be disabled by power supply undervoltage. The reset sources of *FIN_WAKE* bit are the same as *fin_wake* function.

Software can decide to clear the *WK_IN_DET*, *WAKE_UP_EOT_DET* or *WAKE_UP_CAN_DET* before or after the impulse timer timeout, through set *WK_IN_RST*, *WAKE_UP_EOT_RST* or *WAKE_UP_CAN_RST* to 1

FIN_WAK function is stopped by

FIN_WAKE time out,

FIN_WAKE BIT =1,

WK_IN_RST / *WK_EOT_RST* / *WK_CAN_RST* of relevant detect signal.

When *WK_IN_DET* & *WK_EOT_DET* & *WK_CAN_DET* = 0, *FIN_WAK* can be re-triggered by any detection of *WK_IN_DET* / *WK_EOT_DET* / *WK_CAN_DET*. The counter is restarted from 0.

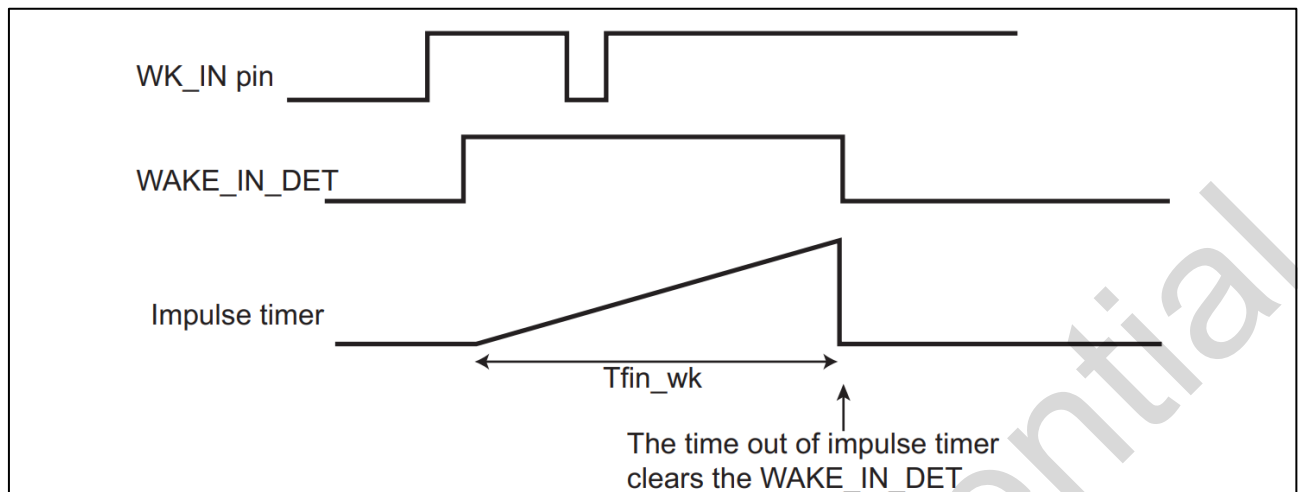


Figure 3-6 Impulse timer clear WAKE_IN_DET after time out

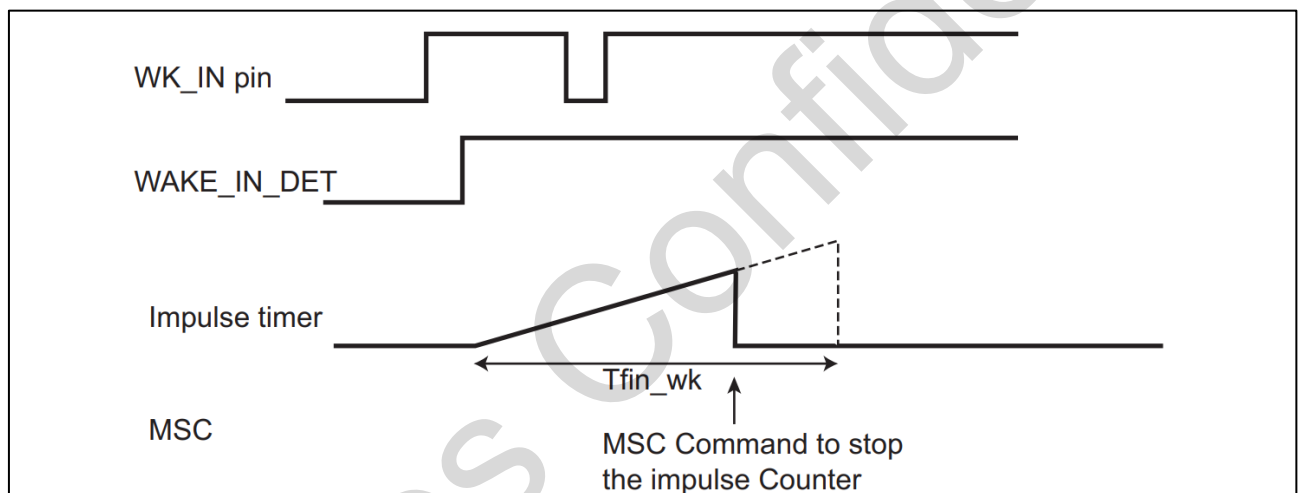


Figure 3-7 Impulse timer disabled before time out

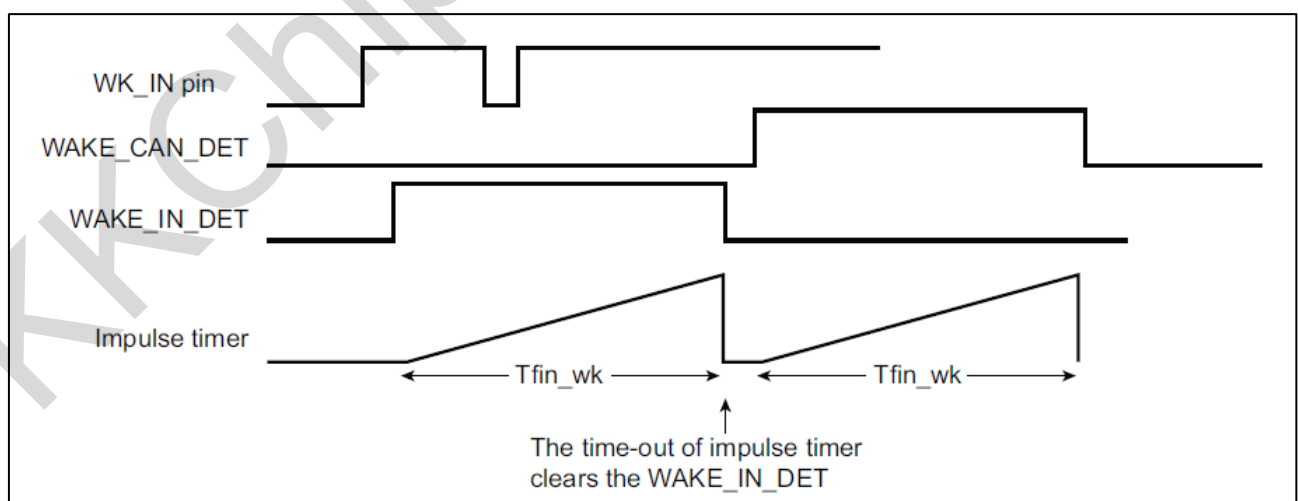


Figure 3-8 Timer enabled for a new detection

Note : Same mechanism applied to WAKE_IN_DET, WAKE_UP_EOT_DET and WAKE_UP_CAN_DET.

Table 3-6 Finish wake timer electrical characteristics

Symbol	Parameter	Min	Typ	Max	Unit	Number
Tfin_wk	Finish wake timer	0.94	1	1.06	s	P_3.1.13.1

3.2 Secure Engine OFF(SEO) function

Secure Engine Off functional switches off, with a certain delay from KEY_OFF, the engine by a hardware path and without any dependence on external micro and no matter what are the other conditions (e.g. the power hold mode).

After the falling edge of the filtered KEY_IN a programmable counter is started. The clock of the timer does not depend on any micro clock or function.

SEO_OUT pin is the output of KEY_IN after a T_SEO_DELAY (CONFIG_REG3 [6:7]) delay. Input pin INJ_ENA enables the internal drivers as listed above (INJ[1:4], is configured), so the configuration of enable/disable of SEO function is realized by hardware in the following way:

For PFI application, SEO_OUT is connected to INJ_ENA, when key goes off internal drivers can be switched off.

For GDI application, SEO_OUT is connected to GDI injector driver enable if present, INJ_ENA is always enabled. When key off, only the external GDI injector driver is switched off.

During SEO time the external micro can switch on/off the drivers

SEO output is implemented with a push pull output stage, the T_SEO_DELAY can be programmed according to Table 3-7.

Table 3-7 SEO timing

Symbol	Parameter	MSC_CONF	Min	Typ	Max	Unit	Pin	Number
T_SEO_DELAY	default	00	100	-	140	ms	SEO_OUT	P_3.2.1
T_SEO_DELAY	-	01	200	-	250	ms	SEO_OUT	P_3.2.2
T_SEO_DELAY	-	10	400	-	480	ms	SEO_OUT	P_3.2.3
T_SEO_DELAY	-	11	800	-	920	ms	SEO_OUT	P_3.2.4

Condition:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, $4.75\text{ V} \leq V_{DD_IO} \leq 5.25\text{ V}$, T_j -40 to 175 °C unless otherwise specified.

Table 3-8 SEO Circuit electrical characteristic

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
--------	-----------	----------------	-----	-----	-----	------	-----	--------

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Operation behavior

SEO_OUT_L	SEO_OUT output low level	VDDIO = 5 V or 3.3 V I_{sink} current= 2 mA	-	-	0.5	V	SEO_OUT	P_3.2.5
SEO_OUT_H	SEO_OUT high level	VDDIO = 5 V or 3.3 V I_{sink} current= 2 mA	VDDIO -0.5	-	-	V	SEO_OUT	P_3.2.6

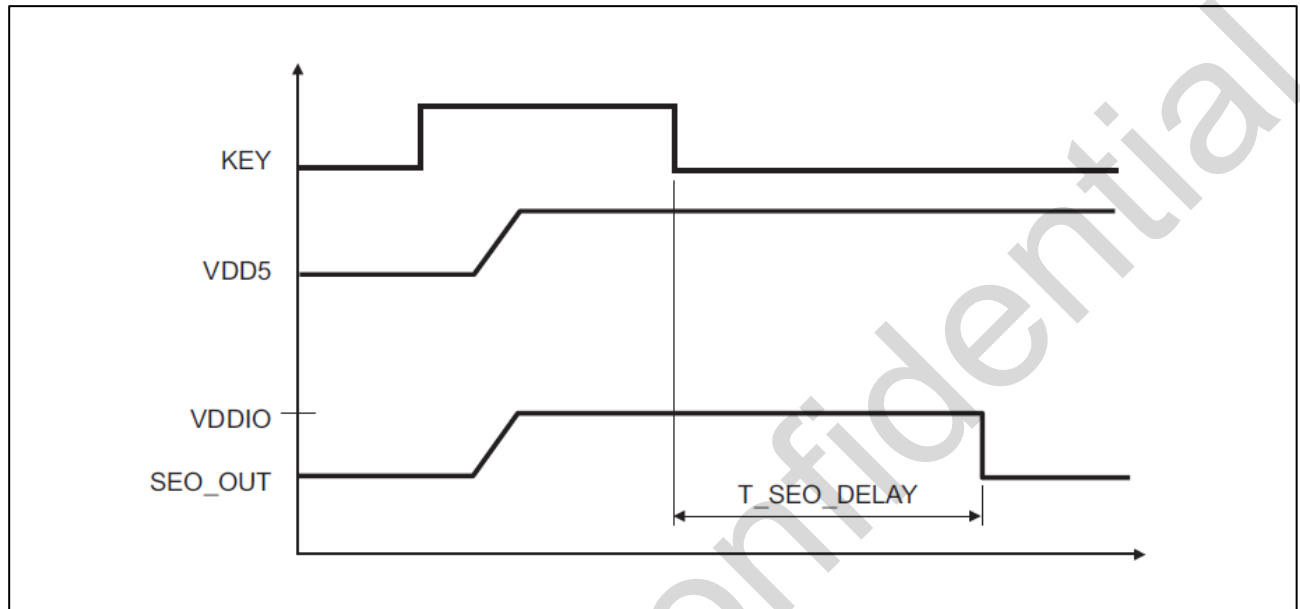


Figure 3-9 SEO delay

3.3 Reset strategies

At the rising edge of reset, the MSC communication is reset. un-send, un-stop frame should be discard.
 SEQ_OUT is triggered by KEY_IN=0
 RV4 and STRG are MSC configured as starters (CONFIG3[5]=CONFIG3[4]=1)
 LIN works as receiving mode only if the disable of the transmitting mode is enabled by MSC)
 UN_LIN works as receiving mode only due to watchdog BOSCH if the WDA_ERR_CNT is reset if also RST_PRL is activated
 CAN logic as receiving only-modify CAN logic as receiving only-modify CAN logic as receiving only-modify
 If the related delay off-function is active the driver starts ON
 CAN works as receiving-only mode
 CAN works as Low-power mode
 excluded WAKE_UP, LIN_START_STOP, EOT_MODE, CAN parameters, WAKE_UP_TIMER, EN_KEY_OC, RERTY_MAX,
 When VDDOS_VDD_SEL = VDDOS depends not depend on VBUS IN if VDDOS_OFF, VDDOS_VDD depends on VBUS IN

Figure 3-10 Reset table

Note: When VB_STBY_UV happens, it stops the EOT counter. The EOT counter will resume when vb_stby_uv goes low (no undervoltage).

3.4 Smart reset circuit

There are two Reset pins:

1. RSTN: reset pin linked to VDD5 supply;
2. RSTC: reset pin linked to VDDIO supply.

3.4.1 Smart reset RSTN pin

The RSTN pin is an input/output active low. As output pin the Smart Reset circuit takes into account several events in the device in order to generate the proper reset signal at RSTN pin for the microcontroller and for portion of the internal logic as well.

The RSTN in output (RSTN(OUT)) is activated by the contributions described by the reset matrix and is intended as reset for the external microcontroller.

The RSTN in input (RSTN(IN)) is filtered Trstflt_RST and stops the internal logic as described in the reset matrix.

The internal RSTN(OUT) has higher priority respect to the external RSTN(IN). The RSTN(IN) is masked until Tmsk_RSTN after RSTN(OUT) has been released.

RSTN(OUT) and RSTN(IN) are recombined on the pad according to the scheme.

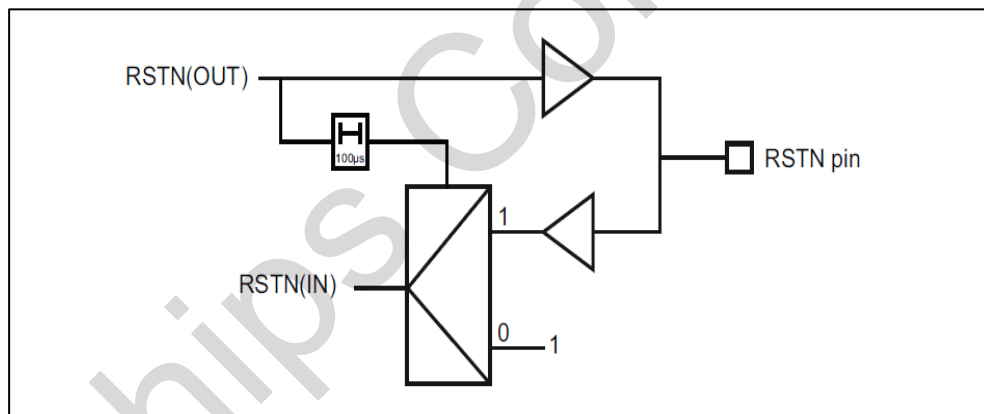


Figure 3-11 Reset circuit

The RSTN activates on the contributions specified by the reset matrix:

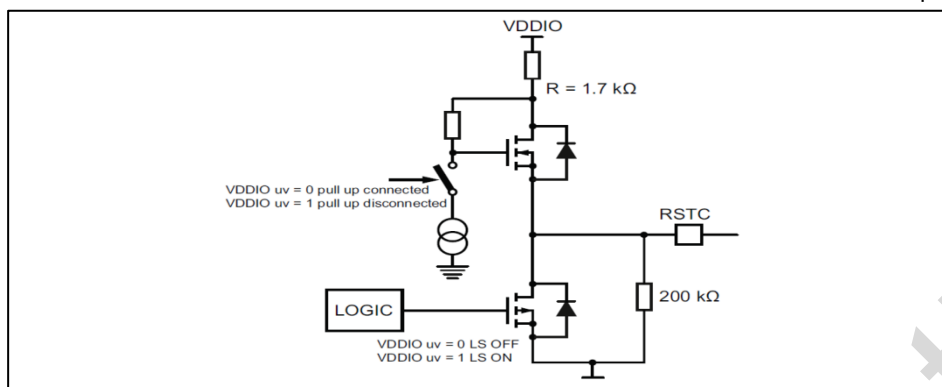


Figure 3-12 Reset matrix

3.4.2 Smart Reset RSTC pin

The RSTC pin is an output active low. As output pin the Smart Reset circuit takes into account several events in the device in order to generate the proper reset signal at RST pin for the microcontroller and for portion of the internal logic as well.

The RSTC activates on the contributions specified by the reset matrix:

3.4.3 After run reset

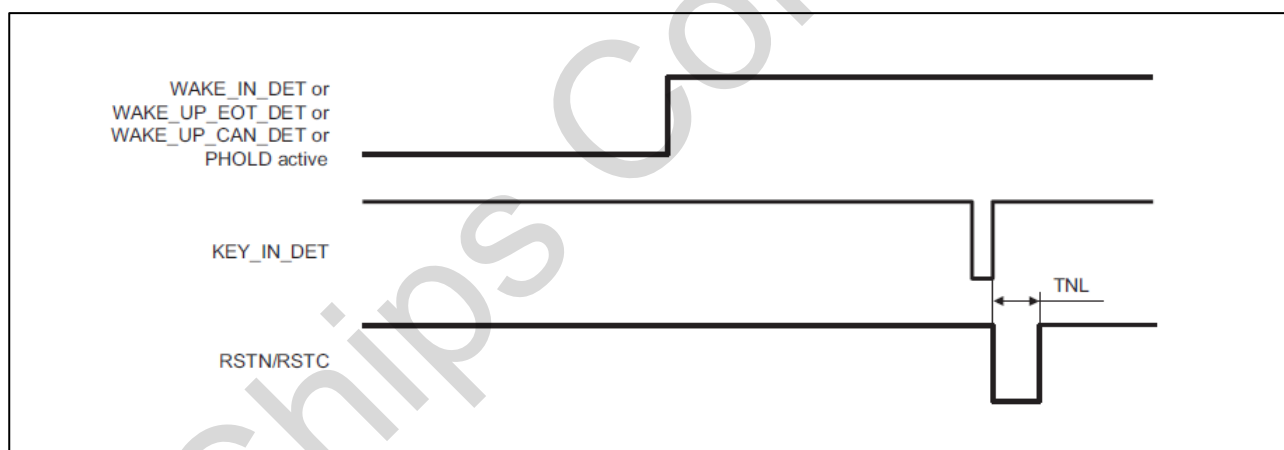


Figure 3-13 RSTN/RSTC after run reset timing

As output:

$4.8 \leq V_{B_IN} \leq 18 \text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 3-9 Smart reset circuit electrical characteristic table 1

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VUV_L0 RSTN	Output low voltage	$1 < VDD5 < VDD_UV$	-	-	0.6	V	RSTN	P_3.4.3.1
VUV_L0 RSTC	Output low voltage	$1 < VDDIO < VDDIO_UV$	-	-	0.6	V	RSTC	P_3.4.3.2
IUVres_max_RSTN	Input current	$VDD5 = VDD_UV$ $V_{UV_reset} = 0.6 \text{ V}$	1	-	-	mA	RSTN	P_3.4.3.3

IUVres_max_RSTC	Input current	VDDIO= VDDIO_UV V _{UV_reset} = 0.6V	1	-	-	mA	RSTC	P_3.4.3.4
-----------------	---------------	---	---	---	---	----	------	-----------

As input:

4.8 ≤ VB_IN ≤ 18 V, 4.5 ≤ VDD_IO ≤ 5.5 V, T_j -40 to 175 °C unless otherwise specified;

Table 3-10 Smart reset circuit electrical characteristic table 2

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
RST_L	RST input low voltage	-	-0.3		1.1	V	RSTN	P_3.4.3.5
RST_H	RST Input high voltage	-	2.3		VDD_IO +0.3	V	RSTN	P_3.4.3.6
Trst_ft_RST	Reset RST filter time	Tested by scan	7.5	10	11.5	μs	RSTN	P_3.4.3.7
RRST_PU	RST pull up resistor +HS Ron	RSTN / RSTC pins to 0V	0.95	-	2.5	kΩ	RSTN RSTC	P_3.4.3.8
RRST_PD	RST pull down resistor	RSTN / RSTC pin to VDDIO	100	-	320	kΩ	RSTN RSTC	P_3.4.3.9
TNL_RST	After run reset time	-	1.4	2	2.6	ms	RSTN RSTC	P_3.4.3.10
TWDA_RST	watchdog RST reset time	-	1.4	2	2.6	ms	RSTN RSTC	P_3.4.3.11
Td_UV_RST	Power on RST reset delay	-	17	-	30	ms	RSTN/RSTC	P_3.4.3.12
Tmsk_RST	Masking time on RST(IN) after RTN(OUT) is released	-	95	100	105	μs	RSTN	P_3.4.3.13

3.5 Thermal shutdown

Table 3-11 Thermal shutdown

Block	Overtemperature flags																
	OVT[1]	OVT[2]	OVT[3]	OVT[4]	OVT[5]	OVT[6]	OVT[7]	OVT[8] or OVT[9]	OVT[18] or OVT[19]	OVT[10]	OVT[11]	OVT[12]	OVT[13]	OVT[14] **	OVT[15]	OVT[16]	OVT[17]
REGULATOR BUCK	▲																
REGULATOR VDD5,																	
REGULATOR TRACK			▲														
DRIVER INJ1				▲													
DRIVER INJ2					▲												
DRIVER INJ3						▲											
DRIVER INJ4							▲										
DRIVER O2H1								▲									
DRIVER O2H2									▲								
DRIVER SOL1										▲							
DRIVER SOL2											▲						
DRIVER RLY[1:2]												▲					
DRIVER RLY[3:5]													▲				
DRIVER MRD														▲			
DRIVER STR[1:3]															▲		
DRIVER LED[1:2]																▲	
LIN																▲	
VRS																	▲

Legenda:

▲ Shutdown

<blank> Not shutdown

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Operation behavior

OVT[1-16] Dedicated temperature sensor for the power

OVT[3] OVT[3] is used as a common thermal sensor for the three Tracking Regulators

OVT[17] Central Temperature sensor

OVT[14]** Over temperature protection MRD is available only when VB_IN is present.

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4 Watchdog

4.1 Basic feature

HE9788N has an internal query and answer (Q&A) watchdog with the aim of monitoring the correctness of microcontroller activity. Via MSC bus a WDA "question" shall be read from a MSC-WDA register. A correct response shall then be written back via MSC in a well defined timing. If response or timing is not correct, then the WDA error counter EC (or ERR_CNT) is increased. If the error counter is increased to values greater than 4, some output functions are shut off. If the error counter reaches values greater than 7 (overflow), then a RST reset may be generated if this is previously configured via MSC. The first sequence of questions and responses starts from VDD5 higher than the UV threshold.

Other way round, with a RST event also the WDA output pin goes to low.

Note that after startup, reset or an RST_CNT overflow the initial value of the error counter is 6.

If WDA resets are enabled via MSC (*WDA_INIT CONFIG-REG 9_1 D3*): the number of RST events generated by an error counter overflow is limited by the reset counter *RST_CNT*. If *RST_CNT* reaches the value of 7, then RST resets via WDA are no longer generated.

With each error counter overflow, the *PWR_CNT* counter is increased. If it reaches a value greater than 7, then the power latch mode is terminated.

Watchdog legenda:

<i>ERR_CNT</i> , EC	error 3-bit counter;
<i>RST_CNT</i>	reset 3-bit counter;
<i>WDA_RST</i>	internal reset generated by Watchdog;
<i>PWR_CNT</i>	power off 3-bit counter;
<i>PWR_INT</i>	internal signal indicating <i>PWR_CNT</i> overflow;
WDA	WDA open drain pin (low active);
<i>WDA_INT</i>	WDA internal signal, when high it drives WDA = low, when low it sets WDA= Hi-z;
Sequencer-run	current state of WDA when it is active.

4.1.1 Monitoring module - WDA Functionality

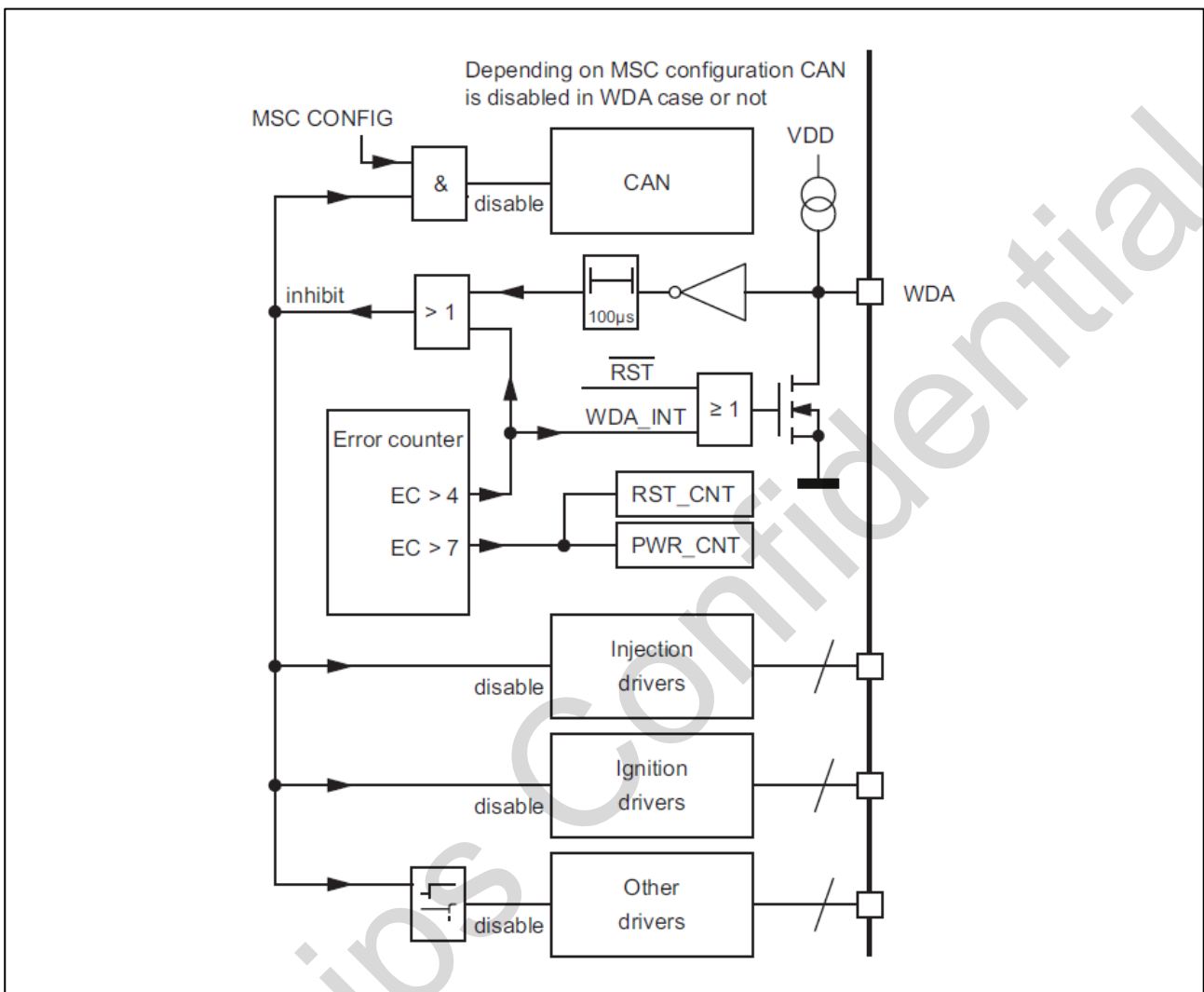


Figure 4-1 WDA block diagram

Note: Delay-off function is not considered in this Block Diagram.

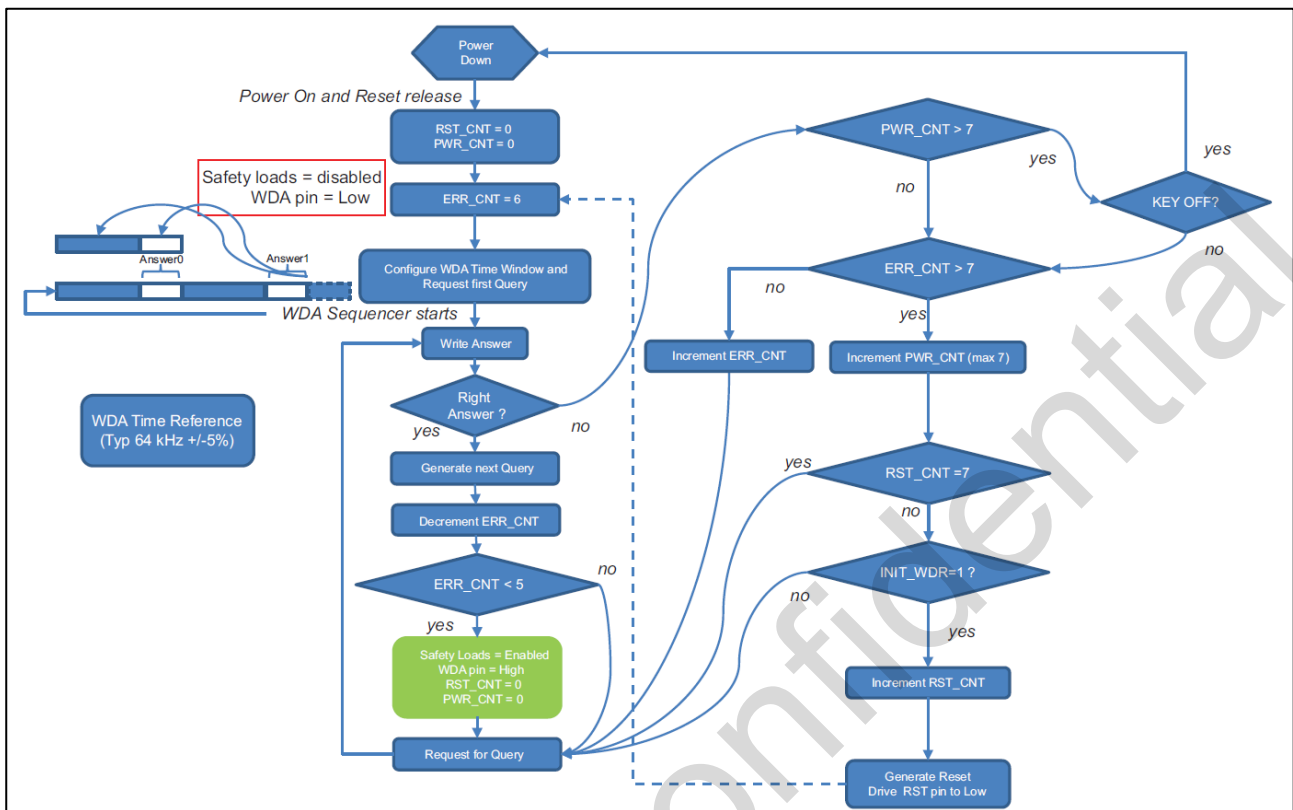


Figure 4-2 WDA state flow

Each time the watchdog error counter is $EC > 7$ counter PWR_CNT counter increases. When this counter is $PWR_CNT=7$ and a further error occurs, the power-latch will be terminated if KEY_IN is low. The PWR_CNT -counter is not cleared when $EC < 7$. PWR_CNT -counter is cleared when $EC < 5$ or by RST_UV .

The monitoring module works independently of the controller functionality. The monitoring module generates various questions, which the controller must fetch and correctly respond to within a defined time window. The monitoring module checks whether the response is returned in a time window and if the response is fully correct.

The question is a 4-bit word. This 4-bit word can be fetched by the controller using a read access to register WDA_REQULO . The monitoring module also calculates the expected correct response, which is compared to the actual response from the controller.

The response is a 32-bit word consisting of the 4 bytes $RESP_BYTE3$, $RESP_BYTE2$, $RESP_BYTE1$ and $RESP_BYTE0$. The 4 bytes are sent to the monitoring module via MSC in the order $RESP_BYTE3 - RESP_BYTE2 - RESP_BYTE1 - RESP_BYTE0$ using four times the command WDA_RESP - once for each answer byte.

The monitoring cycle phase is initialized by (the end of) writing of $RESP_BYTE0$ (least significant response byte) or by a write access to the $RESPTIME$ register. The cycle starts with a variable wait time (response time, set by register $RESPTIME$), followed by a fixed time window. When a monitoring cycle ends (the end of the fixed time window has been reached) a new monitoring cycle is started automatically.

A correct response within the time window (at a response time > 0 ms) decreases an ERROR COUNTER by one. An incorrect response, a response outside the time window or response time = 0 ms leads to the incrementing of the ERROR COUNTER by one.

“... within the time window” means that the end of writing the last answer byte - i.e.

RESP_BYTE0 - falls into the fixed time window mentioned above (see picture below).

Except the last answer byte, the previous answer bytes may also be written earlier than the beginning of the fixed time window.

The question sequence is deterministic. A question will be repeated until it is answered correctly both in value and in time. Then the next question is placed in the sequence.

The ERROR COUNTER (EC) is a 3-bit counter. Various actions are activated depending on the value of the counter.

The result of the comparison of the controller response and the calculated correct response, as well as the next question, are available in registers *REQULO* after receiving the μ C response (LSB of *RESP_BYTE0*) and can be read by the controller.

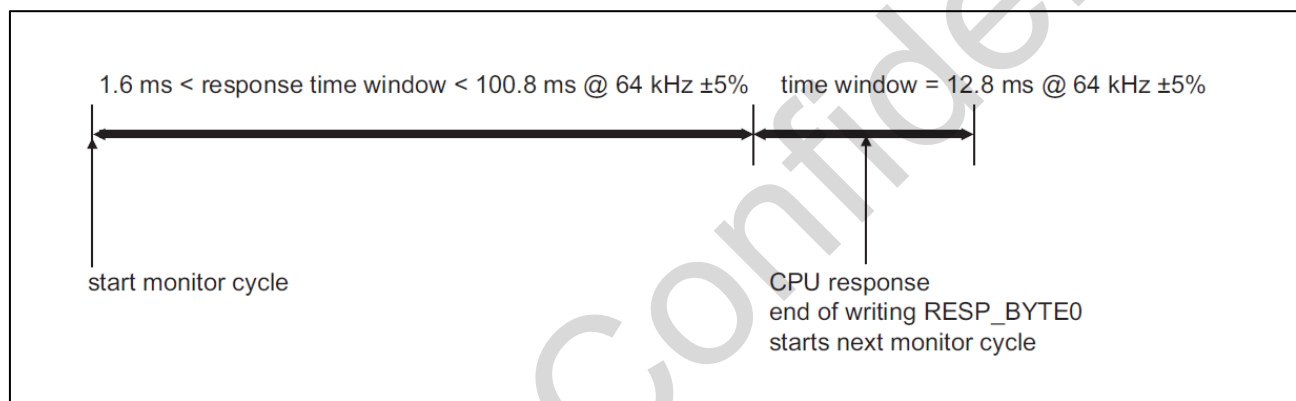


Figure 4-3 Monitoring cycle diagram

4.1.2 ERR CNT (EC) and reactions, PWR COUNTER (PWR_CNT) and generation of the monitoring module reset

Various actions are initiated for specific counter states of the *ERROR COUNTER* EC. The counter reset state is 6.

For *ERROR COUNTER* (EC) > 4, the open drain output WDA is pulled low (active).

Table 4-1 Error counter

ERR_CNT	0 ... 4	5	6 ... 7	Overflow ERR_CNT > 7
WDA	Hi-Z	Low	Low	Low
PWR_CNT	0	unchanged	unchanged	Incremented by 1
Internal PWR_CNT signal	Low (not active)	If PWR_CNT = 7 and ERR_CNT increase => High (active) Else unchanged	If PWR_CNT = 7 and ERR_CNT increase => High (active) Else unchanged	PWR_CNT < 7 Low (inactive) PWR_CNT 6->7 Low (inactive) PWR_CNT 7 -> 7 High (active)

If the ERROR COUNTER reaches the value “7” and a further error occurs the *PWR_CNT* is incremented by one during a sequencer-run.

The state “EC = 7 and a further error occurs” is also called ERROR COUNTER overflow (“EC” > 7).

The counter *PWR_CNT* is a 3 bit counter.

Behaviour of *PWR_CNT*:

asynchronous reset to “000” with *RST_UV*

synchronous reset to “000” IF (EC < 5)

IF (*PWR_CNT* < 7) AND (sequencer-run AND “EC” 7) THEN *PWR_CNT* = *PWR_CNT* + 1
ELSE unchanged.

The counter cannot be decremented and can be only reset to “000” by an active *RST_UV* signal (asynchronous) or *<WDA_INT>* = ‘0’ (synchronous). The signal *PWR_INT* becomes active ‘1’ when *PWR_CNT* = “111” and a further error is detected. When *KEY_IN* = 0, the active *PWR_INT* signal causes a shut-down of the main relay and the voltage regulators.

This function ensures a secure shutdown of the system in an error state of the μC in “power-latch”. Signal *PWR_INT* is set to ‘0’ again only when *<WDA_INT>* = ‘0’.

4.1.3 Generation of a monitoring module reset

The monitoring module may cause a reset at the pin [RSTN] named “monitoring module reset” in conjunction with the internal signal *WDA_RST*. The generation of a monitoring module reset depends on the state of the bit *<WDA_INT>*.

<WDA_INT> = ‘0’ (reset state):

If *<WDA_INT>* = ‘0’, the signal *<WDA_RST>* remains always inactive ‘0’ and the monitoring module can never generate a reset. The error counter can only be decremented via correct responses. If *<WDA_INT>* = ‘0’ the state of the reset counter *<RST_CNT>* remains unchanged when an ERROR COUNTER overflow occurs (description of the reset counter *<RST_CNT>* see below).

<WDA_INT> = ‘1’:

If *<WDA_INT>* = ‘1’, an ERROR COUNTER overflow activates a reset [RST] (signal *<WDA_RST>* becomes active). The signal *<WDA_RST>* becomes active (i.e. ‘1’) due to an ERROR COUNTER overflow when the value of the 3 bit reset counter *<RST_CNT>* (2-0) is 0..6. If the value of *<RST_CNT>* = “111” and an ERROR COUNTER overflow occurs *<WDA_RST>* remains inactive (i.e. ‘0’) and no reset is generated. The “reset counter” *<RST_CNT>* is incremented by one during a sequencer-run due to an ERROR COUNTER overflow when *<WDA_INT>* = ‘1’ and *<RST_CNT>* is between 0 and 6. If *<RST_CNT>* = 7

and an ERROR COUNTER overflow occurs, the counter state remains 7. The counter can not be decremented and can only reset to zero by an active *RST_UV* signal. The occurrence of a monitoring module reset is indicated via the flag *<WDA_RST>* = ‘1’.

Reading the flag via MSC clears it automatically. In effect maximum 7 monitoring module resets can be generated between 2 active *RST_UV* signal. (see also state table for *<WDA_INT>* = ‘1’ below). The state of the “reset counter” *<RST_CNT>* can be read via MSC but cannot be changed.

Table 4-2 Internal states for WDA_INIT=1

RST_CNT old value	EC>7 and sequencer-run	RST_CNT new value	WDA_RST
000 ... 111	False	= RST_CNT old	0, no monitoring module reset
000 ... 110	True	= RST_CNT old + 1	1, monitoring module reset generated
111	True	= 111	0, no monitoring module reset

4.1.4 Question generation

The generation of the 4-bit question (*REQU[3-0]*) is realized as in the table below. The 4-bit counter only changes to the next state during the sequencer-run when the previous question has been answered correctly in value and in time. The state is changed by a sequencer-run because of a write-access to the RESPTIME register or the expiration of the time window.

When RST_UV is active the state is set to 0000b

4.1.5 Response comparison

The 2-bit counter <RESP_CNT (1-0)> counts the received bytes of the 32-bit response and controls the generation of the expected response. Its default value is “11” (corresponds to “waiting for RESP_BYTE3”).

The <RESP_ERR> flag is set '1' when a response byte is incorrect. The flag remains '0' if the 32-bit response is correct. The ERROR COUNTER is updated with the flag. The default state of the flag is '0'.

The 2-bit counter <RESP_CNT (1-0)> and the <RESP_ERR> flag are reset to their corresponding default values at a sequencer-run. The reset condition of the counter <RESP_CNT (1-0)> and the <RESP_ERR> flag are the corresponding default states.

Procedure of the sequential response comparison:

<RESP_CNT (1-0)> = “11”: switch the expected response for RESP_BYTE3 to the comparator

Write access: RESP_BYTE3

Set <RESP_CNT> to “10”, update <RESP_ERR> flag

<RESP_CNT (1-0)> = “10”: switch the expected response for RESP_BYTE2 to the comparator

Write access: RESP_BYTE2

set <RESP_CNT> to “01”, update <RESP_ERR> flag

<RESP_CNT (1-0)> = “01”: switch the expected response for RESP_BYTE1 to the comparator

Write access: RESP_BYTE1

set <RESP_CNT> to “00”, update <RESP_ERR> flag

<RESP_CNT (1-0)> = “00”: switch the expected response for RESP_BYTE0 to the comparator

“update <RESP_ERR> flag” previously described means that it sets <RESP_ERR> flag to 1 if a response byte is incorrect. Once <RESP_ERR> flag is set to 1 it maintains its value until next sequencer run.

Write access: RESP_BYTE0

Start sequencer (SEQU_START signal), set <RESP_CNT> to “11”, update <RESP_ERR> flag, update ERROR COUNTER

Sequencer clears <RESP_ERR> flag to '0

SEQU_START = (RESP_CNT1=0) AND RESP_CNT0=0) AND “response byte write”

Table 4-3 Question and answer

question REQU (3-0)	RESP_BYTE3	RESP_BYTE2	RESP_BYTE1	RESP_BYTE0
0	FF	0F	F0	00
1	B0	40	BF	4F
2	E9	19	E6	16
3	A6	56	A9	59
4	75	85	7A	8A
5	3A	CA	35	C5
6	63	93	6C	9C
7	2C	DC	23	D3
8	D2	22	DD	2D
9	9D	6D	92	62
A	C4	34	CB	3B
B	8B	7B	84	74
C	58	A8	57	A7
D	17	E7	18	E8
E	4E	BE	41	B1
F	01	F1	0E	FE

4.1.6 Reset behavior

All monitoring module registers are reset by RST_UV. The following monitoring module components are also reset by RST_PRL (internal partial reset).

Table 4-4 Watchdog reset behavior

Component	Reset Condition
ERROR COUNTER	110b
Register for “EC>7”	'0'
Register RESPTIME	Maximum value:0011 1111b

RST_PRL is active when at least one of the following signals is active: RSTN or SW_RST.

4.1.7 Access during a sequencer-run

A sequencer-run (which means the same as a monitoring cycle) is initiated by the writing of a response (i.e. all answer bytes `<RESP_BYTE3..0>`) or a write to `<RESPTIME>` or by reaching “end of time window”. It’s not interrupted by a new access, i.e. the monitoring module completes the action already started:

- A sequencer-run was initiated by a “response write”: The sequencer completes its task with the data of the previous access and the new data are ignored.
- A sequencer-run was initiated by a “response-time write”: The sequencer uses the response-time of the previous access, the error counter is correspondingly incremented by one and the `<CHRT>` bit (REQUHI register) is set and the new data are ignored. `<CHRT>` will be reset by reading and by the next start of a sequencer run (not reset by the sequencer run that is started by a “response-time write”).
- A sequencer-run was initiated by “end of time window”: The sequencer finishes the started run, the error counter is incremented by one and the new data are ignored.

The writing of a response-time during a sequencer-run will not set the `<CHRT>` bit (REQUHI register). The new response-time value is also not accepted.

At the end of the sequencer run `W_RESP` bit is set to 0 if the response is correct, it is set to 1 if the response is incorrect.

4.1.8 Clock and time references

The monitoring module works independently by the micro-controller clock so that it can monitor the timing of the micro-controller.

This oscillator is integrated in the U-chip and it provides a clock CLK1 for the monitoring module.

Clocked with CLK1, a divider generates the base time of 1.6 ms @ 64 kHz for the response-time and $8 \times 1.6 \text{ ms} = 12.8 \text{ ms}$ @ 64 kHz for the fixed time window. Accuracy of CLK1 is $\pm 5\%$.

The response-time is adjustable by the controller in the range 0ms to about 100ms (register `RESPTIME`).

The response-time can be calculated with the equation

$$\text{Response_Time} = \text{RESPTIME}[5:0] \times 1.6 \text{ ms}$$

If 39 kHz clock is selected by `wda_win_sel` configuration bit, the base time changes becoming 2.6ms @ 39 kHz. This implies the change of all related timing values: $8 \times 2.6 \text{ ms} = 20.8 \text{ ms}$ @ 39 kHz for the fixed time window and response time becomes `RESPTIME[5:0] * 2.6 ms`.

The `WDA_RESPTIME (CONFIG_REG9_0)` register is set to '0011 1111'b after a reset. The ERROR COUNTER is incremented by one if the controller changes the response-time. If the response-time is set to 0 ms, then the ERROR COUNTER is incremented by one even if a correct response is received within the time window. The maximum error reaction time is given by: maximum response-time, response at the end of a time-window and ERROR COUNTER 0 is $5 \times (100.8 \text{ ms} + 12.8 \text{ ms}) = 568 \text{ ms}$.

In order to assure correct response write, there must be a minimum distance between 2 consecutive writes equal to 5us, this time does not depend on `WDA_WIN_SEL`

(CONFIG_REG9_1 D2).

Note: These clock-tolerances have to be taken into account additionally.

As output (open drain):

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 4-5 WDA circuit electrical characteristic

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
WDA_UV_LO	Output low voltage	$1.3\text{ V} < \text{VDD5} < \text{VDD_UV}$ $R_{\text{reset}} = 4.7\text{ k}\Omega$	-	-	0.6	V	WDA	P_4.1.8.1
WDA_IUVres_max	Input current	$\text{VDD5} = \text{VDD_UV}$ $\text{V}_{\text{UV_reset}} = 0.6\text{ V}$	5	-	-	mA	WDA	P_4.1.8.2
WDA_IikUV_reset	Input leakage current	$\text{V}_{\text{UV_reset}} > \text{VDD_UV}$	-	-	1	μA	WDA	P_4.1.8.3

As input:

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, $4.75\text{ V} \leq \text{VDD_IO} \leq 5.25\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 4-6 WDA reset circuit electrical characteristic

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
WDA_RST_L	WDA input low voltage	-	-0.3	-	1.1	V	WDA	P_4.1.8.4
WDA_RST_H	WDA input high voltage	-	2.3	-	$\text{VDD_IO} + 0.3$	V	WDA	P_4.1.8.5
WDA_Trst_fit	WDA Reset filter time	Tested by scan	-	10	-	μs	WDA	P_4.1.8.6
-	WDA pull up external resistor	Application note, parameter not tested	50	-	250	k Ω	WDA	P_4.1.8.7

5 Main relay driver (low-side driver)

This driver is used to drive main relay only.

The on/off status of the driver depends on the following sources of power on: KEY_IN WK_IN WAKE_UP_TIMER POWER_HOLD (MSC), CAN see [Table 4-2](#). The Main Relay Driver can be also switched on/off by a MSC command protected by lock. This is intended to use in the permanent battery supply scheme only if power up was done by wk_in/EOT/CAN according to [Table 5-1](#). MSC micro commands protected by lock are available for fast on and off diagnosis.

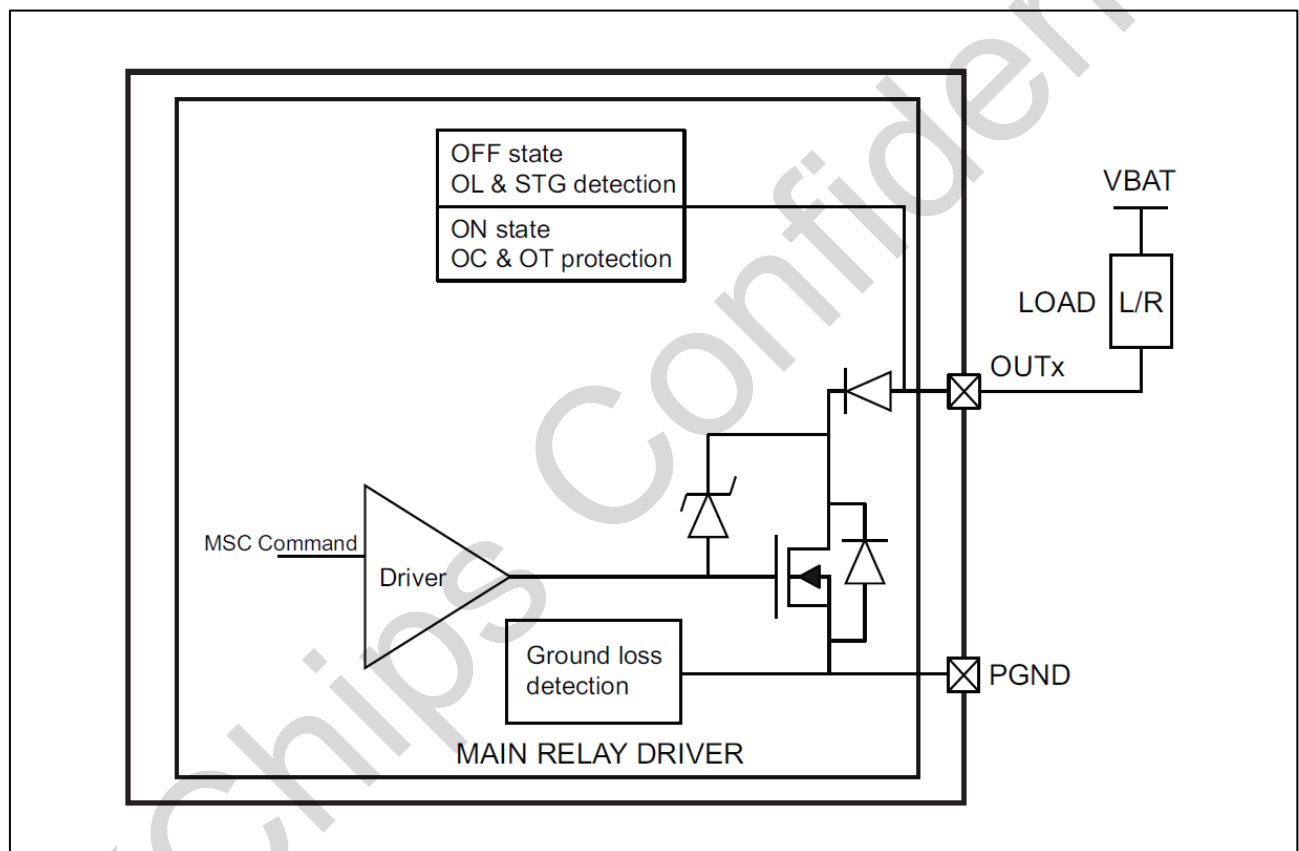


Figure 5-1 Main relay low-side driver stage

The low-side (main relay) can work down to $VB_IN = V_{OFF_VB_IN}$, during cold crank conditions. The driver has integrated diagnosis, with over-current and overtemperature protection circuit during the driver on. The driver turn on/off slew-rate is internally controlled.

The Main Relay Driver on/off status is controlled by the power up/down modes or by the msc commands according to [Table 5-1](#).

These notes are linked to [Table 5-1](#)

1. The battery status can be determined at power up only.

Main relay driver (low-side driver)

2. The key detection switches on always MRD and does not determine the battery status.
3. Once the key detection has switched on MRD, MRD is switched off by one of these events:
 - Clearing of all detections and phold not active (power down),
 - Watchdog power counter timeout,
 - *MSC MRD OFF* command when phold is active (all detections cleared).
4. The *msc_enable_driver* command or *trans_f* error or clock monitor error do not affect the MRD even if it is controlled by MSC commands.

5.1 ON state diagnostic

Table 5-1 ON state diagnostic

		MRD	MSC command		OVC	OVT
			ON	OFF		
KEY_DET=1	Permanent Supply	ON	Y	Y	restart by MSC scenario 19 (see Figure 0-14)	Auto restart without limit of number scenario 21 (see Figure 0-16)
WAKE_IN_DET / WAKE_EOT_DET / WAKE_CAN_DET =1 KEY_DET=0	Permanent Supply	OFF	Y	Y	restart by MSC scenario 19 (see Figure 0-14)	Auto restart without limit of number scenario 21 (see Figure 0-16)
PHOLD_EN =1 KEY_DET / WAKE_IN_DET / WAKE_EOT_DET / WAKE_CAN_DET = 0	Permanent Supply	Keep the status	Y	Y	restart by MSC scenario 20 (see Figure 0-15)	Auto restart without limit of number scenario 29 (see Figure 0-17)

5.1.1 Overcurrent protection

When overcurrent fault is detected the Main Relay Driver MRD is switched off.

The MRD is automatically restarted every T_{res} if KEY_IN is till active in permanent battery supply scheme.

In permanent battery supply scheme in case of power up from WK_IN or WAKE_UP_TIMER or WAKE_UP_CAN the MRD is not switched on at power up. In case of permanent battery supply scheme and OVC protection is activated the driver is restarted by a positive transition on the MSC *MRD_ON* command. It is not restarted by MSC reading.

5.1.2 Thermal protection

The overtemperature fault is detected only when VB_IN is present and the logic is out of power on reset, so when the device is in ON state.

When overtemperature fault is detected the Main Relay Driver is switched off. In permanent battery supply schemes the switching off of the Main Relay Driver does not produce a power down. The MRD is switched on again when temperature decreases to T_SD_LOW.

5.2 OFF state diagnostic

The OFF diagnosis is performed only in the supply scheme with battery permanently connected.

The off diagnosis is the same as other low-side drivers.

There is a bit *lpupd_EN* which is controlled by a MSC bit (CONFIG_REG2 D6), it is used by external μ Controller if it is needed to switch on/off both pull up and pull down diagnosis current.

There is a bit *lpupd_MODE* which is controlled by a MSC bit (CONFIG_REG2 D7), it is used by external μ Controller if it is needed to switch on/off the bigger diagnosis current (I_LS_PU1) in fast mode. It is possible to define the maximum number of retry in case of OVC after wake up by KEY. It can be done setting *KEY_OC_RETRY_MAX_EN* bit in CONFIG-REG 20.

Condition:

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 5-2 Main relay low-side driver electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Main relay								
MRD_out_voltage_low	Low-side out voltage	VB_IN = 4.25 V I _{load} = 0.1 A	-	-	0.9	V	MR D	P_5.2.1
MRD_out_voltage_nom	Low-side out voltage	5.5 V < VB_IN < 18 V I _{load} = 0.3 A	-	-	1.35	V	MR D	P_5.2.2
MRD_lkg	Output leakage current	Output disabled, diagnostic off	-10	-	+10	μ A	MR D	P_5.2.3
rev_MRD	-	Reverse battery	-	-	-16	V	MR D	P_5.2.4
I_load	Output current	Application Info	-	-	0.5	A	MR D	P_5.2.5
SR_ON_20	Voltage slew ON State	From 80% to 30% of VOUT VB_IN = 14 V, R _{load} = 20 Ω	2	-	6	V/ μ s	MR D	P_5.2.6

		C _{load} = 10 nF						
SR_OFF_20	Voltage slew OFF State	From 30% to 80% of V _{OUT} V _{B_IN} = 14 V, R _{load} = 20Ω C _{load} = 10 nF	2	-	6	V/μs	MR D	P_5.2.7
Ton_MRD_20	Propagation Delay from Chip_EN rising edge to 80% output MRD voltage	V _{B_IN} = 14 V, R _{load} = 20Ω C _{load} = 10 nF	-	-	10	μs	MR D	P_5.2.8
Toff_MRD_20	Propagation Delay from Chip_EN falling edge to 20% output MRD voltage	V _{B_IN} = 14V, R _{load} = 20Ω C _{load} = 10 nF	-	-	10	μs	MR D	P_5.2.9
Vclamp_MRD	MRD clamping Voltage	I _{load} = 0.3 A	48	-	55	V	MR D	P_5.2.10
Vclamp_MRD_high_curr	MRD clamping Voltage	I _{load} = 1 A Guaranteed by design, provided single pulse energy limits are not violated by clamping action	48	-	55	V	MR D	P_5.2.11
IOVC_MRD	Over current driver threshold	-	1	-	2	A	MR D	P_5.2.12
I_LS_ocrvflt	LS overcurrent filter time	Guaranteed by scan	4	-	7	μs	MR D	P_5.2.13
Tft_Res	Self retry time after OVC with KEY_IN on or WK_IN active or WKE_UP_EOT active	Guaranteed by scan	15.2	16	16.8	ms	MR D	P_5.2.14
NRes_NN	Maximum number of retry when KEY_OC_RETRY_MAX_EN=1	Guaranteed by scan	-	32	-	-	MR D	P_5.2.15
T_SD_HIGH	Temperature shut down	-	185	-	200	°C	MR D	P_5.2.16
T_SD_LOW	Temperature shut down recover	-	175	-	190	°C	MR D	P_5.2.17
T_SD_hys	Temperature shut down hysteresis	-	5	-	10	°C	MR D	P_5.2.18
tmsd_pre_an	Thermal	Guaranteed	1.5	-	4	μs	MR	P_5.2.19

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Main relay driver (low-side driver)

	shutdown analog filter time	by design					D	
t_SD_deglitch	Digital deglitch filter time on Temperature shut down detection	Guaranteed by scan	30	-	-	µs	MR D	P_5.2.20
OFF state diagnostic								
VLVT	Short to GND threshold voltage	Driver tristate, diag enabled	1.9	-	VOUTOP EN -190mV	V	MR D	P_5.2.21
V_OL	Open load threshold voltage	Driver tristate, diag enabled	VOUTOPEN +160 mV	-	3.0	V	MR D	P_5.2.22
VOUTOPEN	Open load output voltage	Driver tristate, diag enabled	2.3	2.5	2.7	V	MR D	P_5.2.23
I_LS_PU1	Diagnostic pull up current	In OFF condition, OUTx < VLVT, lpupd_EN = ENB, lpupd_MODE = "1"	2.5	3.6	4.7	mA	MR D	P_5.2.24
I_LS_PU2	Diagnostic pull up current	In OFF condition, OUTx < VLVT, lpupd_EN = ENB, lpupd_MODE = "0"	40	70	100	µA	MR D	P_5.2.25
I_LS_PD1	Diagnostic pull down current	In OFF condition, OUTx > VOUTOPEN, lpupd_EN = ENB	60	85	110	µA	MR D	P_5.2.26
Tflt_diagoff1	DIAG Filter time	Filter Mode=0	75	100	125	µs	MR D	P_5.2.27
Tflt_diagoff2	DIAG Filter time	Filter Mode=1	450	600	750	µs	MR D	P_5.2.28
-	Minimum OFF time for correct diagnostic	Application note (esd cap < 10 nF) Filter Mode = 0	175	-	-	µs	MR D	P_5.2.29
-	Fast OFF diagnostic ON→OFF→ON	Filter Mode = 0	175	200	240	µs	MR D	P_5.2.30
-	Fast ON diagnostic OFF→ON→OFF	-	65	-	90	µs	MR D	P_5.2.31
Driver Reliability Data								
EnergyRep MRD	Energy Repetitive	I_OT_n = 0.3 A	-	-	6.5	mJ	MR D	P_5.2.32

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Main relay driver (low-side driver)

	Pulses	Freq = 1 Hz ; 4 Miopulses Tj =150 °C						
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5.3 Error handling

The Main Relay Driver follows the same error handling table as other low-side drivers.

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6 Multi-voltage regulator supplies

The chip has one pre-boost regulator and one pre-buck regulator, which will supply a 5 V regulator with external MOS. Three 5 V tracking regulators are included. One charge pump is used for HS channel and by the 5 V linear regulator to drive the external MOS. See the diagram in Figure 6-1. The chip becomes to be supplied and the regulators are enabled in the sequence: VB_in grows till is higher than VB_IN_uv threshold, then internal 5V supply, charge pump and buck, pre-boost regulator, 5 V linear regulator, 5 V tracking regulator 1, 2 and 3 are switched on. Then the power on reset is released and the chip is working.

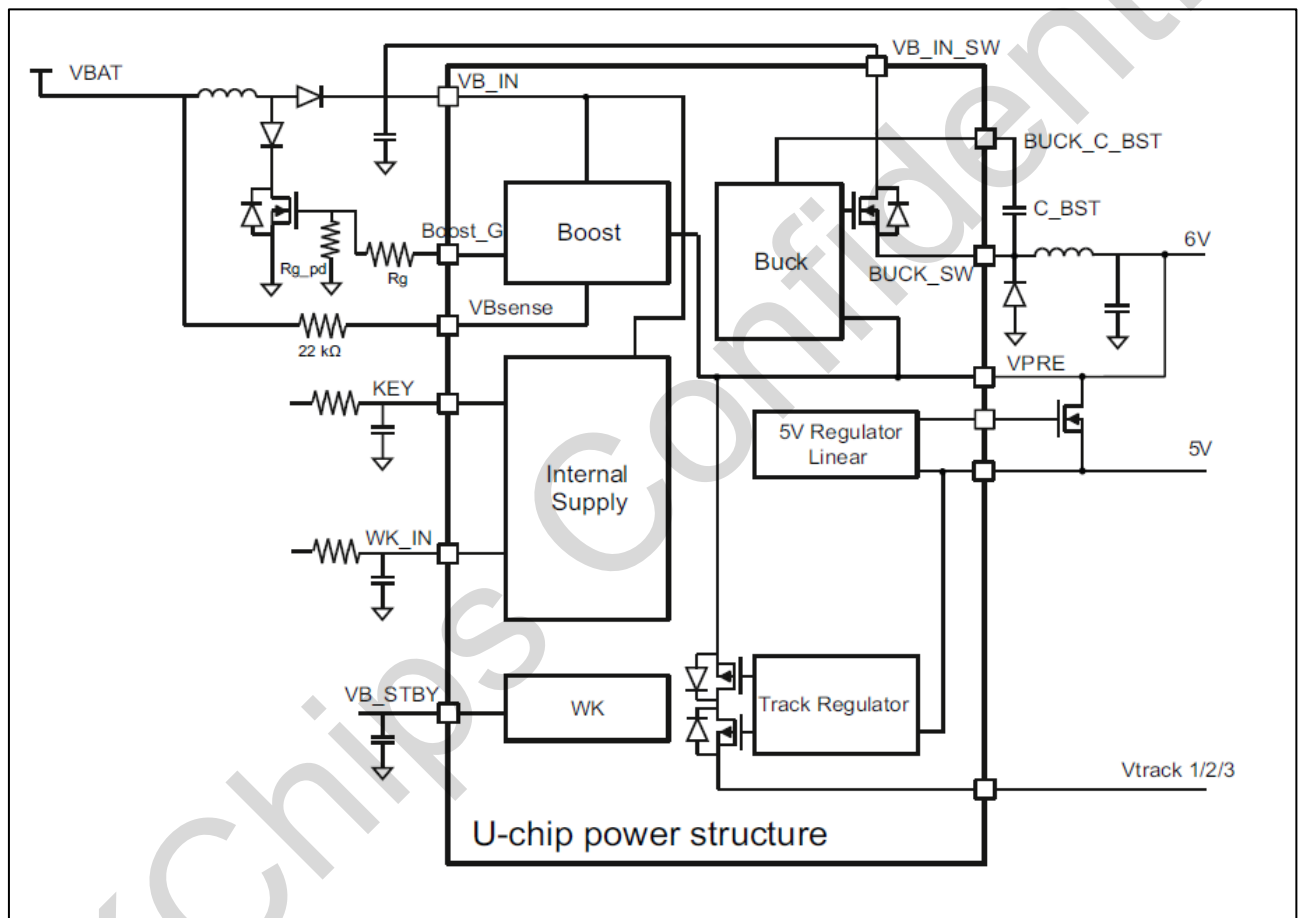


Figure 6-1 Power supply block diagram with pre-boost regulator permanent VBAT

The pre-boost regulator is an optional choice. It can be disabled (BOOST_EN (CONFIGREG 1 D5) in case battery voltage is high enough in cranking phase to supply the pre-buck regulator. See the diagram in Figure 6-2.

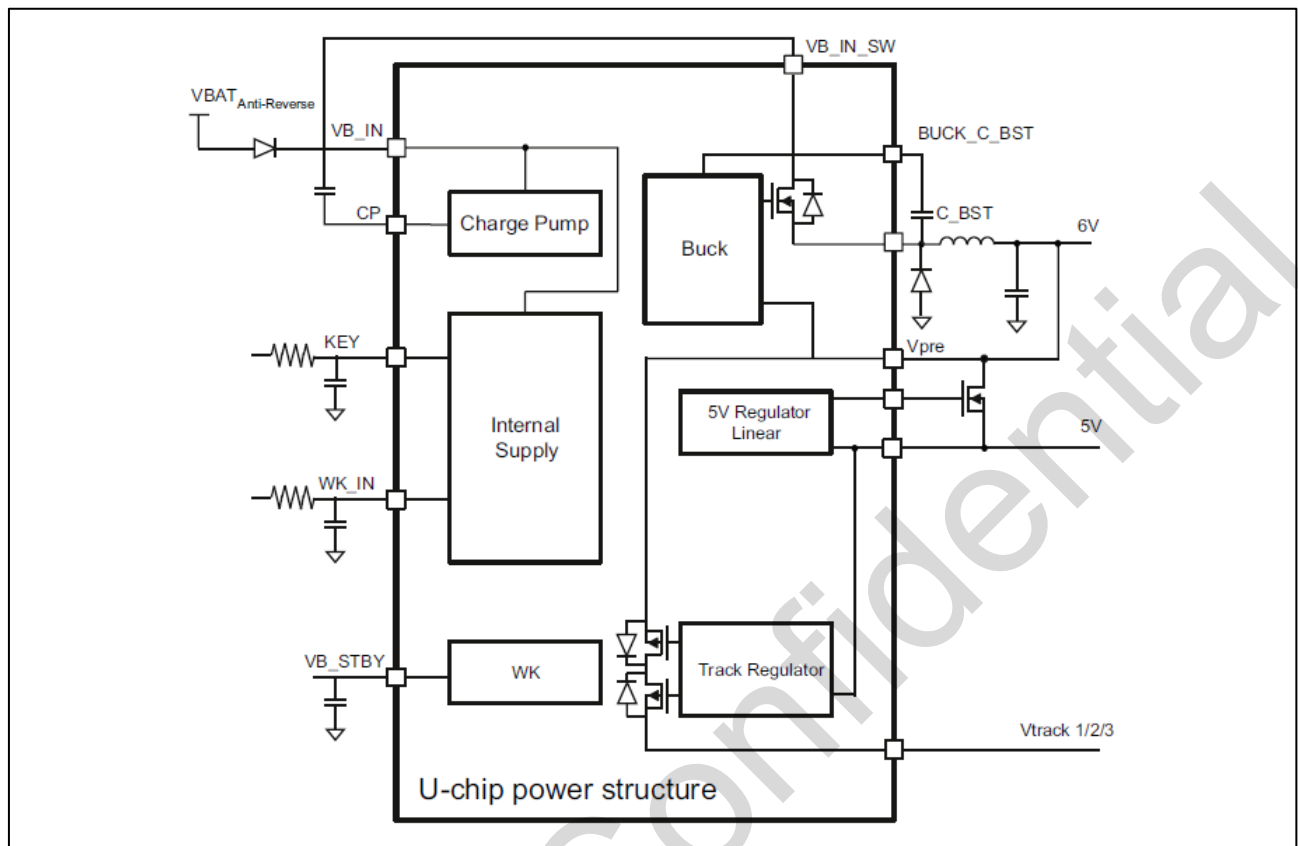


Figure 6-2 Power supply block diagram with charge pump without pre-boost

6.1 Pre-boost regulator

The boost regulator will provide a higher voltage than battery to the buck regulator when battery voltage is too low to maintain 5 V regulator out of reset conditions. Once the output of the pre-boost is higher than VB_IN_th (typ 8.5 V), the pre-boost will stop working.

The boost is disabled at the beginning of the power down sequence. When *Boost_EN* (CONFIG_REG 1 D5) = 0 the internal driver keeps OFF the external MOS.

The parameters of Pre-boost regulator can be guaranteed when Vbat goes down to 3.0 V during cranking phase. The block diagram of the pre-boost is Figure 6-3.

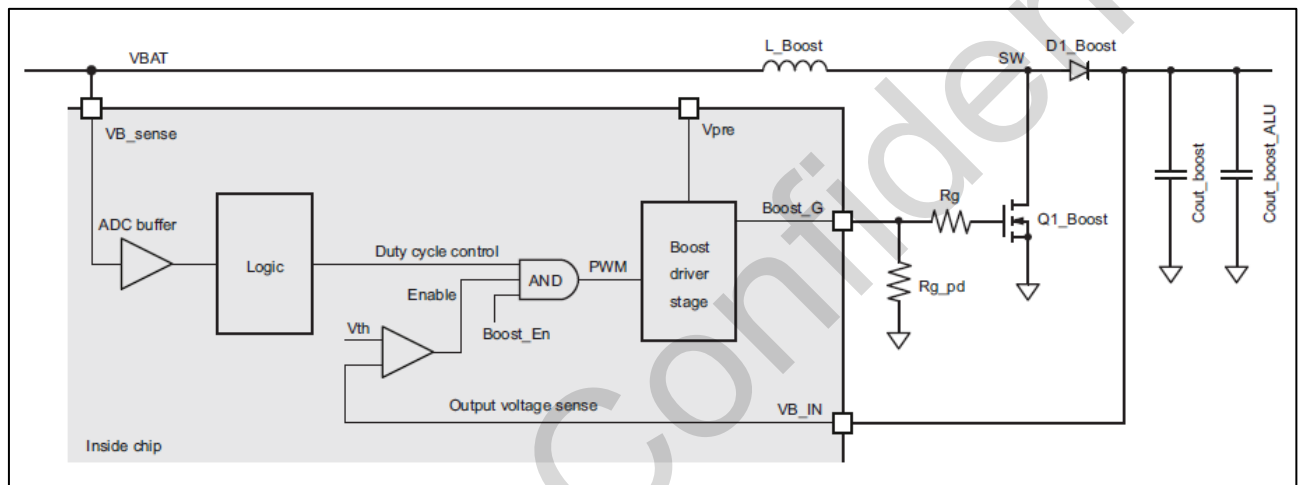


Figure 6-3 Pre-boost block diagram

$3.0\text{ V} \leq \text{VBAT}^{(*)} < 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified.

Note: (*)VBAT is the battery line of Figure 6-3 in front of the limiting 22 k Ω resistor (not shown in the figure).

Table 6-1 Pre-boost regulator output electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VB_IN_th	BOOST enable/disable threshold	When VB_IN < VB_IN_TH, the boost starts for time (ton). When VB_IN ≥ VB_IN_TH, Boost is disabled after ton is completed. ton depends on different duty cycle	8.07	8.5	8.93	V	VB_IN	P_6.1.1
tdBOOST_EN	BOOST Enable/disable	-	0.19	-	2	μs	VB_IN	P_6.1.2

	comparator delay time							
Fsw_BOOST	Switching frequency	43 cycles of 15 MHz main clk, Guaranteed by SCAN	334	352	370	kHz	VB_IN	P_6.1.3
Boost_DC1	Boost Duty cycle 1	0V < VBAT_SENSE < Vcomp_th1 33 out of 43 cycles of 15MHz main clk (100% DC is 43 clk cycles) Guaranteed by SCAN	-	76.74	-	%	VB_IN	P_6.1.4
Vcomp_th1	VBAT_SENSE comparator threshold 1	-	3.95	4.15	4.35	V	VBAT_SENSE	P_6.1.5
Boost_DC2	Boost Duty cycle 2	Vcomp_th1 < VBAT_SENSE < Vcomp_th2 25 out of 43 clk cycles Guaranteed by SCAN	-	58	-	%	VB_IN	P_6.1.6
Vcomp_th2	VBAT_SENSE comparator threshold 2	-	5	5.36	5.7	V	VBAT_SENSE	P_6.1.7
Boost_DC3	Boost Duty cycle 3	Vcomp_th2 < VBAT_SENSE < Vcomp_th3 18 out of 43 clk cycles Guaranteed by SCAN	-	41.86	-	%	VB_IN	P_6.1.8
Vcomp_th3	VBAT_SENSE comparator threshold 3	-	6.25	6.6	6.95	V	VBAT_SENSE	P_6.1.9
Boost_DC4	Boost Duty cycle 4	Vcomp_th3 < VBAT_SENSE < Vcomp_th4 10 out of 43 clk cycles Guaranteed by SCAN	-	23.85	-	%	VB_IN	P_6.1.10
Vcomp_th4	VBAT_SENSE comparator threshold 4	-	7.4	7.8	8.2	V	VBAT_SENSE	P_6.1.11
Boost_DC5	Boost Duty cycle 5	Vcomp_th4 < VBAT_SENSE 4 out of 43 clk cycles Guaranteed by SCAN	-	9.3	-	%	VB_IN	P_6.1.12
tdBAT_SENS	VBAT_SENSE comparators delay time	Vcomp_th1/2/3/4 comparators	0.19	-	2	μs	VBAT_SENSE	P_6.1.13
IBOOST_LS_sink	LS driver sink	BOOST_G = 6 V	30	50	70	mA	BOOST_G	P_6.1.14

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	current							
IBOOST_HS_source	HS driver source current	BOOST_G = 0 V	30	50	70	mA	BOOST_G	P_6.1.15
IBOOST_LS_sink	LS driver sink current	BOOST_G = 0.5 V	15	-	43	mA	BOOST_G	P_6.1.16
IBOOST_HS_source	HS driver source current	BOOST_G = 5.5 V	10	-	43	mA	BOOST_G	P_6.1.17
BOOST_G_Ileak	Leakage current	From 0 V to max pin voltage	-	-	1.2	μA	BOOST_G	P_6.1.18
I_VBAT_Sense_stby	Current leakage on VBAT_sense pin in st-by	-	-	-	1	μA	VBAT_SENSE	P_6.1.19
I_VBAT_Sense	Pull down current on VBAT_sense operative mode	-	-	-	100	μA	VBAT_SENSE	P_6.1.20
R_VBAT_sense	Pull-down resistor from VBAT_sense to GND	-	438	880	1320	kΩ	VBAT_SENSE	P_6.1.21
R1_VBAT_sense	Resistor divider string element	Design info, not tested	311	622	933	kΩ	-	P_6.1.22
R2_VBAT_sense	Resistor divider string element	Design info, not tested	28.75	57.5	86.25	kΩ	-	P_6.1.23
R3_VBAT_sense	Resistor divider string element	Design info, not tested	18.15	36.3	54.45	kΩ	-	P_6.1.24
R4_VBAT_sense	Resistor divider string element	Design info, not tested	12.5	25	37.5	kΩ	-	P_6.1.25
R5_VBAT_sense	Resistor divider string element	Design info, not tested	67.5	135	202.5	kΩ	-	P_6.1.26

Boost application information table:

Table 6-2 Pre-boost regulator external components electrical characteristics

Symbol	Parameter	Min	Typ	Max	Unit	Note	Pin	Number
C _{OUT_BOOST}	Capacitance at VB_IN	36	50	150	μF	Min 5*10 μF, 10%, 25 V, X7R, 1206, CER CAP;	VB_IN	P_6.1.27
C _{OUT_BOOST_ALU}	Option ALU Capacitance at VB_IN	0	470		μF	ALU 35 V ESR 10 mΩ--3 Ω, 470 μF, 20%, 25 V, SMD, AL, ELEC CAP	VB_IN	P_6.1.28
VBoost_ripple	-	-	-	1.2 1.1 1.0 0.8	Vpp	triangular ripple @ 10 kHz and 25 kHz @ 50 kHz @ 100 kHz @ 350 kHz. All characteristics of all blocks supplied by VB_IN	VB_IN	P_6.1.29

						guaranteed		
IBOOST	Load current at VB_IN	0.05	-	2.4	A		VB_IN	P_6.1.30
Q1_Boost	-	-	-	-	-	Choice: STD20NF06L	VB_IN	P_6.1.31
D1_Boost	-	-	-	-	-	Choice: STPS10H100C	VB_IN	P_6.1.32
L_boost	BOOST inductance	0.8	1	1.2	μH	ESR = 10 m Ω	VB_IN	P_6.1.33
L_boost1	BOOST inductance	-	2.5	-	μH	SR = 25 m Ω , $I_{\text{sat}} = 23 \text{ A}$	VB_IN	P_6.1.34
L_boost1	BOOST inductance	-	3.3	-	μH	-	VB_IN	P_6.1.35
Rsense	Resistor for VB_sense pin protection	10	22	30	k Ω	-	VBAT_Sense	P_6.1.36
Rg	-	-	2.2	-	Ω	-	BOOST_G	P_6.1.37
Rg_pd	-	-	150	-	k Ω	-	BOOST_G	P_6.1.38

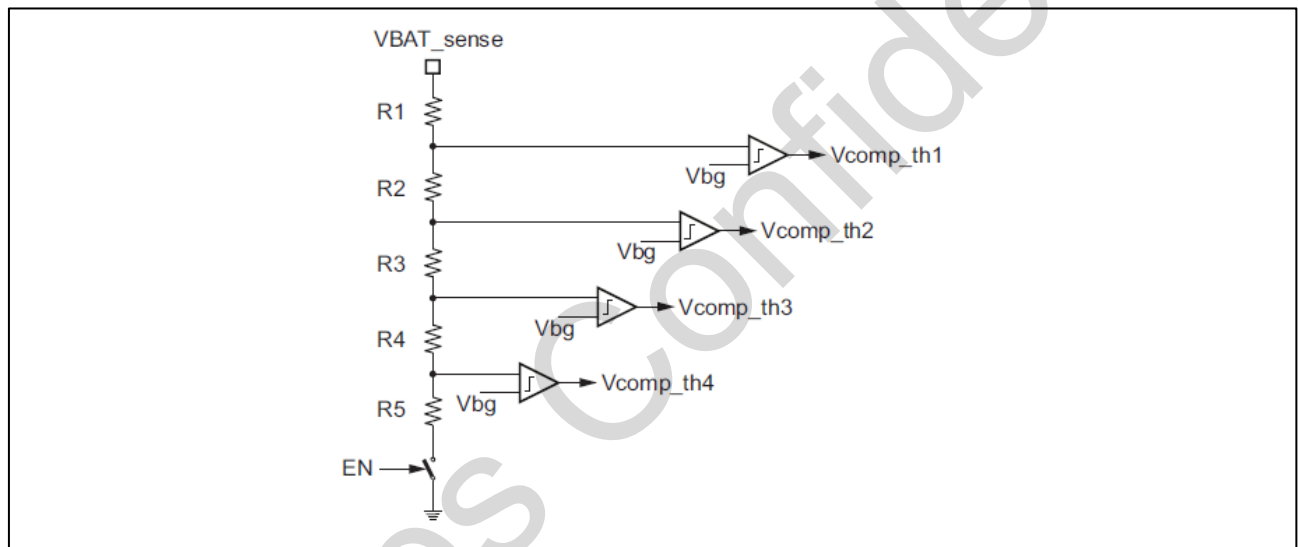


Figure 6-4 Boost threshold comparator's schematic

6.2 Pre-buck regulator with internal MOS

Vpre features:

- 6V regulated output voltage (VPRE);
- 2.2 A maximum output current;
- PWM block (ramp generator, comparator and main latch) with one fixed PWM frequency (Fsw_vpre);
- Internal error amplifier;
- Internal Power FET including Bootstrap;
- dV/dt Control on gate drivers to fulfill EMC requirements;
- Maximum Duty cycle 100%, to allow to charge Bootstrap capacitor;

- Protection by over temperature;
- Soft Start.

The Vpre regulator is a buck regulator which supplies Pre-boost driver/VDD5/Vtrack1.2.3. When the OV of VB_IN is detected, the Vpre(Buck) will be switched off. This function is selectable by a dedicated MSC bit (*VB_IN_OV_RST_EN* in CONFIG_REG12). A bit is also available (*BUCK_SLOW_SR* in CONFIG_REG 21) in order to select slew rate for output voltage; the default value corresponds to fast slew rate, slow slew rate helps to improve emission performance

The Vpre regulator includes a soft start. Range of dV/dt at VPRE is specified in electrical parameters.

Pulse skipping mode: the Vpre regulator can go to pulse skipping mode when the load is too low to avoid that output voltage (VPRE) rises too much. The principle of the pulse skipping mode is to skip some pulses during the switching phase, when the loop can't keep itself the output voltage regulated with a very low load because it is limited by the minimum duty cycle.

VB_IN_OV fault can trigger the turn OFF of Buck regulator, as specified into reset matrix. This effect can be masked with bit *VB_IN_OV_RST_EN* in CONFIG_REG 12.

Conditions:

5.5 V < VB_IN_SW < 18 V, T_j -40 to 175 °C unless otherwise specified.

Table 6-3 Pre-buck regulator output electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VPREdc	VPRE regulated voltage (including aging and static line/load regulation)	VB_IN_SW = 7 V to 18 V, I _{vpre} = 50 mA ~2.2 A, C = 30 µF, L = 22 µH	5.82 (-3%)	6	6.18 (+3%)	V	VPRE	P_6.2.1
Vdrift_VPRE	Total VPRE voltage aging drift	-	-0.8	-	0.8	%	VPRE	P_6.2.2
Vline_VPRE	Line regulation voltage	7 V < VB_IN_SW < 18V I _{vpre} = 50 mA, 2200 mA C=30 µF, L = 22 µH, T = 27 °C, 175 °C	-25	-	25	mV	VPRE	P_6.2.3
Vline_VPRE	Line regulation voltage	7 V < VB_IN_SW < 18V I _{vpre} = 50 mA, 2200 mA C=30 µF, L = 22 µH, T = -40 °C	-60	-	60	mV	VPRE	P_6.2.4
Vload_VPRE	Load regulation voltage	VB_IN_SW = 7 V, 18V, 50 mA<I _{vpre} <2200 mA,	-25	-	25	mV	VPRE	P_6.2.5

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		C = 30 μ F , L = 22 μ H, T = 27 °C, 175 °C						
Vload_VPRE	Load regulation voltage	VB_IN_SW = 7 V , 18 V, 50 mA<Ivpre<2200 mA, C = 30 μ F, L = 22 μ H, T=-40°	-60	-	60	mV	VPRE	P_6.2.6
Vline_VPRE_tr	Transient line regulation accuracy	VB_IN_SW step 1 2 V to 18 V ; 18 V to 12 V, dVB_IN_SW/dt = 3 V/ μ s, Ivpre = 50 mA, 1.8 A, C = 30 μ F, L = 22 μ H, Guaranteed by design(1)	-8	VPRE	8	%	VPRE	P_6.2.7
Vload_VPRE_tr	Transient Load regulation accuracy	VB_IN_SW = 13 V, Iv pre = 50 mA to 1.8 A dI/dt = 500 mA/ μ s, C=30 μ F , L = 22 μ H, Guaranteed by design(1)	-8	VPRE	8	%	VPRE	P_6.2.8
ILIMVPRE	Current limit	-	3.0	-	5	A	VPRE	P_6.2.9
VPRE_coupling	VPRE capacitive coupling during battery transients	VB_IN_SW from 0 to 13 V in 100 μ s, C = 40 μ F , L = 22 μ H, device off Design info, not tested	-	-	3	V	VPRE	P_6.2.10
Vuv_vpre_h	VPRE under voltage hysteresis threshold high; VPRE voltage at Power up to allow Boost activation threshold	Test condition: VB_IN_SW = 5.4 V	4.38	-	4.65	V	VPRE	P_6.2.11
Vuv_vpre_l	VPRE under voltage hysteresis threshold low	Test condition: VB_IN_SW = 5.4 V	4.25	-	4.51	V	VPRE	P_6.2.12
Tuv_filter_vpre	VPRE under voltage filter time (2)		10.1	12	13.3	μ s	VPRE	P_6.2.13
VPRE_slope	Slope control	V B_IN_SW = 13 V	5	-	40	V/ms	VPRE	P_6.2.14
Tmsd_VPRE_H	Thermal shutdown_High	Comparator threshold and functionality are tested in production	185	-	200	°C	VPRE	P_6.2.15
Tmsd_VPRE_L	Thermal shutdown_Low	Comparator threshold and functionality are	175	-	190	°C	VPRE	P_6.2.16

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		tested in production						
Tmsd_hyst	Thermal hysteresis for shutdown	-	5	-	10	°C	VPRE	P_6.2.17
tmsd_pre_an	Thermal shutdown analog filter time	-	1.5	-	4	us	VPRE	P_6.2.18
tmsd_pre_dig	Thermal shutdown digital filter time	-	10	20	30	us	VPRE	P_6.2.19
VPRE_rpd	VPRE pulldown resistor	To keep VPRES below 10 V with Buck switch leakage in OFF state	0.2	-	1.05	MΩ	VPRE	P_6.2.20
Fsw_VPRE	Switching frequency	Fosc from main 15 MHz oscillator, typical 462.5 kHz	450	Fosc/32	500	kHz	BUCK_SW	P_6.2.21
DCVPRE(BUCK)	Duty cycle range	-	3%	-	100%	-	BUCK_SW	P_6.2.22
RVPRE	VPRE (Buck_POWER MOS) transistor RDSON	-	-	-	250	mΩ	BUCK_SW	P_6.2.23
TRVPRE	Rise Time (10-90%)	BUCK_SLOW_SR=0	5	-	25	ns	BUCK_SW	P_6.2.24
TRVPRE_slow	Rise Time (10-90%)	BUCK_SLOW_SR=1	5	-	30	ns	BUCK_SW	P_6.2.25
TFVPRE	Fall Time (10-90%)	Iload = 1 A	5	-	27	ns	BUCK_SW	P_6.2.26
VPRE_SW_Ileak	Leakage current	From 0 V to max pin voltage	-	-	1	μA	BUCK_SW	P_6.2.27

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

2. No VDD5 UV/OV violation is expected during specified transient.

Note: When $18 < V_{B_IN_SW} < V_{OV_VB_IN_L}$, the function of Vpre is guaranteed and no reset happens.

Table 6-4 Pre-buck regulator external components electrical characteristics (Vpre application)

Symbol	Parameter	Min	Typ	Max	Unit	Note	Pin	Number
CVPRE	External capacitor on Vbuck supply	15	40	150	μF	Min 3*10 μF Ceramic X7R 16 V in parallel CER,X7R,10 μF , 10%,16 V,1206, Sn	VPRE	P_6.2.28
LVPRE	Switching regulator external inductor	15	22	29	μH		VPRE	P_6.2.29
CBST	Bootstrap capacitor	-	47	-	nF	Ceramic X7R 16V 47nF, 10%, 16 V, X7R, 0603, TINNED, CER CAP/	BUCK_C_BST	P_6.2.30
DVBUCK_CTRL	Switching regulator	-	-	-	-	Schottky Diode STPS5L60-Y	-	P_6.2.31

	external diode (Schottky or ultrafast)							
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6.3 Tracking regulator for sensors supply

There are three 5V tracking regulators with self protection from over-current, over-voltage and short to battery. STB/OVC/OV don't disable tracking, but set a flag readable by MSC. Those three tracking regulators track VDD5_in voltage with 150mA output. There is one thermal sensor, shared between the three tracking regulators, to provide the thermal shut down protection for those three tracking sensor supplies. The switching off of one internal tracking occurs if that tracking is also in OVC.

Once the thermal shut down has happened, the tracking sensor in over-current condition will be off until temperature decreases to thermal shut down threshold low. Those three tracking regulators can be singularly disabled by dedicated MSC bit (*TRK_EN* *CONFIG_REG7* [3:5] in *CONFIG_REG 7*). *VB_IN_OV* fault can trigger the turn OFF of Tracking regulators, as specified into reset matrix. This effect can be masked with bit *VB_IN_OV_RST_EN* in *CONFIG_REG 12*.

Conditions:

5.5 V < *VB_IN_SW* ≤ 18 V, *T_j* -40 to 175 °C unless otherwise specified.

Table 6-5 T racking sensor supplies electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
ΔV_{TRK}	Output voltage tracking error (including voltage aging and static load/line transient regulation)(1)	$V_{pre} = 5.6 - 18\text{ V}$ $1\text{ mA} < I_{VSENS} < 150\text{ mA}$	VDD5 - 15mV		VDD5 +20mV	V	VSENSE1/2/3	P_6.3.1
I_{MAX_VS}	Output current limitation	$V_{SENS} = -1\text{ V}$	150		300	mA	VSENSE1/2/3	P_6.3.2
I_{TH_VS}	Output over-current threshold	$V_{pre} = 5.6 - 18\text{ V}$	150		I_{max_vs}	mA	VSENSE1/2/3	P_6.3.3
V_{line_VS}	Line regulation voltage	$V_{pre} = 5.6 - 18\text{ V}$ $I_{VSENS} = 50\text{ mA}$ $C_{trk} = 470\text{ nF}$ (2)	-15		30	mV	VSENSE1/2/3	P_6.3.4
V_{load_VS}	Load regulation voltage	$V_{pre} = 6\text{ V}$ $1\text{ mA} < I_{VSENS} < 150\text{ mA}$ $C_{trk} = 470\text{ nF}$ Note(2)	-15		30	mV	VSENSE1/2/3	P_6.3.5
V_{load_tran}	Transient load regulation	$V_{pre} = 6\text{ V}$ $I_{VSENS} = 1\text{ mA}$ to 150 mA and viceversa $dI/dt = 150\text{ mA}/\mu\text{s}$ $C_{trk} = 470\text{ nF}$ Guaranteed by design(3)	-15		+15	%	VSENSE1/2/3	P_6.3.6
V_{line_tran}	Transient	V_{PRE} step 5.6V to 7V	-15		+15	%	VSENSE1/2/3	P_6.3.7

	line regulation	and Viceversa $dV_{PRE}/dt=3V/\mu s$ $I_{VSENS}=1mA, 150mA$ $C_{trk}=470nF$ Guaranteed by design(4),(3)						
I_{sink_VS}	Short circuit reverse current	Output shorted to $V_{B_IN}+2V$	-	-	-4	mA	$V_{SENSE1/2/3}$	P_6.3.8
$V_{SCB_TH_VS}$	Short to battery threshold	-	V_{pre}		$V_{pre}+0.2$	V	$V_{SENSE1/2/3}$	P_6.3.9
$V_{UV_VS_l}$	$V_{SENSE1/2/3}$ undervoltage hysteresis threshold low	-	4.5	4.65	4.75	V	$V_{SENSE1/2/3}$	P_6.3.10
$V_{UV_VS_h}$	$V_{SENSE1/2/3}$ undervoltage hysteresis threshold high	-	4.7	4.8	4.9	V	$V_{SENSE1/2/3}$	P_6.3.11
V_{OV_VS}	$V_{SENSE1/2/3}$ overvoltage threshold	-	5.3	5.4	5.5	V	$V_{SENSE1/2/3}$	P_6.3.12
$PSRR_{VS}$	Supply voltage tracking rejection	$f=10\text{ kHz}$			-35	dB	$V_{SENSE1/2/3}$	P_6.3.13
V_{OS_VS}	Over shoot during power up	-			5.5	V	$V_{SENSE1/2/3}$	P_6.3.14
$T_{UV_filter_VS_running}$	Undervoltage filter time(2)	cover by SCAN TEST	16	24	32	μs	$V_{SENSE1/2/3}$	P_6.3.15
$T_{OV_filter_VS_running}$	Overvoltage filter time(2)	cover by SCAN TEST	16	24	32	μs	$V_{SENSE1/2/3}$	P_6.3.16
$T_{OC_filter_VS_running}$	Overcurrent filter time	cover by SCAN TEST	16	24	32	μs	$V_{SENSE1/2/3}$	P_6.3.17
$T_{msd_V_{PRE_H}}$	Thermal shutdown_High	Comparator threshold and functionality are tested in production	185		200	$^{\circ}C$	$V_{SENSE1,2,3}$	P_6.3.18
$T_{msd_V_{PRE_L}}$	Thermal shutdown_Low	Comparator threshold and functionality are tested in production	175		190	$^{\circ}C$	$V_{SENSE1,2,3}$	P_6.3.19
T_{msd_hyst}	Thermal hysteresis for shutdown	-	5		10	$^{\circ}C$	$V_{SENSE1,2,3}$	P_6.3.20
$t_{msd_pre_an}$	Thermal shutdown analog filter time	Guaranteed by design	1.5		4	μs	$V_{SENSE1,2,3}$	P_6.3.21
$t_{msd_pre_dig}$	Thermal shutdown digital filter time	Implemented in VHDL	10	20	30	μs	$V_{SENSE1,2,3}$	P_6.3.22

1. No oscillation expected with 0mA load current.

2. line and load regulation are guaranteed until the loop is able to regulate after they are defined by low drop parameters; $I_{TH_VS} < I_{MAX_VS}$ which is guaranteed by design; Over shoot during power up should not be triggered over voltage

warning. When Vtrk output is higher than VSCB_TH_VS, The Vtrk output structure will be switched off automatically in order to block the big inrush current into chip through Vtrk circuit. When Vtrk output is higher than VOV_VS, an over voltage flag will be set. UV/OV fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

3. No UV/OV violation is expected during specified transient.

4. Specification valid until Tracking regulator is in regulation range, or, in other words, until Vsense X output voltage is determined by the effect of voltage regulation loop and not by secondary effects on other regulators.

Table 6-6 VT rack external components characteristics

Symbol	Parameter	Min	Typ	Max	Unit	Note	Pin	Number
CVS1/2/3	VSENSE external capacitor	470nF -20%	0.47	20	μF	ESR_max<20 mΩ Suggest part numbers: GCM188R71C105KA49D	VSENSE1/2/3	P_6.3.23

6.4 External tracking regulator monitor

Vsense4_mon is an input pin for monitoring an external tracking regulator output voltage.

Conditions:

5.5 V < VB_IN_SW ≤ 18 V, T_j -40 to 175 °C unless otherwise specified.

Table 6-7 External VT rack monitor

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
V _{UV_VS_L}	VSENSE4 undervoltage threshold low	-	4.55	4.65	4.75	V	VSENSE4_MON	P_6.4.1
V _{UV_VS_h}	VSENSE4 undervoltage threshold high	-	4.7	4.8	4.95	V	VSENSE4_MON	P_6.4.2
V _{OV_VS}	VSENSE4 overvoltage threshold	-	5.3	5.4	5.55	V	VSENSE4_MON	P_6.4.3
T _{UV_filter_VS_running} (1)	Undervoltage filter time	Covered by SCAN TEST	16	24	32	μs	VSENSE4_MON	P_6.4.4
T _{OV_filter_VS_running} (1)	Overvoltage filter time	Covered by SCAN TEST	16	24	32	μs	VSENSE4_MON	P_6.4.5

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

6.5 VDD5 linear regulator with external MOSFET

VDD5 voltage is generated by a linear controller with internal pre-driver and external MOSFET.

Electrical characteristics are available in [Table 6-8](#) and [Table 6-9](#). VB_IN_OV fault can trigger the turn OFF of VDD5 regulator, as specified into reset matrix. This effect can be masked with bit VB_IN_OV_RST_EN in CONFIG_REG 12.

VB_IN_UV fault triggers the turn OFF of VDD5 regulator, as specified into reset matrix. This effect can be masked with bit VDD5_OFF_SEL in CONFIG_REG 11.

VDD5_OV fault has effects into the device specified into reset matrix. These effects can be masked with bit VDD5_OV_RST_EN in CONFIG_REG 12.

Conditions:

5.5 V < VB_IN_SW ≤ 18 V, 5.5 V < VB_IN ≤ 18 V, T_j -40 to 175°C unless otherwise specified.

Table 6-8 VDD5 linear controller pre-driver electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
V _{VDD5}	5 V regulated voltage including static load / line regulation	I _{load} = 10 mA, 1.5 A	4.9	5	5.1	V	VDD5_IN	P_6.5.1
SR _{p-up_VDD5}	Output voltage slew rate at power-up	Cvdd5 = 10 μF; from VDD5*10% to VDD5*90%	2	-	25	V/ms	VDD5_IN	P_6.5.2
I _{VDD5}	Controller stability over VDD5 output current range	-	1.5	-	-	A	VDD5_IN	P_6.5.3
V _{line_VDD5}	Line regulation voltage	5.6 V < VPre < 18 V I _{load} 10mA, 1.5A Cvdd5 = 20 μF ⁽¹⁾	-25	-	25	mV	VDD5_IN	P_6.5.4
V _{load_VDD5}	Load regulation voltage	VPre 5.6 V, 18 V 10 mA < I _{load} < 1.5 A Cvdd5 = 20 μF ⁽¹⁾	-25	-	25	mV	VDD5_IN	P_6.5.5
V _{line_VDD5_tran}	Line transient regulation voltage	Vpre from 5.6 V to 7 V by 1 μs I _{load} = 10 mA, 1200 mA Cvdd5 = 20 μF Ext MOS STD20NF06L Guaranteed by design ⁽¹⁾	-	-	300	mV	VDD5_IN	P_6.5.6
V _{load_VDD5_tran}	Load transient regulation voltage	VPre = 6 V I _{load} from 10 mA to 1200 mA by	-	-	280	mV	VDD5_IN	P_6.5.7

		1.5us Cvdd5=20μF Ext MOS STD20NF06L Guaranteed by design (1)						
V _{drift_Vdd5}	Total VDD5 voltage aging drift	-	-0.8	-	0.8	%	VDD5_IN	P_6.5.8
PSRR _{VDD5}	Supply voltage rejection ratio	Cout = 10 μF, ESR = 0.1Ω Iload = 600 mA f = 10 kHz	-	-	-35	dB	VDD5_IN	P_6.5.9
V _{0S_VDD5}	VDD5 voltage with overshoot admitted at switch on	-	0	0.1	0.2	V	VDD5_IN	P_6.5.10
T _{UV_filter_VDD5}	Undervoltage filter time(1)	-	16	24	32	μs	VDD5_IN	P_6.5.11
T _{OV_filter_VDD5}	Overvoltage filter time(1)	-	16	24	32	μs	VDD5_IN	P_6.5.12
V _{UV_VDD5_L}	VDD5 undervoltage hysteresis threshold low	-	4.5	4.65	4.75	V	VDD5_IN	P_6.5.13
V _{UV_VDD5_H}	VDD5 undervoltage hysteresis threshold high	-	4.7	4.8	4.9	V	VDD5_IN	P_6.5.14
V _{OV_VDD5}	VDD5 overvoltage threshold	-	5.25	5.4	5.5	V	VDD5_IN	P_6.5.15
V _{VDD5_GATE_Min}	Pre-driver gate voltage	VPre = 4.8 V to force regulator in low drop condition	8	-	-	V	VDD5_GATE	P_6.5.16
V _{VDD5_GATE_MAX}	Pre-driver gate voltage	Force 5 mA into VDD5_GATE	-	-	15	V	VDD5_GATE	P_6.5.17
V _{Gs OFF condition}	VDD5_GATEVDD5_IN	Device off	-	-	0.5	V	VDD5_GATE	P_6.5.18

1. line and load regulation are guaranteed until the loop is able to regulate after they are defined by low drop parameters; Over shoot during power up should not be triggered over voltage warning. UV/OV fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase. With external MOS different from STD20NF06L regulated voltage could violate UV threshold but for a time shorter than digital glitch, so it would not be able to trigger UV/OV fault. With STD20NF06L MOS no UV/OV violation is expected on this and other regulators.

Table 6-9 Linear controller pre-driver. External components characteristics

Symbol	Parameter	Note	Min	Typ	Max	Unit	Pin	Number
C _{VDD5}	External VDD5 capacitor	20 μF to guarantee ESR = 50 mΩfor load transient response	5	30	60	μF	VDD5_IN	P_6.5.19
-	External n-MOS	-	-	-	-	-	VDD5_IN, VDD5_GATE	P_6.5.20
-	R _{gate_source}	-	-	1	-	MΩ	-	P_6.5.21
g _{fs}	Forward Transconductance	VDS = 25 V, ID = 10 A	4.5	-	-	S	-	P_6.5.22

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Multi-voltage regulator supplies

VGS(th)	Gate Threshold Voltage	VDS = VGS, ID = 250uA	2.0	-	4.0	V	-	P_6.5.23
Ciss	input capacitance of external FET	-	-	370	1020	pF	VDD5_GATE	P_6.5.24

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6.6 VDD_IO

VDD_IO pin is the external supply pin for internal I/O circuit.

Table 6-10 VDD_IO electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
I _{VDD_IO}	Input current of VDD_IO	VB=14V, all VDD_IO related pin without load	2	4.8	8	mA	VDD_IO	P_6.6.1
V _{VDD_IO}	Operation range	-	3.1	-	5.5	V	VDD_IO	P_6.6.2
VDDIO_UV_LOW	Under voltage threshold on pin VDDIO	-	2.9	3.0	3.1	V	VDD_IO	P_6.6.3
T _{f_VDD_IO_UV}	Under voltage filter time ⁽¹⁾	Tested by scan	3	-	10	μs	VDD_IO	P_6.6.4

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

6.7 VB_IN and VB_IN_SW SUPPLY

VB_IN pin is the power supply pin for internal circuit.

During cranking phase even the VB_IN is = 4.25 V the chip can keep the status, the internal POR is not triggered and also the internal supply under voltage is not triggered (internal 5V regulator undervoltage), the PHOLD timer value is also kept.

Table 6-11 VB_IN electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
I_{VB_IN}	VB_IN current in functional mode	-	-	-	80	mA	VB_IN	P_6.7.1
$I_{VB_IN_STBY}$	VB_IN current in OFF state.	-	-	-	13	μ A	VB_IN	P_6.7.2
$I_{VB_IN_SW}$	VB_IN_SW current in functional mode	VB_IN=VB_IN_S W=13.5V, Vpre load = 500mA	200	-	350	mA	VB_IN_SW	P_6.7.3
$V_{OFF_VB_IN}$	Min VB_IN voltage to keep the MRD, start relay status	-	-	-	4.25	V	VB_IN	P_6.7.4
$V_{UV_VB_IN_L}$	VB_IN undervoltage hysteresis threshold low	-	4.7	4.8	4.9	V	VB_IN	P_6.7.5
$V_{UV_VB_H}$	VB_IN undervoltage hysteresis threshold high	-	4.75	4.9	5.05	V	VB_IN	P_6.7.6
$V_{UV_VB_IN_HYST}$	-	-	-	0.1	-	V	VB_IN	P_6.7.7
$V_{OV_VB_IN_L}$	VB_IN over voltage threshold low	-	28	-	-	V	VB_IN	P_6.7.8
$V_{OV_VB_IN_H}$	VB_IN over voltage threshold high	-	-	-	32	V	VB_IN	P_6.7.9
$V_{OV_VB_IN_HYST}$	-	-	-	2	-	V	VB_IN	P_6.7.10
t_{VBOV1}	Filter time1 for VB_IN overvoltage(1)	Tested by scan	70	85	100	μ s	VB_IN	P_6.7.11
t_{VBOV2}	Filter time2 for VB_IN overvoltage(1)	Tested by scan	11	15	19	ms	VB_IN	P_6.7.12
t_{VBUV1}	Filter time 1 for VB_IN undervoltage $V_{UV_VB_IN_L}$ (1)	Tested by scan	5	-	15	μ s	VB_IN	P_6.7.13
t_{VBUV2}	Filter time 2 for VB_IN undervoltage $V_{UV_VB_IN_H}$ (1)	Tested by scan	5	-	15	μ s	VB_IN	P_6.7.14
$V_{UV_VB_CP_ON}$	VB_IN threshold for	Charge Pump ON	3.75	4	4.15	V	VB_IN	P_6.7.15

	main logic reset							
V _{UV_VB_CP_OFF}	VB_IN threshold for main logic reset	Charge Pump OFF	5.4	5.5	5.7	V	VB_IN	P_6.7.16

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

When VB_IN is higher than 4.25 V (V_{off_vb_in}), the starter relay and MRD can hold the state.

VB_IN_SW pin is the dedicated supply pin for pre-buck regulator.

Note: VB_IN_SW must be shorted to VB_IN in PCB layout.

Conditions:

4.8 V ≤ VB_IN ≤ 18 V; T_j -40 to 175° C unless otherwise specified;

Table 6-12 VB_IN_SW electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
I _{VB_IN_sw_STBY}	VB_IN_SW current in OFF state.	-	-	-	10	μA	VB_IN_SW	P_6.7.17

6.8 Charge pump

Charge pump provides a permanent voltage of at least 4.8 V above V_{B_IN} when V_{B_IN} is higher than 5.5 V, the VDD5 linear regulator, H-side switch and so on.

Once V_{B_IN} overvoltage is detected ($V_{B_IN} > V_{OV_VB_IN_H}$ for $t > t_{VBOV2}$), the charge pump will be switched off automatically.

Charge pump can provide voltage of 4V above V_{B_IN} when V_{B_IN} in low power condition ($4.25V < V_{B_IN} < 5.5V$)

This charge pump is for internal use only.

Note: It is forbidden to use this charge pump pin for external load.

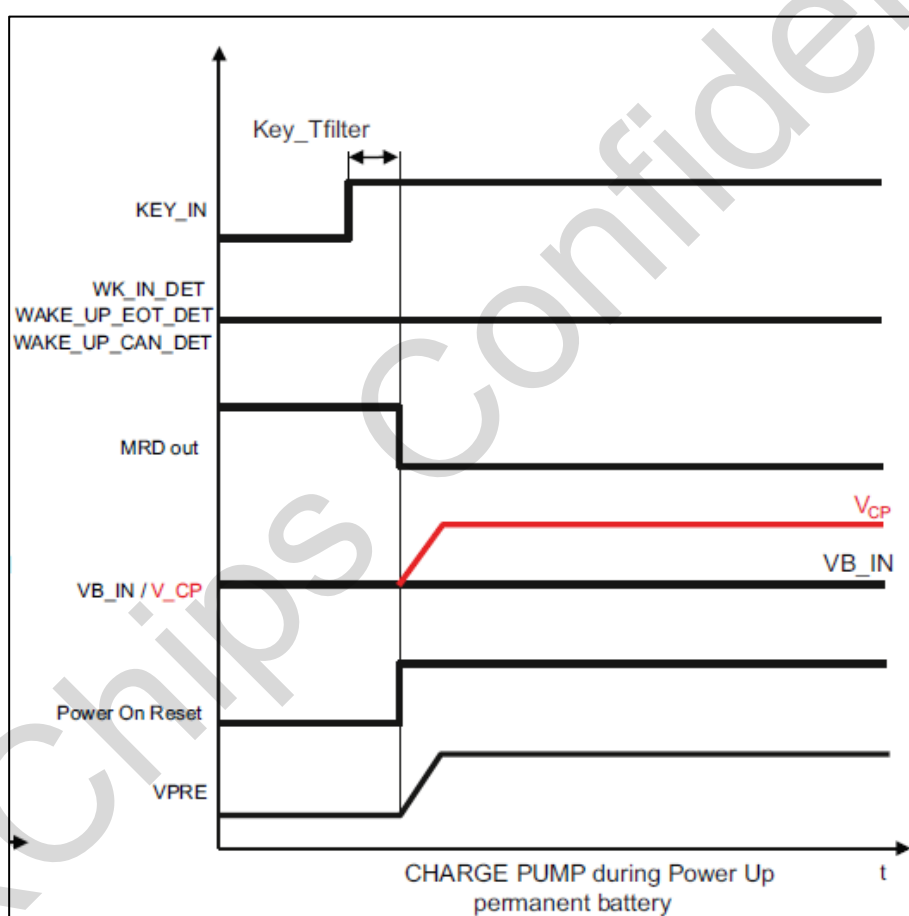


Figure 6-5 Power up charge pump behavior

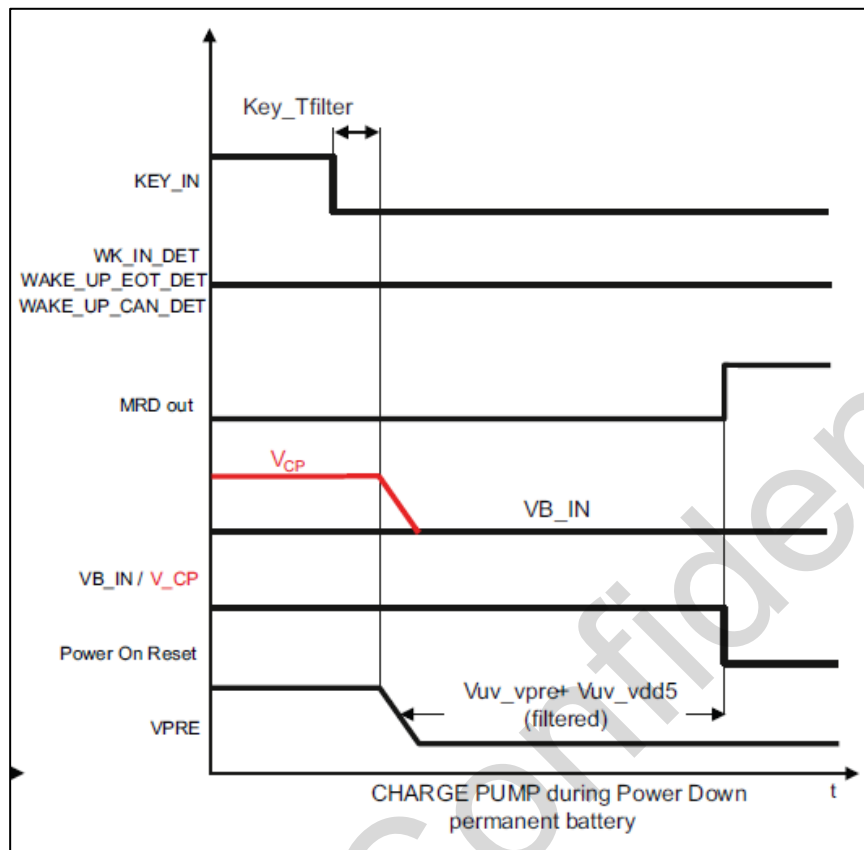


Figure 6-6 Power down charge pump behavior

Conditions:

4.8 V $\leq$ VB_IN $\leq$ 18 V, T_j -40 to 175 °C unless otherwise specified;

Table 6-13 Charge pump characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
CP	External charge Pump capacitor	-	-20%	100	+20%	nF	CP	P_6.8.1
Vcp	Output voltage	VB_IN = 5.5 V-18 V Driver Starter 1 & 2 & 3 configured as HS and switching	VB_IN+4.8V	-	VB_IN+8V	V	CP	P_6.8.2
		VB_IN_OFF(4.25V)<VB_IN<5.5 V with 2 Driver Starter configured as HS and fully ON	VB_IN+4V	-	VB_IN+8V	V	CP	P_6.8.3
Icp_leakage	Leakage current in off state	CP = VB_IN; chip off	-3	-	3	μA	CP	P_6.8.4

6.9 Main oscillator

The device uses an internal oscillator, called MAIN Oscillator, to synchronize internal logic and internal charge pump. A second oscillator is used to check internally if the Main oscillator is working correctly. The Main Oscillator output is continuously verified by the Auxiliary Oscillator output. The two clocks frequency are measured and compared so that out of range frequency values can be detected. In case of frequency error greater than 20% a flag “*freq_err*” (Upstream Read11) is set and read and cleared by MSC.

In case of frequency error greater than 30% another flag “*clk_mon_fault*” (Upstream Read11) is set and read and cleared by MSC, drivers are disabled when this error occurs. The monitor circuit is able also to detect a stuck condition in the reference clock, moreover the check is done on both the main and the auxiliary oscillator so that a failure in any of the two oscillators is able to set the fault flag.

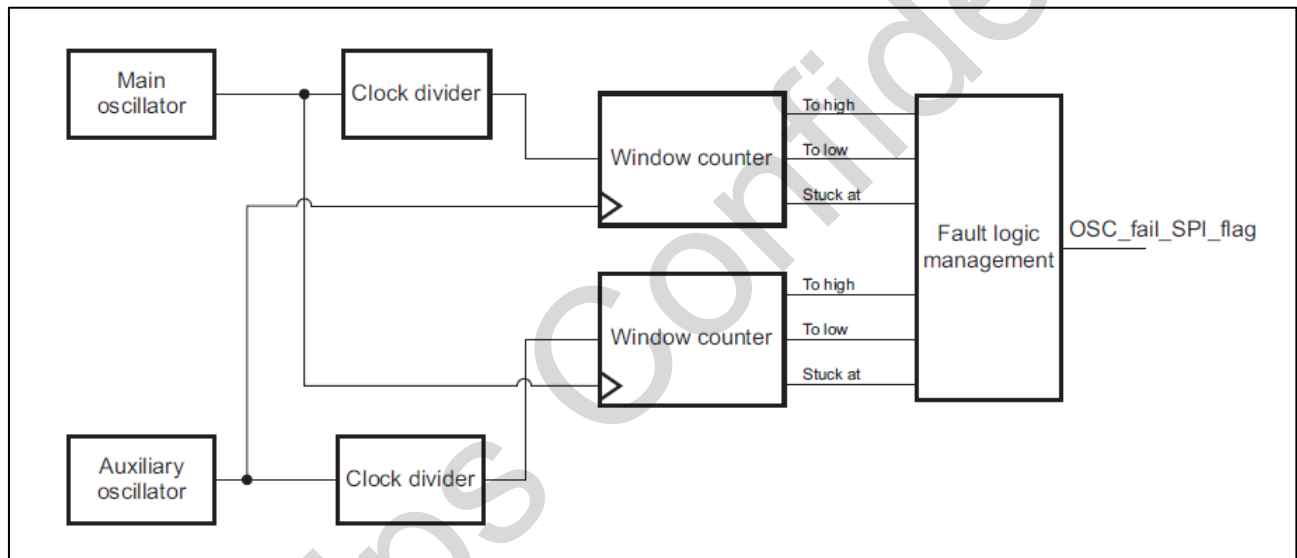


Figure 6-7 Clock monitor block diagram

The Main and Auxiliary oscillators are kept independent using redundant circuits, biased by independent current sources, and layouted in different floor plan regions kept isolated thanks to deep trench usage. The only common point between the oscillators is the digital supply that is however monitored by the independent mechanisms.

Aux oscillator is used just for safety purpose and connected to a small digital portion (no impact on EMC performance). The main oscillator has spread spectrum solution applied by default to reduce emissions; it means that oscillator frequency has an average value equal to MainOSCIL with a triangular modulation applied, so final frequency is $\text{MainOSCIL} \pm 3.5\%$ and a triangular variation of about 111 kHz frequency.

Conditions:

$4.8 \leq V_{B_IN} \leq 18V$, $T_j -40$ to $175\text{ }^{\circ}\text{C}$ unless otherwise specified;

Table 6-14 Clock monitor characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
MainOSCIL	Internal MAIN oscillator frequency	VB_IN = 4.8 V-18 V	14.25	15	15.75	MHz	-	P_6.9.1
MainOSCIL	Internal redundant oscillator frequency	VB_IN = 4.8 V-18 V	-	15	-	MHz	-	P_6.9.2
-	Delta 1 Oscillator CHECK	VB_IN = 4.8 V-18 V "freq_err= 1"	20	-	-	%	-	P_6.9.3
-	Delta 2 Oscillator CHECK	VB_IN=4.8V-18V "FREQ_ERR=1"& "clk_mon_fault=1"(drivers OFF)	30	-	-	%	-	P_6.9.4

7 Low-side drivers

Low-side drivers are used for power stages driving and they include:

- 4-channel Low-side Injectors INJ[1:4];
- 2-channel Low-side O2 Heaters O2H[1:2];
- 2-channel Low-side Solenoids SOL[1:2];
- 5-channel Low-side Relays RLY[1:5];
- 2-channel Low-side LED[1:2].

All drivers support the same failure diagnosis.

7.1 Diagnosis

The driver has the following fault diagnosis:

Overcurrent (Short To Vbat) protection in On Phase (OVC);

Open Load in Off phase (DIAGOL);

Short to Ground in Off Phase (DIAGLV);

Over Temperature Protection in On phase (OT).

When an overcurrent fault is detected the driver switches off with higher slew rate (FASTSR) to reduce the power dissipation.

7.1.1 ON state - overcurrent protection

An overcurrent protection is present for LS transistors of each driver channel. Once the overcurrent fault is detected the faulted power MOS is switched off and a fault bit is set. This bit can be reset by every Read Diag Communication to re-activate the power stage in this two condition:

- The MSC command still High, when the Fault OVC Bit is read by MSC communication.
- The MSC command becomes Low and then again High.

To increase EMC robustness and avoid unwanted fault triggering a small deglitch filter is applied.

In order to avoid false OVC detection during the turn on slew rate phase, an analog blanking time filtering strategy is implemented. This means that the OVC comparator output is masked until the driver gate voltage has reached a threshold value which guarantees that the slew rate phase is ended.

7.1.2 ON state - thermal protection

To protect power stages from temperature overheat (high battery range of operation, soft short conditions, etc.) a dedicated thermal sensor placed close to power MOS is present in the layout sensing FET temperature; One threshold is implemented (T_SD_H) with Hysteresis (T_SD_hys). When the T_SD_H threshold is detected the Driver switches OFF.

Once the power stage has been switched-off for over-temperature detection, it will be able to switch on again when temperature is decreased below thermal shut down threshold plus hysteresis value to avoid high frequency on-off cycling.

7.1.3 ON/OFF state - Error in on status diagnosis

To avoid unwanted driving of power stages the consistency of gate driving signal with digital command is always monitored inside the device. Each time a turn-on/turn-off signal is output by the logic core its effectiveness is verified comparing it with the status of the output voltage.

The main logic can check the consistence between inner logic and actual signal at output pin reading the OpenLoad comparator or Short-To-Ground Comparator according to [Table 7-2](#).

The driver status diagnosis (STA) is activated by the *MSC on command*, the relative comparator is filtered for Tsta filter time to validate the diagnosis. In case of MOS pre-driver the selected blanking time Tblank is applied before filtering.

Table 7-1 Driver status diagnosis

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Tsta	Driver status diagnosis filter time	Tested by scan	30	32.5	35	µs	-	P_7.1.3.1

Table 7-2 STA diagnosis for drivers

STA diagnosis for drivers				
-	INJ,SOL,O2H, RLY,STR, in Low-side configuration	STR in high-side configuration.	MOS pre-driver	IGN
ON state	Detected by STG comparator, if the output pin voltage is higher than VLVT, STA fault is detected	Detected by Open load comparator, if the output pin voltage is lower than V_OL, STA fault is detected.	Detected by STG comparator, if the MOS_DRN pin voltage is higher than VLVT, STA fault is detected.	Same information as STG bit.
OFF state (for LS/HS/PDRV) LS_ON state (for IGN)	Detected by Open load comparator or Short To Ground comparator. If the output pin voltage is lower than V_OL or lower than VLVT STA fault is detected.	Detected by STG comparator, if the output pin voltage is higher than VLVT, STA fault is detected.	Detected by Open load comparator, if the MOS_DRN pin voltage is lower than V_OL, STA fault is detected.	Not able to detect.
STA processing	filtering	filtering	blanking+filtering	filtering

Note: In case of activation of OVT[17] the STA diagnosis is not available.

In case of activation of relative driver OVC the STA diagnosis is available.

7.1.4 OFF state - short load and open load

The device provides off-state diagnosis for each driver channel. Simplified schematic of implemented diagnosis is shown in [Figure 7-1](#).

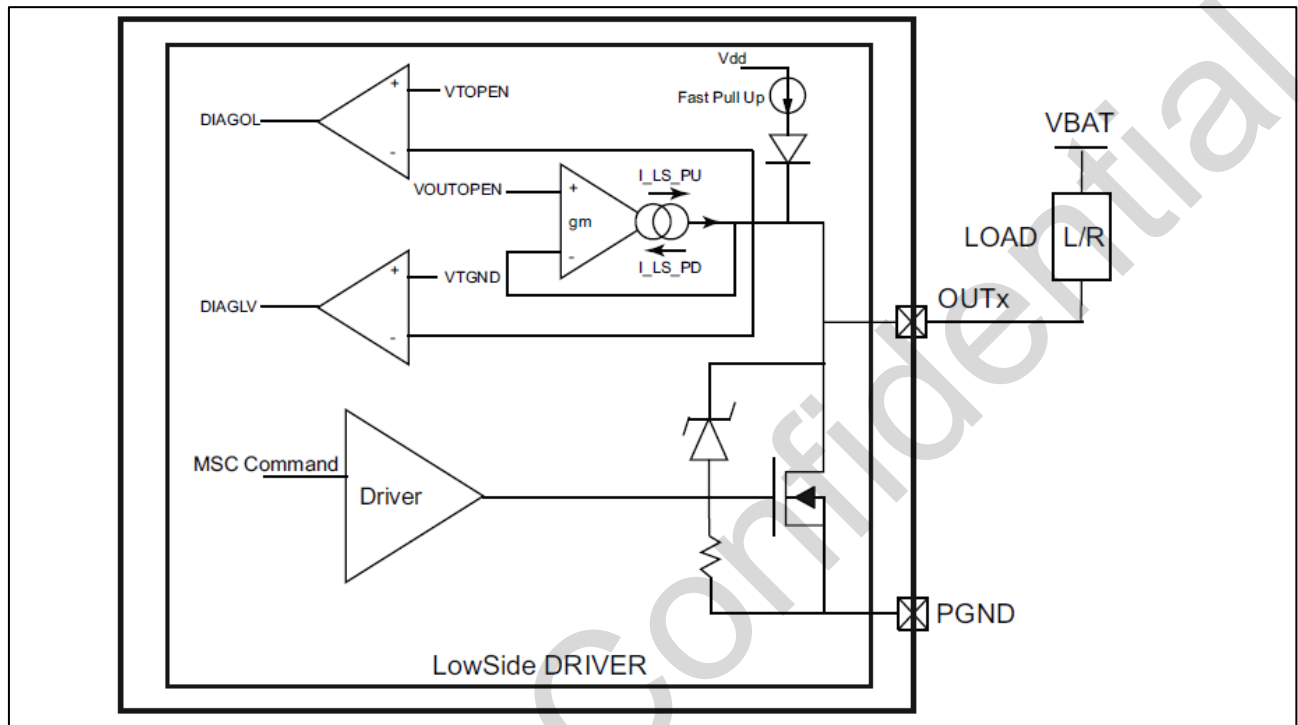


Figure 7-1 Low-side driver OFF state diagnosis schematic

In the Low-Side Driver we can have three different load conditions as shown in [Figure 7-2](#):

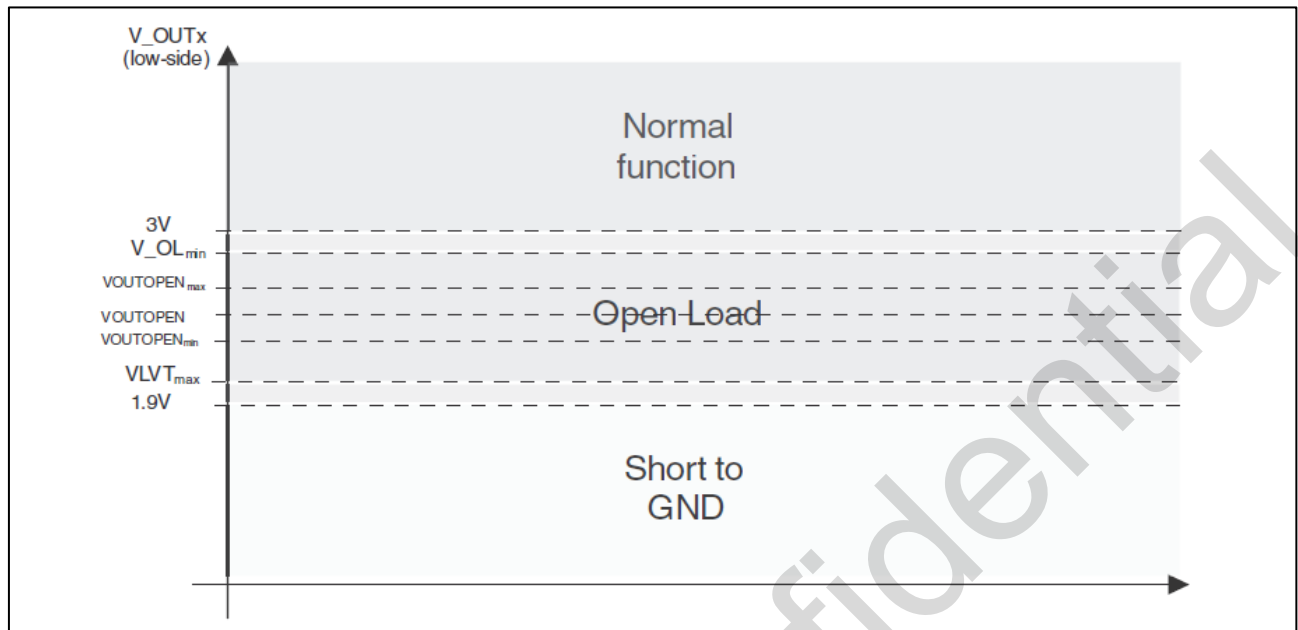


Figure 7-2 Low-side driver OFF state diagnosis load thresholds

Normal Load: it means output driver connected to the load, No Fault Present, $V_{OUT} \geq V_{OL}$ Threshold

Open Load: It means output driver disconnected from the load, Open Load Fault Present, $VLVT \leq V_{OUT} \leq V_{OL}$ Threshold

Short To Ground: it means output driver shorted to GND voltage, Short To Gnd Fault Present, $V_{OUT} \leq VLVT$ Threshold There is a bit *IPUPD_EN* in *CONFIG_REG2* which is controlled by a MSC bit, that it is used

by external μ Controller if it is needed to switch on/off both pull up and pull down diagnosis current.

The Diagnostic in Off state is done by the following blocks:

- An internal Buffer using the pull up/down currents is able to force the *VOUTOPEN* voltage on the Low-Side Driver Output when the driver is in Open Load Condition.
- A comparator to detect the Open load Condition. The Open Load fault is detected when the Low-Side output pin is $VLVT \leq V_{OUT} \leq V_{OL}$ Threshold for a time longer than Diag Filter Time.
- A comparator to detect the Short to GND Condition. The Short to GND fault is detected when the Low-Side output pin is $V_{OUT} \leq VLVT$ Threshold for a time longer than Diag Filter Time.

There is a bit *IPUPD_MODE* in *CONFIG_REG2* which is controlled by a MSC bit, it is used by external

μ Controller if it is needed to switch on/off the bigger diagnosis current (*I_LS_PU1*) in fast mode.

The Fast pull up current is used to avoid the false short to tGND diagnosis when the driver

Low-side drivers switches off in Open load condition. This current switches on after the falling edge of MSC command (driver in off state) and remains on until the Diag Filter Time is expired or until the VOUT is > VLVT Threshold.(note: if fast pull up current is selected, the filter time Tflt_diagoff1 is suggested. Otherwise, the filter time Tflt_diagoff2 is suggested)

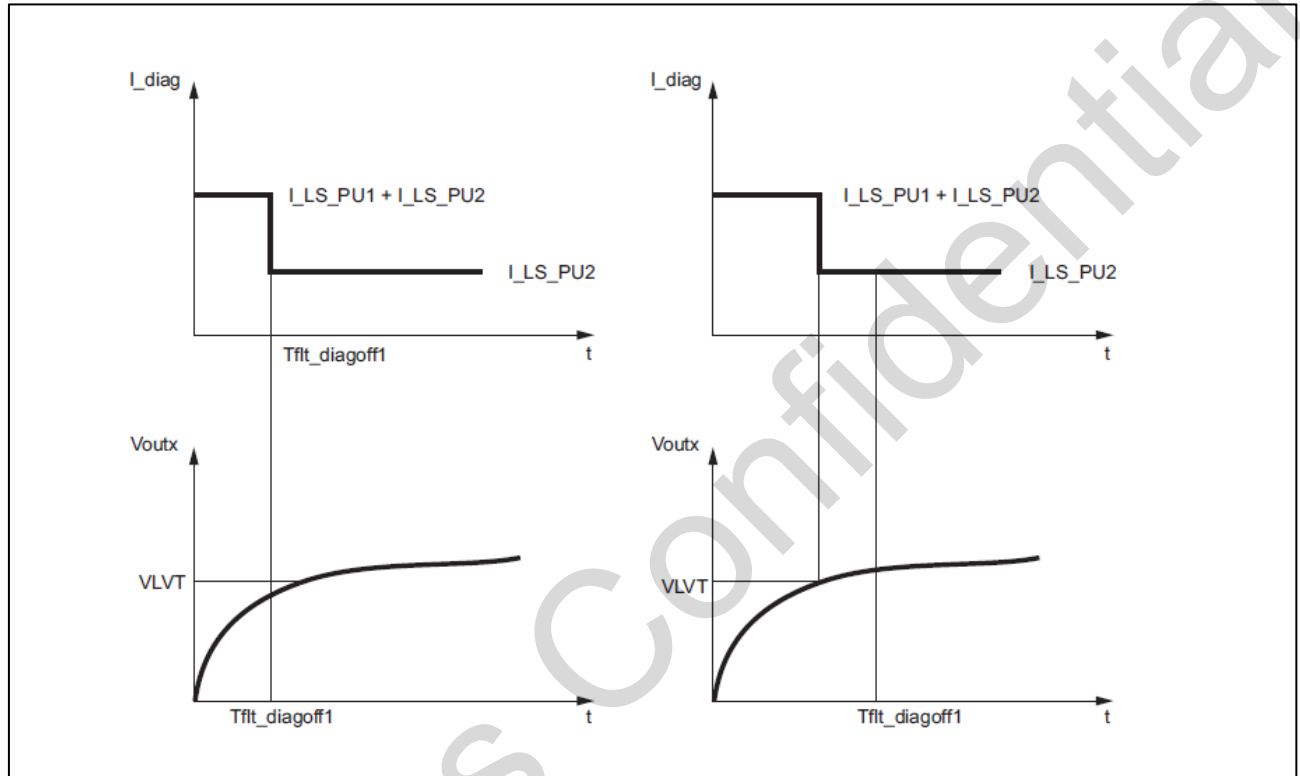


Figure 7-3 Low-side driver OFF state fast pull up current behavior

Despite the fast mode of all drivers in off diagnostic condition (*lpupd_MODE* bit), for Injector and Solenoid drivers there is a dedicated MSC bit (*IDIAG_HIGH_SOL/IDIAG_HIGH_INJ* in *CONFIG_REG13*) for each kind of driver to select off diagnostic high or low pull up/down current

By default diagnostic pull up/down currents are disabled and comparator outputs are masked by internal logic, to enable OFF state diagnostic the channel must be put first in tristate condition and then (if not already done with previous MSC frames) diagnostic must be enabled.

A Filter Time (Diag Filter Time) is implemented in order to avoid to detect false diagnosis as shown in [Figure 7-4](#).

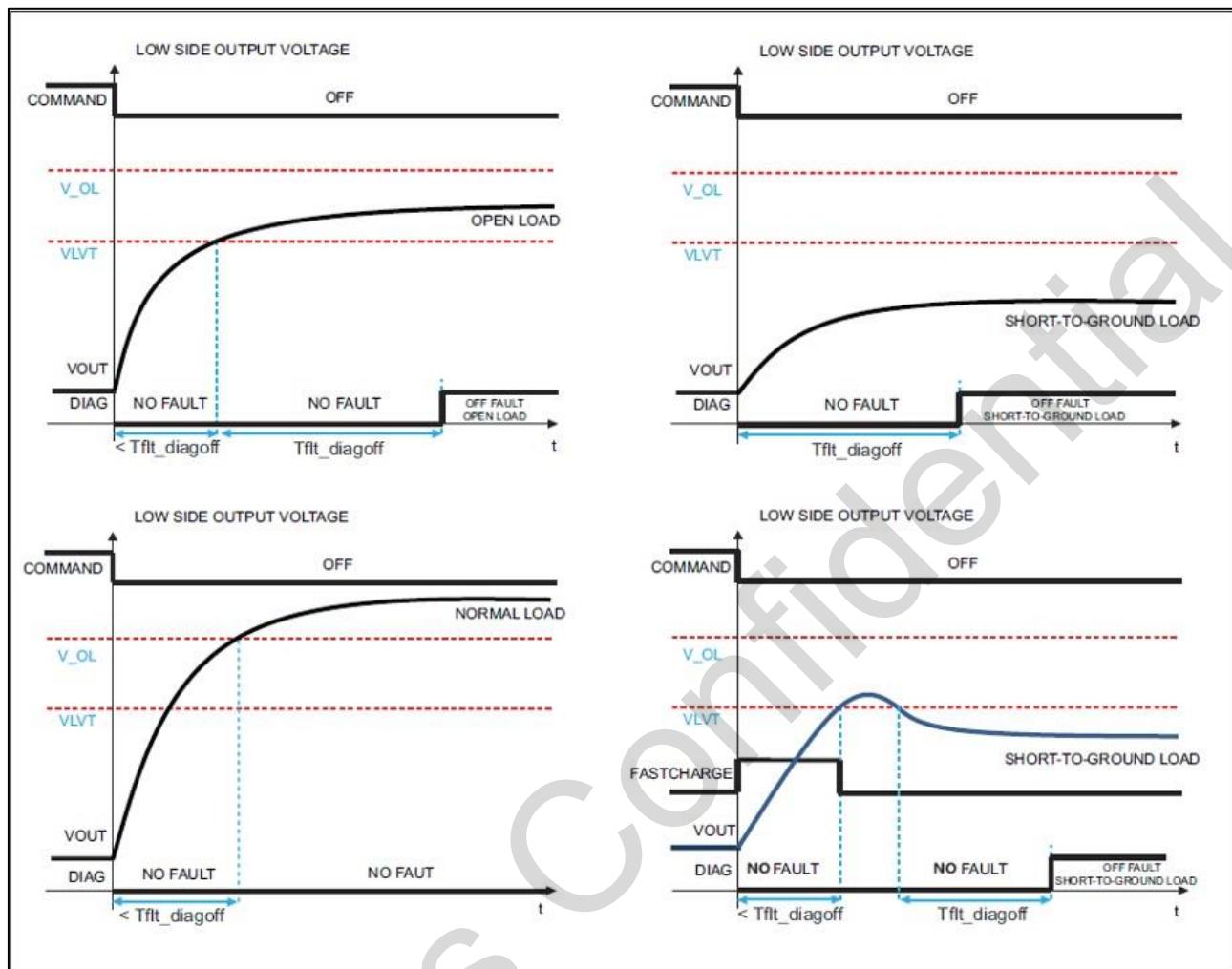


Figure 7-4 Low-side driver OFF state diagnosis timings

If DiagOL or DiagLV signals remain high for a time higher than Diag Filter Time the fault bit is setted and can be read by Read Diag Communication.

This bit is reset by every Read Diag Communication.

7.1.5 Error handling

Table 7-3 Low-side driver error handling

Type of error	Detection condition	Action	Clear MSC flag	Restart condition
Driver Diagnostic				
Overcurrent (OVC)	Driver on	Driver is put in off state. The fault is latched in MSC diagnosis register.	On read	At MSC read of diagnosis register if MSC command is still high. If MSC command is low and high again.
Short to ground	Driver off	The fault is latched	On read	-

(STG)		in MSC diagnosis register.		
Open load (OPL)	Driver off	The fault is latched in MSC diagnosis register.	On read	-
Driver Status Error (STA)	Driver on/off When the driver output level is not aligned with the command on/off	The fault is latched in MSC diagnosis register.	On read	-
Overttemperature (OVT) Shutdown	Driver on	Driver is put in off state. The fault is latched in MSC diagnosis register.	On read	The driver restarts when the temperature decreases. The fault is cleared at MSC read

7.2 Low-side driver - INJECTOR INJ[1:4]

These 4 low-side drivers are designed to driver Injector Load. They are driven by *MSC command*. The output voltage is clamped to voltage limit by internal clamp circuit.

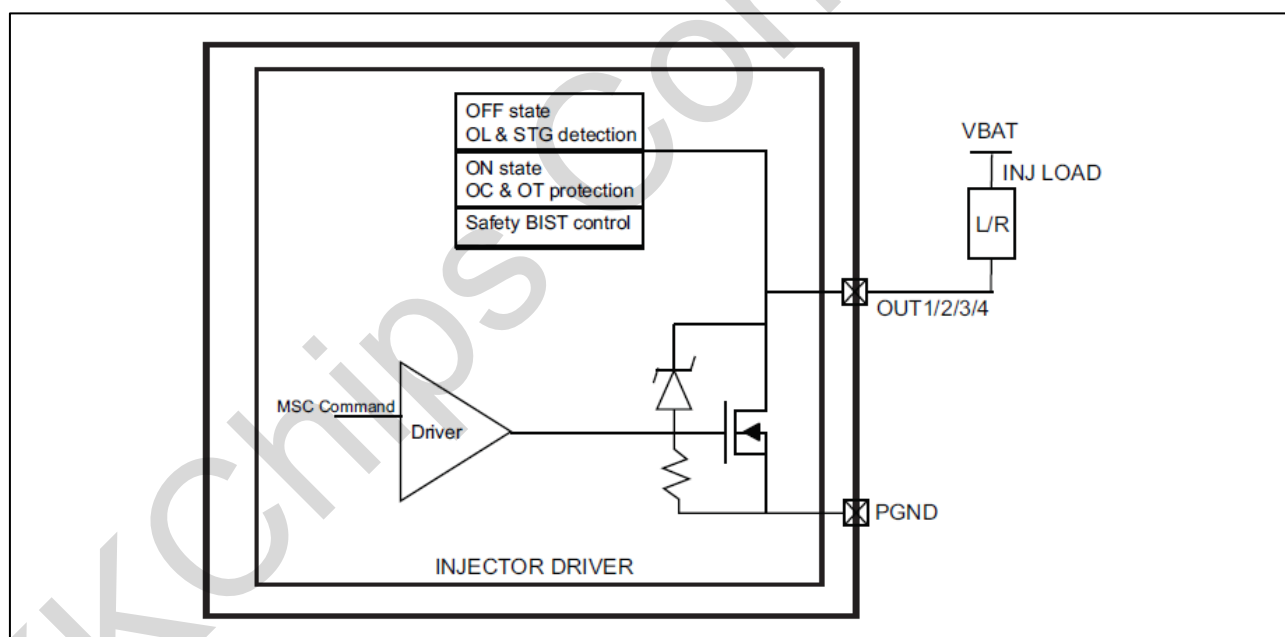


Figure 7-5 INJECTOR LowSide driver stage

Conditions:

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$; T_j -40 to 175 °C unless otherwise specified;

Table 7-4 Low-side driver - INJECTOR electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
INJECTOR								
LS_RdsON	Low-side RdsON	$T_j = 150\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.3	0.44	0.6	Ω	INJ1/2/3/4	P_7.2.1
LS_RdsON	-	$T_j = 25\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.22	0.3	0.45	Ω	INJ1/2/3/4	P_7.2.2
LS_RdsON	-	$T_j = -40\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.15	0.2	0.3	Ω	INJ1/2/3/4	P_7.2.3
OUTx_ikg	Output leakage current	Output disabled, diagnostic off	-10	-	10	μA	INJ1/2/3/4	P_7.2.4
Irev_OUTx	Body Diode reverse current voltage drop	$I_{\text{LOAD}} = -2\text{ A}$	-	-	2	V	INJ1/2/3/4	P_7.2.5
SR_ON	Voltage slew ON State	From 80% to 30% of V_{OUT} , $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 15\Omega$, $C_{\text{load}} = 10\text{ nF}$	0.6	-	1.84	$\text{V}/\mu\text{s}$	INJ1/2/3/4	P_7.2.6
SR_OFF	Voltage slew OFF State	From 80% to 30% of V_{OUT} , $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 15\Omega$, $C_{\text{load}} = 10\text{ nF}$	0.6	-	1.84	$\text{V}/\mu\text{s}$	INJ1/2/3/4	P_7.2.7
S/RGkill_LSH	FAST VS/R off when and OVC fault happens	From 30% to 80% of V_{OUT} , $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 15\Omega$, $C_{\text{load}} = 10\text{ nF}$	5	-	20	$\text{V}/\mu\text{s}$	INJ1/2/3/4	P_7.2.8
Ton_OUTx	Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage	$V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 15\Omega$, $C_{\text{load}} = 10\text{ nF}$	-	-	8	μs	INJ1/2/3/4	P_7.2.9
Toff_OUTx	Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage	$V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 15\Omega$, $C_{\text{load}} = 10\text{ nF}$	-	-	8	μs	INJ1/2/3/4	P_7.2.10
clamp_OUTx	OUTx clamping Voltage	$I_{\text{LOAD}} = 1.3\text{ A}$	50	55	60	V	INJ1/2/3/4	P_7.2.11
Vclamp_OUTx_high_curr	OUTx clamping Voltage	$I_{\text{LOAD}} = 3\text{ A}$ Guaranteed by design, provided single pulse energy limits are not violated by clamping action	50	55	60	V	INJ1/2/3/4	P_7.2.12
IOVC_OUTx	Over Current Driver Threshold	-	3	-	6	A	INJ1/2/3/4	P_7.2.13
I_LS_ocrvflt	LS overcurrent filter time	Guaranteed by scan	4	-	7	μs	INJ1/2/3/4	P_7.2.14
T_SD_HIGH	Temperature shut down	-	185	-	200	$^{\circ}\text{C}$	INJ1/2/3/4	P_7.2.15
T_SD_LOW	Temperature shut down recover	-	175	-	190	$^{\circ}\text{C}$	INJ1/2/3/4	P_7.2.16
T_SD_hys	Temperature shut down hysteresis	-	5	-	10	$^{\circ}\text{C}$	INJ1/2/3/4	P_7.2.17
tmsd_pre_an	Thermal shutdown analog filter time	Guaranteed by design	1.5	-	4	μs	INJ1/2/3/4	P_7.2.18
t_SD_deglitch	Digital deglitch	Guaranteed by design	-	10	-	μs	INJ1/2/3/4	P_7.2.19

	filter time on Temperature shut down detection						4	
Driver Reliability Data								
EnergyRep INJ	Energy repetitive pulse	T _{case} = 30 °C; I _{OUT_n} = 1.8 A, 18 Miopulses	-	-	7.5	mJ	INJ1/2/3/ 4	P_7.2.20
EnergyRep INJ	Energy repetitive pulse	T _{case} = 115 °C; I _{OUT_n} = 1.4 A 648 Miopulses	-	-	4	mJ	INJ1/2/3/ 4	P_7.2.21
EnergyRep INJ	Energy repetitive pulse	T _{case} = 130 °C; I _{OUT_n} = 1 A 96 Miopulses	-	-	3	mJ	INJ1/2/3/ 4	P_7.2.22
EnergyRep INJ	Energy repetitive pulse	T _{case} = 140 °C; I _{OUT_n} = 1 A 4 Miopulses	-	-	3	mJ	INJ1/2/3/ 4	P_7.2.23
Driver Reliability Data Generator Defect								
EnergyRep INJ	Energy repetitive pulse	T _{case} = 25 °C ; I _{OUT_n} = 2 A 0.5 Miopulses	-	-	9	mJ	INJ1/2/3/ 4	P_7.2.24
EnergyRep INJ	Energy repetitive pulse	T _{case} = 135 °C; I _{OUT_n} = 1.5 A 0.5 Miopulses	-	-	8	mJ	INJ1/2/3/ 4	P_7.2.25
Driver Reliability Data jump start								
EnergyRep INJ	Energy repetitive pulse	T _{case} = 25 °C; I _{OT_n} = 3.0 A MAX.0.021Mio cycles 10 jumps starts over lifetime, each start < 2min	-	-	17.5	mJ	INJ1/2/3/ 4	P_7.2.26
EnergyRep INJ	Energy repetitive pulse	T _c = 75 °C ; I _{OT_n} = 2.3A MAX.0.021Mio cycles 10jumps starts over lifetime, each start < 2min	-	-	10	mJ	INJ1/2/3/ 4	P_7.2.27
OFF state diagnostic								
VLVT	Short to GND threshold voltage	Driver tristate, diag enabled	1.9	-	V _{OUT} OPEN -180 mV	V	INJ1/2/3/ 4	P_7.2.28
V_OL	Open load threshold voltage	Driver tristate, diag enabled	V _{OUT} OPEN +160 mV	-	3	V	INJ1/2/3/ 4	P_7.2.29
VOUTOPEN	Open load threshold voltage	Driver tristate, diag enabled	2.3	2.5	2.7	V	INJ1/2/3/ 4	P_7.2.30
I_LS_PU1	Diagnostic pull up current	In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "1"	2.5	3.6	4.7	mA	INJ1/2/3/ 4	P_7.2.31
I_LS_PU2	Diagnostic pull up low current	In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "0" and IDIAG_HIGH_INJ = "0"	40	70	100	μA	INJ1/2/3/ 4	P_7.2.32
I_LS_PU3	Diagnostic pull up high current	In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "0" and IDIAG_HIGH_INJ = "1"	100	-	200	μA	INJ1/2/3/ 4	P_7.2.33
I_LS_PD1	Diagnostic pull	In OFF condition,	60	85	110	μA	INJ1/2/3/	P_7.2.34

Low-side drivers

	down high current	OUTx > VOUTOPEN, Ipupd_EN = ENB IDIAG_HIGH_INJ = "0"					4	
I_LS_PD2	Diagnostic pull down high current	In OFF condition, OUTx > VOUTOPEN, Ipupd_EN = ENB IDIAG_HIGH_INJ = "1"	325	-	550	μA	INJ1/2/3/ 4	P_7.2.35
Tflt_diagoff1	DIAG Filter time	Filter Mode = 0	75	100	125	μs	INJ1/2/3/ 4	P_7.2.36
Tflt_diagoff2	DIAG Filter time	Filter Mode = 1	450	600	750	μs	INJ1/2/3/ 4	P_7.2.37
-	Minimum OFF time for correct diagnostic	Application note (esd cap < 10nF) Filter Mode=0	175	-	-	μs	ESVx	P_7.2.38

7.2.1 Enable pin INJ_ENA

The Pin INJ_ENA is a digital input pin that is used as enable of Injector driver. If this pin is at low digital level the internal logic drives all injector drivers in OFF state. If this pin is at high digital, all injector drivers can be switched on by *dedicated MSC bit*. This pin can be used also as a SEO function for injector driver. If the application doesn't use this enable function, the INJ_ENA pin have to be pulled up externally to VDD5 by a resistor of 10 kΩ

Conditions:

5.5 V ≤ VB_IN ≤ 18 V, T_j -40 to 175 °C unless otherwise specified.

Table 7-5 INJ_ENA electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VIH_Tx	HIGH_level input voltage	-	1.75	-	-	V	INJ_ENA	P_7.2.1
VIL_Tx	LOW_level input voltage	-	-	-	0.75	V	INJ_ENA	P_7.2.2
VHYS_Tx	Input voltage hysteresis	-	0.1	0.5	-	V	INJ_ENA	P_7.2.3
PDOWN_Tx	Pull-down resistance	-	50	100	200	kΩ	INJ_ENA	P_7.2.4

7.3 Low-side driver - O2 HEATER O2H[1:2]

These 2 LowSide driver are designed to driver O2 heater Load. They are driven by *MSC command*. The Output voltage is clamped to voltage limit by internal clamp circuit. O2 channel can be configured as valve driver with faster slew rate and lower over current threshold by MSC commands. Current sense generates on Curr_sense_O2H pin a reduced copy of the current flowing through O2H Power. Current sense current generator is done with a P-channel current mirror connected to VDD5_IN, for this reason voltage on Curr_sense_O2H1 and Curr_sense_O2H2 pins can never exceed VDD5_IN.

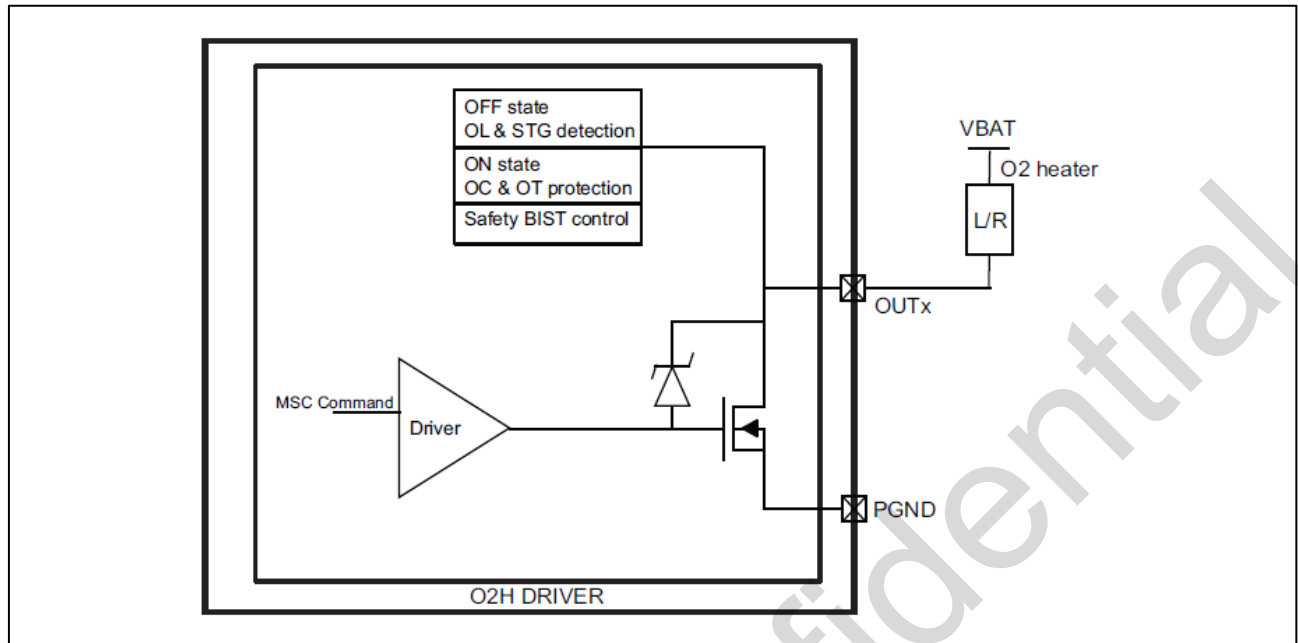


Figure 7-6 O2 heater LowSide driver stage

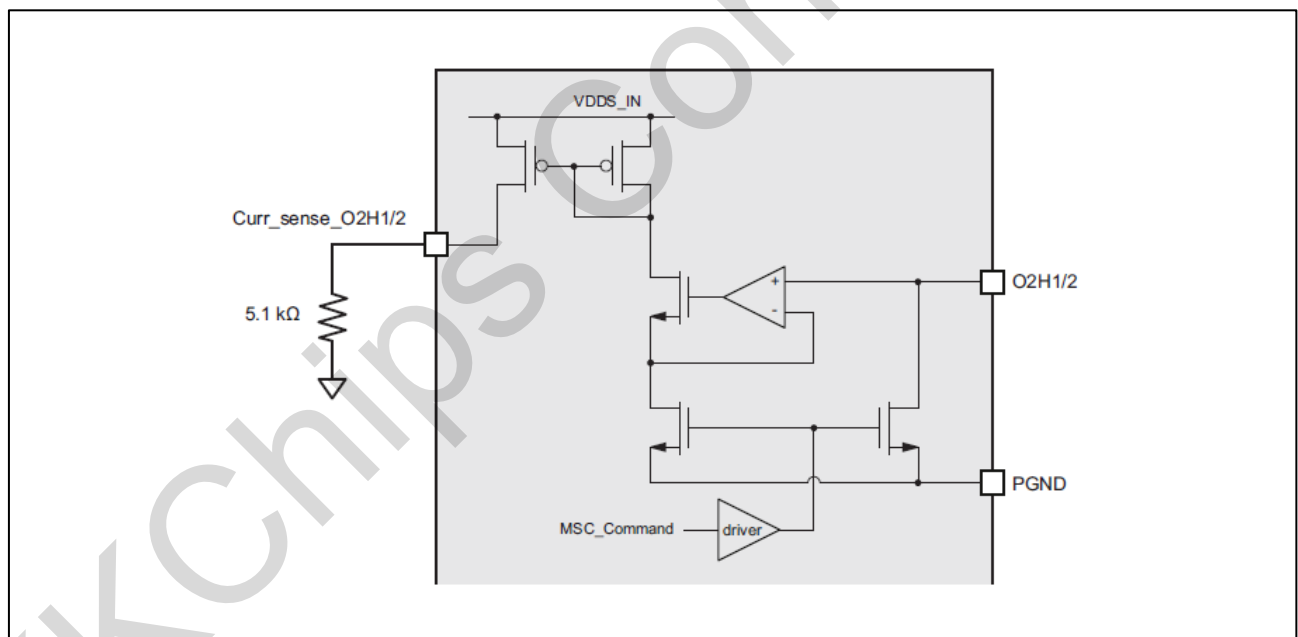


Figure 7-7 O2 Heater LowSide driver current sense block

Conditions:

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified.

Table 7-6 Low-side driver - O2 HEATER electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
O2H								
LS_RdsON	Low-side RdsON	$T_j = 150\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.04	0.16	0.23	Ω	O2H1/2	P_7.3.1
		$T_j = 25\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.04	0.12	0.2	Ω	O2H1/2	P_7.3.2
		$T_j = -40\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.03	0.9	0.13	Ω	O2H1/2	P_7.3.3
OUTx_lkg	Output leakage current	Output disabled, diagnostic off $V_{\text{pin}} = 13.5\text{ V}$	-10	-	+10	μA	O2H1/2	P_7.3.4
Irev_OUTx	Body diode reverse current voltage drop	$I_{\text{LOAD}} = -2\text{ A}$	-	-	2	V	O2H1/2	P_7.3.5
SR_ON	Voltage slew ON State	From 80% to 30% of VOUT $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 3\text{ }\Omega$, $C_{\text{load}} = 10\text{ nF}$; configure as O2	0.2	-	0.8	V/ μs	O2H1/2	P_7.3.6
SR_ON_fast	Fast voltage slew ON State	From 80% to 30% of VOUT $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 3\text{ }\Omega$, $C_{\text{load}} = 10\text{ nF}$; configure as sol, fast slew rate	2	-	6	V/ μs	O2H1/2	P_7.3.7
SR_OFF	Voltage slew OFF state	From 30% to 80% of VOUT $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 3\text{ }\Omega$, $C_{\text{load}} = 10\text{ nF}$; configure as O2	0.2	-	0.8	V/ μs	O2H1/2	P_7.3.8
SR_OFF_fast	Fast Voltage slew OFF State	From 30% to 80% of VOUT $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 3\text{ }\Omega$, $C_{\text{load}} = 10\text{ nF}$; configure as sol, fast slew rate	2	-	6	V/ μs	O2H1/2	P_7.3.9
S/RGkill_LSH	FAST VS/R off when an OVC fault happens	From 30% to 80% of VOUT $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 3\text{ }\Omega$, $C_{\text{load}} = 10\text{ nF}$	5	-	20	V/ μs	O2H1/2	P_7.3.10
Ton_OUTx	Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage	$V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 3\text{ }\Omega$, $C_{\text{load}} = 10\text{ nF}$	-	-	12	μs	O2H1/2	P_7.3.11
Toff_OUTx	Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage	$V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 3\text{ }\Omega$, $C_{\text{load}} = 10\text{ nF}$	-	-	10	μs	O2H1/2	P_7.3.12
Vclamp_OUTx	OUTx clamping Voltage	$I_{\text{LOAD}} = 1.3\text{ A}$	45	50	55	V	O2H1/2	P_7.3.13
Vclamp_OUTx_high_curr	OUTx clamping Voltage	$I_{\text{LOAD}} = 1.3\text{ A}$ Guaranteed by design, provided single pulse energy limits are not violated by clamping action	45	50	55	V	O2H1/2	P_7.3.14
I_ovc1_o2h	Over current driver	configure as sol	3	4.5	6	A	O2H1/2	P_7.3.15

Low-side drivers

	threshold 1							
I_ovc2_o2h	Over current driver threshold 2	T _j = -40 °C configured as O2	8.6	-	12.4	A	O2H1/2	P_7.3.16
I_ovc2_o2h	Over current driver threshold 2	T _j = +25 °C configured as O2	8.0	-	11.2	A	O2H1/2	P_7.3.17
I_ovc2_o2h	Over current driver threshold 2	T _j = +150 °C configured as O2	7.8	-	10.5	A	O2H1/2	P_7.3.18
I_LS_ocvflt	LS overcurrent filter time	Guaranteed by scan	4	-	7	µs	O2H1/2	P_7.3.19
T_SD_HIGH	Temperature shut down	-	185	-	200	°C	O2H1/2	P_7.3.20
T_SD_LOW	Temperature shut down recover	-	175	-	190	°C	O2H1/2	P_7.3.21
T_SD_hys	Temperature shut down hysteresis	-	5	-	10	°C	O2H1/2	P_7.3.22
tmsd_pre_an	Thermal shutdown analog filter time	Guaranteed by design	1.5	-	4	µs	O2H1/2	P_7.3.23
t_SD_deglitch	Digital deglitch filter time on Temperature shut down detection	Guaranteed by scan	-	10	-	µs	O2H1/2	P_7.3.24
Driver reliability data								
EnergyRep O2H	Energy repetitive pulses	T _{case} = 25 °C ; I _{OT_n} = 1.5A 18 Miopulses	-	-	15	mJ	O2H1/2	P_7.3.25
EnergyRep O2H	Energy repetitive pulses	T _{case} = 115 °C ; I _{OT_n} = 1.2A 648 Miopulses	-	-	10	mJ	O2H1/2	P_7.3.26
EnergyRep O2H	Energy repetitive pulses	T _{case} = 130 °C ; I _{OT_n} = 1.0A 96 Miopulses	-	-	9	mJ	O2H1/2	P_7.3.27
EnergyRep O2H	Energy repetitive pulses	T _{case} = 140 °C ; I _{OT_n} = 1.0A 4 Miopulses	-	-	9	mJ	O2H1/2	P_7.3.28
Driver Reliability Data jump start								
EnergyRep O2H	Energy repetitive pulses	T _{case} = 25 °C ; I _{OT_n} = 2.2A 0.021 Miopulses 10 jumps starts over lifetime, each start < 2	-	-	30	mJ	O2H1/2	P_7.3.29
EnergyRep O2H	Energy repetitive pulses	T _{case} = 75 °C ; I _{OT_n} = 1.8A 0.021 Miopulses 10 jumps starts over lifetime, each start < 2	-	-	18	mJ	O2H1/2	P_7.3.30
Driver reliability data generator defect								
EnergyRep O2H	Energy repetitive pulses	T _{case} = 25 °C ; I _{OT_n} = 1.6A 0.5 Miopulses	-	-	17.5	mJ	O2H1/2	P_7.3.31
EnergyRep O2H	Energy repetitive pulses	T _{case} = 135 °C ; I _{OT_n} = 1.1 A 0.5 Miopulses	-	-	10	mJ	O2H1/2	P_7.3.32
OFF state diagnostic								
VLVT	Short to GND threshold voltage	Driver tristate, diag enabled	1.9	2.1	VOU TOPE N -180 mV	V	O2H1/2	P_7.3.33
V_OL	Open load threshold voltage	Driver tristate, diag enabled	VOU TOPE N +160 mV	2.9	3.0	V	O2H1/2	P_7.3.34
VOL	Open load voltage	Driver tristate, diag enabled	2.3	2.5	2.7	V	O2H1/2	P_7.3.35
I_LS_PU1	Diagnostic pull up current	In OFF condition, OUTx < VLVT,	2.5	3.6	4.7	mA	O2H1/2	P_7.3.36

Low-side drivers

		lpupd_EN = ENB, lpupd_MODE = "1"						
I_LS_PU2	Diagnostic pull up current	In OFF condition, OUTx < VLVT, lpupd_EN = ENB, lpupd_MODE = "0"	40	70	100	μA	O2H1/2	P_7.3.37
I_LS_PD1	Diagnostic pull down current	In OFF condition, OUTx > VOUTOPEN, lpupd_EN = ENB	60	85	105	μA	O2H1/2	P_7.3.38
Tflt_diagoff1	DIAG Filter time	Filter Mode = 0	75	100	125	μs	O2H1/2	P_7.3.39
Tflt_diagoff2	DIAG Filter time	Filter Mode = 1	450	600	750	μs	O2H1/2	P_7.3.40
-	Minimum OFF time for correct diagnostic	Application note (esd cap < 10 nF,) Filter Mode = 0	175	-	-	μs	O2H1/2	P_7.3.41
Current sensing characteristics								
Gain_sense	Transresistance sense gain	-		0.8		V/A	Curr_Sense O2H1/2	P_7.3.42
V_OUT_0p5A	Output voltage AD for Ipower = 0.5 A	Rext = 5.1 kΩ	0.3	0.4	0.55	V	Curr_Sense O2H1/2	P_7.3.43
V_OUT_1A	Output voltage AD for Ipower = 1 A	Rext = 5.1 kΩ	0.64	0.8	0.96	V	Curr_Sense O2H1/2	P_7.3.44
V_OUT_2A	Output voltage AD for Ipower = 2 A	Rext = 5.1 kΩ	1.44	1.6	1.76	V	Curr_Sense O2H1/2	P_7.3.45
V_OUT_3A	Output voltage AD for Ipower = 3 A	Rext = 5.1 kΩ	2.16	2.4	2.64	V	Curr_Sense O2H1/2	P_7.3.46
V_OUT_4A	Output voltage AD for Ipower = 4 A	Rext = 5.1 kΩ	2.88	3.2	3.52	V	Curr_Sense O2H1/2	P_7.3.47

7.4 Low-side driver - SOLENOID SOL [1:2]

These 2 LowSide drivers are designed to driver Solenoid Valves Load. They are driven by *MSC command*. The Output voltage is clamped to voltage limit by internal clamp circuit.

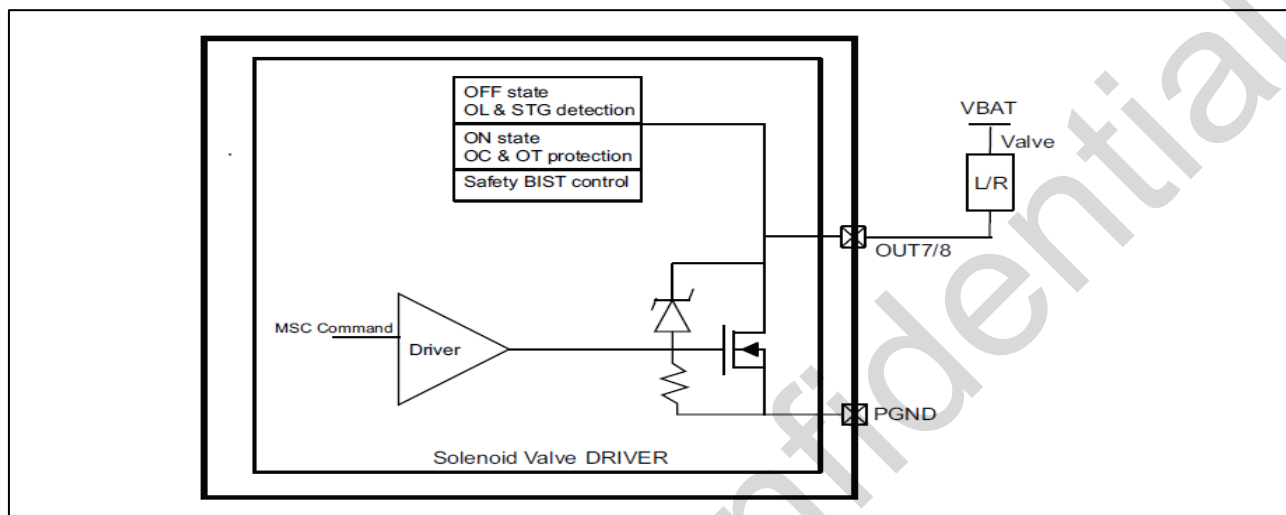


Figure 7-8 Solenoid valves LowSide driver stage

Conditions:

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified.

Table 7-7 Low-side driver - SOLENOID VALVE electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Solenoid valves								
LS_RdsON	Low-side RdsON	$T_j = 150\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.24	0.35	0.47	Ω	SOL1/2	P_7.4.1
		$T_j = 25\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.17	0.22	0.35	Ω	SOL1/2	P_7.4.2
		$T_j = -40\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 3\text{ A}$	0.12	0.17	0.23	Ω	SOL1/2	P_7.4.3
OUTx_lkg	Output leakage current	Output disabled, diagnostic off $V_{\text{pin}} = 13.5\text{ V}$	-10	-	+10	μA	SOL1/2	P_7.4.4
Irev_OUTx	Body diode reverse current voltage drop	$I_{\text{LOAD}} = -2\text{ A}$	-	-	2	V	SOL1/2	P_7.4.5
SR_ON	Voltage slew ON State	From 80% to 30% of VOUT $\text{VB_IN} = 14\text{ V}$, $R_{\text{load}} = 15\Omega$, $C_{\text{load}} = 10\text{ nF}$	0.6	-	1.75	V/ μs	SOL1/2	P_7.4.6
SR_OFF	Voltage slew OFF state	From 30% to 80% of VOUT $\text{VB_IN} = 14\text{ V}$, $R_{\text{load}} = 15\Omega$, $C_{\text{load}} = 10\text{ nF}$	0.6	-	1.75	V/ μs	SOL1/2	P_7.4.7
S/RGkill_LSH	FAST VS/R off	From 30% to 80% of	5	-	20	V/ μs	SOL1/2	P_7.4.8

Low-side drivers

	when an OVC fault happens	VOUT VB_IN = 14 V, Rload = 15Ω, Cload = 10 nF							
Ton_OUTx	Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage	VB_IN = 14 V, Rload = 15Ω, Cload = 10 nF	-	-	8	μs	SOL1/2	P_7.4.9	
Toff_OUTx	Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage	VB_IN = 14 V, Rload = 15Ω, Cload = 10 nF	-	-	8	μs	SOL1/2	P_7.4.10	
Vclamp_OUTx	OUTx clamping Voltage	ILOAD = 1.3 A	50	55	60	V	SOL1/2	P_7.4.11	
Vclamp_OUTx_high_curr	OUTx clamping Voltage	ILOAD = 3 A Guaranteed by design, provided single pulse energy limits are not violated by clamping action	50	55	60	V	SOL1/2	P_7.4.12	
I_ovc_sol	Over current driver threshold	-	3	4.5	6	A	SOL1/2	P_7.4.13	
I_LS_ovcflt	LS overcurrent filter time	Guaranteed by scan	4	-	7	μs	SOL1/2	P_7.4.14	
T_SD_HIGH	Temperature shut down	-	185	-	200	°C	SOL1/2	P_7.4.15	
T_SD_LOW	Temperature shut down recover	-	175	-	190	°C	SOL1/2	P_7.4.16	
T_SD_hys	Temperature shut down hysteresis	-	5	-	10	°C	SOL1/2	P_7.4.17	
-	Thermal shutdown analog filter time	Guaranteed by design	1.5	-	4	μs	SOL1/2	P_7.4.18	
t_SD_deglitch	Digital deglitch filter time on Temperature shut down detection	Guaranteed by scan	-	10	-	μs	SOL1/2	P_7.4.19	
Driver reliability data									
Energy_Rep_SOL	Energy repetitive pulses	Tcase = 25 °C ; I_OT_n = 1.5A 18 Miopulses	-	-	15	mJ	SOL1/2	P_7.4.20	
Energy_Rep_SOL	Energy repetitive pulses	Tcase = 115 °C ; I_OT_n = 1.2A 648 Miopulses	-	-	10	mJ	SOL1/2	P_7.4.21	
Energy_Rep_SOL	Energy repetitive pulses	Tcase = 130 °C ; I_OT_n = 1.0A 96 Miopulses	-	-	9	mJ	SOL1/2	P_7.4.22	
Energy_Rep_SOL	Energy repetitive pulses	Tcase = 140 °C ; I_OT_n = 1.0A 4 Miopulses	-	-	9	mJ	SOL1/2	P_7.4.23	
Driver Reliability Data jump start									
Energy_Rep_SOL	Energy repetitive pulses	Tcase = 25 °C ; I_OT_n = 2.2A 0.021 Miopulses 10 jumps starts over lifetime, each start < 2 min	-	-	30	mJ	SOL1/2	P_7.4.24	
Energy_Rep_SOL	Energy repetitive pulses	Tcase = 75 °C ; I_OT_n = 1.8A 0.021 Miopulses 10 jumps starts over lifetime, each start < 2 min	-	-	18	mJ	SOL1/2	P_7.4.25	
Driver reliability data generator defect									
EnergyRep SOL	Energy repetitive pulses	Tcase = 25 °C ; I_OT_n = 1.6A 0.5 Miopulses	-	-	17.5	mJ	SOL1/2	P_7.4.26	

EnergyRep SOL	Energy repetitive pulses	T _{case} = 135 °C ; I _{OT_n} = 1.1 A 0.5 Miopulses	-	-	10	mJ	SOL1/2	P_7.4.27
OFF state diagnostic								
VLVT	Short to GND threshold voltage	Driver tristate, diag enabled	1.9	2.1	VOU TOPE N - 180m V	V	SOL1/2	P_7.4.28
V_OL	Open load threshold voltage	Driver tristate, diag enabled	VOU TOPE N +160 mV	2.9	3.0	V	SOL1/2	P_7.4.29
VOL	Open load voltage	Driver tristate, diag enabled	2.3	2.5	2.7	V	SOL1/2	P_7.4.30
I_LS_PU1	Diagnostic pull up current	In OFF condition, OUTx < VLVT, lpupd_EN = ENB, lpupd_MODE = "1"	2.5	3.6	4.7	mA	SOL1/2	P_7.4.31
I_LS_PU2	Diagnostic pull up current	In OFF condition, OUTx < VLVT, lpupd_EN = ENB, lpupd_MODE ="0" and IDIAG_HIGH_SOL ="0"	40	70	100	μA	SOL1/2	P_7.4.32
I_LS_PU3	Diagnostic pull up high current	In OFF condition, OUTx < VLVT, lpupd_EN=ENB,lpupd_MO DE="0" and IDIAG_HIGH_SOL ="1"	100		200	μA	SOL1/2	P_7.4.33
I_LS_PD1	Diagnostic pull down current	In OFF condition, OUTx > VOUTOPEN, lpupd_EN = ENB and IDIAG_HIGH_SOL ="0"	60	85	105	μA	SOL1/2	P_7.4.34
II_LS_PD2		In OFF condition, OUTx > VOUTOPEN, lpupd_EN = ENB IDIAG_HIGH_SOL ="1"	325	-	550	μA	SOL1/2	P_7.4.35
Tflt_diagoff1	DIAG Filter time	Filter Mode = 0	75	100	125	μs	SOL1/2	P_7.4.36
Tflt_diagoff2	DIAG Filter time	Filter Mode = 1	450	600	750	μs	SOL1/2	P_7.4.37
-	Minimum OFF time for correct diagnostic	Application note (esd cap < 10 nF,) Filter Mode = 0	175	-	-	μs	SOL1/2	P_7.4.38

7.5 Low-side driver - RELAY RLY [1:5]

These 5 LowSide drivers are optimized for relay or generic low current loads driving. They are driven by MSC command. The Output voltage is clamped to voltage limit by internal clamp circuit.

Each relay driver is separately put in on or off state by the *MSC CONFIG-REG 14 [0:4]* just for the time required to perform on or off diagnosis (Fast Diagnosis). At the end of the diagnosis it is automatically put in the state determined by the current value of MSC data frame.

The micro command is activated by writing 1 from 0 in the MSC registers (both for ON and OFF).

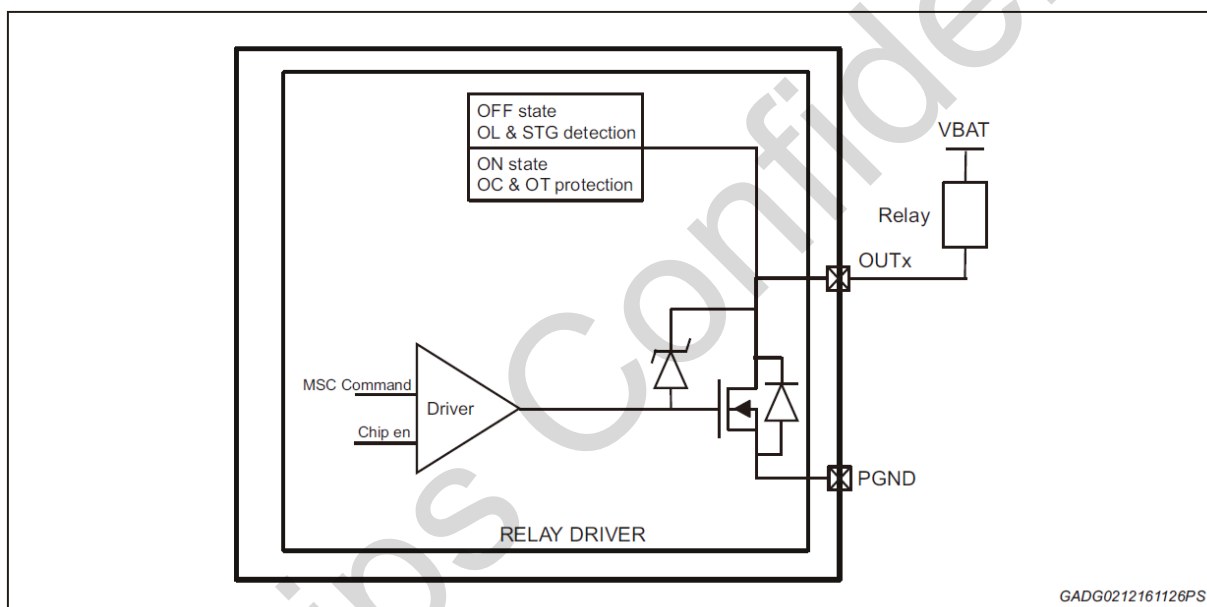


Figure 7-9 Relay LowSide driver stage

Conditions:

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified; the RLY4 can be associated to DELAY_OFF function if the *MSC CONFIG_REG_3 [0] = RLY4_DLY_OFF_EN* is set. In this case the RLY4 is configured as starter and can work down to $\text{VB_IN} = 4.25\text{ V}$ for THOLD.

Table 7-8 Low-side driver - RELAY electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Relay								
LS_RdsON	Low-side RdsON	$T_j = 150\text{ °C}$, $I_{\text{LOAD}} = 1\text{ A}$	-	-	1.5	Ω	RLY/1/2/3/4/5	P_7.5.1
		$T_j = 25\text{ °C}$, $I_{\text{LOAD}} = 1\text{ A}$	-	-	0.82	Ω	RLY/1/2/	P_7.5.2

HE9788N

Multifunction IC for automotive engine management system



Low-side drivers

							3/4/5	
		$T_j = -40\text{ }^{\circ}\text{C}$, $I_{\text{LOAD}} = 1\text{ A}$	-	-	0.63	Ω	RLY/1/2/3/4/5	P_7.5.3
LS_RdsON_RL4	Low-side RdsON	$V_{\text{B_IN}} = 4.25\text{ V}$, $I_{\text{LOAD}} = 0.6\text{ A}$	-	-	3	Ω	RLY4	P_7.5.4
OUTx_ikg	Output leakage current	Output disabled, diagnostic off	-10	-	+10	μA	RLY/1/2/3/4/5	P_7.5.5
Irev_OUTx	Body diode reverse current voltage drop	$I_{\text{LOAD}} = -0.6\text{ A}$	-	-	2	V	RLY/1/2/3/4/5	P_7.5.6
SR_ON	Voltage slew ON State	From 80% to 30% of V_{OUT} , $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 68\Omega$ $C_{\text{load}} = 10\text{ nF}$	0.4	-	2.1	$\text{V}/\mu\text{s}$	RLY/1/2/3/4/5	P_7.5.7
SR_OFF	Voltage slew OFF State	From 30% to 80% of V_{OUT} , $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 68\Omega$ $C_{\text{load}} = 10\text{ nF}$	0.4	-	2.1	$\text{V}/\mu\text{s}$	RLY/1/2/3/4/5	P_7.5.8
S/RGkill	FAST VS/R off when an OVC fault happens	From 30% to 80% of V_{OUT} , $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 68\Omega$ $C_{\text{load}} = 10\text{ nF}$	5	-	20	$\text{V}/\mu\text{s}$	RLY/1/2/3/4/5	P_7.5.9
Ton_OUTx	Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage	V_{OUT} , $V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 68\Omega$ $C_{\text{load}} = 10\text{ nF}$	-	-	10	μs	RLY/1/2/3/4/5	P_7.5.10
Toff_OUTx	Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage	$V_{\text{B_IN}} = 14\text{ V}$, $R_{\text{load}} = 68\Omega$ $C_{\text{load}} = 10\text{ nF}$	-	-	10	μs	RLY/1/2/3/4/5	P_7.5.11
Vclamp_OUTx	OUTx clamping Voltage	$I_{\text{LOAD}} = 0.6\text{ A}$	45	-	55	V	RLY/1/2/3/4/5	P_7.5.12
Vclamp_OUTx_high_curr	OUTx clamping Voltage	$I_{\text{LOAD}} = 1\text{ A}$ Guaranteed by design, provided single pulse energy limits are not violated by clamping action	45	-	55	V	RLY/1/2/3/4/5	P_7.5.13
I_ovc_rly	Over current driver threshold	-	1	1.5	2	A	RLY/1/2/3/4/5	P_7.5.14
I_LS_ocr_ft	LS overcurrent filter time	Guaranteed by scan	4	-	7	μs	RLY/1/2/3/4/5	P_7.5.15
T_SD_HIGH	Temperature shut down	-	185	-	200	$^{\circ}\text{C}$	RLY/1/2/3/4/5	P_7.5.16
T_SD_LOW	Temperature shut down recover	-	175	-	190	$^{\circ}\text{C}$	RLY/1/2/3/4/5	P_7.5.17
T_SD_hys	Temperature shut down hysteresis	-	5	-	10	$^{\circ}\text{C}$	RLY/1/2/3/4/5	P_7.5.18
-	Thermal shutdown analog filter time	Guaranteed by design	1.5		4	μs	RLY/1/2/3/4/5	P_7.5.19
t_SD_deglitch	Digital deglitch filter time on Temperature shut down detection	Guaranteed by scan	-	10	-	μs	RLY/1/2/3/4/5	P_7.5.20
Driver reliability data								

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Low-side drivers

EnergyRep RLY	Energy repetitive pulses	T _{case} = 25 °C; I _{OT_n} = 0.45 A 1.1 Miopulses	-	-	9	mJ	RLY/1/2/ 3/4/5	P_7.5.21
EnergyRep RLY	Energy repetitive pulses	T _{case} = 115 °C; I _{OT_n} = 0.3 A 40 Miopulses	-	-	6.5	mJ	RLY/1/2/ 3/4/5	P_7.5.22
EnergyRep RLY	Energy repetitive pulses	T _{case} = 130 °C; I _{OT_n} = 0.3 A 9 Miopulses	-	-	6.5	mJ	RLY/1/2/ 3/4/5	P_7.5.23
EnergyRep RLY	Energy repetitive pulses	T _{case} = 140 °C; I _{OT_n} = 0.3 A 1 Miopulses	-	-	6.5	mJ	RLY/1/2/ 3/4/5	P_7.5.24
Driver reliability data generator defect								
EnergyRep RLY	Energy repetitive pulses	T _{case} = 25 °C; I _{OT_n} = 0.5 A 0.02 Miopulses	-	-	11	mJ	RLY/1/2/ 3/4/5	P_7.5.25
EnergyRep RLY	Energy repetitive pulses	T _{case} = 135 °C; I _{OT_n} = 0.35 A 0.02 Miopulses	-	-	8	mJ	RLY/1/2/ 3/4/5	P_7.5.26
Driver reliability data JUMP START								
EnergyRep RLY	Energy repetitive pulses	T _{case} = 25 °C; I _{OT_n} = 0.75 A 0.001 Miopulses, 10 jumps starts over lifetime, each start < 2 min	-	-	25	mJ	RLY/1/2/ 3/4/5	P_7.5.27
EnergyRep RLY	Energy repetitive pulses	T _c =75°C ; I _{OT_n} =0.5A 0.001Miopulses, 10 jumps starts over lifetime, each start < 2 min	-	-	17	mJ	RLY/1/2/ 3/4/5	P_7.5.28
OFF state diagnostic								
VLVT	Short to GND threshold voltage	Driver tristate, diag enabled	1.9		VOU TOPE N - 180m V	V	RLY/1/2/ 3/4/5	P_7.5.29
V_OL	Open load threshold voltage	Driver tristate, diag enabled	VOU TOPE N +160 mV		3.0	V	RLY/1/2/ 3/4/5	P_7.5.30
VOUTOPEN	Open load output voltage	Driver tristate, diag enabled	2.3	2.5	2.7	V	RLY/1/2/ 3/4/5	P_7.5.31
I_LS_PU1	Diagnostic pull up current	In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "1"	2.3	3.6	4.7	mA	RLY/1/2/ 3/4/5	P_7.5.32
I_LS_PU2	Diagnostic pull up current	In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "0"	40	70	100	μA	RLY/1/2/ 3/4/5	P_7.5.33
I_LS_PD1	Diagnostic pull down current	In OFF condition, OUTx > VOUTOPEN, Ipupd_EN = ENB	60	85	105	μA	RLY/1/2/ 3/4/5	P_7.5.34
Tflt_diagoff1	DIAG Filter time	Filter Mode=0	75	100	125	μs	RLY/1/2/ 3/4/5	P_7.5.35
Tflt_diagoff2	DIAG Filter time	Filter Mode=1	450	600	750	μs	RLY/1/2/ 3/4/5	P_7.5.36

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Low-side drivers

-	Minimum OFF time for correct diagnostic	Application note (esd cap < 10 nF,) Filter Mode = 0	175	-	-	μs	RLY/1/2/3/4/5	P_7.5.37
-	Fast OFF diagnostic ON→OFF→ON	Filter Mode=0	175	200	225	μs	RLY/1/2/3/4/5	P_7.5.38
-	Fast OFF diagnostic OFF→ON→OFF	-	72		90	μs	RLY/1/2/3/4/5	P_7.5.39

7.6 Low-side driver - LED[1:2]

These 2 LowSide drivers are designed to drive LED Load. They are driven by *MSC command*. The Output voltage is clamped to voltage limit by internal clamp circuit.

The pull down diagnostic current of LED driver is configurable by MSC (*LEDx_PD_EN CONFIG_REG1 D3;D4*). Overcurrent and Short to Ground diagnostic function are available, open load diagnostic function is not available when pull down diagnostic current is disabled.

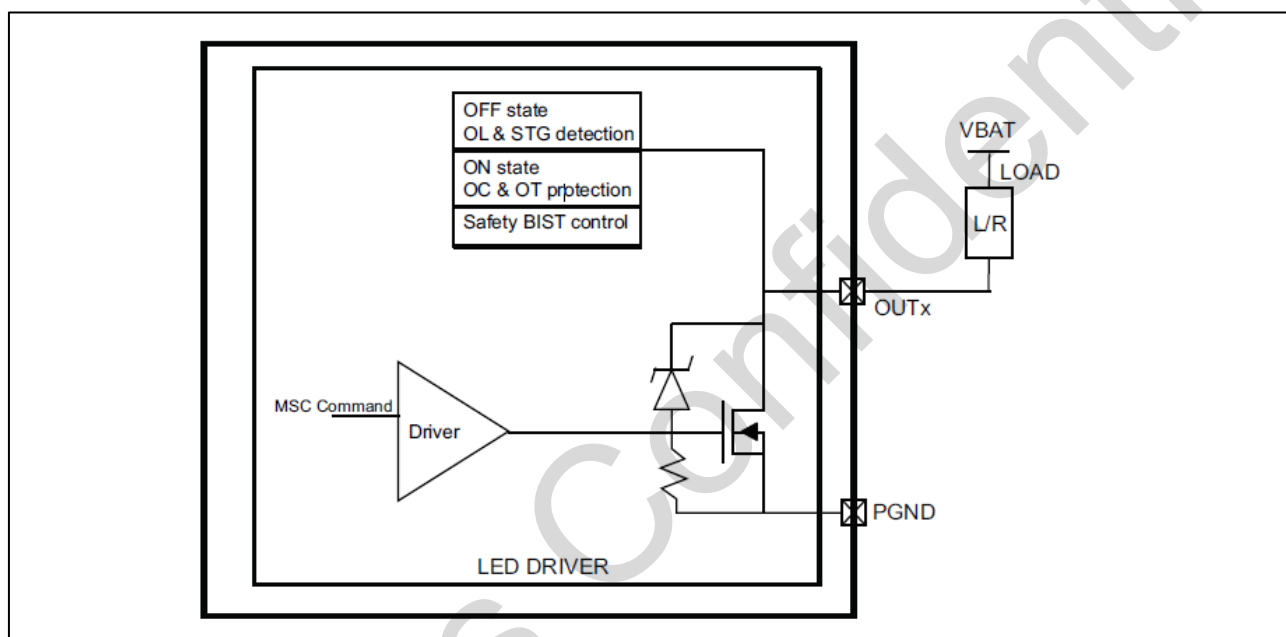


Figure 7-10 LED LowSide driver stage

Conditions:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 7-9 Low-side driver - LED electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
LED								
LS_RdsON	Low-side RdsON	$T_j = 150\text{ }^{\circ}\text{C}$, $I_{LOAD} = 70\text{ mA}$	10	14	20	Ω	LED1/2	P_7.6.1
		$T_j = 25\text{ }^{\circ}\text{C}$, $I_{LOAD} = 70\text{ mA}$	6	8	12	Ω	LED1/2	P_7.6.2
		$T_j = -40\text{ }^{\circ}\text{C}$, $I_{LOAD} = 70\text{ mA}$	4.9	6	9	Ω	LED1/2	P_7.6.3
OUTx_lkg	Output leakage current	Output disabled, diagnostic off	-10	-	+10	μA	LED1/2	P_7.6.4
Irev_OUTx	Body diode reverse current	$I_{LOAD} = -50\text{ mA}$	-	-	2	V	LED1/2	P_7.6.5

Low-side drivers

	voltage drop							
SR_ON	Voltage slew ON state	From 80% to 30% of VOUT VB_IN = 14 V, Rload = 270Ω,	10	-	25	V/μs	LED1/2	P_7.6.6
SR_OFF	Voltage slew OFF state	From 30% to 80% of VOUT VB_IN = 14 V, Rload = 270Ω	14	-	21	V/μs	LED1/2	P_7.6.7
Ton_OUTx	Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage	VB_IN = 14 V, Rload=270Ω,	-	-	5	μs	LED1/2	P_7.6.8
Toff_OUTx	Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage	VB_IN = 14 V, Rload=270Ω,	-	-	5	μs	LED1/2	P_7.6.9
Vclamp_OUTx	OUTx clamping Voltage	ILOAD = 50 mA	40	45	50	V	LED1/2	P_7.6.10
Vclamp_OUTx_high_curr	OUTx clamping Voltage	ILOAD = 70 mA Guaranteed by design, provided single pulse energy limits are not violated by clamping action	40	45	50	V	LED1/2	P_7.6.11
IOVC_OUTx	Over current driver threshold	-	70	-	110	mA	LED1/2	P_7.6.12
I_LS_ocvflt	LS overcurrent filter time	Guaranteed by scan	4	-	7	μs	LED1/2	P_7.6.13
VLVT	Short to GND threshold voltage	Driver tristate, diag enabled	1.9	-	VOU TOPE N - 180mV	V	LED1/2	P_7.6.14
V_OL	Open load threshold voltage	Driver tristate, diag enabled	VOU TOPE N +160 mV	-	3	V	LED1/2	P_7.6.15
VOUTOPEN	Open load voltage	Driver tristate, diag enabled	2.3	2.5	2.7	V	LED1/2	P_7.6.16
I_LS_PU1	Diagnostic pull up current	In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "1"	2.5	3.6	4.7	mA	LED1/2	P_7.6.17
I_LS_PU2	Diagnostic pull up current	In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "0"	40	70	100	μA	LED1/2	P_7.6.18
I_LS_PD1	Diagnostic pull down current	In OFF condition, OUTx > VOUTOPEN, Ipupd_EN = ENB LEDx_PD_EN = 1	60	85	105	μA	LED1/2	P_7.6.19
Tflt_diagoff1	DIAG filter time	Filter Mode = 0	75	100	125	μs	LED1/2	P_7.6.20
Tflt_diagoff2	DIAG filter time	Filter Mode = 1	450	600	750	μs	LED1/2	P_7.6.21

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Low-side drivers

-	Minimum OFF time for correct diagnostic	Application note (esd cap < 10nF,) Filter Mode=0	175	-	-	μs	LED1/2	P_7.6.22
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8 High/low-side drivers – STARTER STR[1:3]

These 3 high/low-side drivers with diagnosis and overcurrent protection have a floating architecture and can be used in high-side or low-side mode.

The configuration is defined by MSC bit (*LSD/HSD_DRVx-CFG CONFIG_REG3 D1~D3*).

The driver is optimized for relay and low-current loads, and it can be associated to smart starter functional block.

They are driven by MSC command. The Output voltage is clamped to voltage limit by internal clamp circuit. In order to guarantee a negligible output current in case of LED application an internal dedicated comparator is present. This comparator, in case of highside configuration, Diagoff disabled and channel OFF, determines the disabling of internal driver in case of output voltage (STR_SRC) pin in the range of $-0.4V_{typ} < STR_SRC < 1.8V_{typ}$. This implementation guarantees a negligible output current down to 0V on STR_SRC pin (where internal compensation is no more effective) and at the same time the correct reenabling of the driver when the output voltage goes negative, in order to guarantee the correct clamping functionality during recirculation. In any other condition (Low-side configuration, or Diagoff enabled, or channel ON) the effect of this comparator is masked and an output current will be observed.

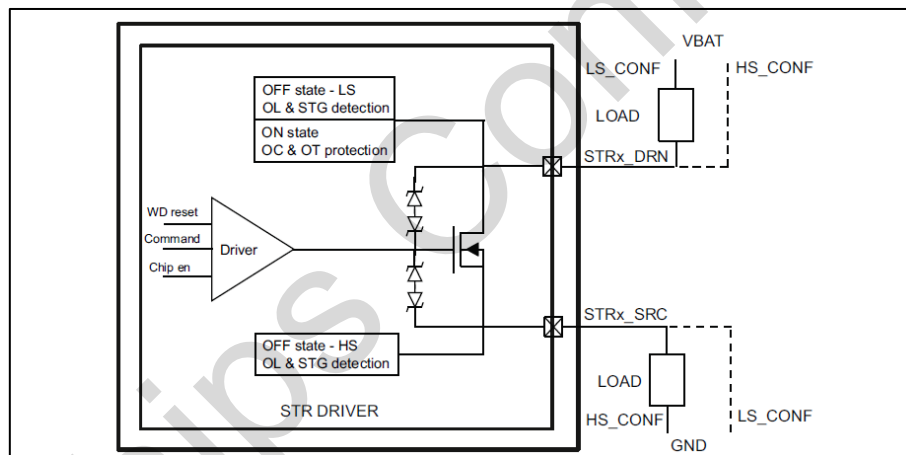


Figure 8-1 Configurable high/low-side Driver block schematic

The driver has the following detection diagnosis in low-side mode:

- Overcurrent (Short to Vbat) protection in On Phase (OVC);
- Open Load in Off phase (DIAGOL);
- Short to Ground in Off Phase (DIAGLV);
- Over Temperature Protection in On phase (OT);
- The driver has the following detection diagnosis in high-side mode:
- Overcurrent (Short To GND) protection in On Phase (OVC);
- Open Load in Off phase (DIAGOL);
- Short to battery in Off Phase (DIAGHV);
- Over Temperature Protection in On phase (OT);

High/low-side drivers - STARTER STR[1:3]

In low-side configuration, if they are configured as starter `CONFIG_REG3[4]=STR2_EN=1` and `CONFIG_REG3[5]=STR3_EN=1`, the driver STR2 and STR3 can work down to `VB_IN = 4.25 V` for THOLD.

In high-side configuration, if they are configured as starter `CONFIG_REG3[4]=STR2_EN=1` and `CONFIG_REG3[5]=STR3_EN=1`, the driver STR2 and STR3 can work down to `VB_IN = 4.25 V` for THOLD.

Furthermore, the driver can latch its ON state for THOLD even if the main microcontroller is temporarily not working due to low-battery conditions.

HS/LS are intended to be used as starter relay driver. One for high-side, one for low-side. STR2/3 should keep the status till `VOFF_VB_IN` (4.25 V).

Each starter driver is separately put in on or off state by the *MSC CONFIG-REG 14[5:7]* just for the time required to perform on or off diagnosis (Fast Diagnosis). At the end of the diagnosis it is automatically put in the state determined by the current value of *MSC data frame*.

The micro command is activated by writing 1 from 0 in the MSC registers (both for ON and OFF).

8.1 ON state diagnosis

The ON state diagnosis has the same features as the low-side driver ON state diagnosis described in chapter [Section 7.1.1](#) and [Section 7.1.2](#).

8.2 ON/OFF state - Error in on status diagnosis

Refer to paragraph [Section 7.1.3](#)

8.3 OFF state diagnosis LS mode

The OFF state diagnosis in LS mode has the same features as the low-side driver OFF state diagnosis described in paragraph [Section 7.1.4](#).

8.4 OFF state diagnosis HS mode

The device provides off-state diagnostic for HS mode, simplified schematic of implemented diagnostic is shown in [Figure 8-2](#).

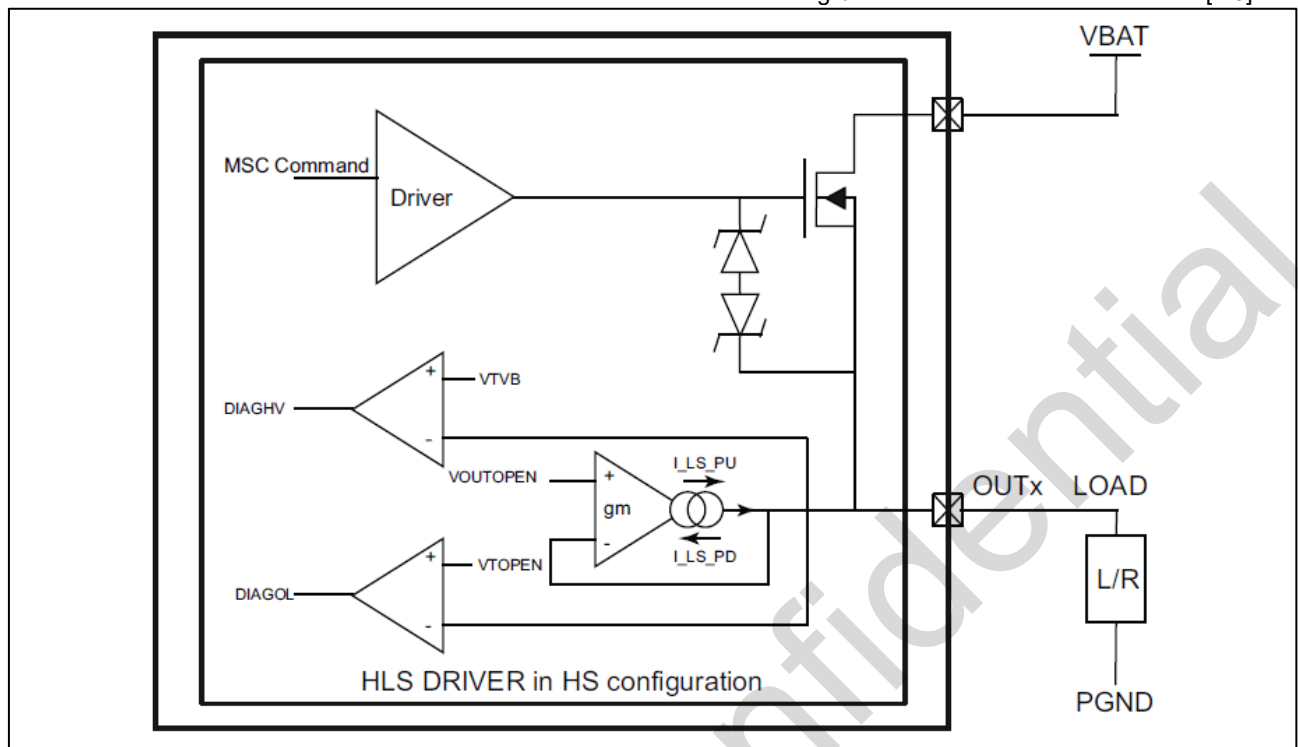


Figure 8-2 OFF state diagnostic HS config block schematic

In the HS mode we can have three different load conditions:

- Normal Load, it means output driver connected to the load, No Fault Present, $V_{OUT} \leq V_{OL}$ Threshold;
- Open Load, It means output driver disconnected to the load, Open Load Fault Present, $V_{HVT} \geq V_{OUT} \geq V_{OL}$ Threshold;
- Short To VB, it means output driver shorted to VB voltage, Short To VB Fault Present, $V_{OUT} > V_{HVT}$.

There is a bit *IPUPD_EN_STRx* in CONFIG_REG13 which is controlled by a MSC bit, it is used by external μ Controller if it is needed to switch on/off both pull up and pull down diagnosis current.

There is a bit *IPUPD_MODE* in CONFIG_REG2 which is controlled by a MSC bit, it is used by external

μ Controller if it is needed to switch on/off the bigger diagnosis current (I_{HS_PD1}) in fast mode.

The Diagnostic in Off state is done by the following block:

- An internal Buffer using the pull up/down currents is able to force the VOUTOPEN voltage on the High-Side Driver Output when the driver is in Open Load Condition.
- A comparator to detect the Open load Condition. The Open Load fault is detected when the OUTx pin is $V_{HVT} \geq V_{OUT} \geq V_{OL}$ Threshold for a time longer than Diag Filter Time.
- A comparator to detect the Short to VB Condition. The Short To VB fault is detected

High/low-side drivers - STARTER STR[1:3]

when the OUTx pin is $V_{OUT} \geq V_{HVT}$ Threshold for a time longer than Diag Filter Time.

- Fast pull down current, this current is used to avoid the false short to VB diagnosis when the driver switches off in Open load condition. This current switches on after the falling edge of *MSC command* (driver in off state) and remain on until the Diag Filter Time is expired or until the V_{OUT} is $< V_{HVT}$ Threshold. (Note: if fast pull down current is selected, the filter time $T_{flt_diagoff1}$ is suggested. Otherwise, the filter time $T_{flt_diagoff2}$ is suggested).

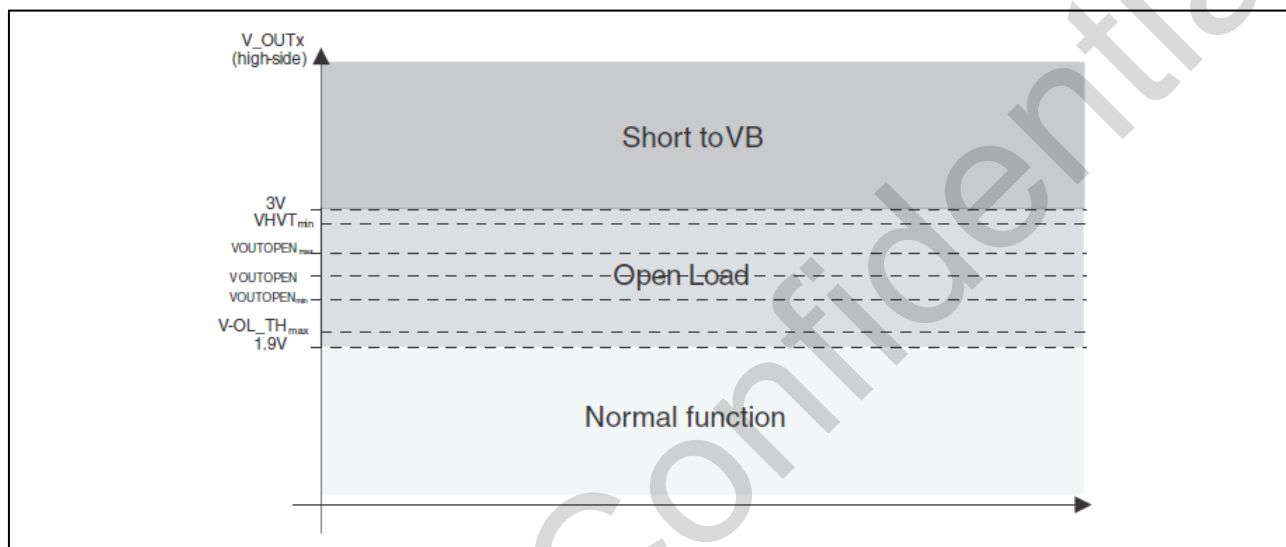


Figure 8-3 OFF state diagnostic HS config

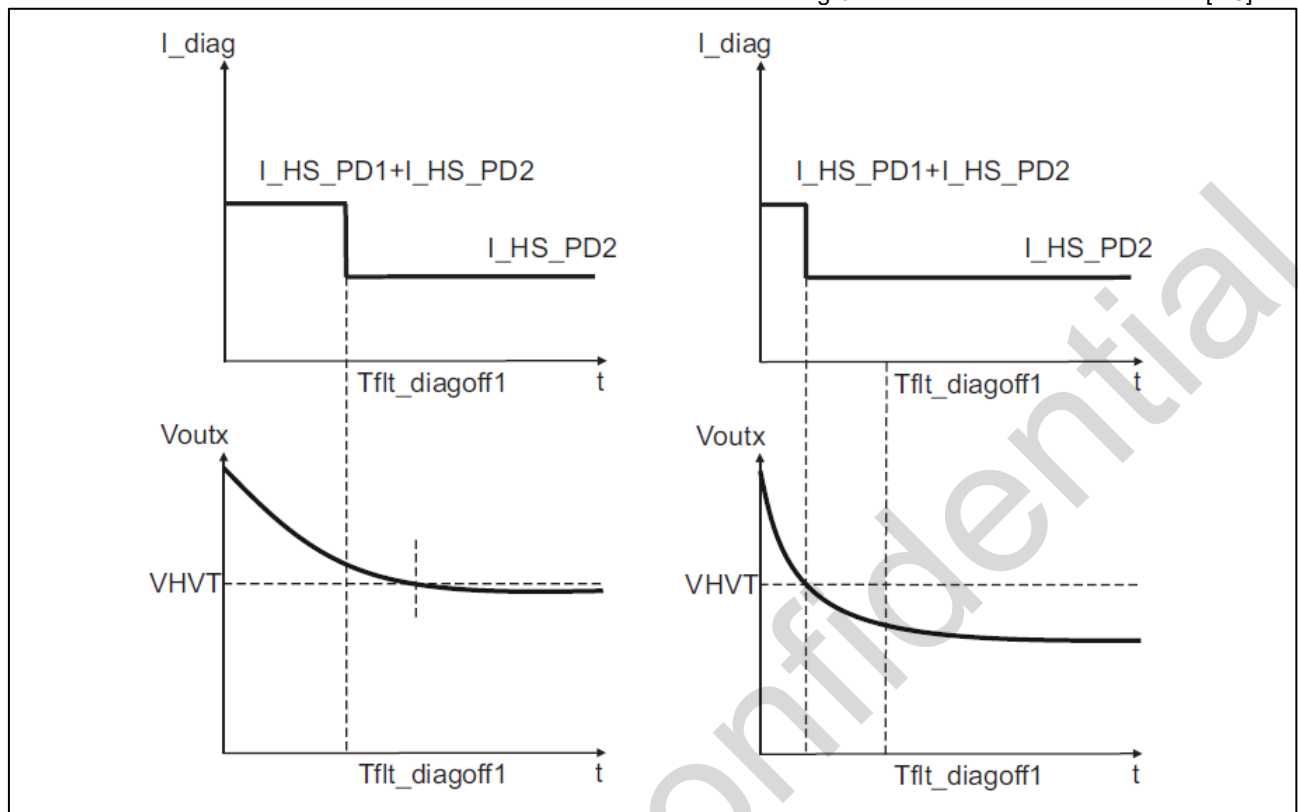


Figure 8-4 High-side driver OFF state fast pull down current behavior

By default diagnostic pull up/down currents are disabled and comparator outputs are masked by internal logic, to enable OFF state diagnostic the channel must be put first in tristate condition and then (if not already done with previous MSC frames) diagnostic must be enabled. Once the desired channel is put in tristate a false diagnostic can be sensed due to the time needed to discharge output voltage through the load.

A Filter Time (Diag Filter Time) is implemented in order to avoid to detect false diagnosis as shown in Figure 8-5.

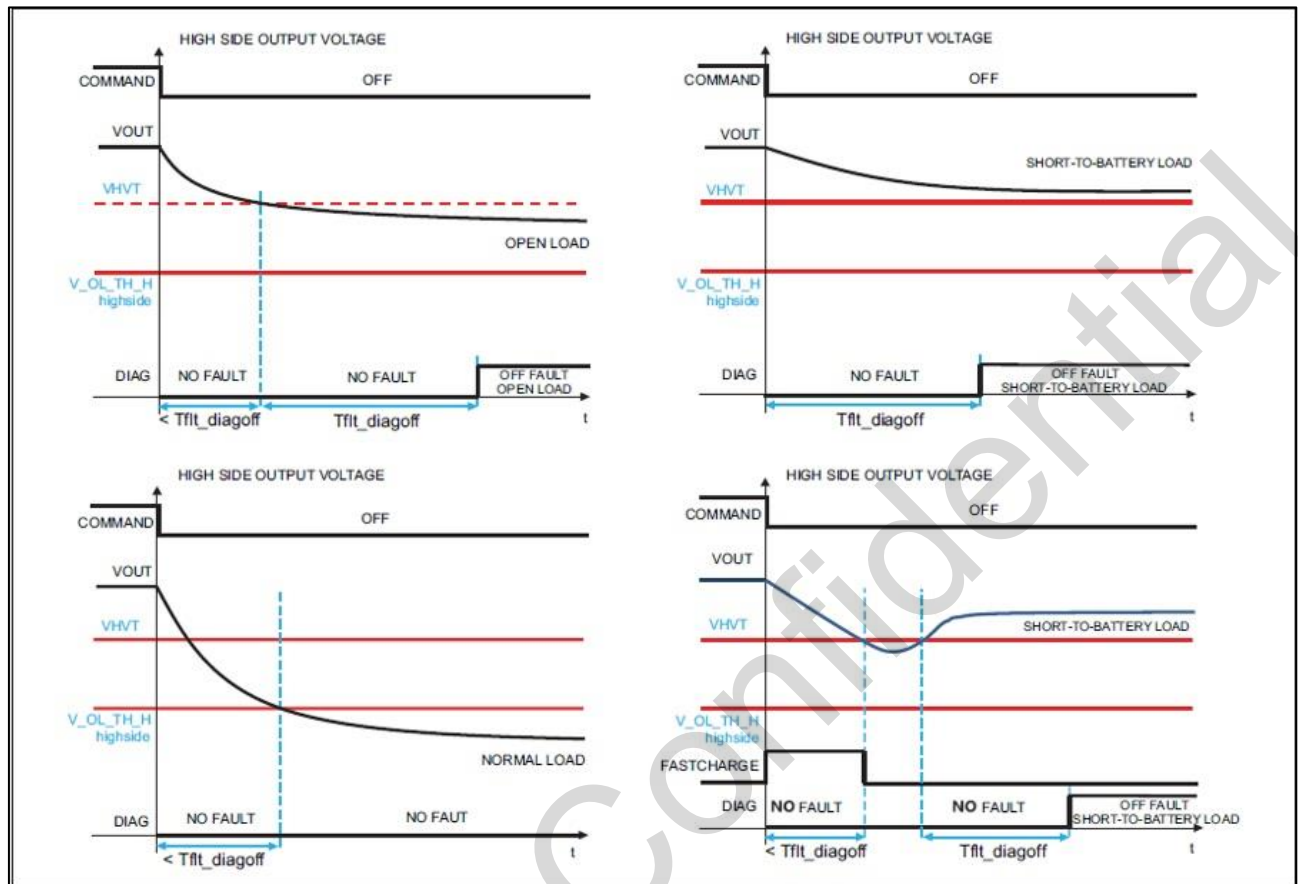


Figure 8-5 OFF state diagnostic high-side, timing

If DiagOL or DiagHV signal remains high for a time higher than Diag Filter Time the fault bit is set and can be read by Read Diag Communication.

This bit can be reset by every Read Diag Communication.

Conditions:

$5.5\text{ V} \leq \text{VB_IN} \leq 18\text{ V}$, $T_j -40$ to $175\text{ }^\circ\text{C}$ unless otherwise specified;

Table 8-1 High/low-side driver - STARTER electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Irev_OUTx	Body diode reverse current voltage drop	I _{LOAD} = -0.6 A	-	-	1.5	V	STR1/2/3	P_8.4.1
Relay HS configuration								
RdsON	Drain-source resistance	HS configuration VB_IN = 13.5 V; T _j = 150 °C, I _{load} = 1 A	-	-	1.5	Ω	STR1/2/3	P_8.4.2
RdsON	Drain-source resistance	HS configuration VB_IN = 13.5 V; T _j = 25 °C, I _{load} = 1 A	-	-	0.81	Ω	STR1/2/3	P_8.4.3

High/low-side drivers - STARTER STR[1:3]

RdsON	Drain-source resistance	HS configuration VB_IN = 13.5 V; T _j = 40 °C, I _{load} = 1 A	-	-	0.63	Ω	STR1/2/3	P_8.4.4
RdsON	Drain-source resistance	HS configuration VB_IN = 4.25 V I _{load} = 1 A	-	-	3	Ω	STR1/2/3	P_8.4.5
STR_SRC_I _{kg}	Output leakage current	Driver disabled, diagnostic off ; HSide configuration	-10	-	+10	μA	STR1/2/3	P_8.4.6
SR_ON	HS configuration Voltage slew ON State	From 30% to 80% of VOUT VB_IN = 14 V, R _{load} = 68Ω	0.7	-	2.1	V/μs	STR1/2/3	P_8.4.7
SR_OFF	HS configuration Voltage slew OFF State	From 80% to 30% of VOUT VB_IN = 14 V, R _{load} = 68Ω	0.7	-	2.1	V/μs	STR1/2/3	P_8.4.8
S/RGkill_HS	FAST VS/R off when an OVC fault happens high-side configuration	From 80% to 30% of VOUT VB_IN = 14 V, R _{load} = 68Ω	5	-	20	V/μs	STR1/2/3	P_8.4.9
Ton_OUTx_Hside	Propagation Delay from MSC_EN rising edge to 20% output OUTx voltage	VB_IN = 14 V, R _{load} = 68Ω	-	-	10	μs	STR1/2/3	P_8.4.10
Toff_OUTx_Hside	Propagation Delay from MSC_EN falling edge to 80% output OUTx voltage	VB_IN = 14 V, R _{load} = 68Ω	-	-	10	μs	STR1/2/3	P_8.4.11
Vclamp_OUTx_HS	OUTx clamping voltage HS configuration	I _{LOAD} = 0.6 A	-4.1	-	-2.5	V	STR1/2/3	P_8.4.12
I _{ovc_str}	Over current driver threshold HS configuration	-	1	1.5	2	A	STR1/2/3	P_8.4.13
I _{HS_ocvflt}	HS overcurrent filter time	Guaranteed by scan	4	-	7	μs	STR1/2/3	P_8.4.14
Relay LS configuration								
RdsON	Drain-source resistance	LS configuration VB_IN = 13.5 V; T _j = 150 °C, I _{load} = 1 A	-	-	1.5	Ω	STR1/2/3	P_8.4.15
RdsON	Drain-source resistance	LS configuration VB_IN = 13.5 V; T _j = 25 °C, I _{load} = 1 A	-	-	0.81	Ω	STR1/2/3	P_8.4.16
RdsON	Drain-source resistance	LS configuration VB_IN = 13.5 V; T _j = -40 °C, I _{load} = 1 A	-	-	0.63	Ω	STR1/2/3	P_8.4.17
RdsON	Drain-source resistance	LS configuration VB_IN = 4.25 V I _{load} = 1 A	-	-	3	Ω	STR1/2/3	P_8.4.18
STR_DRN_I _{kg}	Output leakage current	Output disabled, diagnostic off ; Low-Side configuration, DRN1=DRN2=DRN3=1 8 V, total current from 3 pins, T = -40 °C, 27 °C	-10	-	+18	μA	STR1/2/3	P_8.4.19
STR_DRN_I _{kg}	Output leakage	Output disabled,	-10	-	+48	μA	STR1/2/3	P_8.4.20

High/low-side drivers - STARTER STR[1:3]

	current	diagnostic off ; Low-Side configuration, DRN1=DRN2=DRN3=1 8 V, total current from 3 pins, T = 175 °C						
SR_ON	LS configuration voltage slew ON State	From 80% to 30% of VOUT VB_IN = 14 V, Rload = 68Ω	0.8	-	2.5	V/μs	STR1/2/3	P_8.4.21
SR_OFF	LS configuration voltage slew OFF State	From 30% to 80% of VOUT VB_IN = 14 V, Rload = 68Ω,	0.8	-	2.5	V/μs	STR1/2/3	P_8.4.22
S/RGkill_LS	FAST VS/R off when an OVC fault happens in low-side configuration	From 30% to 80% of VOUT VB_IN = 14 V, Rload = 68Ω,	5	-	20	V/μs	STR1/2/3	P_8.4.23
Ton_OUTx_Lside	Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage	VB_IN = 14 V, Rload = 68Ω	-	-	10	μs	STR1/2/3	P_8.4.24
Toff_OUTx_Lside	Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage	VB_IN = 14 V, Rload = 68Ω	-	-	10	μs	STR1/2/3	P_8.4.25
Vclamp_OUTx_LS	OUTx clamping voltage LS configuration	ILOAD = 0.6 A	40	-	50	V	STR1/2/3	P_8.4.26
Vclamp_OUTx_LS_ high_curr	OUTx clamping voltage LS configuration	ILOAD = 1 A Guaranteed by design, provided single pulse energy limits are not violated by clamping action	40	-	50	V	STR1/2/3	P_8.4.27
I_ovc_str	Over current driver threshold LS configuration	-	1	1.5	2	A	STR1/2/3	P_8.4.28
I_LS_ovcflt	LS overcurrent filter time	Guaranteed by scan	4	-	7	μs	STR1/2/3	P_8.4.29
Driver reliability data as Low-Side								
EnergyRep	Energy repetitive pulses	Tcase = 25 °C; I_OT_n = 0.45 A 1.1 Miopulses	-	-	9	mJ	STR1/2/3	P_8.4.30
EnergyRep	Energy repetitive pulses	Tc = 115 °C; I_OT_n = 0.3 A 40 Miopulses	-	-	6.5	mJ	STR1/2/3	P_8.4.31
EnergyRep	Energy repetitive pulses	Tc = 130 °C; I_OT_n = 0.3 A 9 Miopulses	-	-	6.5	mJ	STR1/2/3	P_8.4.32
EnergyRep	Energy repetitive pulses	Tc = 140 °C; I_OT_n = 0.3 A 1 Miopulses	-	-	6.5	mJ	STR1/2/3	P_8.4.33
Driver reliability data as High-Side								
EnergyRep	Energy repetitive pulses	Tcase = 25 °C; I_OT_n = 0.45 A 1.1 Miopulses	-	-	12.5	mJ	STR1/2/3	P_8.4.34
EnergyRep	Energy repetitive pulses	Tc = 115 °C; I_OT_n = 0.3 A	-	-	9	mJ	STR1/2/3	P_8.4.35

High/low-side drivers - STARTER STR[1:3]

		40 Miopulses						
EnergyRep	Energy repetitive pulses	T _c = 130 °C; I _{OT_n} = 0.3 A 9 Miopulses	-	-	9	mJ	STR1/2/3	P_8.4.36
EnergyRep	Energy repetitive pulses	T _c = 140 °C; I _{OT_n} = 0.3 A 1 Miopulses	-	-	9	mJ	STR1/2/3	P_8.4.37
Driver reliability data generator defect as Low-Side								
EnergyRep STR	Energy repetitive pulses	T _c = 25 °C; I _{OT_n} = 0.5 A 0.02 Miopulses	-	-	11	mJ	STR1/2/3	P_8.4.38
EnergyRep STR	Energy repetitive pulses	T _c = 135 °C; I _{OT_n} = 0.35 A 0.02 Miopulses	-	-	8	mJ	STR1/2/3	P_8.4.39
Driver reliability data generator defect as High-Side								
EnergyRep STR	Energy repetitive pulses	T _c = 25 °C; I _{OT_n} = 0.5 A 0.02 Miopulses	-	-	15	mJ	STR1/2/3	P_8.4.40
EnergyRep STR	Energy repetitive pulses	T _c = 135 °C; I _{OT_n} = 0.35 A 0.02 Miopulses	-	-	11	mJ	STR1/2/3	P_8.4.41
Driver reliability data JUMP START as Low-Side								
EnergyRep STR	Energy repetitive pulses	T _c = 25 °C; I _{OT_n} = 0.75A 0.001 Miopulses, 10 jumps starts over lifetime, each start < 2 min	-	-	25	mJ	STR1/2/3	P_8.4.42
EnergyRep STR	Energy repetitive pulses	T _c = 75 °C; I _{OT_n} = 0.5 A 0.001 Miopulses, 10 jumps starts over lifetime, each start < 2 min	-	-	17	mJ	STR1/2/3	P_8.4.43
Driver reliability data JUMP START as High-Side								
EnergyRep STR	Energy repetitive pulses	T _c = 25 °C; I _{OT_n} = 0.75A 0.001 Miopulses, 10 jumps starts over lifetime, each start < 2 min	-	-	31	mJ	STR1/2/3	P_8.4.44
EnergyRep STR	Energy repetitive pulses	T _c = 75 °C; I _{OT_n} = 0.5 A 0.001 Miopulses, 10 jumps starts over lifetime, each start < 2 min	-	-	21	mJ	STR1/2/3	P_8.4.45
OFF state diagnostic - HS configuration								
VHVT	Short to VB threshold voltage	HS configuration	VOU TOPE N +160 mV	-	3.0	V	STR1/2/3	P_8.4.46
V_OL_TH_Highside	Open load threshold voltage	HS configuration	1.9	-	VOU TOPE N - 180m V	V	STR1/2/3	P_8.4.47
VOUTOPEN_High side	Open load threshold	HS configuration	2.3	2.5	2.7	V	STR1/2/3	P_8.4.48

High/low-side drivers - STARTER STR[1:3]

	voltage							
I_HS_PD1	Diagnostic pull down current in high-side configuration	In OFF condition, OUTx STRx_SRC > VHVT, Ipupd_EN IPUPD_EN_STRx = ENB, Ipupd_MODE = "1"	2.5	3.6	4.7	mA	STR1/2/3	P_8.4.49
I_HS_PD2		In OFF condition, OUTx STRx_SRC > VHVT, Ipupd_EN IPUPD_EN_STRx = ENB, Ipupd_MODE = "0"	40	70	100	μA	STR1/2/3	P_8.4.50
I_HS_PU1	Diagnostic pull up current in high-side configuration	In OFF condition, OUTx STRx_SRC < VOUTOPEN, Ipupd_EN IPUPD_EN_STRx = ENB STR_SRCx = 1.9 V	35	-	75	μA	STR1/2/3	P_8.4.51
I_HS_PU2		In OFF condition, OUTx STRx_SRC < VOUTOPEN, Ipupd_EN IPUPD_EN_STRx = ENB STR_SRCx = 0 V	100	-	180	μA	STR1/2/3	P_8.4.52
I_HS_OUT_Dis1	Output current in high-side configuration (measured on STR_SRC)	OFF condition, Diagoff disabled, STR_SRC > or = 0 V	-15	-	+15	μA	STR1/2/3	P_8.4.53
I_HS_OUT_Dis2		OFF condition, Diagoff disabled, STR_SRC < 0 V	-190	-	+10	μA	STR1/2/3	P_8.4.54
V_negative_threshold	Output voltage of STR_SRC below which the output current is measured again	OFF condition, Diagoff disabled Guaranteed by design	-	-	-0.2	V	STR1/2/3	P_8.4.55
V_positive_threshold	Positive threshold of internal comparator responsible of remove output current on STR_SRC pin (over this voltage negligible current is guaranteed by compensation circuit)	OFF condition, Diagoff disabled Guaranteed by design	-	1.8	-	V	STR1/2/3	P_8.4.56
Tflt_diagoff2	-	Guaranteed by scan Cload = 15 nF	1000	1200	1500	us	STR1/2/3	P_8.4.57
Tflt_diagoff1	-	Guaranteed by scan Cload = 15 nF	130	160	200	us	STR1/2/3	P_8.4.58
OFF state diagnostic - LS configuration								
VLVT	Short to GND threshold voltage	LS configuration	1.9	-	VOU TOPE N	V	STR1/2/3	P_8.4.59

High/low-side drivers - STARTER STR[1:3]

					- 180m V			
V_OL_TH_Lowside	Open load threshold voltage	LS configuration	VOU TOPE N +160 mV	-	3.0	V	STR1/2/3	P_8.4.60
VOUTOPEN_Lowside	Open load voltage	LS configuration	2.3	2.5	2.7	V	STR1/2/3	P_8.4.61
I_LS_PU1	Diagnostic pull up current in Ls configuration	In OFF condition, STRx_DRN < VLVT, IPUPD_EN_STRx =ENB, Ipupd_MODE ="1"	2.5	3.6	4.7	mA	STR1/2/3	P_8.4.62
I_LS_PU2		In OFF condition, STRx_DRN < VLVT, IPUPD_EN_STRx =ENB, Ipupd_MODE ="0"	40	70	100	μA	STR1/2/3	P_8.4.63
I_LS_PD1	Diagnostic pull down current in Ls configuration	In OFF condition, STRx_DRN > VOUTOPEN, IPUPD_EN_STRx =ENB	55	85	100	μA	STR1/2/3	P_8.4.64
Tflt_diagoff2	DIAG Filter time in Ls configuration Filter Mode = 1	Guaranteed by scan Cload = 15 nF	450	600	750	μs	STR1/2/3	P_8.4.65
Tflt_diagoff1	DIAG Filter time in Ls configuration fast discharge enable Filter Mode = 0 (fast diag off)	Guaranteed by scan Cload = 15 nF	75	100	125	μs	STR1/2/3	P_8.4.66
Common diagnostic - LS/HS configuration								
T_SD_HIGH	Temperature shut down	-	185	-	200	°C	STR1/2/3	P_8.4.67
T_SD_LOW	Temperature shut down recover	-	175	-	190	°C	STR1/2/3	P_8.4.68
T_SD_hys	Temperature shut down hysteresis	-	5	-	10	°C	STR1/2/3	P_8.4.69
-	Thermal shutdown analog filter time	Guaranteed by design	1.5	-	4	μs	STR1/2/3	P_8.4.70
t_SD_deglitch	Digital deglitch filter time on Temperature shut down detection	Guaranteed by scan	-	10	-	μs	STR1/2/3	P_8.4.71
-	Fast OFF diagnostic ON->OFF->ON	Filter Mode=0	220	245	270	μs	STR1/2/3	P_8.4.72
-	Fast ON diagnostic OFF->ON->OFF	-	65	-	90	μs	STR1/2/3	P_8.4.73

8.5 Delay-off function

The Delay-off function involves the RLY4 and STR[2:3] drivers when they are configured as starters by the bit `MSC_CONFIG_REG3[0]=RLY4_DELAY_OFF_EN=1`, `CONFIG_REG3[4]=STR2_EN=1`, `CONFIG_REG3[5]=STR3_EN=1`. These drivers follow the reset matrix as other drivers when they are not configured as starters.

If the drivers is configured as starter and it is in on state its delay-off timer starts when one of the following events occurs:

- under-voltage of the main supply VDD5 is detected,
- over-voltage of the main supply VDD5 is detected,
- the MSC time out occurs,
- an active signal ("0") at pin WDA(IN).
- an active signal ("0") at pin RSTN(IN).

During the delay off time THOLD

- The driver is ON regardless of whether the ON/OFF command is reset.
- The starter configuration is held.
- The HSD/LSD configuration bit is held.
- The driver is ON even in case of Watchdog `ERR_CNT>4` and `ERR_CNT>7`
- The driver is ON even in case of `VDDIO_UV`.
- Once it has started, the delay_off is not restarted by a new triggering event before THOLD has expired.

The delay-off timer can be terminated by one of the following events:

THOLD time expires.

Driver is switched off by *MSC command off*.

After DELAY_OFF time the RLY4, STR[2:3] take the status of the on/off data command. If before the delay off time expires, the above triggering sources are recovered, the delay-off function remains valid.

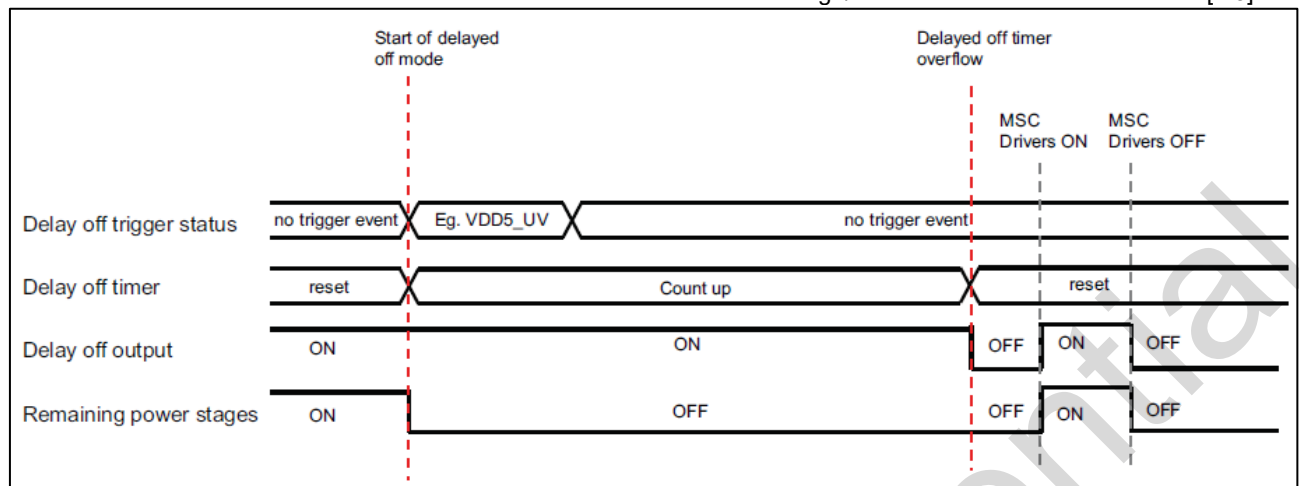


Figure 8-6 DELAY_OFF timing

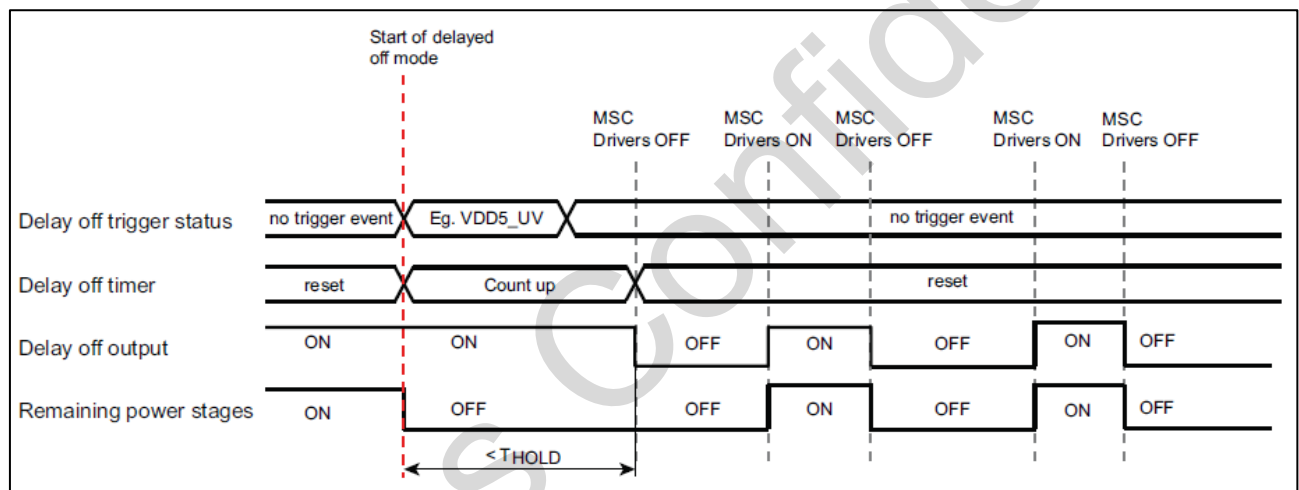


Figure 8-7 DELAY_OFF timing terminated by MSC off command

Conditions:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 8-2 High/low-side driver delay-off function electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
THOLD	Delay off time	-	400	-	800	ms	RLY4 STR2/3	P_8.5.1

8.6 Starter control

To prevent unintended activation of starter relays at the same time a dual control mechanism is implemented inside the logic by means of independent MSC commands for on/off for each starter (*RLY4_ON RLY4_OFF STR2_ON STR2_OFF STR3_ON STR3_OFF*). If the RLY4, STR2, STR3 are not configured as starter they are controlled by data frame on/off as the other drivers

The starter control function comes with a dedicated failsafe management circuit to always provide a safety switch off path able to disconnect the main supply in case of fault conditions. Starter switch off path is implemented with dual polarity command acting on drivers STR[2:3] and RLY4: if the EN_P pin is high or the EN_N pin is high or if the FAULT_WARN is activated. RLY4, STR2, STR3 are switched off by the safety switched off path independently if they are configured as starter.

Safety switch off path can be triggered both by the micro-controller through EN pins or by the HE9788N detection of an internal fault as described in the [Figure 8-8](#). The functionality of the switch off path is guaranteed through physical isolation from the rest of the device: the circuit is kept layouted in a dedicated area surrounded by trench isolations, avoiding top routing over this cell and keeping a safe distance respect to the nearby functional top routing to ensure complete independency.

The switch off path generates locally its supply voltage to drive the safety turn-off switch in order to avoid interactions with the rest of the HE9788N; the supply voltage is created starting directly from battery line. All the circuits used inside the switch-off path are able to sustain high voltage to prevent unwanted functionality in case of local failure, moreover no external reference is routed to this area and the EN_N and EN_P control logic is located inside this area as well to complete isolation and remove any dependency from the rest of the HE9788N.

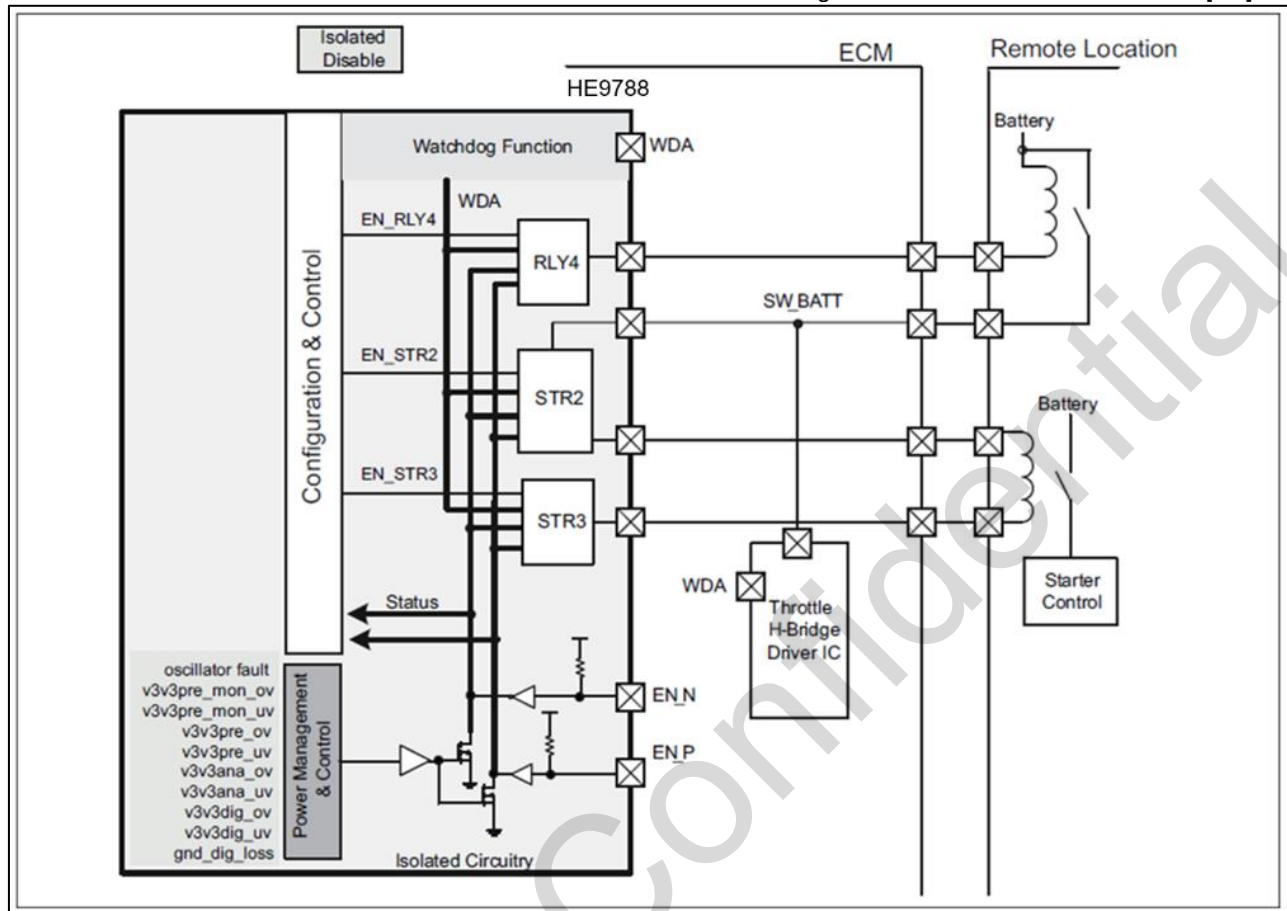


Figure 8-8 Safety switch off

Conditions:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 8-3 EN_P and EN_N electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VIH_TX	HIGH_level input voltage	-	1.75	-	-	V	EN_P	P_8.6.1
VIL_TX	LOW_level input voltage	-	-	-	0.75	V	EN_P	P_8.6.2
VHYS_TX	Input voltage hysteresis	-	0.06	0.5	-	V	EN_P	P_8.6.3
PULL_TX	Pull-up resistance	-	50	100	200	kΩ	EN_P	P_8.6.4
VIH_TX	HIGH_level input voltage	-	1.75	-	-	V	EN_N	P_8.6.5
VIL_TX	LOW_level input voltage	-	-	-	0.75	V	EN_N	P_8.6.6
VHYS_TX	Input voltage hysteresis	-	0.06	0.5	-	V	EN_N	P_8.6.7
PULL_TX	Pull-up resistance	-	50	100	200	kΩ	EN_N	P_8.6.8

Note: For this application the function is implemented, but kept disabled using external resistor to pull EN_N and EN_P to low.

9 Pre-drivers - ExtFET (MOS_Gate1/2/3/4/5) - PRD[1:5]

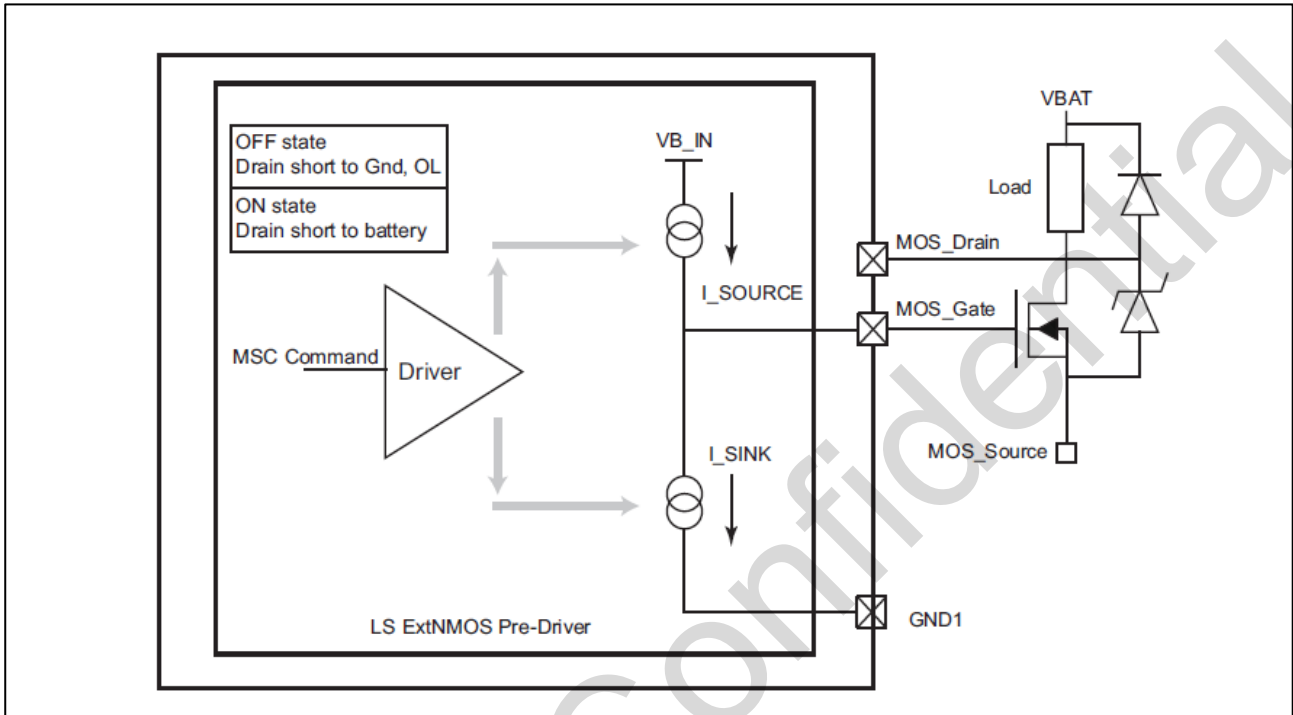


Figure 9-1 LS External MOSFET Pre-Driver Stage

9.1 LS external MOSFET pre-driver

These 5 pre-drivers are designed to drive LS external MOSFET Load. They are driven by *MSC command*.

The push-pull stage is made up of limited source current and sink current. The voltage of MOS_Drain is monitored for diagnosis.

The driver has the following detection diagnosis:

- ☐ Drain short to battery @ pre-driver = On state
- ☐ Drain short to ground @ pre-driver = Off state
- ☐ Drain open load @ pre-driver = Off state

Predriver1 & Predriver3 can be used to drive ExtMos for O2H Load.

For these two predrivers there is the possibility to select a lower gate current and sink current (typ = 250 μ A).

A dedicated MSC bit is used to select the predriver O2H function (Bit *O2H_PDRV* = *CONFIG_REG_16_1[7:6]*). Considering that this sink/source current is very low the logic will switch to the selected higher sink/source current after the cmd delay time (typ 150 μ s for *CONFIG_REG20[7]=PDRV_O2H_DLY=0* or 300 μ s

Pre-drivers - ExtFET (MOS_Gate1/2/3/4/5) - PRD[1:5]

$CONFIG_REG20[7]=PDRV_O2H_DLY=1$). After the CMD delay time the current used is the one selected by $CONFIG_REG4=PRDx_IDRV[1:0]$.

When the Predriver is used for O2H load the overcurrent detection can't be done using a VDS comparator due to slow voltage slew-rate, for this reason has to be used and external Rshunt.

When the Predriver is used for O2H load no blanking time is applied to OVC detection. The OVC detection is done reading the voltage drop on this external Rshunt by the VDS comparator of another predriver (not used).

This means that when the Predriver 1 and Predriver 3 are used for O2H load the Predriver 2 and Predriver 4 can't be used.

Bit $O2H_PDRV = CONFIG_REG_16_1[7:6] = 00$: all 5 predriver channels can be used

Bit $O2H_PDRV = CONFIG_REG_16_1[7:6] = 11$:

predriver 2 & Predriver 4 can't be used;

predriver 1 & predriver3 work with low gate current and sink current (typ = 250 uA);

it is possible read the Overcurrent on external Rshunt of Predriver1&3 using the Vds comparator of Drain2 and Drain4;

Predriver 5 works as in $O2H_PDRV = CONFIG_REG_16_1[7:6] = 00$.

There is a dedicated ground pin PDR_GND used for source pin of External Mos or ground connection of External shunt resistor. The internal voltage reference for VDS comparator threshold ($V_{th_VDS_xxx}$ parameter) is connected to this ground pin PDR_GND. Application has to connect this PDR_GND pin to the source of external MOS or ground of external shunt resistor in order to obtain a good accuracy of overcurrent threshold.

Pre-driver after power up is OFF, output LOW.

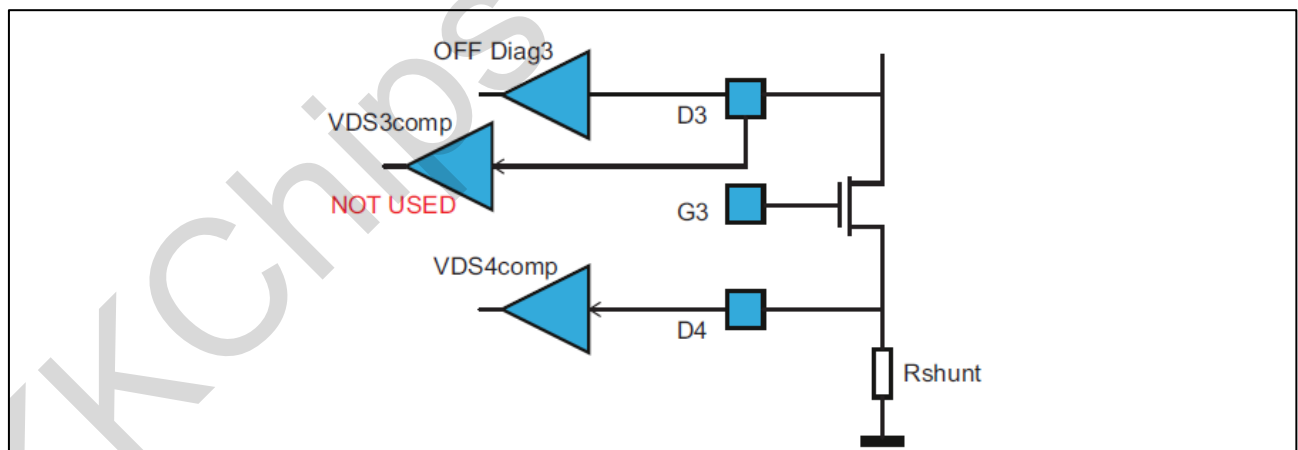


Figure 9-2 PreDriver3 configured for O2H load bit $O2H_PDRV_3 = CONFIG_REG_16_1[7] = 1$:

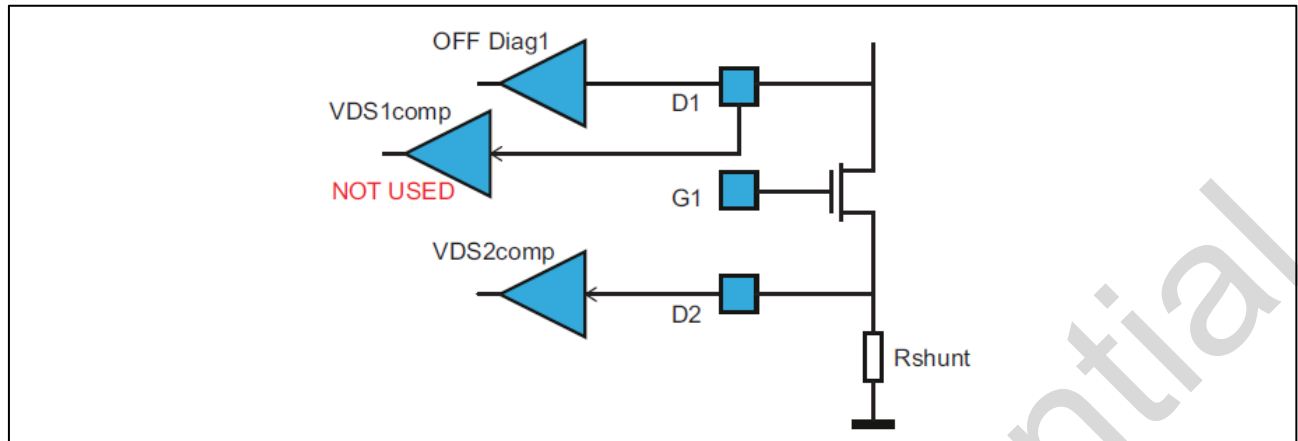


Figure 9-3 PreDriver1 configured for O2H load bit O2H_PDRV_1 = CONFIG_REG_16_1[6] = 1:

9.2 ON state diagnostic

In the On state of pre-driver, the diagnosis will compare the Drain-source voltage with the threshold generated internal. If $MOS_DRAIN_X > V_{th_DS_XXX}$ ($PRD_x_VDS[0:2]$ CONFIG_REG5/ CONFIG_REG6 / CONFIG_REG7) for a filter time, the Drain short to battery fault will be confirmed. The internal comparator reads the voltage on drain pin versus a dedicated GND sense pin (PIN X) that at application should be connected on the source of external mos. Starting from the condition ON state a programmable blanking time (MSC CONFIG_REG16=PRD_x_BLK[1:0]) is applied to prevent the detection of the short-to-battery

fault. The blanking time is not applied when the driver is already ON. The pre-driver will switch OFF and the fault will be latched until the Read_Diag_CMD action. This bit can be reset by every Read Diag Communication. To re-activate ON the power stage in these two following conditions:

The MSC command still High, when the Fault OVC Bit is read by MSC communication.

The MSC command becomes Low and then again High.

When Bit MSC_O2HPredr_1 = 1 the VDS comparator of predriver 1 is ignored.

When Bit MSC_O2HPredr_3 = 1 the VDS comparator of predriver 3 is ignored.

When Bit MSC_O2HPredr = 1 the VDS comparator of predriver 2 (or 4) is enabled immediately with the turn ON command of Predriver (and checked by logic after the blanking time), while when Bit MSC_O2HPredr = 0 the VDS comparator of corresponding channel is enabled after selected blanking time.

The over current protection can be done reading the voltage on external Rshunt using the VDS comparator of predriver 2&4.

9.3 ON/OFF state - Error in status diagnosis

Refer to [Section 7.1.3](#)

9.4 Error handling

Table 9-1 Pre-driver - ExtFET error handling

Type of error	Detection condition	Action	Clear MSC flag	Restart condition
Driver diagnostic				
Overcurrent (OVC)	Driver on	Driver is put in off state. The fault is latched in MSC diagnosis register	On read	At MSC read of diagnosis register if MSC command is still high. If MSC command is low and high again.
Short to ground (STG)	Driver off	The fault is latched in MSC diagnosis register.	On read	
Open load (OPL)	Driver off	The fault is latched in MSC diagnosis register.	On read	
Driver Status Error (STA)	Driver on/off When the driver output level is not aligned with the cmd on/off	The fault is latched in MSC diagnosis register	On read	

9.5 OFF state diagnostic

The Predriver follows the same off diagnosis as low-side driver. But despite the fast mode of all drivers in off diagnostic condition (*lpupd_MODE* bit), for all pre_Driver, there is a dedicated MSC bit (*IDIAG_HIGH_PDRV*) to select off diagnostic high or low pull up/down current

Conditions:

$5.5\text{ V} \leq \text{VB}_{\text{IN}} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 9-2 Pre-driver - ExtFET electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Pre-driver								
Vgate_on	On state MOS_Gate voltage	I_source_00	5	-	6	V	MOS_Gate 1/2/3/4/5	P_9.5.1
Vgate_off	Off state MOS_Gate voltage	I_sink_00	0	-	0.1	V	MOS_Gate 1/2/3/4/5	P_9.5.2
Ileak_off	Leakage current of MOS_Gate @ driver in tri-state		-1	-	1	μA	MOS_Gate 1/2/3/4/5	P_9.5.3
I_source_00	Pre-driver source current 1	MOS_Gate1/2/3/4/5 short to gnd	14	20	26	mA	MOS_Gate 1/2/3/4/5	P_9.5.4
I_source_01	Pre-driver source current 2	MOS_Gate1/2/3/4/5 short to gnd	7	10	13	mA	MOS_Gate 1/2/3/4/5	P_9.5.5
I_source_10	Pre-driver	MOS_Gate1/2/3/4/5	2.8	4	5.2	mA	MOS_Gate	P_9.5.6

Pre-drivers - ExtFET (MOS_Gate1/2/3/4/5) - PRD[1:5]

	source current 3	short to gnd					1/2/3/4/5	
I_source_11	Pre-driver source current 4	MOS_Gate1/2/3/4/5 short to gnd	1.3	2	2.6	mA	MOS_Gate 1/2/3/4/5	P_9.5.7
I_source_02 H	Pre-driver source current 5	MOS_Gate1/3/ short to gnd	0.355	0.25	0.155	mA	MOS_Gate 1/3	P_9.5.8
I_sink_00	Pre-driver sink current 1	MOS_Gate1/2/3/4/5 short to 5 V	-26	-20	-14	mA	MOS_Gate 1/2/3/4/5	P_9.5.9
I_sink_01	Pre-driver sink current 2	MOS_Gate1/2/3/4/5 short to 5 V	-13	-10	-7	mA	MOS_Gate 1/2/3/4/5	P_9.5.10
I_sink_10	Pre-driver sink current 3	MOS_Gate1/2/3/4/5 short to 5 V	-5.2	-4	-2.8	mA	MOS_Gate 1/2/3/4/5	P_9.5.11
I_sink_11	Pre-driver sink current 4	MOS_Gate1/2/3/4/5 short to 5 V	-2.6	-2	-1.4	mA	MOS_Gate 1/2/3/4/5	P_9.5.12
I_sink_O2H	Pre-driver sink current 5	MOS_Gate1/3 short to 5V	0.155	0.25	0.355	mA	MOS_Gate 1/3	P_9.5.13
ON/OFF state diagnostic								
Vth_DS_000	Diagnosis Vds threshold 1		120	150	190	mV	MOS_Drain1/2/3/4/5	P_9.5.14
Vth_DS_001	Diagnosis Vds threshold 2		220	245	285	mV	MOS_Drain1/2/3/4/5	P_9.5.15
Vth_DS_010	Diagnosis Vds threshold 3		300	325	360	mV	MOS_Drain1/2/3/4/5	P_9.5.16
Vth_DS_011	Diagnosis Vds threshold 4		380	405	445	mV	MOS_Drain1/2/3/4/5	P_9.5.17
Vth_DS_100	Diagnosis Vds threshold 5		500	525	560	mV	MOS_Drain1/2/3/4/5	P_9.5.18
Vth_DS_101	Diagnosis Vds threshold 6		620	660	710	mV	MOS_Drain1/2/3/4/5	P_9.5.19
Vth_DS_110	Diagnosis Vds threshold 7		900	950	1000	mV	MOS_Drain1/2/3/4/5	P_9.5.20
T_filter_ovc	Over current filtering time	Tested by scan	5	-	7	μs	MOS_Drain1/2/3/4/5	P_9.5.21
V_OL	Open load threshold voltage	Driver tristate, diag enabled	VOUTOPEN +160mV	-	3	V	MOS_Drain1/2/3/4/5	P_9.5.22
VOUTOPEN	Open load output voltage	Driver tristate, diag enabled	2.3	2.5	2.7	V	MOS_Drain1/2/3/4/5	P_9.5.23
VLVT	Output short circuit to GND Voltage threshold	Driver tristate, diag enabled	1.9	-	VOUTOPEN -180mV	V	MOS_Drain1/2/3/4/5	P_9.5.24
I_LS_PU1	Diagnostic pull up current	In OFF condition, MOS_DRAINx < VLVT, lpupd_EN = ENB, lpupd_MODE = "1"	2.5	3.6	4.7	mA	MOS_Drain1/2/3/4/5	P_9.5.25
I_LS_PU2	Diagnostic pull up current	In OFF condition, MOS_DRAINx < VLVT, lpupd_EN = ENB, lpupd_MODE = "0" and IDIAG_HIGH_PDRV = "0"	40	70	100	μA	MOS_Drain1/2/3/4/5	P_9.5.26
I_LS_PU3	Diagnostic pull up high current	In OFF condition, MOS_Drainx < VLVT, lpupd_EN = ENB,	100	-	200	μA	MOS_Drain1/2/3/4/5	P_9.5.27

		lpupd_MODE ="0" and IDIAG_HIGH_PDRV ="1"						
I_LS_PD1	Diagnostic pull down current	In OFF condition, MOS_DRAINx > VOUTOPEN, lpupd_EN = ENB and IDIAG_HIGH_PDRV ="0"	60	85	105	μA	MOS_Drai n1/2/3/4/5	P_9.5.28
I_LS_PD2	Diagnostic pull down high current	In OFF condition, OUT MOS_Drainx > VOUTOPEN, lpupd_EN = ENB IDIAG_HIGH_PDRV ="1"	350	-	550	μA	MOS_Drai n1/2/3/4/5	P_9.5.29
I_leakage	MOS_DRAIN leakage current	Output disable diagnostic OFF condition	-10		10	μA	MOS_Drai n1/2/3/4/5	P_9.5.30
Tflt_diagoff1	DIAG Filter time	Filter Mode=0	75	100	125	μs	MOS_Drai n1/2/3/4/5	P_9.5.31
Tflt_diagoff2	DIAG Filter time	Filter Mode=1	450	600	750	μs	MOS_Drai n1/2/3/4/5	P_9.5.32
Tblank_00	Blanking time on ON diag	PRD1_BLK[0:1]=00	5.1	6	6.7	μs	MOS_Drai n1/2/3/4/5	P_9.5.33
Tblank_01	Blanking time on ON diag	PRD1_BLK[0:1]=01	11.1	12	13.5	μs	MOS_Drai n1/2/3/4/5	P_9.5.34
Tblank_10	Blanking time on ON diag	PRD1_BLK[0:1]=10	17.1	18	20	μs	MOS_Drai n1/2/3/4/5	P_9.5.35
Tblank_11	Blanking time on ON diag	PRD1_BLK[0:1]=11	23.1	24	26.7	μs	MOS_Drai n1/2/3/4/5	P_9.5.36
cmd_dly_0	Command delay for MSC_CONFIG_ REG20[7]=PDR V_O2H_DLY=0	Tested by scan	140	150	160	μs	MOS_Drai n1/2/3/4/5	P_9.5.37
cmd_dly_1	Command delay for MSC_CONFIG_ REG20[7]=PDR V_O2H_DLY=1	Tested by scan	280	300	320	μs	MOS_Drai n1/2/3/4/5	P_9.5.38

10 Pre-drivers – IGNITER IGN[1:6]

These 6 Ignition pre-drivers are designed to drive ignition load. They are driven by *MSC command*. The push-pull stage is made up of a high-side current generator and a low-side driver.

When an overcurrent fault is detected the driver switches off with higher slew rate (FASTSR) to reduce the power dissipation.

There is a bit selection for disabling the LSD in case of external IGBT. It is suggested to configure the MSC bit (*IGN_LSD_DIS CONFIG_REG6 D7*) disabling the LSD of igniter before the *msc_driver_enable* command.

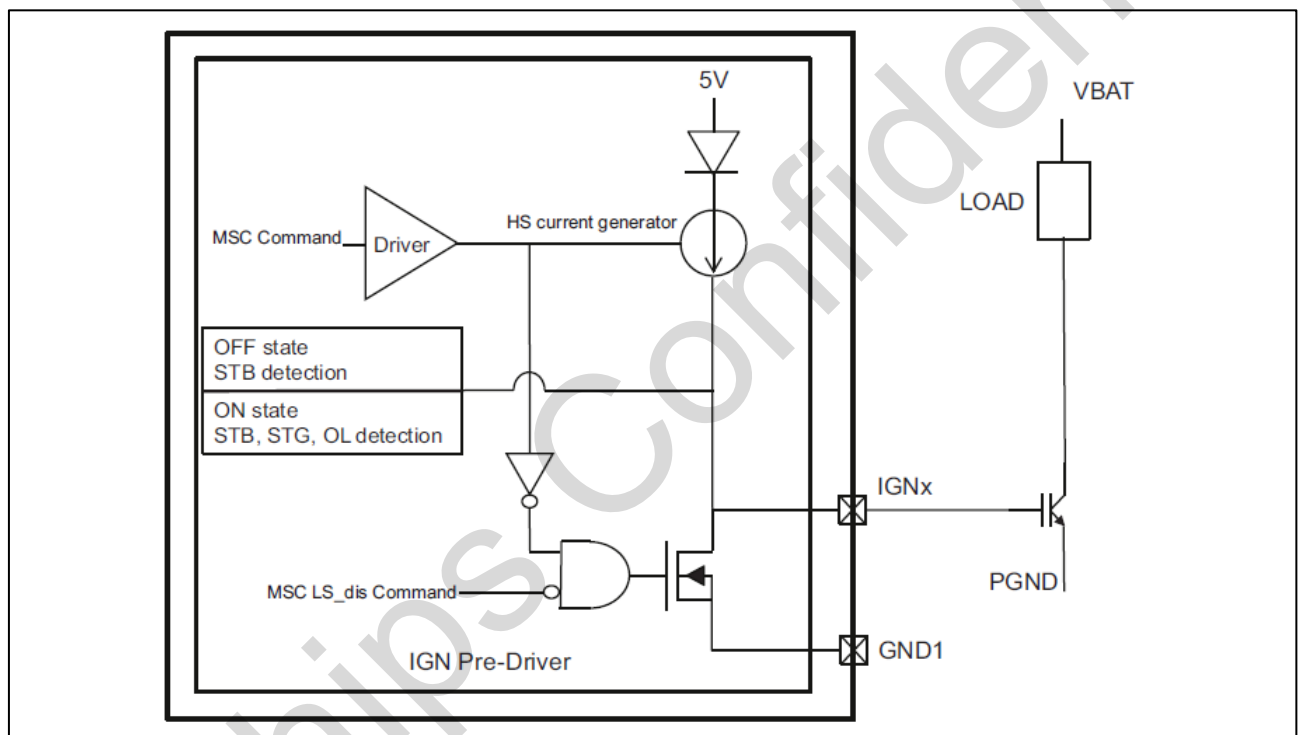


Figure 10-1 Ignition pre-drivers block schematic

10.1 Ignition pre_drivers diagnosis

The driver has the following detection diagnosis:

- Short To Vbat protection by checking the IGNx pin voltage higher than $V_{th_STB_IGN}$, when the driver is enabled high (high-side on) or low (low-side on);
- Open Load when the driver is enabled high (high-side on) (DIAGOL);
- Short to Ground when the driver is enabled high (high-side on) (DIAGLV).

Table 10-1 Pre-driver - IGNITER diagnosis

Controlled load	Diagnosis & protection type	Detected faults @ pre-driver enabled HIGH	Detected faults @ pre-driver enabled LOW
Internal IGBT/smart IGBT (IGN_DIAG (CONFIG_REG2 D5) = "0")	The controlled device is module internal. No diagnosis and protection required	-	-
External IGBT Load (IGN_DIAG = "1")	IGNx voltage and current monitoring	_ IGNx short to GND _ IGNx open load _ IGNx short to VBAT	_ IGNx short to VBAT

10.2 Short to GND

When IGNx pin voltage is lower than $V_{th_STG_IGN}$ for a filter time at pre-driver is enabled high, short to ground fault is detected. So the fault is latched until Read_Diag_CMD action.

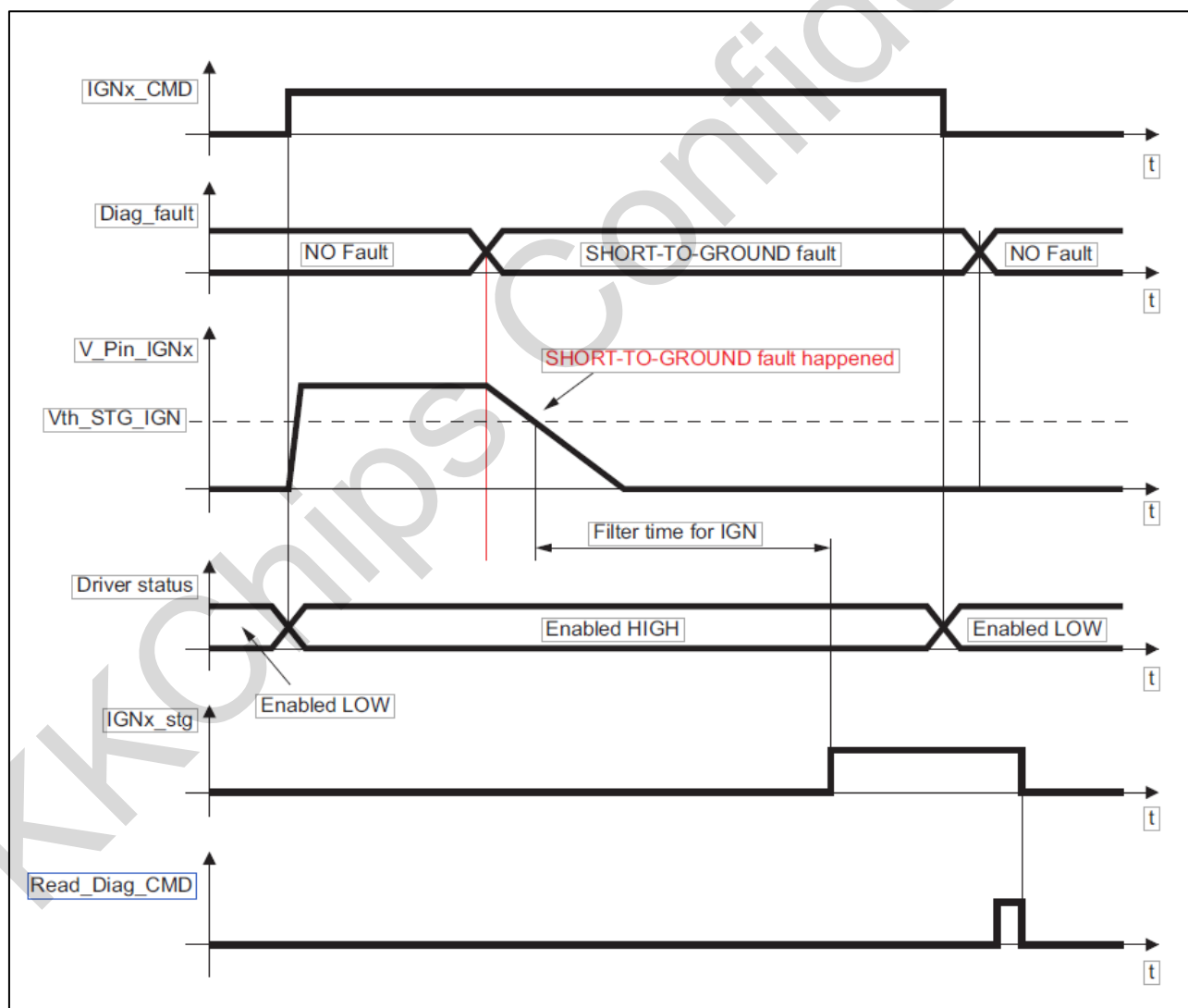


Figure 10-2 IGN pre-drivers diagnosis timing diagram at short to GND

10.3 Short to BAT

When IGNx pin voltage is higher than $V_{th_STB_IGN}$ for a filter time at pre-driver is enabled high or enabled low, short to battery fault is detected and the driver is disabled. So the fault is latched.

During Short-to-Bat fault present the driver is auto-restart with these modalities:

- The high-side restart during OVC is on read diagnostic or $IGNx_CMD$ goes to low and high again.
- The low-side restart during OVC is on read diagnostic.

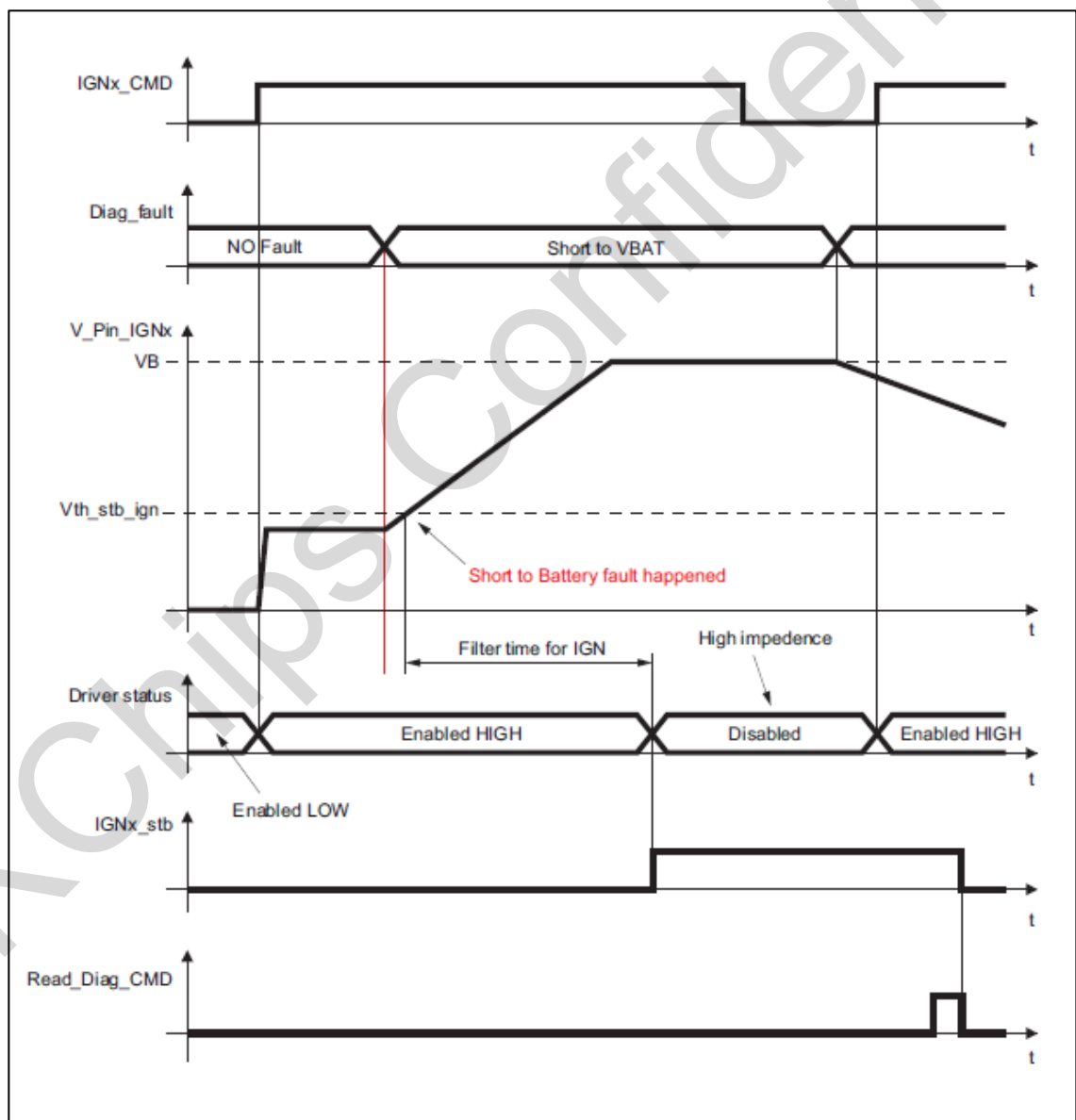


Figure 10-3 IGN pre-drivers diagnosis timing diagram at short to VBAT

10.4 Open load

When IGNx pre-driver is enabled HIGH, if IGNx current is lower than OL current detection $I_{th_OL_IGN}$ for a filter time, open load fault is detected and the driver is not disabled. Open load function can be disabled by a MSC bit. Open Load fault is latched until Read_Diag_cmd action. OPL threshold is selectable between 2 values, the default (between 0.4 and 0.8mA) and a lower one.

OPL threshold reduction is available just in low current limitation condition, so when *OL_RED* and *IGN_CURRENT_CFG* (*CONFIG_REG2 D4*) bit are both high.

10.5 Error in driver status

Refer to [Section 7.1.3](#)

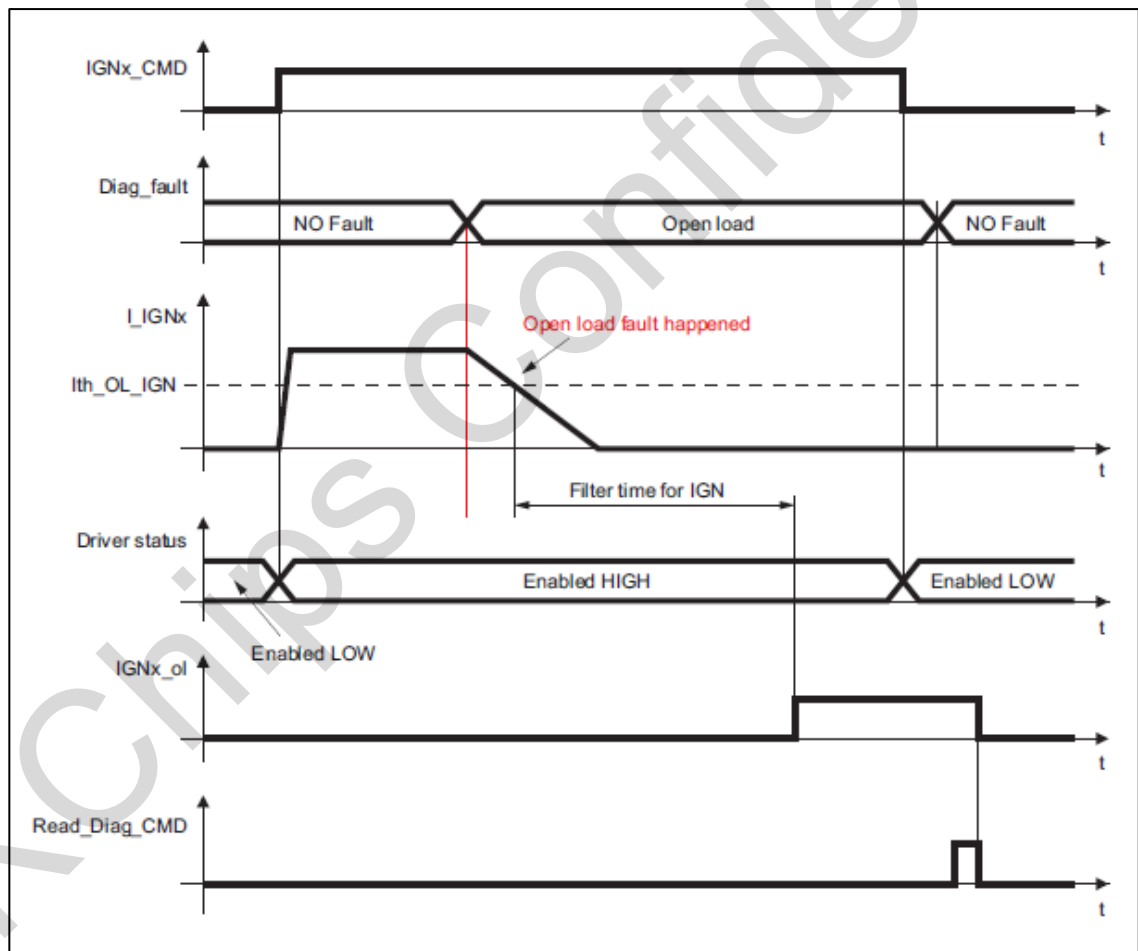


Figure 10-4 Low-side drivers ON diagnosis timing diagram at open load

Conditions:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 10-2 Pre-driver - IGNITER electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
High-side current generator parameters								
IOH1_HS_IGN	High-side output current 1	VIGNX = 4 V VDD5 = 5 V MSC Bit IGN_CURRENT CFG = 1, OL_RED = 0	4.7	-	15.3	mA	IGN1/2/3/ 4/5/6	P_10.5.1
IOH1_HS_IGN_OL_RED	High-side output current 1	VIGNX = 4 V VDD5 = 5 V MSC Bit IGN_CURRENT CFG = 1, OL_RED = 1	2	-	15	mA	IGN1/2/3/ 4/5/6	P_10.5.2
IOH2_HS_IGN	High-side output current 2	VIGNX = 4 V VDD5 = 5 V MSC Bit IGN_CURRENT CFG = 0 (Default Value)	14.7	-	30.3	mA	IGN1/2/3/ 4/5/6	P_10.5.3
VOH1_HS_IGN	Output voltage 1 @ high-side on	Isource = 5 mA MSC Bit IGN_CURRENT CONFIG = 1, OL_RED = 0	VDD5 -0.4	-	VDD5	V	IGN1/2/3/ 4/5/6	P_10.5.4
VOH1_HS_IGN_OL_RED	Output voltage 1 @ high-side on	Isource = 2 mA MSC Bit IGN_CURRENT CONFIG = 1, OL_RED = 1	VDD5 -0.4	-	VDD5	V	IGN1/2/3/ 4/5/6	P_10.5.5
VOH2_HS_IGN	Output voltage 2 @ high-side on	Isource = 15mA MSC Bit IGN_CURRENT CONFIG = 0 (Default Value)	VDD5 -0.6	-	VDD5	V	IGN1/2/3/ 4/5/6	P_10.5.6
IOFF_LK_IGN	High-side leakage current @ off condition	IGNx = 2.5 V in off	-	-	10	μA	IGN1/2/3/ 4/5/6	P_10.5.7
Low-side MOSFET parameters								
RDS-on_LS_IGN	Drain-source resistance	I _{load} = 0.1A @ T = -40 °C	1.8	2.2	3.1	Ω	IGN1/2/3/ 4/5/6	P_10.5.8
RDS-on_LS_IGN	Drain-source resistance	I _{load} = 0.1 A @ T = 27 °C	2.4	3.0	3.91	Ω	IGN1/2/3/ 4/5/6	P_10.5.9
RDS-on_LS_IGN	Drain-source resistance	I _{load} = 0.1 A @ T = 150 °C	3.8	5.0	7.0	Ω	IGN1/2/3/ 4/5/6	P_10.5.10
VOL_LS_IGN	Output voltage @ LS low-side on	IOL = 20 μA sunked	-	-	0.1	V	IGN1/2/3/ 4/5/6	P_10.5.11
VOL_LS_IGN	Output voltage @ LS low-side on	IOL = 100 mA sunked	-	-	0.7	V	IGN1/2/3/ 4/5/6	P_10.5.12
Irev_LS_IGN	Body diode reverse current voltage drop	I _{load} = 0.3 mA sourced	-	-	1	V	IGN1/2/3/ 4/5/6	P_10.5.13
General IGN block parameters								
TDON_IGN	Turn-on delay time	CLOAD = 10 nF Isource = 10 mA	-	-	10	μs	IGN1/2/3/ 4/5/6	P_10.5.14
TDOFF_IGN	Turn-off delay time	C LOAD = 10 nF	-	-	10	μs	IGN1/2/3/ 4/5/6	P_10.5.15
TR_IGN	Output rise time	CLOAD = 10 nF	-	-	9	μs	IGN1/2/3/	P_10.5.16

HE9788N

Multifunction IC for automotive engine management system



Pre-drivers – IGNITER IGN[1:6]

		Isource = 10 mA C LOAD = 10 nF					4/5/6	
TF_IGN	Output fall time		0.05	-	2	μs	IGN1/2/3/ 4/5/6	P_10.5.17
IOUT_LK_IGN	Output leakage current	Vpin = 13.5 V	-	-	10	μA	IGN1/2/3/ 4/5/6	P_10.5.18
Vth_STB_IGN	Short circuit to VBAT voltage threshold	IGNx = VBAT	5.3	-	6.5	V	IGN1/2/3/ 4/5/6	P_10.5.19
Vth_STG_IGN	Short circuit to GND threshold	-	2.1	-	2.7	V	IGN1/2/3/ 4/5/6	P_10.5.20
Ileak_STG_IGN	Leakage current @ short circuit to GND, IGN in tristate	IGNx = GND	-	-	10	μA	IGN1/2/3/ 4/5/6	P_10.5.21
Ileak_STB_IGN	Leakage current @ short circuit to VBAT,IGN in tristate	IGNx = VBAT	-1	-	1	mA	IGN1/2/3/ 4/5/6	P_10.5.22
Ileak_STB_IGN	Leakage current @ short circuit to VBAT	GND = VB_IN = 0V IGNx = 18 V	-	-	10	μA	IGN1/2/3/ 4/5/6	P_10.5.23
TFLT_SCGOL IGN	Open load and short circuit to GND filter time	-	72	-	90	μs	IGN1/2/3/ 4/5/6	P_10.5.24
TFLT_SCB IGN	Short circuit to VBAT switch-off delay	-	5.9	-	7.2	μs	IGN1/2/3/ 4/5/6	P_10.5.25
Ith_OL_IGN	Open load current threshold	-	0.18	-	1.25	mA	IGN1/2/3/ 4/5/6	P_10.5.26
Ith_OL_IGN_OL_RE ED	Open load current threshold	OL_RED = 1	0.03	-	0.2	mA	IGN1/2/3/ 4/5/6	P_10.5.27

11 Inductive sensor interface

The interface handles signals coming from magnetic pick-up sensors or Hall Effect sensors.

The interface feeds the digital signal to microcontroller that extracts flying wheel rotational position, angular speed and acceleration.

11.1 VRS interface

VRS interface runs either in normal mode to convert the input differential voltage or in diagnostic mode to detect eventual short to ground, short to battery or open condition at sensor pins. Figure 11-1 shows the main parts inside VRS block and, in red, the internal connection in case of diagnostic mode active.

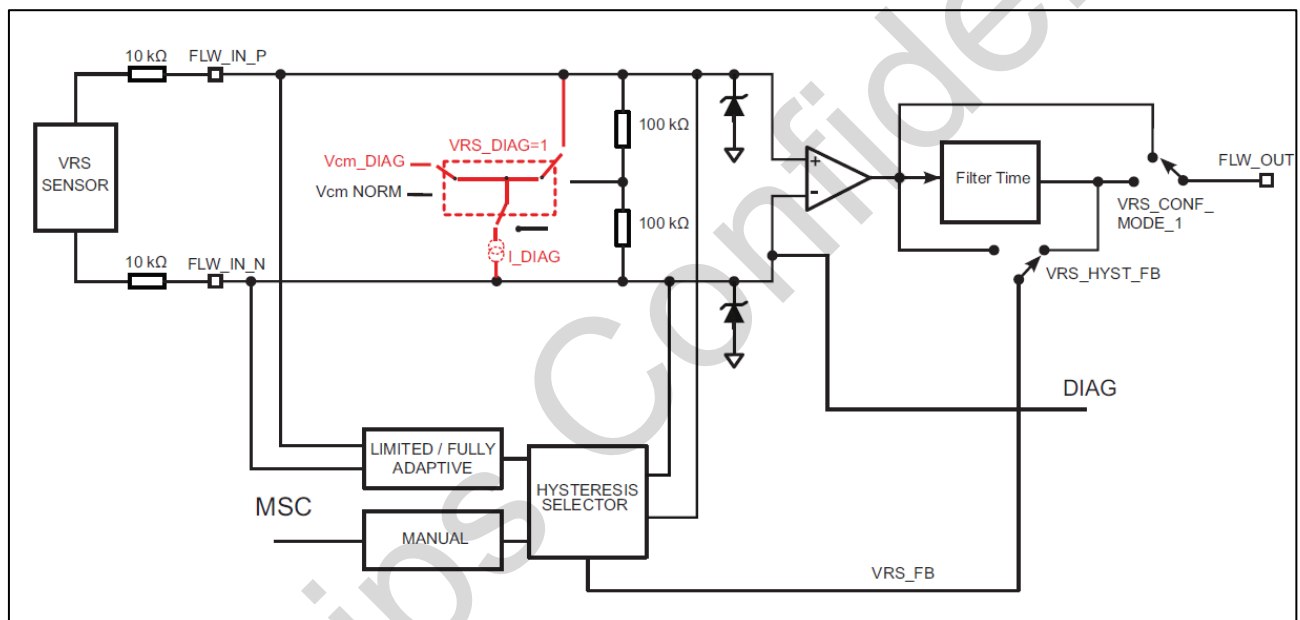


Figure 11-1 VRS interface block diagram

Operating mode is defined in VRS register through *VRS_DIAG* in *CONFIG_REG8* bit: when *VRS_DIAG*=0, VRS block is set in normal mode; when *VRS_DIAG*=1 the VRS diagnosis mode is activated.

If HE9788N is supplied and VRS is running, in case *FLW_IN_P* or *FLW_IN_N* voltage rises their values are clamped at *VclpH*.

In case of activation of clamp both on *FLW_IN_P* and *FLW_IN_N*, it is guaranteed by design that *FLW_IN_P* voltage is higher than *FLW_IN_N*.

11.2 VRS – Normal mode

The VRS normal mode is set with *VR_DIAG*=0

In Normal mode, the circuit is configured as the one reported in [Figure 11-2](#). It allows decoding the VRS signal while the wheel is in rotation.

Due to the high variability of the input signal (± 200 V), the input is clamped in the range $[V_{clpH}:V_{clpL}]$ in order to allow the analog circuitry to process the signal itself. Moreover, the sensor input pins have an input common mode, VCM.

The preconditioned input signal is then processed by a zero-crossing comparator, which toggles at each transition of the input signal.

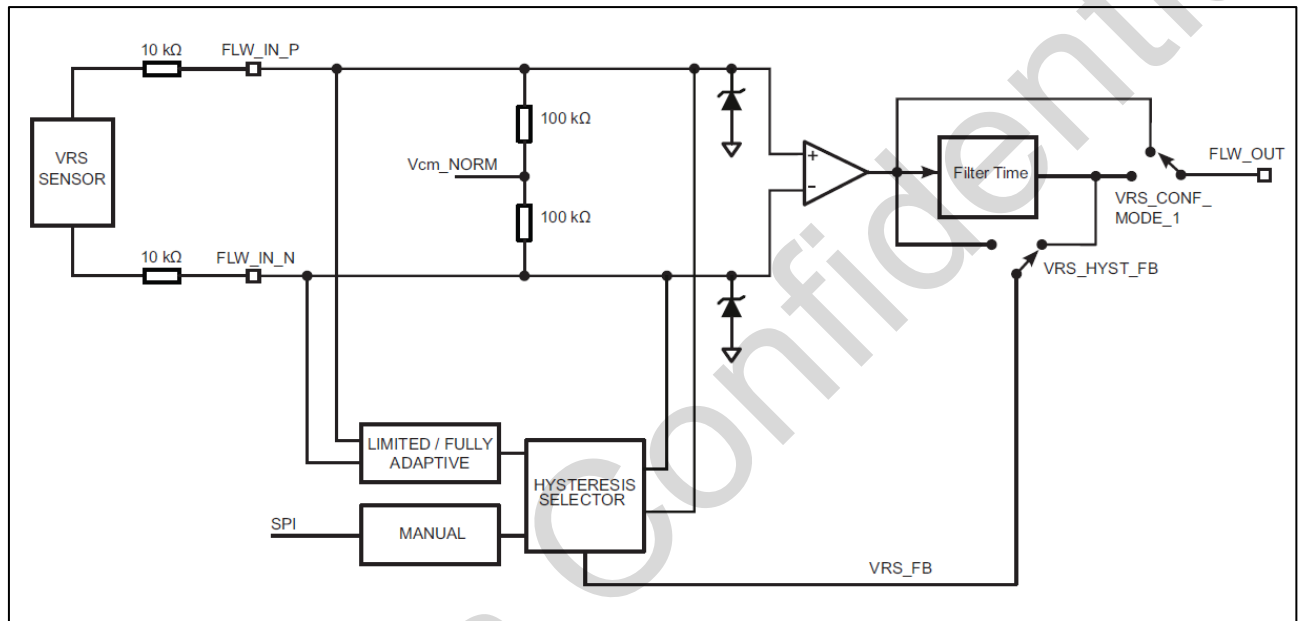


Figure 11-2 VRS block diagram - Normal operating mode

To avoid spurious commutations of the zero crossing comparator, a hysteresis mechanism is implemented. HE9788N is able to sink a hysteresis current which generates a voltage drop across the external resistors. The voltage levels related to the hysteresis function shown hereafter are calculated considering an external series resistance of 10 kΩ on FLW_IN_P and 10 kΩ on FLW_IN_N pins.

As reported in [Figure 11-3](#), the V_{diff} ($FLW_IN_P - FLW_IN_N$) input differential signal exhibits

some steps at each zero crossing:

- when the output of the zero crossing comparator is high, the hysteresis current is kept OFF;
- when the output of the zero crossing comparator is low, the hysteresis current is switched ON.

This approach applies the hysteresis current only on the transition L-H of the VRS_FB signal.

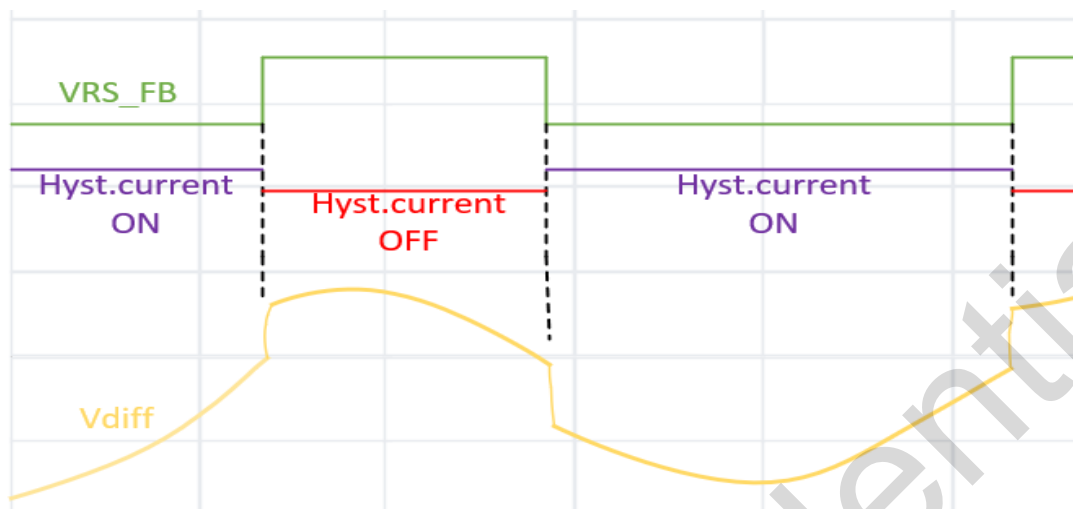


Figure 11-3 Hysteresis application

The output of the zero crossing comparator can be further processed by a filtering circuit or directly routed to FLW_OUT.

11.2.1 VRS normal mode configurations

HE9788N integrates two main configurable architectures: VRS_A and VRS_B. These architectures are selected in VRS register, *VRS_MODE_SEL* bit in CONFIG_REG 2.

Once VRS_A (*VRS_MODE_SEL* =1) or VRS_B (*VRS_MODE_SEL* =0) has been configured, hysteresis and filtering strategy are defined through *VRS_MODE[1:0]* bit in the same CONFIG_REG 8 register:

VRS_MODE[1] defines filtering function (OFF/ON and if ON, its time value)

VRS_MODE[0] defines the hysteresis (manual or adaptive)

Table 11-1 summarizes the parameters of VRS_A and VRS_B architecture; the next paragraphs detail the two configurations.

Table 11-1 VRS_A, VRS_B hysteresis and filter time definition

VRS_MO DE_SEL	VRS _MODE[1:0]	Filter		Hyst.	
VRS_A configuration					
1	00	OFF	0 μs	Manual	Ref to VRS_A - Manual Hysteresis
1	01	OFF	0 μs	Full adaptive	Ref. to VRS_A - Fully Adaptive Hysteresis
1	10	ON	T(n-1)/32	Manual	Ref to VRS_A - Manual Hysteresis
1	11	ON	T(n-1)/32	Full adaptive	Ref. to VRS_A - Fully Adaptive Hysteresis
VRS_B configuration					
0	00	OFF	0 μs	Manual	Ref. to VRS_B - Manual Hysteresis
0	01	OFF	0 μs	Limited adaptive	Ref. to VRS_B -Limited Adaptive Hysteresis
0	10	ON	4 μs	Manual	Ref. to VRS_B - Manual Hysteresis
0	11	ON	4 us	Limited adaptive	Ref. to VRS_B -Limited Adaptive Hvsteresis

In case a change of *VRS_MODE_SEL* bit within the normal operating mode occurs (1->0 or 0->1) with hysteresis current active, this leads to the change of the hysteresis (to HI1 or HI3, according the new selection programmed) not synchronized with any *VRS_FB* zero crossing.

11.2.2 VRS_A - Manual Hysteresis

To set the manual hysteresis on *VRS_A* configuration, bit *VRS_MODE[0]* has to be configured at '0'. Hysteresis value is manually set through *VRS_HYST[2:0]* of CONFIGREG 8 according to *Table 11-2*. Such hysteresis is fixed until a new SPI programming occurs.

Default hysteresis current after exiting reset is HI3.

New MSC current value is updated during HYST CURRENT OFF phase that means the output comparator is high.

Table 11-2 VRS_A hysteresis value

Hysteresis current [H]	Min	Typ	Max	Unit	Correspondent value on 20 kΩ ext. resistor	Unit	Number
HI1	3	5	7	μA	100	mV	P_11.2.2.1
HI2	7.5	10	13.5	μA	200	mV	P_11.2.2.2
HI3	13	17	23	μA	347	mV	P_11.2.2.3
HI4	23	32	40	μA	644	mV	P_11.2.2.4
HI5	35	51	60	μA	1020	mV	P_11.2.2.5
No Hyst	-	-	-	-	-	-	P_11.2.2.6

11.2.3 VRS_A - Fully Adaptive Hysteresis

To set the adaptive hysteresis on *VRS_A* configuration, bit *VRS_MODE[0]* has to be set to '1'. In this configuration, *VRS* input differential signal is fed into a peak detector circuit and then quantized on 5 different voltage levels, based on 4 *PVi* thresholds (see *Table 11-3*). Default hysteresis current after exiting reset is HI3.

Table 11-3 Peak voltage value ranges

Peak voltage [PVi]	Min	Typ	Max	Unit	Number
PV1	600	930	1300	mV	P_11.2.3.1
PV2	1200	1600	1950	mV	P_11.2.3.2
PV3	2000	2300	2650	mV	P_11.2.3.3
PV4	2600	3000	3300	mV	P_11.2.3.4

The quantized output is sent to a logic block (Hysteresis Selection Table) that chooses the proper hysteresis value (HIi) depending on the input peak voltage (PVi), see *Table 11-4*.

Table 11-4 Input peak voltage range

Input peak voltage range	Selected hysteresis (Hli)
0 - PV1	HI1
PV1 - PV2	HI2
PV2 - PV3	HI3
PV3 - PV4	HI4
> PV4	HI5

Peak detector and Hysteresis Selection Table circuits are enabled by VRS_FB signal according to *HYS_FB_SEL* in CONFIG-REG 8 bit value that establishes if the feedback signal is before or after the filter time.

VRS input differential voltage is continuously acquired: its max value, reached during timeframe VRS_FB signal is asserted (hysteresis current is off), is latched through a peak detector; such a peak defines a specific value of hysteresis current, turned on as soon as the VRS_FB falls to zero and switches OFF when next rising edge occurs.

Based on the hysteresis current, the signal is processed by a squaring circuit which processes the output signal of the comparator, see Figure 11-4.

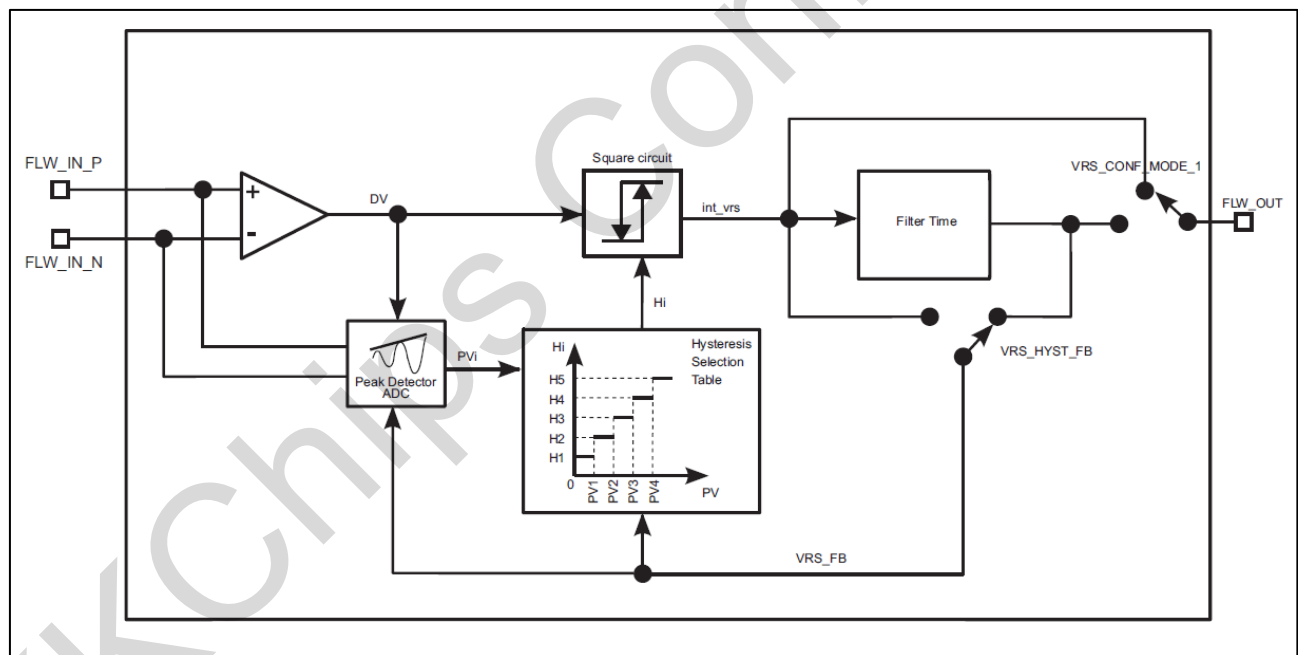


Figure 11-4 VRS_A fully adaptive hysteresis

11.2.4 VRS_A - Adaptive filter time

In VRS_A mode, it is possible to enable the filter time on the output of the zero crossing comparator through the bit *VRS_MODE[1]* of CONFIG-REG 8.

Once enabled, the most suitable internal filter based on the input signal frequency is determined.

According to VRS previous output period, filter time value is updated as follows:

$$T_{\text{filter}(n)} = \frac{T_{\text{period}(n-1)}}{32}$$

If the value of the previous period is lower than 128 μs , the filter time would be saturated at 4 μs fixed value. After reset $T_{\text{filter}} = 200 \mu\text{s}$ (TYP)

Through *EN_FALLING_FILT* CONFIG-REG8 bit in VRS register, it is possible to configure two different strategies for the filtering algorithm.

VRS_OUT rising edge: the transition depends on the hysteresis crossing of differential signal V_{diff} ; VRS output is set if V_{diff} remains asserted and stable for a period longer than T_{filter} .

VRS_OUT falling edge: the transition depends on the zero crossing of differential signal V_{diff} :

EN_FALLING_FILT = 1: VRS_OUT is deasserted when the signal is low and remains stable for at least T_{filter} ; see [Figure 11-5](#).

EN_FALLING_FILT = 0: VRS_OUT is de-asserted at first zero crossing transition of differential signal and next eventual commutations are ignored for T_{filter} time. See [Figure 11-6](#).

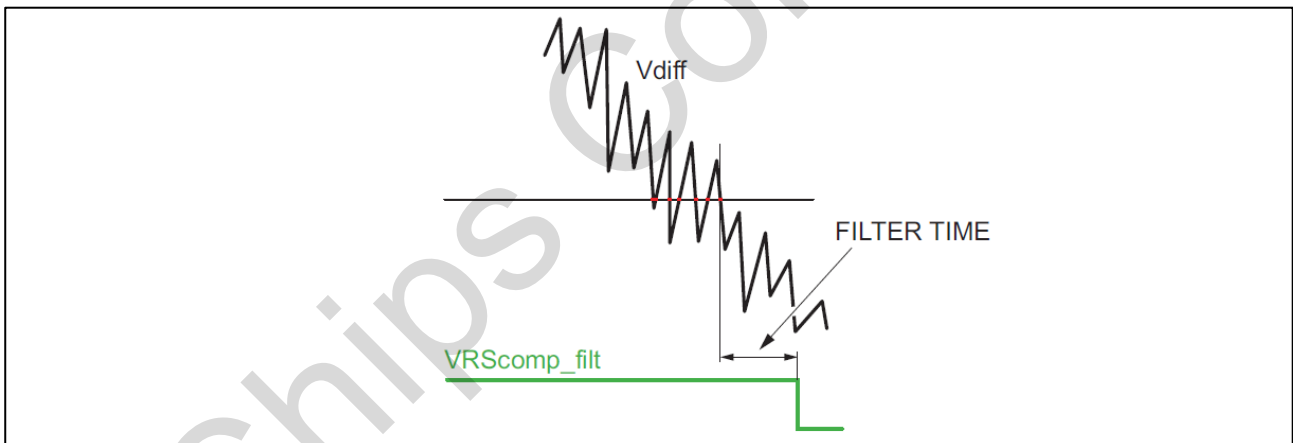


Figure 11-5 *EN_FALLING_FILT* = 1

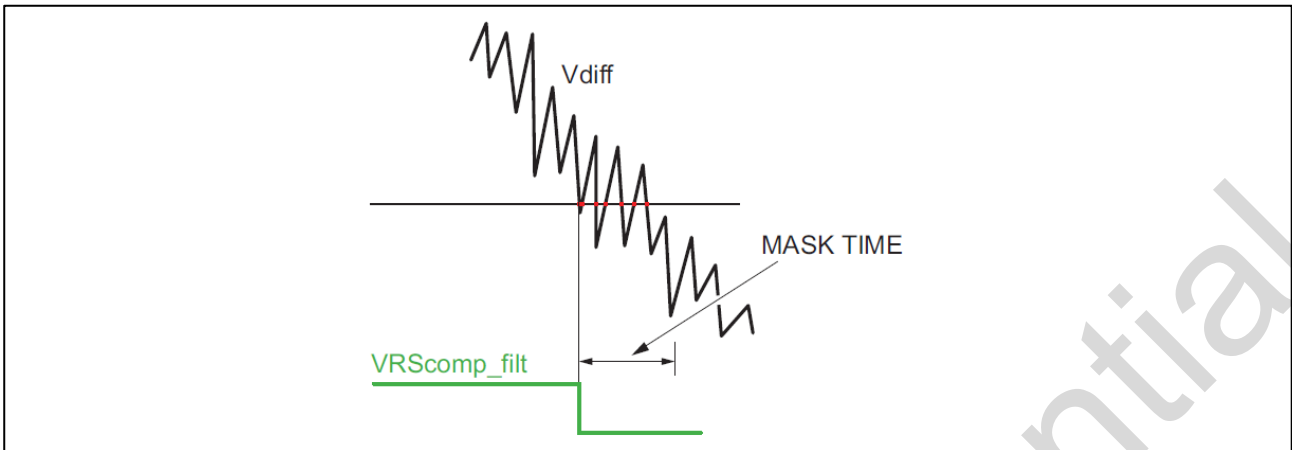


Figure 11-6 EN_FALLING_FILT = 0

11.2.5 VRS_B - Manual Hysteresis

To set the manual hysteresis on VRS_B configuration, bit *VRS_MODE* [0] has to be set to '0'. Hysteresis value is manually set through *VRS_HYST* [2:0] of CONFIG-REG 8 according to Table 11-2. Such hysteresis is fixed until a new SPI programming occurs.

Once a new value is defined, new hysteresis threshold is applied after the second VRS_FB H-L transition and until the next rising edge of the VRS input differential voltage occurs.

11.2.6 VRS_B - Limited Adaptive Hysteresis

To set the limited adaptive hysteresis on VRS_B configuration, bit *VRS_MODE* [0] has to be set to '1'. In this mode, user programs a hysteresis threshold through *VRS_HYST* [2:0] bit in VRS register and the internal logic selects a hysteresis based on input signal peak value: the maximum of these two values is actually applied.

Once a new value is defined, new hysteresis threshold is applied after the second VRS_FB H-L transition and until the next rising edge of the VRS input differential voltage occurs.

11.2.7 VRS_B - Fixed Filter Time

In VRS_B configuration, it is possible to enable the filter time on the output of the zero crossing comparator through the bit *VRS_MODE* [1] of VRS register. This configuration allows defining the internal filter time at a fixed value of 4 μ s, active on both rising and falling edges of VRS output.

As per VRS_A architecture, *EN_FALLING_FILT* allows configuring the same two different strategies for the filtering algorithm.

11.3 VRS diagnostic mode

The diagnostic mode is selected through *VRS_DIAG* = '1' in CONFIG-REG8. This mode provides feedback to detect faulty conditions either on FLW_IN_P or FLW_IN_N.

To be noted that diagnostic results are not reliable while the flying wheel is rotating. If a fault is detected in DIAG mode, VRS correct functionality is not guaranteed. Fault bit *VRS_DIAG*

of VRS register is consequently set.

Figure 11-7 shows the circuit used in Diagnostic mode. When VRS diagnostic mode is activated, FLW_IN_P is fixed at V_{cm_DIAG} and IDIAG current generator is enabled, the current path is that one in green in Figure 11-7.

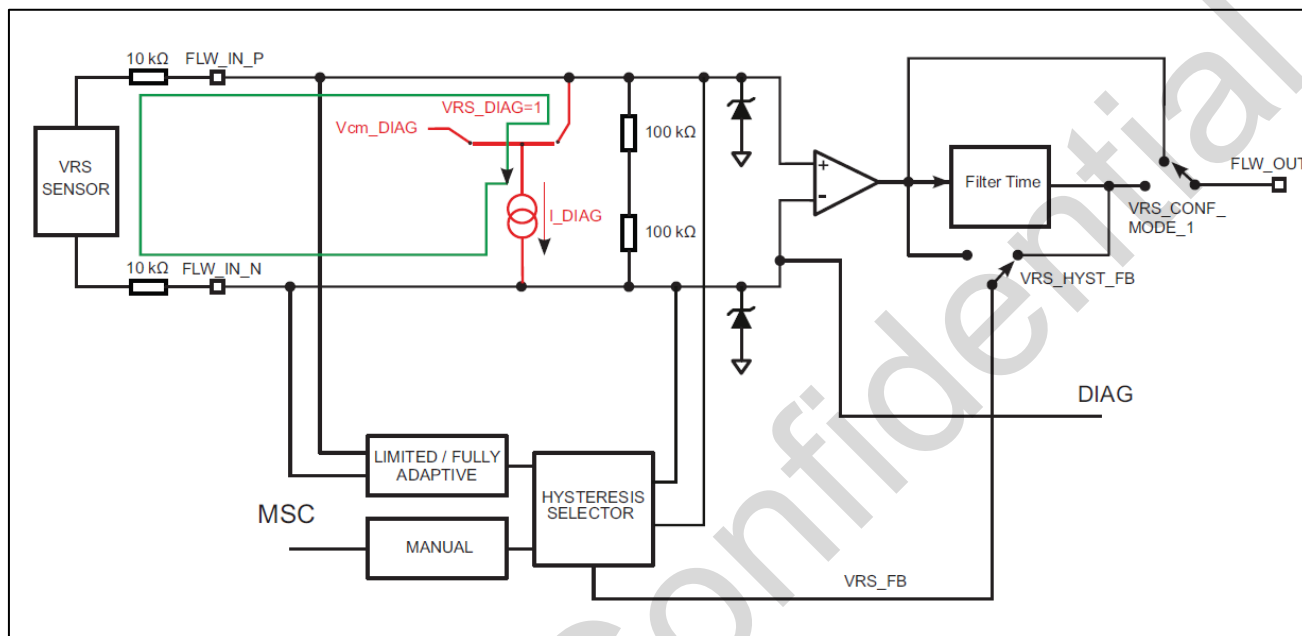


Figure 11-7 VRS block diagram - Diagnostic operating mode - Current path

11.4 Application circuit

Sensor sketch and parameters are reported in Figure 11-8 and Table 11-5 VRS sensor parameters.

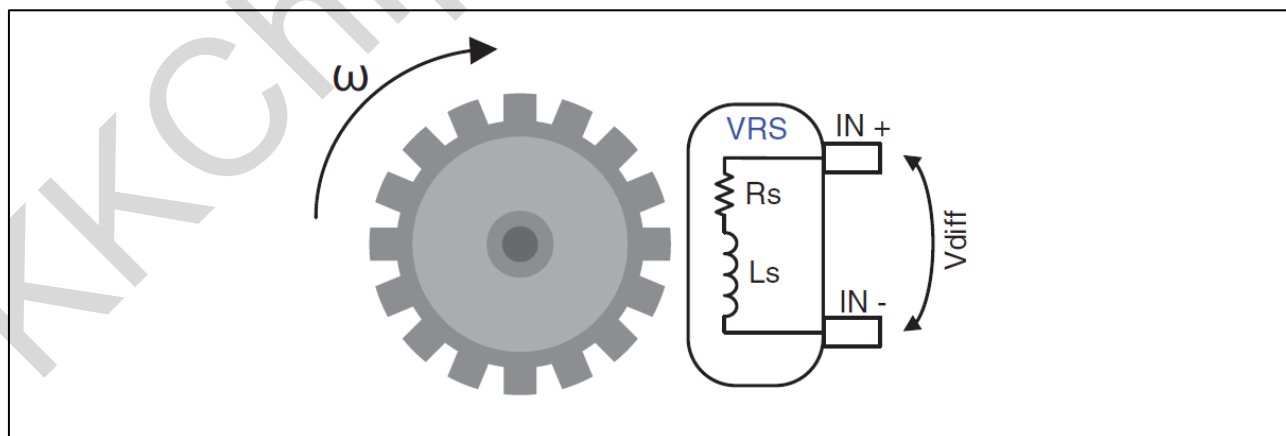


Figure 11-8 Sensor sketch

Table 11-5 VRS sensor parameters

Symbol	Parameter	Min	Typ	Max	Unit	Number
Rs	Sensor resistance	300	600	1000	Ω	P_11.4.1
Ls	Sensor inductor	-	250	-	mH	P_11.4.2
Vdiff	Sensor output voltage	-200	-	+200	V	P_11.4.3
Tout	Output period	100	-	5000	μ s	P_11.4.4

The interface handles signals coming from magnetic pick-up sensors, see [Figure 11-9](#) Variable reluctance sensor (VRS), or Hall Effect sensors with two possible configurations, as per [Figure 11-10](#) and [Figure 11-11](#).

The interface feeds the digital signal to microcontroller that extracts flying wheel rotational position, angular speed and acceleration.

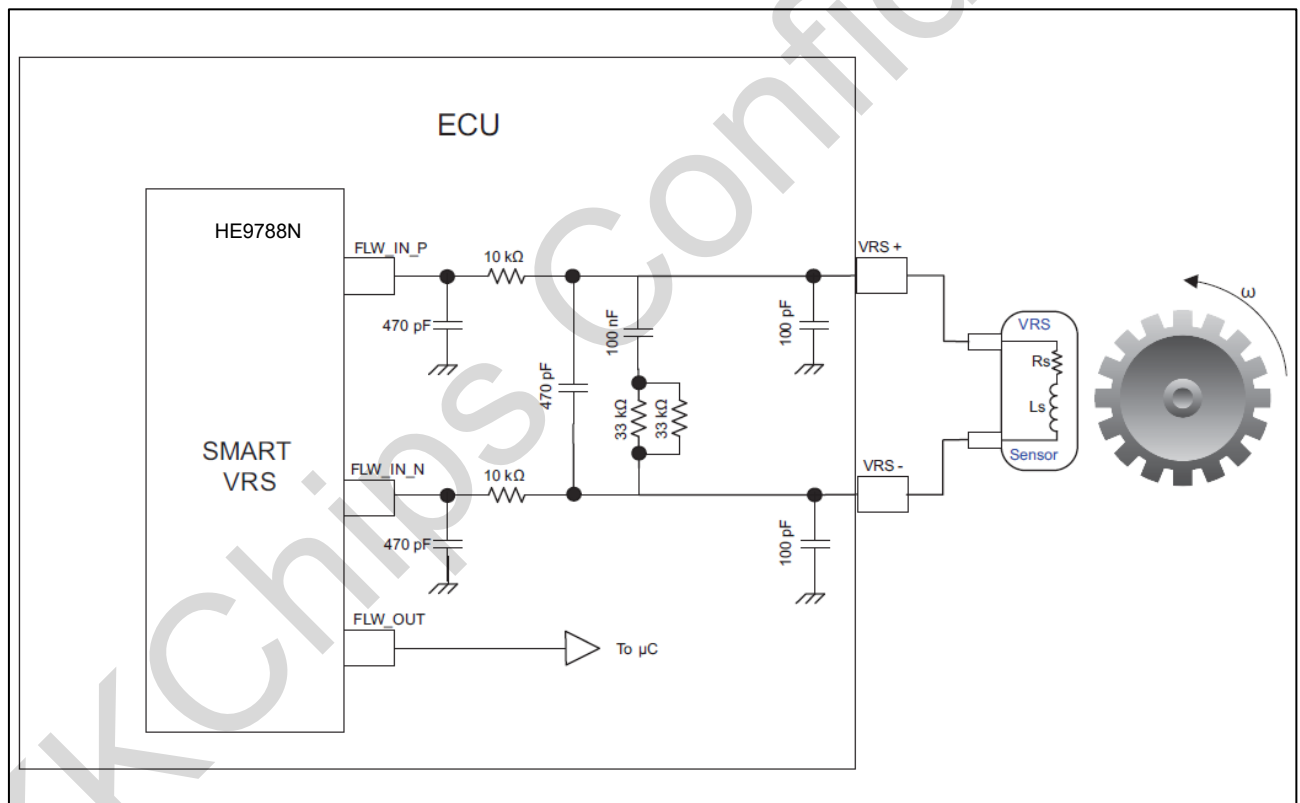


Figure 11-9 Variable reluctance sensor (VRS)

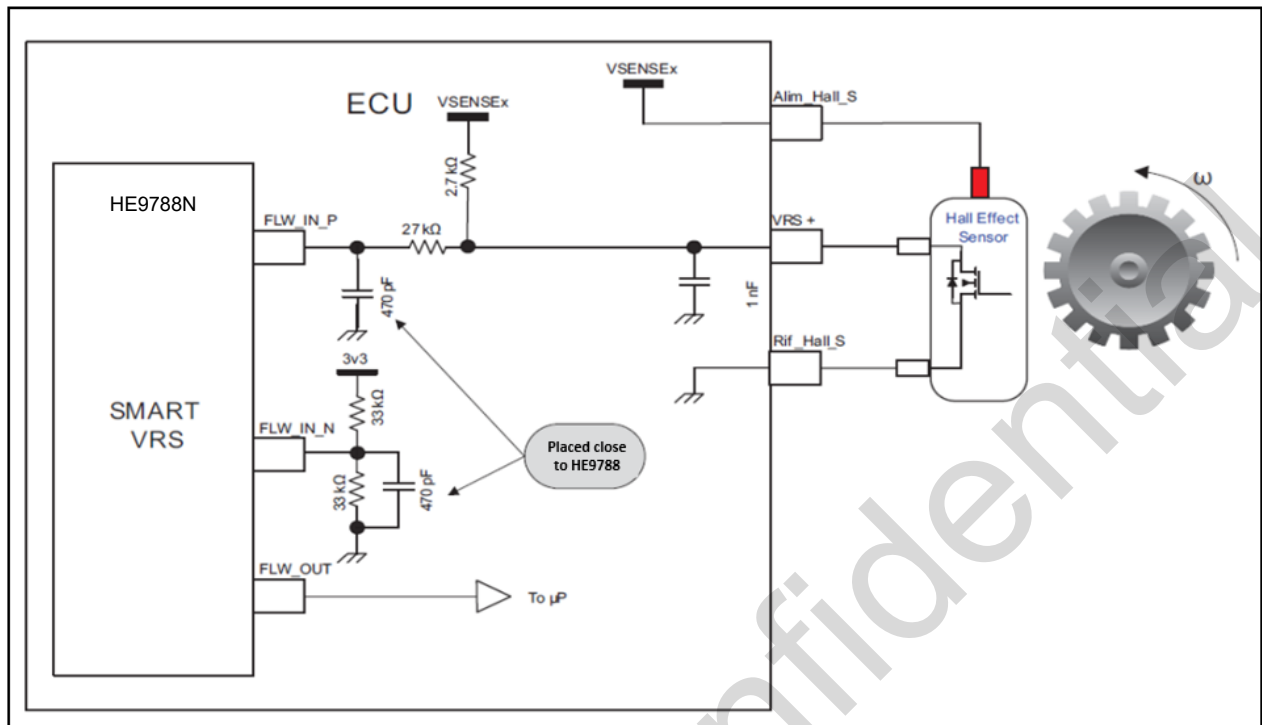


Figure 11-10 Hall effect sensor configuration 1

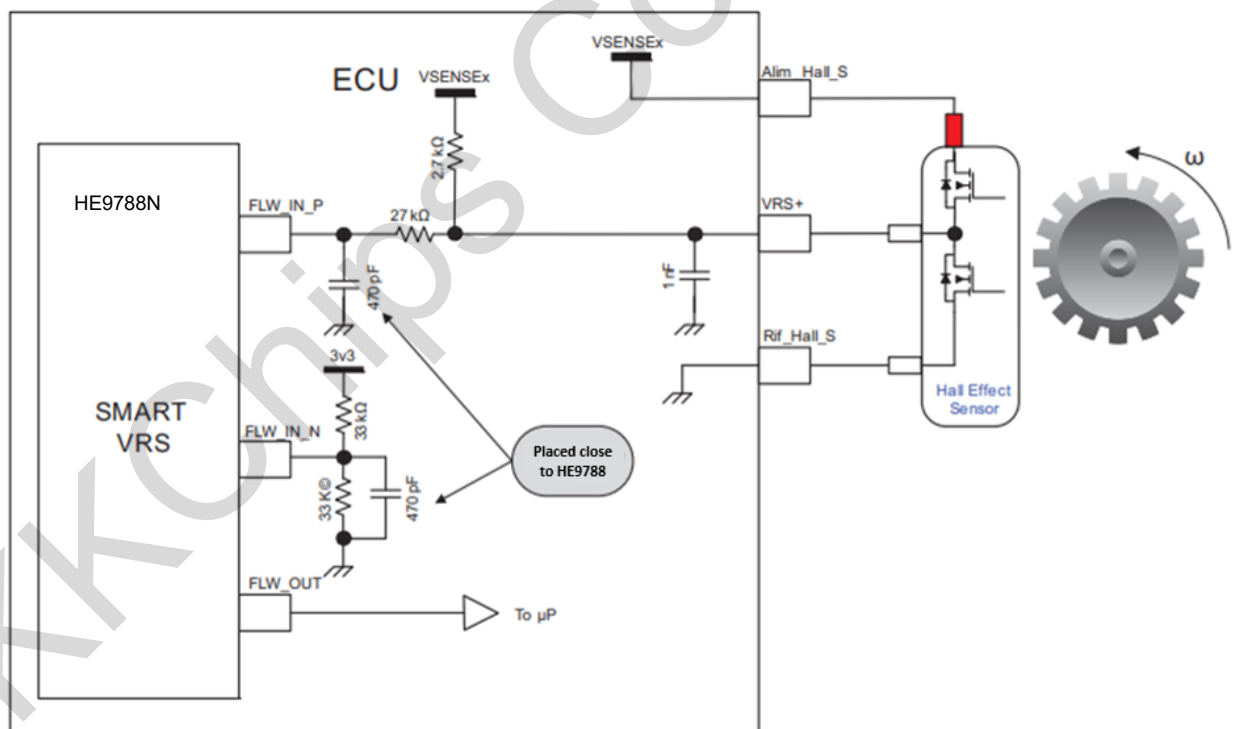


Figure 11-11 Hall effect sensor configuration 2

Conditions:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, T_j -40 to 175 °C unless otherwise specified;

Table 11-6 Flying wheel interface function - Diagnosis test

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
V_{iThL}	Input high-to-low differential threshold voltage	-	-50	0	50	mV	FLW_IN_P FLW_IN_N	P_11.4.5
V_{CM}	Common mode operating range	Not to be tested. It is an application note.	0	1.65	3	V	FLW_IN_P FLW_IN_N	P_11.4.6
V_{clpH}	Input high clamping voltage	$ I_{FLW_IN_P} = I_{FLW_IN_N} = 20\text{ mA}$, device ON	5.0-0.5	-	5.0+0.5	V	FLW_IN_P FLW_IN_N	P_11.4.7
V_{clpH}	Input high clamping voltage	$ I_{FLW_IN_P} = I_{FLW_IN_N} = 20\text{ mA}$, device OFF	1	-	2.8	V	FLW_IN_P FLW_IN_N	P_11.4.8
V_{clpL}	Input low clamping voltage	$ I_{FLW_IN_P} = I_{FLW_IN_N} = 20\text{ mA}$	-1.5	-	-0.3	V	FLW_IN_P FLW_IN_N	P_11.4.9
$V_{openload}$	Output open load voltage	FLW_IN_P = FLW_IN_N = $V_{openload}$	1.5	(3.3)/2	1.8	V	FLW_IN_P FLW_IN_N	P_11.4.10
I_{bvrsp}	Input bias current V_{rsp}	VRS_INP → FLW_IN_P	-	-	2	μA	FLW_IN_P FLW_IN_N	P_11.4.11
I_{bvrsn}	Input bias current V_{rsm}	FLW_IN_N = $V_{openload}$	-	-	2	μA	FLW_IN_P FLW_IN_N	P_11.4.12
V_{OL}	Output Low Voltage	VDD5 = 5 V or 3.3 V I_{sink} current = 2 mA	-	-	0.5	V	FLW_OUT	P_11.4.13
V_{OH}	Output High Voltage	VDD5 = 5 V or 3.3 V I_{source} current = 2 mA	VDD5-0.5	-	-	V	FLW_OUT	P_11.4.14
I_{lk_outvrs}	Input leakage current to GND	-	-	-	1	μA	FLW_OUT	P_11.4.15
I_{lk_outvrs}	Input leakage current to VDD5	-	-	-	8	μA	FLW_OUT	P_11.4.16
$T_{d_off_outvrs}$	Delay on falling edge	Test Ext cap = 300 pF	-	-	2	μs	FLW_OUT	P_11.4.17
$T_{d_on_outvrs}$	Delay on rising edge	Input signal=4ms	-	-	150	μs	FLW_OUT	P_11.4.18
$T_{r_OUT_vrs}$	MRX Rise Time	Test Ext cap = 300 pF	-	-	250	ns	FLW_OUT	P_11.4.19
$T_{f_OUT_vrs}$	MRX Fall Time	Test Ext cap = 300 pF	-	-	150	ns	FLW_OUT	P_11.4.20
$V_{outdiag}$	Output diag voltage	FLW_IN_P = open; diag mode	0.9	(3.3)/3	1.3	V	FLW_OUT	P_11.4.21
$I_{outdiag}$	Output diag Current	FLW_IN_P = open; FLW_IN_N = GND; diag mode	50	65	80	μA	FLW_OUT	P_11.4.22
V_{outsh} $V_{bdiagth}$	Output Short-circuit range to VBAT Open Load threshold	FLW_IN_P = open; FLW_IN_N = V_{ramp} ; diag mode	2.8	3	3.2	V	FLW_OUT	P_11.4.23
$V_{outshgnddiagth}$	Output Short-to GND range threshold	FLW_IN_P = open; FLW_IN_N = V_{ramp} ; diag mode	1.1	1.3	1.5	V	FLW_OUT	P_11.4.24

Note: When FLW_IN_P and FLW_IN_N are both in input high clamping condition, the clamp voltage of FLW_IN_N is 30 mV (typical) higher than FLW_IN_P.

12 CAN FD interface

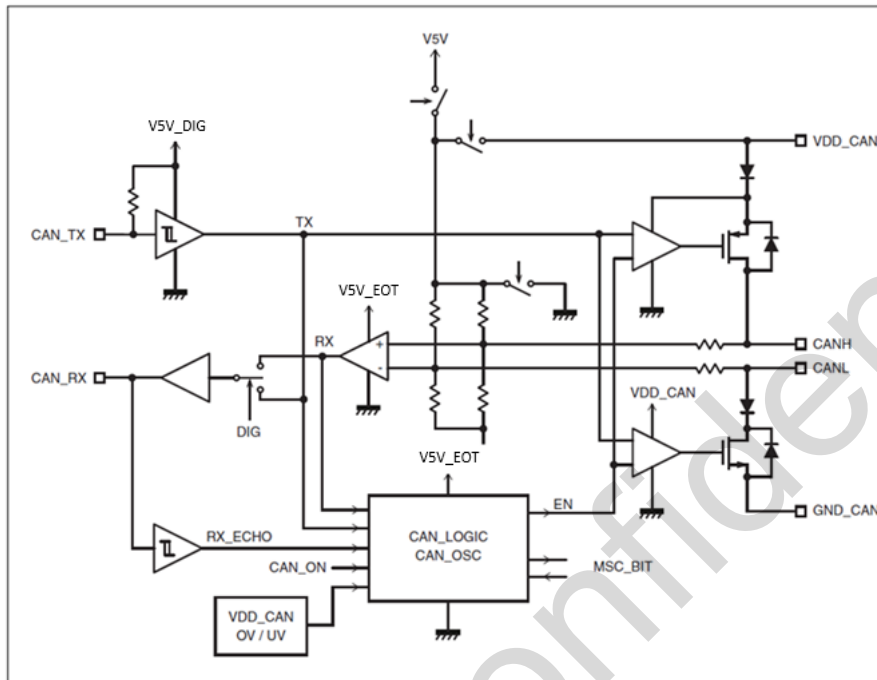


Figure 12-1 CAN FD interface diagram

The pins related to this block are:

- CANH;
- CANL;
- CANTX;
- CANRX;
- GND_CAN;
- VB_CAN;
- VDD_CAN

VDD_CAN= 5V is the supply for the CAN transceiver block. The voltage on this pin is monitored and if the voltage is below VDD_CAN_SUP_LOW for a $t_{cansuplow}$ filter time an undervoltage fault on VDD_CAN is detected. The fault is latched in a dedicated MSC flag CAN_SUP_LOW until an MSC Upstream Read14 is given. While the undervoltage fault is detected on VDD_CAN the CAN transmitter stays disabled (not MSC read clear access is necessary to restore the transmitter just the fault detection expiration).

If VDD_CAN is above VDD_CAN_SUP_OV for a $t_{cansuphigh}$ filter time an overvoltage fault on VDD_CAN is detected. The fault is latched in a dedicated MSC flag VDD_CAN_OV until an MSC Upstream Read14 is given during the MSC flag VDD_CAN_OV stays set the CAN transmitter is disabled

During a VDD_CAN overvoltage or undervoltage detected fault the biasing voltage is switched from VDD_CAN to a pre-regulated internal voltage (2.5V) supplied by VB_STBY.

When the fault ends the biasing voltage switches again on VDD_CAN. The msc flag *VDD_CAN_OV* or *CAN_SUP_LOW* remains latched until an MSC Upstream Read14 is given.

The biasing voltage is switched from VDD_CAN to a pre-regulated internal voltage (2.5V) supplied by VB_STBY also in CAN_BIAS_ON and CAN_BIAS_OFF.

VB_CAN is the signal that enables CAN cell; if this pin is connected to VB_STBY the CAN is enabled while if it's connected to GND the CAN is disabled

12.1 Functional description

General requirements:

- Compliant with C&S CAN FD certification(ISO16845-2:2015);
- Communication Speed up to 2 Mbit/s. (up to 5Mbit/s in "programming" mode);
- Function range from -27 V to +40 V DC at CAN Pins;
- GND disconnection fail safe at module level;
- GND shift operation at system level;
- Microcontroller interface with CMOS compatible I/O Pins;
- Matched output slopes and propagation delay;
- Receive-only mode available.

In order to further reduce the current consumption in low-power mode, the integrated CAN bus interface offers an ultra low current consumption state when the u-chip is off called Sleep mode.

12.2 CAN state diagram

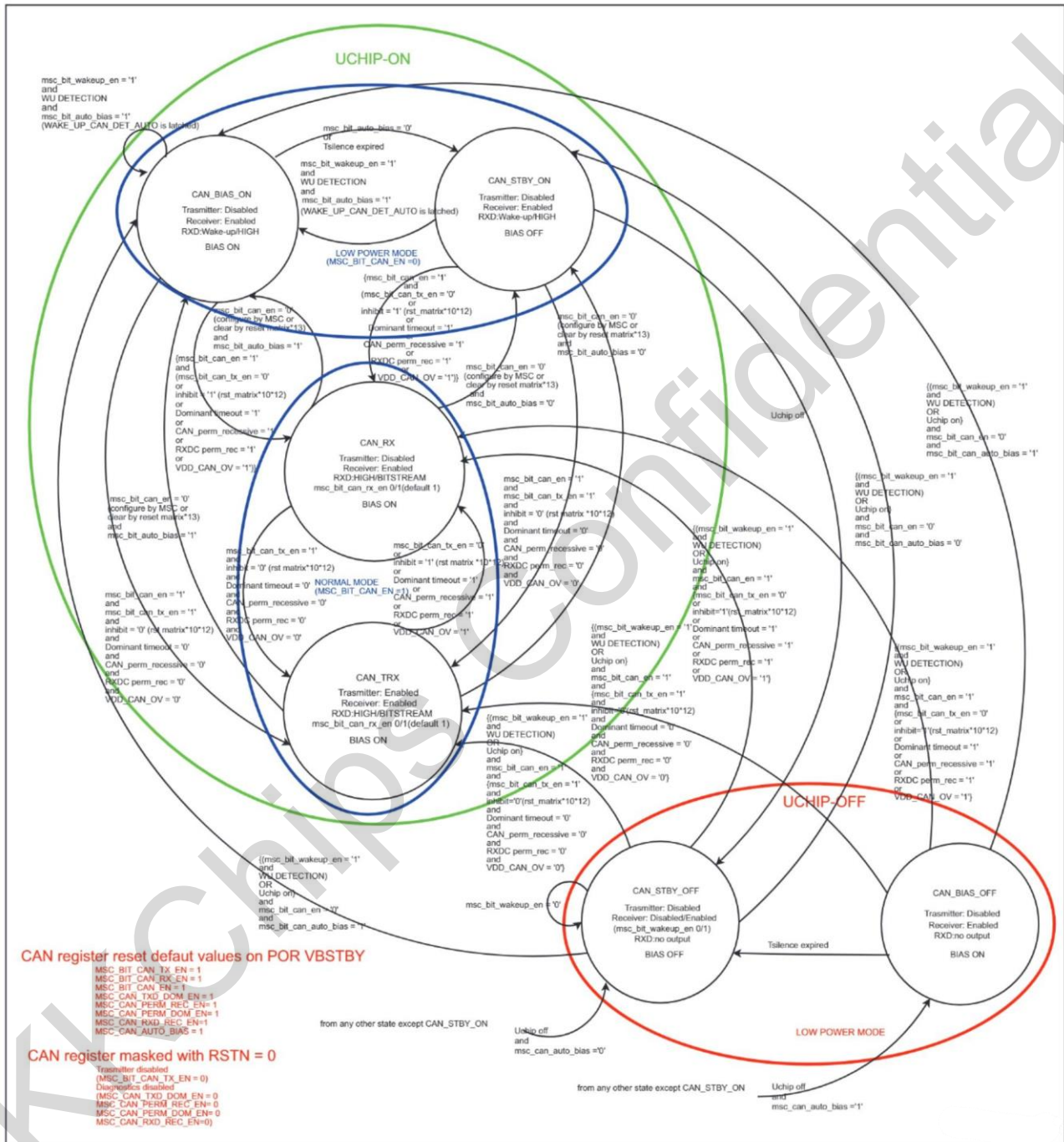


Figure 12-2 CAN state diagram

12.3 CAN normal mode

If `CAN_EN` bit `MSC_CONFIG-REG17-1[2]` is set and the U-chip is in ON state the CAN transceiver is in normal mode (referring to state diagram, `CAN_RX` or `CAN_TRX` state). In this state the transmitter and receiver can be configured by `MSC` (`CAN_TX_EN`, `CAN_RX_EN`) as follows:

TRX Standby: transmitter disabled and `CAN_RX` pin masked to 1 (`CAN_TX_EN` = '0', `CAN_RX_EN` = '0');

TRX Listen: transmitter disabled, receiver enabled (`CAN_TX_EN` = '0', `CAN_RX_EN` = '1');

TRX transmit: transmitter enabled `CAN_RX` pin masked to 1 (`CAN_TX_EN` = '1', `CAN_RX_EN` = '0');

TRX Normal (default): transmitter enabled receiver enabled (`CAN_TX_EN` = '1', `CAN_RX_EN` = '1')

In normal mode the CAN biasing is always active

Wake up by CAN disabled option is allowed only for the application that does not require the transceiver (`CANH/L` not connected to any bus).

12.4 CAN low-power mode

When HE9788N is ON it is possible to deactivate the CAN transceiver setting the `MSC` bit `CAN_EN` (`MSC_CONFIG-REG17_1[2]`) to zero, sending the CAN in Low-power Mode. The CAN transceiver remains deactivated until it will be activated again setting the `CAN_EN` bit. When HE9788N is OFF the CAN transceiver is always in Low-power Mode independently from any other configuration bit except `CAN_WAKEUP_EN`, which if set to zero, sends it to Sleep mode once the transceiver goes to `CAN_STBY_OFF` state. In Low-power mode the biasing can be active or inactive depending on `CAN_AUTO_BIAS` bit value and from the CAN bus activity (see state diagram), furthermore the transmitter is always disabled and the `CAN_RX`

pin is always masked to recessive. When biasing is disabled the receiver input termination will be biased at zero volt.

12.5 CAN Sleep Mode

When the CAN transceiver goes in `CAN_STBY_OFF` with `CAN_WAKEUP_EN` = '0' it falls in an ultra low current consumption state called Sleep mode. In this state for the HE9788N is impossible to detect Wake-up activity on CAN bus and consequently no power-up is done for a CAN detection (pattern or no pattern).

12.6 CAN error handling

CAN Driver HS and LS are supplied by a dedicated pin (`VDD_CAN`) to avoid disturbances of CAN activity on main supply lines. After internal power on reset of `VB_STBY` the CAN transceiver is configured according to the configuration register bit default values. When the

CAN transceiver is in normal mode, the transmitter is disabled and its state changes to CAN_RX in case of at least one of the following events:

- Dominant TxDC time out and *CAN_TXD_DOM_EN* MSC bit = 1;
- CAN permanent recessive *CAN_PERM_REC_EN* MSC bit = 1;
- RxDC permanent recessive *CAN_RXD_REC_EN* MSC bit = 1;
- VDD_CAN overvoltage detection (MSC bit *VDD_CAN_OV* = 1);
- VDD_CAN undervoltage detection active;
- MSC bit *CAN_TX_EN* = 0;
- Reset matrix Conditions *10*12 (see Reset Matrix section).

The CAN receiver is not disabled in case of any failure condition.

The device provides the following 4 error handling features; the CAN error handling function can be disabled by setting the dedicated corresponding MSC bit. The error handling must be also enabled by setting the *CAN_EN* bit MSC_CONFIG-REG17-1[2].

12.6.1 Dominant CAN_TX time out

If CAN_TX is in dominant state (low) for $t > t_{TXDC_DOM_TO}$ the transmitter will be disabled, CAN_TXD_DOM status bit will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared or until the state returns recessive according to the setting of the bit *CAN_TX_DOM_ERR_CFG* MSC_CONFIGREG21 [0]. The detection of this error is enabled by the *CAN_TXD_DOM_EN* bit MSC_CONFIG_REG21[3] and by the *CAN_TX_EN* bit MSC_CONFIG_REG17_1[3].

12.6.2 CAN permanent recessive

If CAN_TX changes to dominant (low) state but CAN bus does not follow for 4 times, the transmitter will be disabled, *CAN_PERM_REC* status bit will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared.

The detection of this error is enabled by the *CAN_PERM_REC_EN* bit MSC_CONFIG_REG21[4] and by the *CAN_TX_EN* bit MSC_CONFIG_REG17_1[3].

12.6.3 CAN permanent dominant

If the bus state is dominant (low) for $t > t_{dom}$ a permanent dominant status will be detected. *CAN_PERM_DOM* status bit is latched and can be read and cleared by MSC. The transmitter will not be disabled. The detection of this error is enabled by the *CAN_PERM_DOM_EN* bit MSC_CONFIG_REG21[5].

12.6.4 CAN_RX permanent recessive

If CAN_RX Pin is clamped to recessive (high) state, the controller is not able to recognize a bus dominant state and could start messages at any time, which results in disturbing the overall bus communication. Therefore, if CAN_RX does not follow CAN_TX for 4 times the transmitter will be disabled. *CAN_RXD_REC* status bit will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared. The detection of this error is enabled by the *CAN_RXD_REC_EN* bit MSC_CONFIG_REG21[6] and by the *CAN_TX_EN* bit MSC_CONFIG_REG17_1[3].

12.6.5 Smart reset RSTN effects

In order to avoid spurious interference on CAN Bus and false CAN error detection, during the power-up sequence, as long the Smart Reset RSTN pin signal stays low the following CAN configuration registers are masked to zero:

- `CAN_TX_EN` (transmitter disabled);
- `CAN_TXD_DOM_EN` (Dominant TxDC Time Out disabled);
- `CAN_PERM_REC_EN` (CAN Permanent Recessive disabled);
- `CAN_PERM_DOM_EN` (CAN Permanent Dominant disabled);
- `CAN_RXD_REC_EN` (CAN RXDC Permanent Recessive disabled).

12.7 Wake up With U-Chip OFF

When the U-chip is off with CAN wake up option enabled (`CAN_WAKEUP_EN = '1'`), the CAN transceiver can detect wake up activity on bus and then wake up the device. In this situation the MSC bit `WAKE_UP_CAN_DET` is latched (MSC UPSTREAM READ 12, Frame3). If the U-chip is off with CAN wake up option disabled (`CAN_WAKEUP_EN = '0'`) and automatic voltage biasing disabled (`CAN_AUTO_BIAS = '0'`) the transceiver goes in Sleep mode. Otherwise if the U-chip is off with CAN wake up option disabled

(`CAN_WAKEUP_EN = '0'`) and automatic voltage biasing enabled (`CAN_AUTO_BIAS = '1'`) the transceiver goes in `CAN_BIAS_OFF` state and after a period greater than T_{silence} without activity on bus goes in Sleep mode.

12.8 Wake up with U-Chip ON

When the U-chip is on and the CAN transceiver is in low power mode (`CAN_EN` bit = '0') and the wake up option is enabled (`CAN_WAKEUP_EN = '1'`), the CAN Transceiver can detect wake up activity on bus. In this situation the MSC bit `WAKE_UP_CAN_DET_AUTO` is latched (MSC UPSTREAM READ 13, Frame3) and a dominant pulse of $t_{\text{dom_pulse}}$ is send on pad `CAN_RX`. if the `CAN_AUTO_BIAS` bit MSC_CONFIG_REG21[7] is set and the CAN transceiver in the state `CAN_STBY_ON` the detection of a wake up activity will bring the transceiver in the state `CAN_BIAS_ON`

12.8.1 Wake up options

For the wake up feature the device logic differentiates different Wake-up detection conditions.

12.8.2 Normal pattern wake up

Normal pattern wake up can occur when CAN pattern wake up option is enabled (`CAN_PATTERN_EN = '1'`), the wake up enable bit is set (`CAN_WAKEUP_EN = '1'`) and the CAN transceiver is in low-power mode. In order to have a Wake-up detection, the following criteria must be fulfilled:

- The CAN interface wake up receiver must receive a series of two consecutive valid

dominant pulses, each of which must be longer than t_{filter} ;

- The distance between 2 pulses must be longer than t_{filter} ;
- The two pulses must occur within a time frame of t_{wake} .

12.8.3 No pattern wake up

If the CAN pattern wake up option is disabled ($\text{CAN_PATTERN_EN} = '0'$) and the wake up enable bit is set ($\text{CAN_WAKEUP_EN} = '1'$) and the transceiver is in Low-power mode, a dominant state on CAN BUS for more than t_{filter} will cause a Wake-up detection.

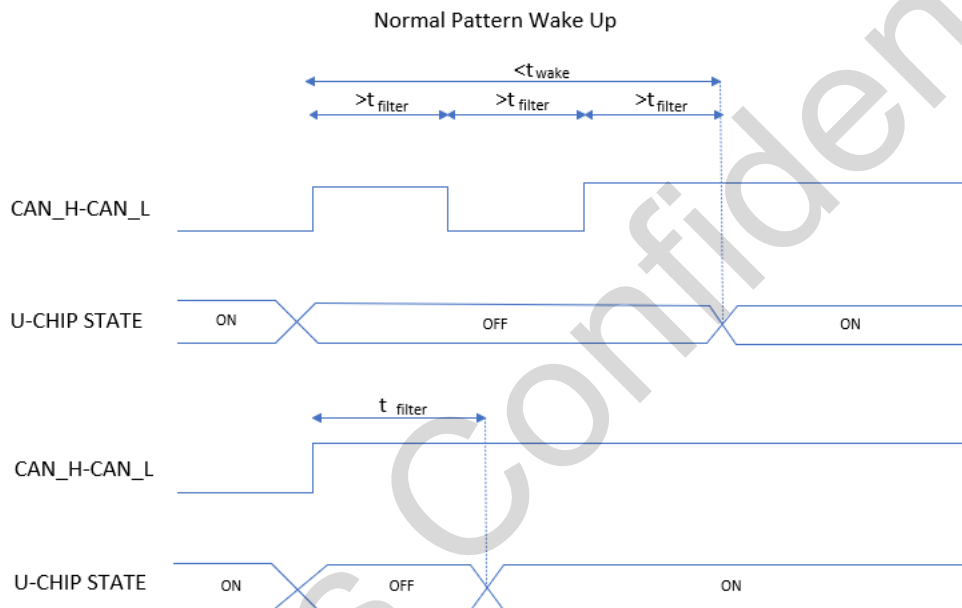


Figure 12-3 CAN wake up options

12.9 Automatic voltage biasing

If the CAN transceiver is in Low-power mode ($\text{CAN_EN} = 0$ or U-CHIP OFF) and the automatic voltage biasing is active ($\text{CAN_AUTO_BIAS} = '1'$, MSC_CONFIG-REG21[7]) if there has been no activity on the bus for longer than t_{SILENCE} , the bus lines are biased towards 0 V via the receiver input resistors RCANH/RCANL (biasing disabled) and the MSC flag CAN_SILENT is set to 1. If the wake up option is enabled ($\text{CAN_WAKEUP_EN} = '1'$) and a wake-up activity on the bus lines is detected (wake-up pattern or no pattern), the bus lines are biased to VCANHrec respectively VCANLrec via the internal receiver input resistors RCANH/RCANL also the MSC bit $\text{WAKE_UP_CAN_DET_AUTO}$ is latched (MSC UPSTREAM READ 12, Frame3). The biasing is activated not later than t_{Bias} .

12.10 CAN reset matrix

In the reset matrix table there is a row dedicated to CAN.

Reset matrix *12 source means that at least one of the following faults is active:

- VDD5_OV;
- OSC FAULT;
- TNL;
- RST_N.

When the CAN transceiver is in normal mode and at least one of the above faults occurs the transmitter is disabled.

Reset matrix *10 source means that at least one of the following fault is active and the *CAN_TDI* bit is set:

- *ERR_CNT*>4;
- *ERR_CNT*>7;
- RST_PRL;

When the CAN transceiver is in normal mode and at least one of the above faults occurs the transmitter is disabled.

Reset matrix *13 source means that at least one of the following fault is active:

1. VB_STBY_UV
2. VB_OV $t > TBOV2$
3. VB_UV
4. V5VA_OV, V5VA_UV, V5VD_OV and V5VD_UV
5. VDD5_UV
6. VDDIO_UV

When the CAN transceiver is in normal mode and at least one of the above faults occurs the MSC bit *CAN_EN* is cleared sending the CAN transceiver in Low-power Mode.

12.11 CAN looping mode

If the proper configuration bit (*CAN_LOOP_EN*) in control register is set the TxDC input is mapped directly to the RxDC Pin. This mode can be used in combination with the CAN receive-only mode, to run diagnosis for the CAN protocol handler of the microcontroller.

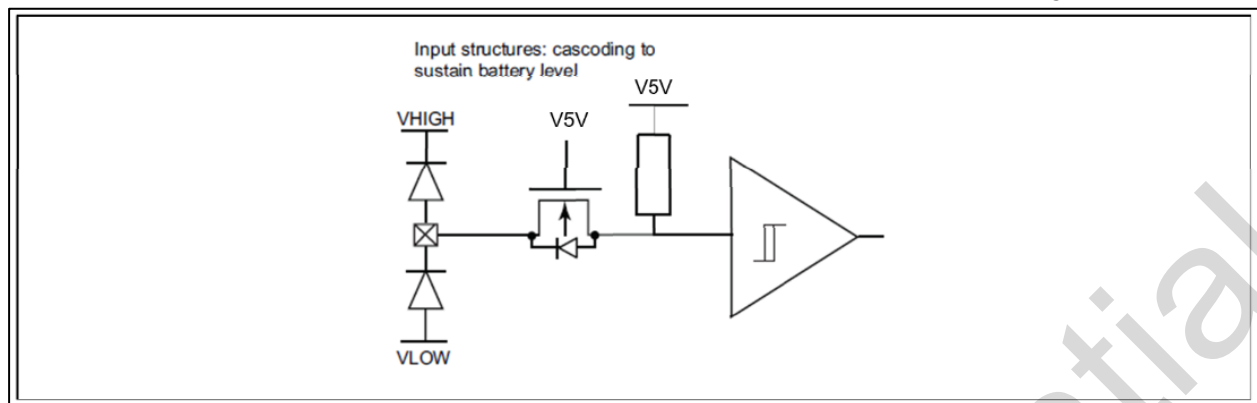


Figure 12-4 CAN_TX input structure

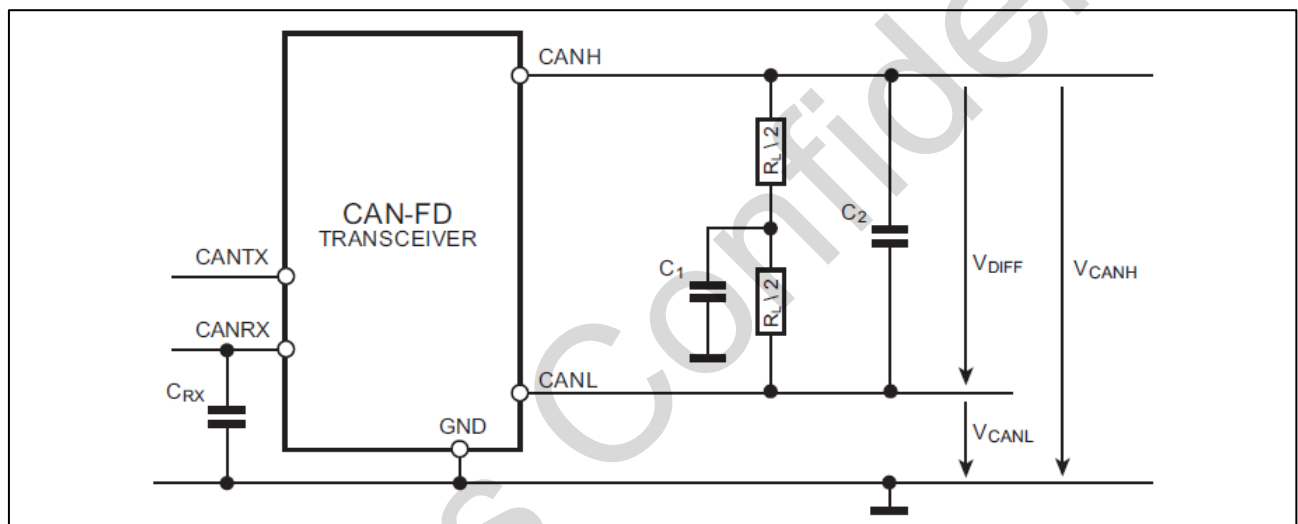


Figure 12-5 CAN transceiver test circuit

Conditions:

5.5 V $\leq$ VB_IN $\leq$ 18 V, 4.5 V $\leq$ VDD_CAN $\leq$ 5.5 V, 5 V $\leq$ VB_STBY $\leq$ 18 V, T_j -40 to 175 °C unless otherwise specified;

Table 12-1 CAN threshold related Low supply voltage flag

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VDD_CAN_SUP_LOW	Threshold to detect CAN Supply low	VDD_CAN decreasing	4	4.65	4.8	V	-	P_12.11.1
VB_CAN_TH(1)	Threshold for CAN cell enable	Not tested For information only	0.8	-	2.2	V	-	P_12.11.2
VDD_CAN_SUP_OV	Threshold to detect CAN Supply over-voltage	VDD_CAN increasing	6.1	6.85	7.5	V	-	P_12.11.3

1. Below this threshold all CAN functions are disabled. In case CAN functionality is requested by application VB_CAN has to be shorted to VB_stby, otherwise it should be put to GND to disable CAN block.

Table 12-2 CAN communication operating range

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VDD_CAN	Supply Voltage operating range for CAN transceiver	-	4.5	-	5.5	V	-	P_12.11.4
VCANHL,CM	Common mode Bus voltage (VCANH + VCANL) / 2	Measured with respect to the ground of the CAN transceiver	-12	-	12	V	-	P_12.11.5
ITRCV_REC ⁽¹⁾	Transceiver current consumption during normal mode from VDD_CAN, Recessive State	VTXDC = VTXDCHIGH	-	-	10	mA	-	P_12.11.6
ITRCV_DOM ⁽¹⁾	Transceiver current consumption during normal mode from VDD_CAN Dominate State	VTXDC = 0 V	-	-	60	mA	-	P_12.11.7
ITRCV_short	Transceiver current consumption during output short from VDD_CAN	RL = 50Ω to 65Ω; VCANH = -3 V or VCANL = 40 V	-	-	120	mA	-	P_12.11.8
ITRCVLPbias	Transceiver current consumption during lowpower mode; biasing active from VDD_CAN	RL = 50Ω to 65Ω; VTXDC = VTXDCHIGH;	-	-	350	μA	-	P_12.11.9
ITRCVLP	Transceiver current consumption during lowpower mode; biasing inactive from VDD_CAN	RL = 50 Ω to 65Ω; VTXDC = VTXDCHIGH;	-	-	50	μA	-	P_12.11.10
ITR_VB	Transceiver current consumption from VB_CAN	-	-	-	5	μA	-	P_12.11.11
BR	Supported Bit-rates	Supported bit-rates at which all	5	-	-	Mb/s	-	P_12.11.12

		requirements are fulfilled Application info						
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1. To be confirmed after ATE measurements.

Table 12-3 CAN transmit data input: Pin TxDC

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Ileak_cantx	Input leakage current	CAN_TX = VDDIO - 1 V	-	-	10	μA	CAN_TX	P_12.11.13
VTXDCLOW	Input voltage dominant level	-	-	-	0.9	V	CAN_TX	P_12.11.14
VTXDCHIGH	Input voltage recessive level	-	2	-	-	V	CAN_TX	P_12.11.15
VTXDCCHYS	VTXDCHIGH-VTXDCLOW	-	0.1	-	0.4	V	CAN_TX	P_12.11.16
RTXDCPU	TxDC pull up resistor	-	20	50	100	kΩ	CAN_TX	P_12.11.17
td,TXDC(dom-rec)	TxDC - CANH,L Delay Time dominant - recessive	RL = 60 Ω (±1%); C2 = 100 pF (±1%); 70% VTXD - VDIFF = 0.5 V; TXDC rise time = 10 ns (10% - 90%) Guaranteed by bench correlation	0	-	140	ns	-	P_12.11.18
td,TXDC(rec-diff)	TxDC - CANH,L Delay Time recessive - dominant	RL = 60Ω(±1%); C2=100pF (±1%); 30% VTXD - VDIFF = 0.9 V; TXDC fall time = 10ns (90% - 10%) Guaranteed by bench correlation	0	-	120	ns	-	P_12.11.19
tTXDC_DOM_TO	TxDC dominant timeout	-	0.8	2	5	ms	-	P_12.11.20

Table 12-4 CAN transmit data output: Pin RxDC

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VRXDCLOW	Output voltage dominant level	Active mode, IRXDC = 2 mA	(0)	0.2	0.5	V	CAN_RX	P_12.11.21
VRXDCHIGH	Output voltage recessive level	Active mode, IRXDC = -2 mA	VDDIO-0.5	VDDIO-0.2	VD DIO	V	CAN_RX	P_12.11.22
tr, RXDC	RxDC rise time	CRX = 15 pF, 30% - 70% VRXDC Guaranteed by bench correlation	0	-	25	ns	-	P_12.11.23
tf, RXDC	RxDC fall time	CRX = 15 pF, 70% - 30% VRXDC Guaranteed by bench correlation	0	-	25	ns	-	P_12.11.24
td, RXDC (dom-rec)	CANH,L ^ RxDC Delay Time dominant - recessive	CRX = 15 pF, VDIFF = 0.5V - 70% VRXDC Guaranteed by bench correlation	0	-	120	ns	-	P_12.11.25

td, RXDC (rec - dom)	CANH,L ^ RxDC Delay Time recessive - dominant	CRX = 15 pF, VDIFF = 0.9 V - 30% VRXDC	0	-	120	ns	-	P_12.11.26
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Table 12-5 CAN transmitter dominant output characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VCANHdom	Single Ended CANH voltage level in dominant state	VTXDC = VTXDCLOW; RL = 50Ω to 65Ω;	2.75	3.5	4.5	V	-	P_12.11.27
VCANLdom	Single Ended CANL voltage level in dominant state	VTXDC = VTXDCLOW; RL = 50Ω to 65Ω;	0.5	1.5	2.25	V	-	P_12.11.28
VDIFF, dom	Differential output voltage in dominant state: VCANHdom-VCANLdom	VTXDC = VTXDCLOW; RL = 50Ω to 65Ω;	1.5	2.0	3	V	-	P_12.11.29
VDIFF, dom_extR	Differential output voltage in dominant state: VCANHdom-VCANLdom at extended termination resistor range	VTXDC = VTXDCLOW; RL = 45Ω to 70Ω;	1.4	-	3.3	V	-	P_12.11.30
VDIFF, dom_Arb	Differential output voltage in dominant state: VCANHdom-VCANLdom during arbitration	VTXDC = VTXDCLOW; RL = 2240 Ω;	1.5	-	5	V	-	P_12.11.31
VSYM	Driver Symmetry: VSYM = (VCANHdom + VCANLdom)/VDD_CAN VDD_CAN = 5V	Measured over one 1 MHz period (1 μs) RL = 60Ω(±1%) fTXDC = 1MHz (square wave, 50% duty cycle); C1 = 4.7nF (±5%)	0.9	1	1.1	V	-	P_12.11.32
IOCANH, dom (-3 V)	CANH output current in dominant state	VTXDC = VTXDCLOW; VCANH = -3 V;	-115	-		mA	-	P_12.11.33
IOCANL, dom (16 V)	CANL output current in dominant state	VTXDC = VTXDCLOW; VCANL = 16 V;	-	-	115	mA	-	P_12.11.34
IOCANH, dom (40 V)	CANH output current in dominant state	VTXDC = VTXDCLOW; VCANH = 40 V; Vs = 40 V	0	-	5	mA	-	P_12.11.35
IOCANL, dom (40 V)	CANL output current in dominant state	VTXDC = VTXDCLOW; VCANL = 40 V; Vs = 40 V	(0)	-	115	mA	-	P_12.11.36

Table 12-6 CAN transmitter recessive output characteristics, normal mode

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VCANHrec	CANH voltage level in recessive state (Normal mode)	VTXDC = VTXDCHIGH; No Load	2	2.5	3	V	-	P_12.11.37
VCANLrec	CANL voltage level in recessive state (Normal mode)	VTXDC = VTXDCHIGH; No Load	2	2.5	3	V	-	P_12.11.38
VDIFF, recOUT	Differential output voltage in recessive state (Normal mode): VCANHrec-VCANLrec	VTXDC = VTXDCHIGH; No Load	-50	-	50	mV	-	P_12.11.39

Table 12-7 CAN transmitter recessive output characteristics, low power mode, biasing active

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VCANHrecLPbias	CANH voltage level in recessive state	VTXDC = VTXDCHIGH; No Load $5.5\text{ V} \leq \text{VB_STBY} \leq 18\text{ V}$	2	2.5	3	V	-	P_12.11.40
VCANLrecLPbias	CANL voltage level in recessive state		2	2.5	3	V	-	P_12.11.41
VDIFF, recOUTLPbias	Differential output voltage in recessive state VCANHrec-VCANLrec		-50	-	50	mV	-	P_12.11.42

Table 12-8 CAN transmitter recessive output characteristics, low-power mode, biasing inactive

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VCANHrecLP	CANH voltage level in recessive state	VTXDC = VTXDCHIGH; No Load	-0.1	0	0.15	V	-	P_12.11.43
VCANLrecLP	CANL voltage level in recessive state		-0.1	0	0.15	V	-	P_12.11.44
VDIFF, recOUTLP	Differential output voltage in recessive state VCANHrec-VCANLrec		-50	-	50	mV	-	P_12.11.45

Table 12-9 CAN Receiver input characteristics during normal mode

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VTHdom	Differential receiver threshold	$-12\text{ V} \leq \text{VCANH} \leq +12\text{ V}$, $-12\text{ V} \leq \text{VCANL} \leq +12\text{ V}$	0.4	-	1.1	V	-	P_12.11.46

	voltage recessive to dominant state							
VTHrec	Differential receiver threshold voltage dominant to recessive state	- 12 V ≤ V _{CANH} ≤ +12 V, - 12 V ≤ V _{CANL} ≤ +12 V	0.4	-	0.95	V	-	P_12.11.47

Table 12-10 CAN Receiver input characteristics during low power mode, biasing active

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VTHdomLPbias	Differential receiver threshold voltage recessive to dominant state	- 12 V ≤ V _{CANH} ≤ +12 V, - 12 V ≤ V _{CANL} ≤ +12 V	0.4	-	0.95	V	-	P_12.11.48
VTHrecLPbias	Differential receiver threshold voltage dominant to recessive state	- 12 V ≤ V _{CANH} ≤ +12 V, - 12 V ≤ V _{CANL} ≤ +12 V	0.4	-	0.95	V	-	P_12.11.49

Table 12-11 CAN Receiver input characteristics during low power mode, biasing inactive

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VTHdomLP	Differential receiver threshold voltage recessive to dominant state	- 12 V ≤ V _{CANH} ≤ +12 V, - 12 V ≤ V _{CANL} ≤ +12 V	0.4	-	0.95	V	-	P_12.11.50
VTHrecLP	Differential receiver threshold voltage dominant to recessive state	- 12 V ≤ V _{CANH} ≤ +12 V, - 12 V ≤ V _{CANL} ≤ +12 V	0.4	-	0.95	V	-	P_12.11.51

Table 12-12 CAN Receiver input resistance

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Rdiff	Differential internal resistance	V _{TXDC} = V _{TXDCHIGH} ; No Load	12	-	100	kΩ	-	P_12.11.52
RCANH, CANL	Single Ended Internal resistance	V _{TXDC} = V _{TXDCHIGH} ; No Load	6	-	50	kΩ	-	P_12.11.53
mR	Internal Resistance matching RCANH,CANL	Biasing active; V _{TXDC} =V _{TXDCHIGH} ; no load $R = 2 \times (RCAN_H - RCAN_L) / (RCAN_H + RCAN_L)$	-0.03	-	0.03	-	-	P_12.11.54
Cin	Internal capacitance	Guaranteed by design	-	-	66	pF	-	P_12.11.55
Cin, diff	Differential internal capacitance	Guaranteed by design	-	-	33	pF	-	P_12.11.56

Table 12-13 CAN transceiver delay

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
tLOOP,hl	Loop delay TXDC to RXDC (High to Low)	RL = 60Ω(±1%); C2 = 100pF (±1%); 30% VTXDC ^ 30% VRXDC; TXDC fall time = 10ns (90% - 10%); CRX = 15 pF	-	-	255	ns	-	P_12.11.57
tLOOP,lh	Loop delay TXDC to RXDC (Low to High)	RL = 60Ω(±1%); C2 = 100pF (±1%); 70% VTXD ^ 70% VRXD; TXDC fall time = 10ns (10% - 90%); CRX = 15 pF	-	-	255	ns	-	P_12.11.58
TBit(RXD)≤1 Mb/s	Recessive bit symmetry at RXDC	RL = 60Ω(±1%); VDIFF: 0.5 V(falling) - 0.9 V(rising); C2 = 100pF (±1%); CRX = 15 pF; TXD rise and fall time = 10 ns (10% - 90%, 90% - 10%); Test signal to be applied on the TXD input of the implementation is a square wave signal with a positive duty cycle of 1/6 and a period of six times the nominal recessive bit width. Rectangular pulse signal TTXDC= 6000ns, high pulse 1000ns, low pulse 5000ns	745	1000	1255	ns	-	P_12.11.59
TBit(RXD)≤2Mb/s	Recessive bit symmetry at RXDC	Rectangular pulse signal TTXDC= 3000ns, high pulse 500ns, low pulse 2500ns	400	500	550	ns	-	P_12.11.60
TBit(RXD)≤5Mb/s	Recessive bit symmetry at RXDC	Rectangular pulse signal TTXDC = 1200 ns, high pulse 200 ns, low pulse 1000 ns	120	200	250	ns	-	P_12.11.61
TBit(BUS)≤1Mb/s	Recessive bit symmetry at CAN-Bus	RL = 50 Ω to 65Ω; VDIFF: 0.5 V(falling) - 0.9 V(rising); CL = 100 pF; CRXD = 15 pF; TXD rise and fall time =	800	1000	1100	ns	-	P_12.11.62

		10 ns (10% - 90%, 90% - 10%); Test signal to be applied on the TXD input of the implementation is a square wave signal with a positive duty cycle of 1/6 and a period of six times the nominal recessive bit width Rectangular pulse signal TTXDC = 6000 ns, high pulse 1000 ns, low pulse 5000 ns						
TBit(BUS) ≤ 2Mb/s	Recessive bit symmetry at CAN-Bus	Rectangular pulse signal TTXDC = 3000 ns, high pulse 500 ns, low pulse 2500 ns	435	500	530	ns	-	P_12.11.63
TBit(BUS) ≤ 5Mb/s	Recessive bit symmetry at CAN-Bus	Rectangular pulse signal TTXDC = 1200 ns, high pulse 200 ns, low pulse 1000 ns; Guaranteed by bench correlation	155	200	210	ns	-	P_12.11.64
ΔtREC " ≤ 2Mb/s	Receiver Timing Symmetry (TBit(RXD) - TBit(BUS))	Rectangular pulse signal TTXDC = 3000 ns, high pulse 500 ns, low pulse 2500 ns; Guaranteed by bench correlation	-65	-	40	ns	-	P_12.11.65
ΔtREC " ≤ 5Mb/s	Receiver Timing Symmetry (TBit(RXD) - TBit(BUS))	Rectangular pulse signal TTXDC = 1200 ns, high pulse 200 ns, low pulse 1000 ns; Guaranteed by bench correlation	-45	-	15	ns	-	P_12.11.66
Tdom	CAN permanent dominant time-out	Guaranteed by scan	500	700	1000	μs	-	P_12.11.67
tWUP-RXD	time between WUP ₍₁₎ on the CAN bus until RXD is active (i.e. the CAN signal is represented at the RXD output)	Wake-Up according to ISO11898-5 RXD output enabled; Guaranteed by scan	0	-	1	ms	-	P_12.11.68

1. Time starts with the end of last dominant phase of the WUP.

Table 12-14 Maximum leakage currents on CAN_H and CAN_L, unpowered

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
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CAN FD interface

I _{Leakage} , CANH	Input leakage current CANH	Unpowered device; V _{CANH} = 5V ; V _{CANL} = 5V ; V _{DD_CAN} , V _{B_STBY} connected via 0 Ω to GND V _{DD_CAN} , V _{B_STBY} connected via 47 k Ω to GND	-10	-	10	μ A	-	P_12.11.69
I _{Leakage} , CANL	Input leakage current CANL	Unpowered device; V _{CANH} = 5V ; V _{CANL} = 5V ; V _{DD_CAN} , V _{B_STBY} connected via 0 Ω to GND V _{DD_CAN} , V _{B_STBY} connected via 47 k Ω to GND	-10	-	10	μ A	-	P_12.11.70

Table 12-15 Biasing control timings

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
t _{filter}	CAN activity filter time	Guaranteed by scan	0.5		1.8	μ s		P_12.11.71
t _{wake}	Wake-up time out	Guaranteed by scan	0.35	1	5	ms		P_12.11.72
t _{Silence}	CAN timeout	Guaranteed by scan	600	700	1200	ms		P_12.11.73
t _{BIAS}	Bias Reaction Time	R _L = 60 Ω ($\pm 1\%$); C ₂ = 100 pF ($\pm 1\%$); C ₁ = 100 pF ($\pm 1\%$); V _{TXDC} = V _{TXDCLOW} ; 50% V _{DIFF} ; V _{SYM} = 0.1(1); Transition bias inactive to bias active; Guaranteed by scan	-	-	220	μ s	-	P_12.11.74
t _{canhigh}	CAN_SUP_LOW filter time	Guaranteed by scan	8.5	10	11.5	μ s	-	P_12.11.75
t _{canlow}	V _{DD_CAN_OV} filter time	Guaranteed by scan	8.5	10	11.5	μ s	-	P_12.11.76
t _{dom_pulse}	Dominant interrupt pulse	Guaranteed by scan	47.6	56	64.4	μ s	-	P_12.11.77

1. Measured from the start of a dominant-recessive-dominant sequence (each phase 6 μ s) until V_{SYM} $\geq$ 0.1

Table 12-16 Standby current consumption (B attery line @RT)

Total quiescent current consumption(1)	Max value	Unit
EOT disabled and CAN block enabled, CAN wake up disabled	60	μ A
EOT disabled and CAN block enabled, CAN wake up enabled	70	μ A
EOT enabled and CAN block enabled, CAN wake up disabled	70	μ A
EOT enabled and CAN block enabled, CAN wake up enabled	80	μ A

1. HE9788N Total quiescent current consumption at pins V_{B_STBY}, V_{B_IN}, V_{B_IN_SW}, V_{B_SENSE}, CP (through 100 nF capacitor), V_{B_CAN}, MRD, RLY1, LED2, STR2_DRN, PDR1_DRN connected to battery. The total current was measured at battery line @RT.

Table 12-17 Standby current consumption (Battery line @HT)

Total quiescent current consumption(1)	Max value	Unit
EOT disabled and CAN block enabled, CAN wake up disabled	140	μA

1. HE9788N Total quiescent current consumption at pins VB_STBY, VB_IN, VB_IN_SW, VB_SENSE, CP (through 100 nF capacitor), VB_CAN, MRD, RLY1, LED2, STR2_DRN, PDR1_DRN connected to battery. The total current was measured at battery line @HT.

Expected additional current due to bias ON is:

Table 12-18 can_auto_bias = 1 and can_wakeup_en = 1

-	27 °C, I(VB_STBY)	150 °C, I(VB_STBY)
TYP	548 μA	576 μA
MAX	650 μA	675 μA

Table 12-19 can_auto_bias = 1 and can_wakeup_en = 0

-	27 °C, I(VB_STBY)	150 °C, I(VB_STBY)
TYP	563 μA	591 μA
MAX	660 μA	690 μA

13 LIN/K-LINE interface

13.1 Functional description

This interface is adapted to “LIN Specification Rev.2.1”, and it is also adapted to “K-LINE Specification (ISO9141)”.

The Chip has over-voltage protection, loss of ground protection from -27 V to 40 V. -27 V AMR is guaranteed at room & hot temperature. +40 V AMR is guaranteed in all temp range. Short Circuit to Battery or Ground Protection is provided. If the bus line is short circuit to battery, the over current protection is protected by current limitation function. Upon return of connection, normal operation resumes without any intervention on the LIN bus line.

In case of power loss (VB_IN pin disconnected or power off) or ground loss (GND pin disconnected) or LIN block in receive only mode (Transmission function disabled), the LIN driver will not disturb the communication of the remaining transceivers connected to LIN or KLINE bus. Upon return of connection, normal operation will resume without any intervention on the LIN bus line.

LIN-TX is internally pulled up to internal 3.3 V.

All values in this section apply over the entire operating temperature and life of the U-chip.

The LIN block MAX baudrate is 20 Kbit/s.

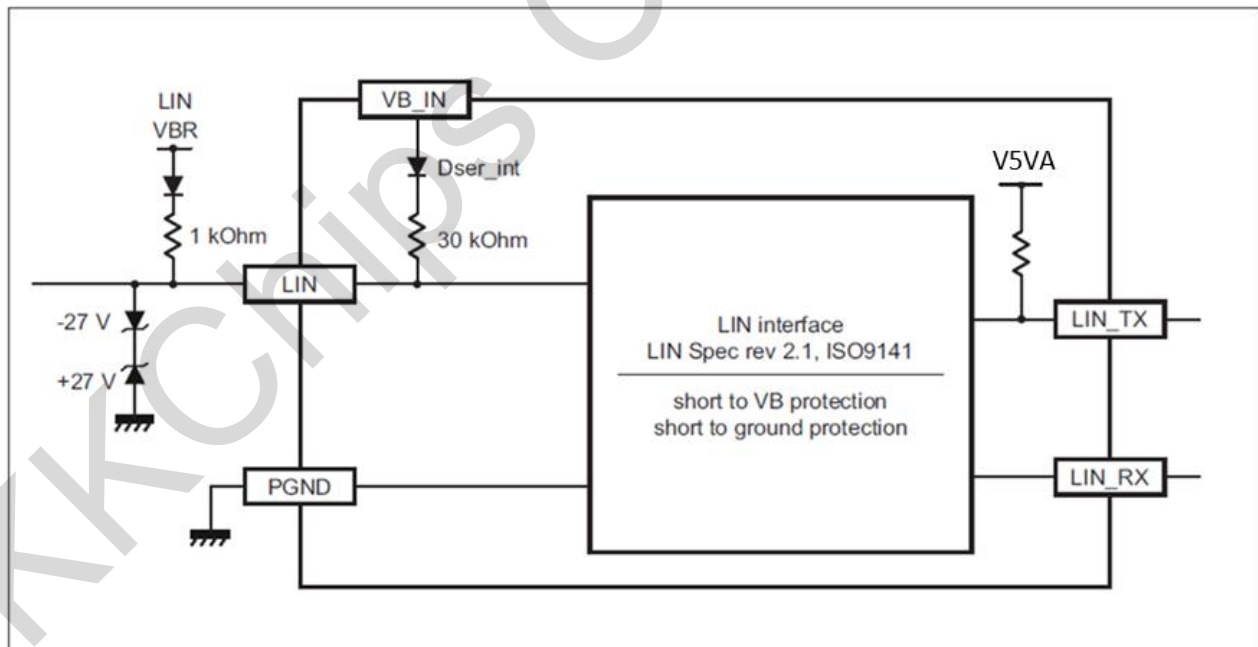


Figure 13-1 Block diagram of LIN interface

13.2 LIN receive only mode

There is a MSC bit (*LIN_TX_EN* CONFIG-REG 11) to put the LINE block in receive only mode (Transmission function disabled). Refer to reset matrix for the conditions to enter receive only mode.

13.3 LIN thermal shut down

The LIN block has a dedicated thermal sensor to protect itself.

To protect LIN block from temperature overheat (high battery range of operation, soft short conditions, etc.) a dedicated thermal sensor placed close to power mos is present in the layout sensing FET temperature; One threshold is implemented (*T_SD_H*) with Hysteresis (*T_SD_hys*). When the *T_SD_H* threshold is detected the Driver switches OFF. Once the power stage has been switched-off for over-temperature detection, it will be able to switch on again when temperature is decreased below thermal shut down threshold plus hysteresis value to avoid high frequency on-off cycling.

13.4 LIN error handling

The Chip provides the following 3 error handling features which are not described in the standard LIN Spec Rev2.1, but are realized in different stand alone LIN Interface / microcontrollers to switch the application back to normal operation mode.

The error handling features can be disabled through the *LIN_ERR_EN[2:0]* (CONFIG-REG 11 D1~D3)bit.

13.4.1 LIN dominant TXD timeout

If TXDC is in dominant state (low) for $t > t_{TXDC_DOM_TO}=12\text{ms}(\text{typ})$ the transmitter will be disabled, status bit *LIN_TXD_DOM* (Upstream Bit Map Read15 frame 1) will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared or until the state returns recessive, according to MSC configuration bit *LIN_TX_DOM_ERR_CFG*.

13.4.2 LIN Permanent Dominant

In case the bus state is dominant (low level) for more than $t_{LIN2}: 12\text{ ms}(\text{typ})$ a permanent dominant status is detected. The status bit *LIN_PERM_DOM* (Upstream Bit Map Read15 frame 1) is latched and can be read and cleared by MSC. The transmitter is not switched off.

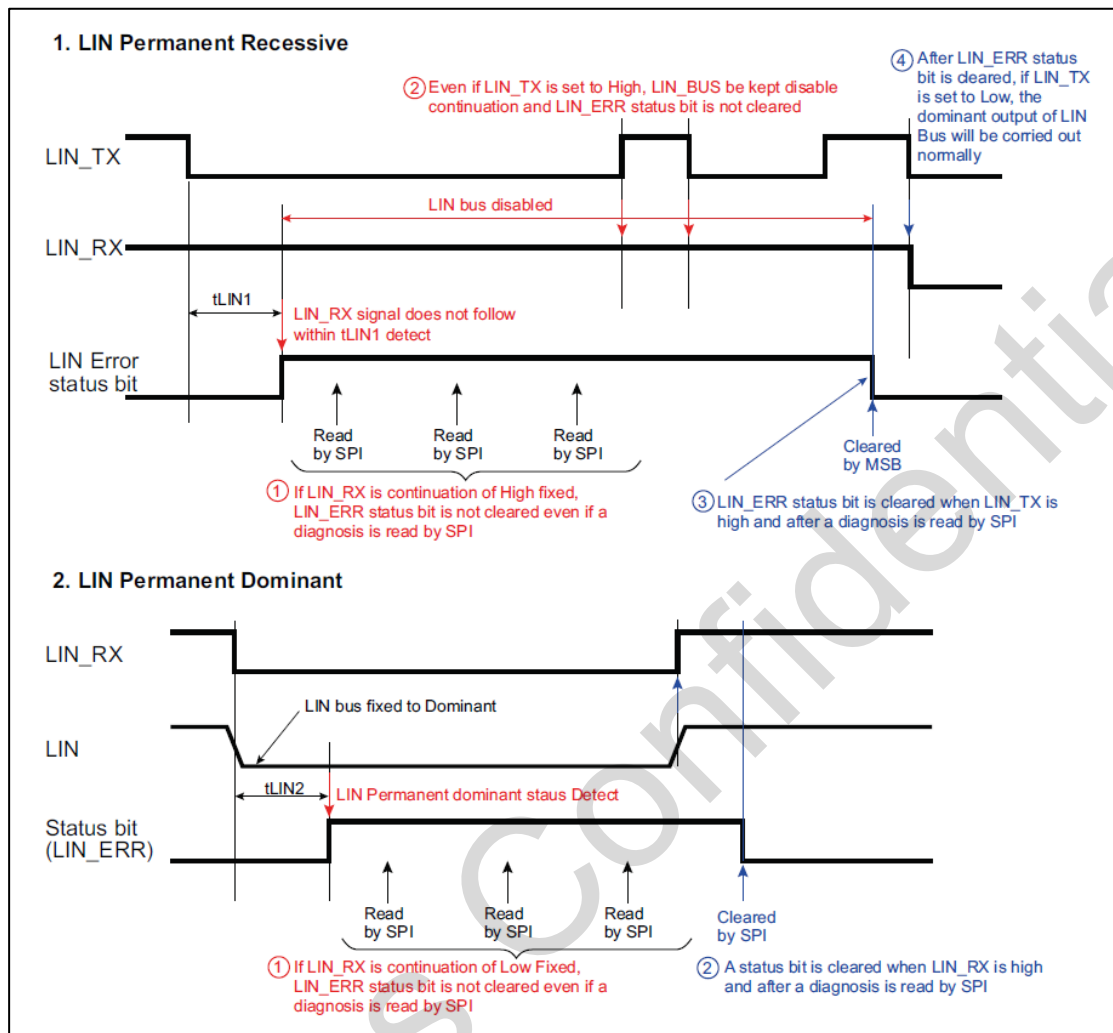


Figure 13-2 LIN TimeOut function

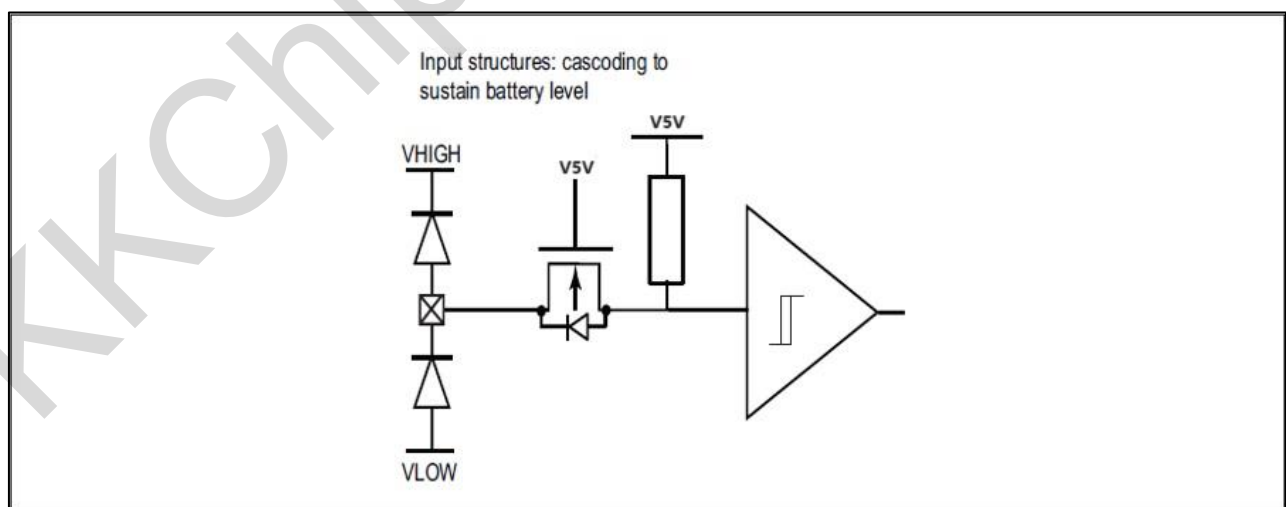


Figure 13-3 LIN_TX input structure

Conditions:

7 V ≤ VB_IN ≤ 18 V, T_j -40 to 175 °C unless otherwise specified; VB_IN = 18 to 27 V, T_j ≤ 50°C, all specs in limit. The rest of supply range, parameter are degraded or reset.

If an ECU is not intending to transmit on the LIN bus (e.g. transmit input of a LIN transceiver is recessive), the LIN driver will not drive the LIN bus to dominant state. If the LIN bus is in recessive state, the LIN receiver output will provide a recessive state.

Table 13-1 Electrical characteristics of LIN interface

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
I _{leak_lintx}	Input leakage current	LIN_Tx = VDDIO-1 V	-	-	10	μA	LIN_Tx	P_13.4.2.1
VIH_Tx	HIGH_level Input voltage	-	2	-	-	V	LIN_Tx	P_13.4.2.2
VIL_Tx	LOW_level Input voltage	-	-	-	0.9	V	LIN_Tx	P_13.4.2.3
VHYS_Tx	Input voltage hysteresis	-	0.1	-	0.4	V	LIN_Tx	P_13.4.2.4
PULL_Tx	Pull-up resistance	L in_Tx = 0 V	50	100	200	kΩ	LIN_Tx	P_13.4.2.5
VOH_Rx	HIGH-level output voltage	I _{source} = 2 mA	VDDIO -0.6	VDDIO -0.2	-	V	LIN_Rx	P_13.4.2.6
VOL_Rx	LOW-level output voltage	I _{sink} = 2 mA	-	0.2	0.5	V	LIN_Rx	P_13.4.2.7
VOL_LIN	LOW-level output voltage	Pull up =500 Ω, LIN_Tx = 0 V	-	-	0.2 VB_IN	V	LIN	P_13.4.2.8
VOH_LIN	HIGH-level output voltage	Pull up =500Ω, LIN_Tx = HIGH	0.8 VB_IN	-	VB_IN	V	LIN	P_13.4.2.9
I _{bus_dom}	Dominant Source Current	LIN_Tx = HIGH, VB_IN = 12 V, LIN = 0V	-600	-	0	μA	LIN	P_13.4.2.10
I _{bus_res}	Resessive sink current	LIN_Tx=HIGH, VB_IN=8~18V, LIN=8~18V, LIN≥VB_IN	0	-	20	μA	LIN	P_13.4.2.11
I _{bus_stby}	Standby current	VB_IN=8~18V, LIN=8~18V, LIN≥VB_IN	0	-	20	μA	LIN	P_13.4.2.12
IOVC	Current limitation	LIN=18V, LIN_Tx=0V	70	-	150	mA	LIN	P_13.4.2.13
R _{I-g_iso}	LIN bus to Ground isolation resistance	-	500K	-	-	Ω	LIN	P_13.4.2.14
ILK_NG1	Ground loss leakage current	VB_IN=12V, GND = Open, LIN=0~18 V	-1	-	1	mA	LIN	P_13.4.2.15
ILK_NB1	VB_IN Loss Leakage Current	VB_IN = Open, LIN = 0~18 V	-23	-	23	μA	LIN	P_13.4.2.16
t _{TXDC_DOM_TO}	TXDC dominant time-out	-	9	12	15	ms	LIN	P_13.4.2.17
t _{LIN1 (Time Out)}	LIN permanent recessive	-	30	40	50	μs	LIN	P_13.4.2.18

tLIN2	LIN permanent dominant	-	9	12	15	ms	LIN	P_13.4.2.19
ViD_LIN	Input voltage for dominant state	Application information	-	-	0.4 VB_IN	V	LIN(Bus Receiver)	P_13.4.2.20
Vth_DOM	Input voltage for dominant threshold	Application information	0.4 VB_IN	0.45 VB_IN	0.5 VB_IN	V	LIN(Bus Receiver)	P_13.4.2.21
ViR_LIN	Input voltage for recessive state	Application information	0.6 VB_IN	-	VB_IN	V	LIN(Bus Receiver)	P_13.4.2.22
Vth_REC	Input voltage for recessive threshold	-	0.5 VB_IN	0.55 VB_IN	0.6 VB_IN	V	LIN(Bus Receiver)	P_13.4.2.23
$Vth_CNT = (Vth_REC + Vth_DOM) / 2$	Input receiver tolerance center voltage	-	0.475 VB_IN	0.5 VB_IN	0.525 VB_IN	V	LIN(Bus Receiver)	P_13.4.2.24
$Vth_HYS = Vth_REC - Vth_DOM$	Input voltage hysteresis voltage	-	0.07 VB_IN	0.1 VB_IN	0.175 VB_IN	V	LIN(Bus Receiver)	P_13.4.2.25
tdly(LIN)HL	Output delay time HtoL	Cbus = 10 nF, Pull up = 500Ω	-	-	50	μs	LIN(Bus Receiver)	P_13.4.2.26
tdly(LIN)LH	Output delay time LtoH	Cbus = 10 nF, Pull up = 500Ω	-	-	50	μs	LIN(Bus Receiver)	P_13.4.2.27
tDLY_HL	Input delay time HtoL	CRXD=20pF	-	-	6	μs	LIN(Bus Receiver)	P_13.4.2.28
tDLY_LH	Input delay time LtoH	CRXD=20pF	-	-	6	μs	LIN(Bus Receiver)	P_13.4.2.29
tDLY	Input delay time	tDLY_HL - tDLY_LH	-2	-	2	μs	LIN(Bus Receiver)	P_13.4.2.30
D1	Duty Cycle	Thresmax = 0.744VB_IN, Thdommax = 0.581VB_IN, VB_IN = 7-18 V, tbit = 50 μs, D1 = tbus_resmin/(2xtbit)	0.396	-	-	-	LIN(Bus Receiver)	P_13.4.2.31
D2	Duty Cycle	Thresmin = 0.422VB_IN, Thdommin = 0.284VB_IN, VB_IN = 7.6-18V,	-	-	0.581	-	LIN(Bus Receiver)	P_13.4.2.32

		$t_{bit} = 50 \mu s$, $D2 =$ $t_{bus_resmax}/(2 \times t_{bit})$						
D3	Duty Cycle	$T_{hresmax} =$ $0.778 V_{B_IN}$, $T_{hdommax} =$ $0.616 V_{B_IN}$, $V_{B_IN} = 7-18 V$, $t_{bit} = 96 \mu s$, $d3 =$ $t_{bus_resmin}/(2 \times t_{bit})$	0.417	-	-	-	LIN(Bu s Receiv er)	P_13.4.2.33
D4	Duty Cycle	$T_{hresmin} = 0.389$ V_{B_IN} , $T_{hdommin} = 0.251$ V_{B_IN} , $V_{B_IN} = 7.6-18$ V , $t_{bit}=96\mu s$, $D4 =$ $t_{bus_resmax}/(2 \times t_{bit})$	-	-	0.59		LIN(Bu s Receiv er)	P_13.4.2.34
trise_LIN	Slew Rate(LIN)	$C_{bus} = 2.2 nF$, $R_{bus} = 1 k\Omega$, 40- 60%, $V_{B_IN}=13.$ 5V	0.8	2	3	V/ μ s	LIN(Bu s Receiv er)	P_13.4.2.35
Tfall_LIN	Slew Rate(LIN)	$C_{bus} = 2.2 nF$, $R_{bus} = 1 k\Omega$, 40- 60%, $V_{B_IN}=13.$ 5V	0.8	2	3	V/ μ s	LIN(Bu s Receiv er)	P_13.4.2.36
trise_K_LINE	Slew Rate(K_LINE)	$C_{bus} = 7.2 nF +$ 2 nF, $R_{bus} = 510 \Omega$, 20-80%, V_{B_IN} = 18 V	0.8	-	5	V/ μ s	LIN(Bu s Receiv er)	P_13.4.2.37
Tfall_K_LINE	Slew Rate(K_LINE)	$C_{bus} = 7.2 nF +$ 2 nF, $R_{bus} = 510 \Omega$, 20-80%, V_{B_IN} = 18 V	1.2	-	11	V/ μ s	LIN(Bu s Receiv er)	P_13.4.2.38
T_SD_HIGH	Temperature shut down	-	185	-	200	°C	LIN	P_13.4.2.39
T_SD_LOW	Temperature shut down recover	-	175	-	190	°C	LIN	P_13.4.2.40
T_SD_hys	Temperature shut down hysteresis	-	5	-	10	°C	LIN	P_13.4.2.41
-	Thermal shutdown analog filter time	Guaranteed by design	1.5	-	4	μs	LIN	P_13.4.2.42
-	Digital deglitch filter time on Temperature shut down detection	Guaranteed by scan	-	10	-	μs	LIN	P_13.4.2.43
Rslave ⁽¹⁾	-	-	20	30	60	k Ω	LIN	P_13.4.2.44
VserDiode ⁽¹⁾	Diode drop + pullup resistor drop	$I_{bias} = 10 \mu A$	0.7	1	1.3	V	LIN	P_13.4.2.45
VserDiode_dr op	Diode drop	$I_{bias} = 550 \mu A$ Guaranteed by design	0.7	1	1.3	V	LIN	P_13.4.2.46

C_{LINIO}	LINIO input capacitance	Guaranteed by design	-	-	25	pF	LIN	P_13.4.2.47
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13.4.3 Timing diagram

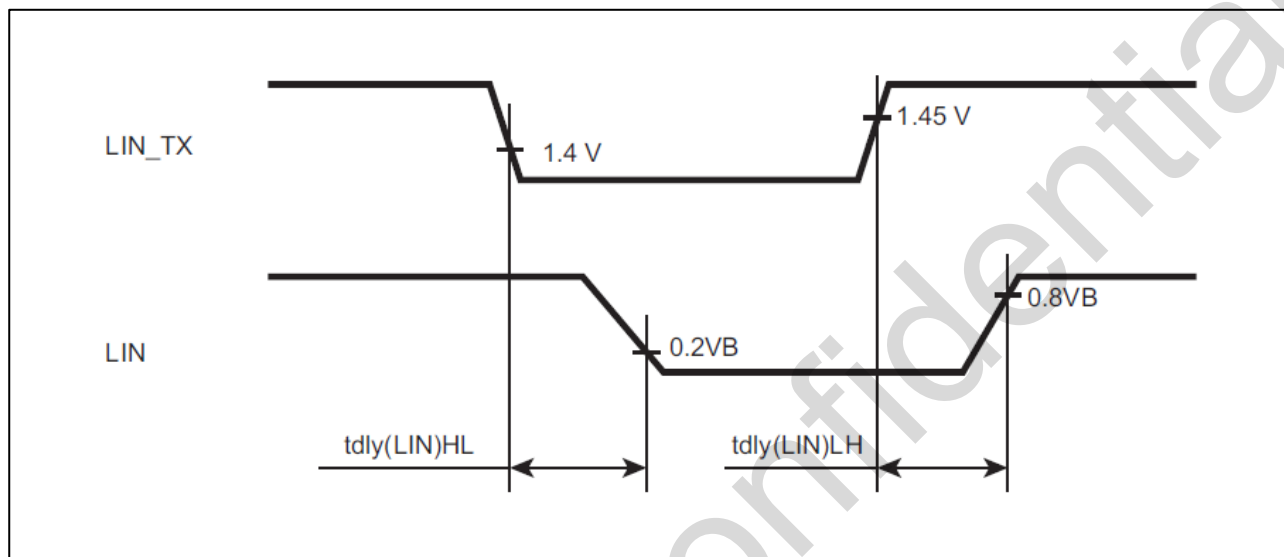


Figure 13-4 LIN transmission timing chart

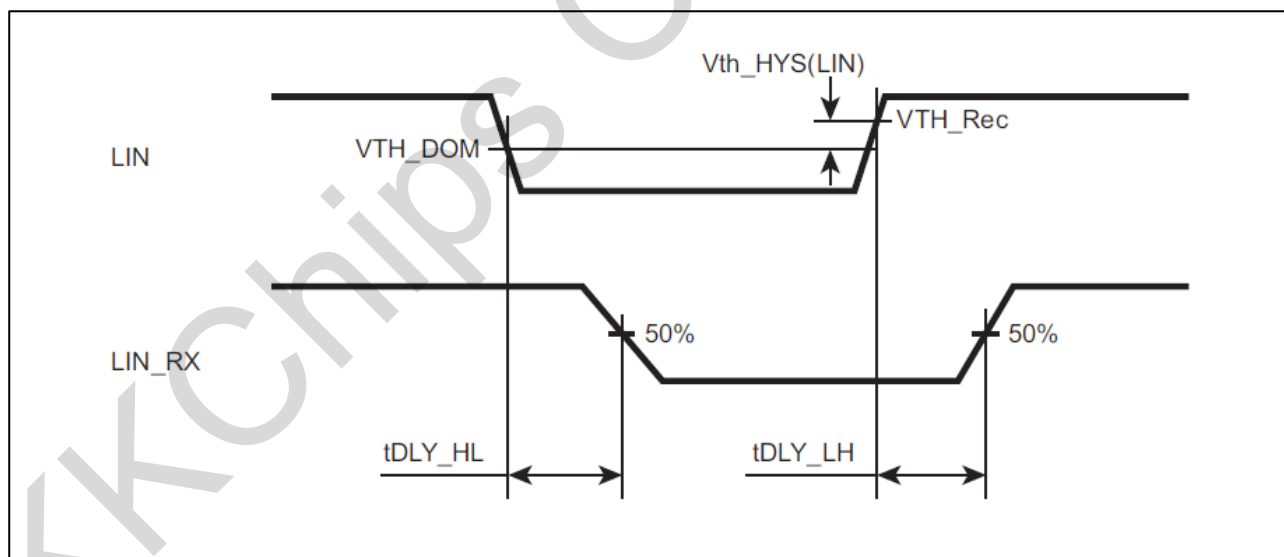


Figure 13-5 LIN reception timing chart

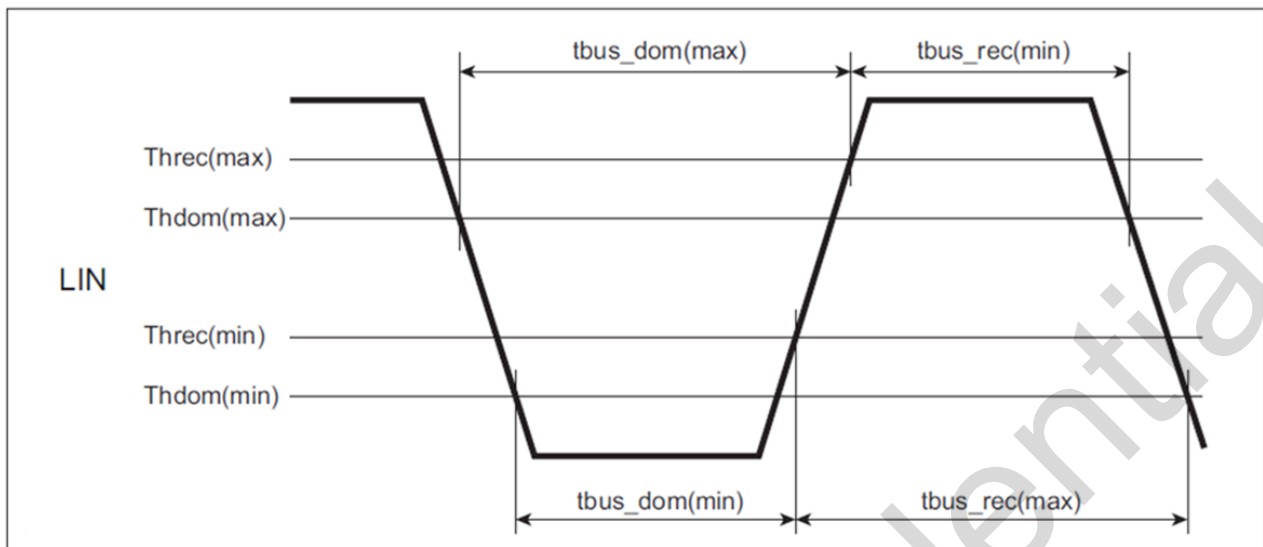


Figure 13-6 LIN duty cycle timing chart

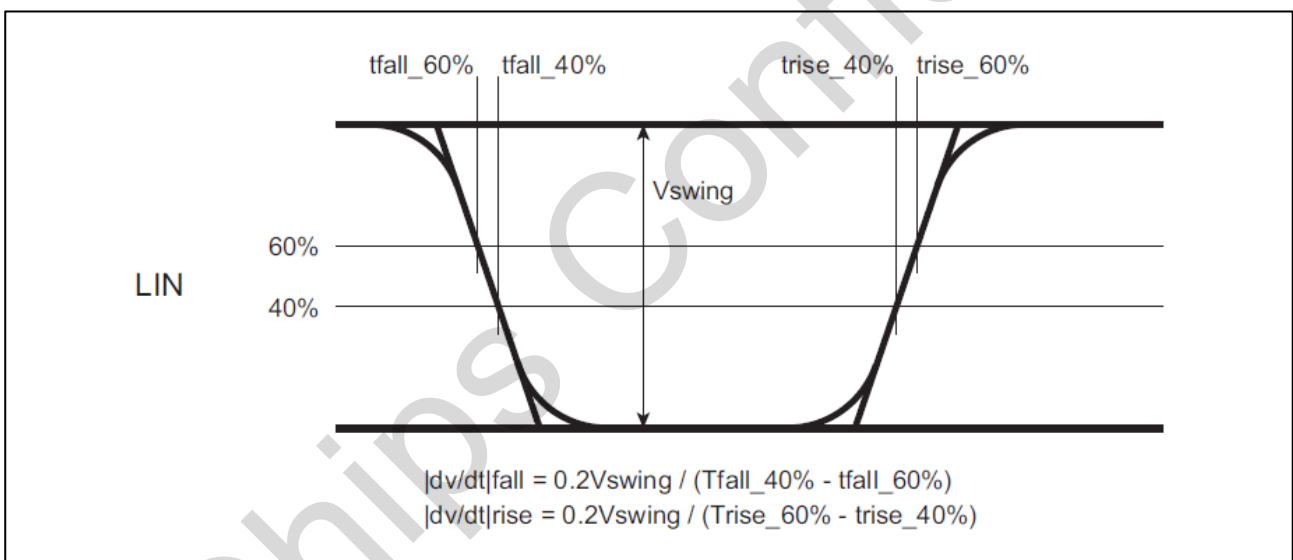


Figure 13-7 LIN slew rate timing chart

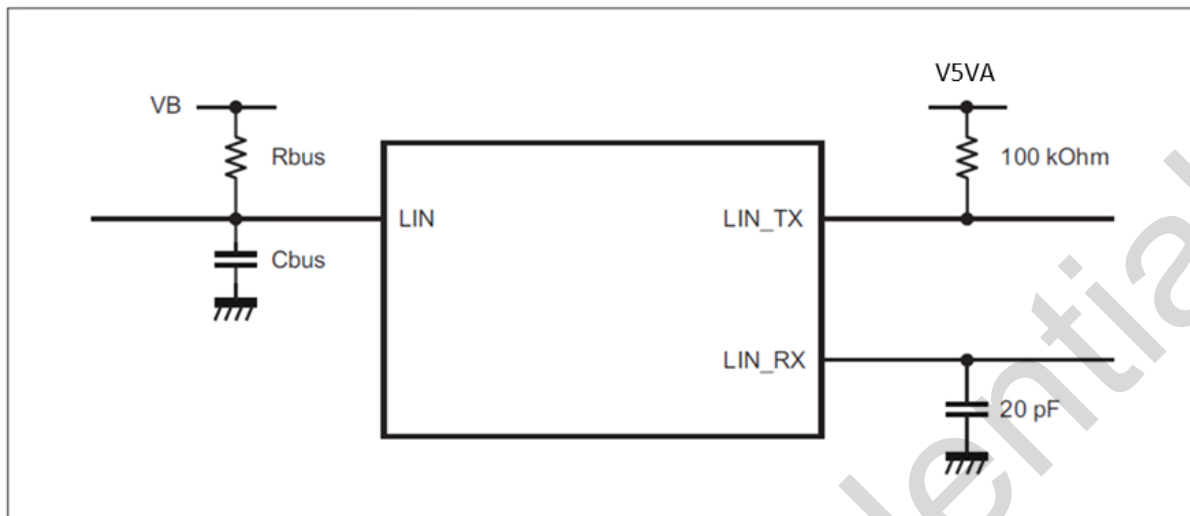


Figure 13-8 Test circuit for measurement of slew rate

13.4.4 LIN permanent recessive

In case TXD changes to dominant (low level) state but RXD signal does not follow within tLIN1: 40 us (typ), the transmitter is disabled. The status bit *LIN_PERM_REC* (Upstream Bit Map Read15 frame 1) is latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared.

14 Built in self test

14.1 Power supply independency and voltage monitors

In order to guarantee independence between monitor functions of the HE9788N and monitored functions of the same HE9788N two independent band-gap references are used; a simple field of application of this concept is the under/over voltage detection of MCU power supply generated by the same HE9788N.

The basic schematic of the band-gap used to generate the main function and of the bandgap used to monitor it, including their own supply architecture, is shown in [Figure 14-1](#).

Two band-gaps VBG1 and VBG2 are generated starting from protected battery line: VBG1 is used as reference voltage to generate internal analog and digital supply lines and to generate external power supply voltages; VBG2 is used as reference for the uv/ov monitors of these supplies.

The concept to guarantee independence between the two references is that both of them have independent supply and ground architectures to ensure that a single point failure in one of the two structures will not affect the other; independence is achieved by means of physical isolation between the two band-gaps and their supply circuits (A and B areas), isolation is done layouting the circuits in different floor plan areas and by usage of deep trench isolation in between.

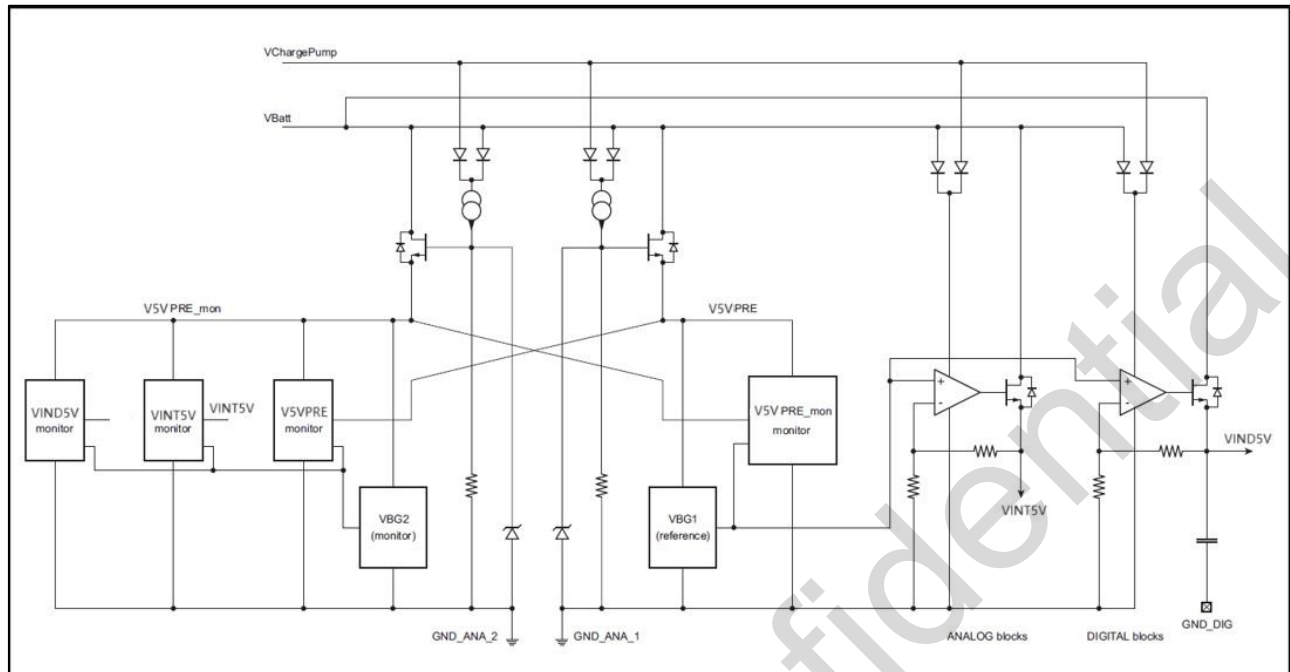


Figure 14-1 Band-gap supply architecture

A cross-referenced monitor is used to detect eventual failures in any of the band-gap supply and enter a safe state: VBG1 is used as reference for the monitor of the supply of VBG2 and VBG2 is used as monitor of the supply of VBG1.

The concept is needed to solve dependencies of band-gap voltage from the supply line: let us suppose that V5VPRE_mon in the above picture has a fault impacting VBG2 voltage reference, since V5VPRE_mon is also the supply for monitor circuits a failure in this region will cause the loss of monitor functionality with an undetected latent fault; the insertion of a monitor supplied from V5VPRE and referenced to VBG1 allows the detection of this latent.

The same concept applies on the vice-versa.

The remaining common points of the internal architecture can be summarized as:

- protected battery line,
- charge pump supply,
- V5VPRE_mon monitor input (netA),
- V5VPRE monitor input (netB)

The above points can be tolerated because:

- Protected battery line is the main supply of the HE9788N, used to generate supply for analog and digital core. All the circuits connected to this rail are capable of high voltage operation, in case the connection is lost no logic supply can be generated and the device will be stuck in reset condition.
- Charge pump is used to guarantee proper functionality during battery cranking only and is useless in normal mode. All the circuits connected to this line are capable of high voltage, in case the line is open it can be detected by dedicated monitor (not shown here) during normal operation or at power-up of the device.

- In case netA is shorted to ground there is fault detection while short to supply line is normal condition, short to VBG1 through V5VPRE_mon monitor inputs is not possible since the inputs are cascaded; in case a double fault is present (shorted cascade) it will cause an increase in analog and digital supply lines (VINT5V and VIND5V) detected by their monitors using VBG2 reference and triggering safety switch-off path for safety relevant outputs.
- In case netB is shorted to ground there is fault detection while short to supply line is normal condition, short to VBG2 through V5VPRE monitor inputs is not possible since the inputs are cascaded; in case a double fault is present (shorted cascade) it will cause a failure on the monitor circuit only, without affecting normal functionality until a third fault occurs.

14.2 Analog comparators BIST

For all the safety relevant monitors a self test is implemented to verify that the comparator can detect a variation of the monitored signal applied at its input. The basic implementation scheme is shown in Figure 14-2: comparator inputs are chopped by a diagnostic clock and the output of the comparator is xored with diagnostic clock itself: if the comparator is not able to toggle due to fault condition (internal or external) its output will start toggling following diagnostic clock and causing a self test failure.

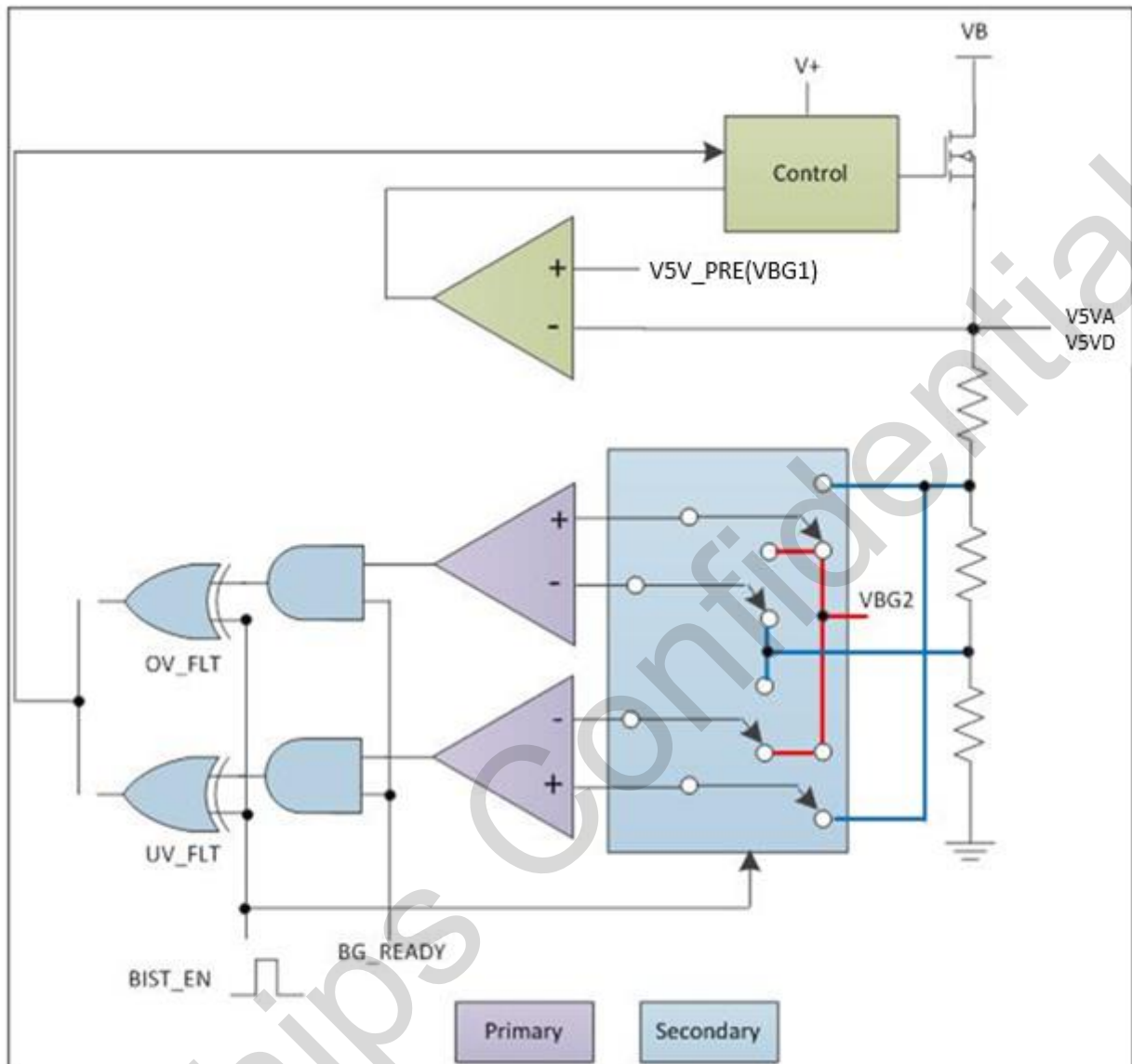


Figure 14-2 Analog bist implementation

The analog bist is applied to the following analog comparators:

V5VA_OV, V5VA_UV, V5VD_OV, V5VD_UV (internal supply monitors) VDD5_OV, VDD5_UV (VDD5 monitors)

The bist is run on MSC command *BIST_EN*. The result is *read and clear by MSC*.

- On driver section BIST is implemented for diagnostic of main outputs:
- OVC, OPL, STG comparators (injectors)
- STG comparator (igniter)
- The bist is run on MSC command. The result is read and clear by MSC.

15 Stand-by memory

This memory is a general purpose memory registers array for microcontroller-data which is supposed to be saved during standby when ECU is switched off. Intention is to reduce the number of write cycles in microcontroller-flash.

In case of standby-power-supply failure the data is no longer valid. This condition can be detected reading flag *VB_STBY_UV* MSC_READ13.FRAME2[0].

As long as the standby-power-supply is valid, no data is lost.

There are 15 memory-registers which can be used for application data. The 16th register is an address-register which cannot be used for application data.

Before writing application data into one of the memory-registers, the address of the memory-register must be written into the address-register with the *STBY_NVM_ADD_REG* command.

After setting the address a *STBY_NVM_ADD_REG* command can write the corresponding array byte.

Using the *STBY_NVM_ADD_REG* command and thus updating the address-register is not necessary in case the previously written address is still correct, e.g. when application writes consecutively to same register.

The memory-register-addresses can have values from 0 to 14. Value 15 is not used for write commands. This means *STBY_NVM_ADD_REG* commands will have no effect when writing to address 15.

The application data is written with mentioned *STBY_NVM_ADD_REG* command which writes one byte (8bit) of application data into the memory-register. To write more than one register, *STBY_NVM_ADD_REG* and *STBY_NVM_ADD_REG* sequences are necessary.

As the other registers above, the upstream reading of the registers is not done register-wise but in blocks of four registers. *MEM_REG1 ... 4*, *MEM_REG5 ... 8*, *MEM_REG9 ... 12* and *MEM_REG13 ... 15* are grouped in the four upstream blocks of Read16.

The address-register *MEM_ADR_REG16* is located at the end of the memory and is the 16th register. As mentioned, it cannot be used for application data. It is the fourth byte of the upstream Read16 and can be read back with a read command for this block.

Before upstream-reading, the register-address must be set with the *STBY_NVM_ADD_REG* command. If the previously written address is already the correct one, the *STBY_NVM_ADD_REG* command is not necessary.

For upstream reading of the *MEM_REGx* and *MEM_ADR_REG16* registers, the command *RD_COMMAND16* is used. The *RD_COMMAND16* command is the same for all four upstream blocks as the block to be read is defined via the address-register. The read command will not only read the addressed register, but will read the complete four-byte upstream-block which contains the addressed register.

The *STBY_NVM_ADD_REG* command allows to write a *MEM_VALID* bit (D6) that can be set to 1 by the microcontroller to signal that the memory has been written and validated reading back data. If standby-power-supply is removed *MEM_VALID* bit is reset.

16 DAC

Device integrates a 2 bit DAC, with 4 possible output voltages between 0V and VDD5_IN. Selected level is determined by 2 bits DAC[0:1] into *CONFIG_REG_7*.

Conditions:

5.5 V ≤ VB_IN ≤ 18 V, 4.75 V ≤ VDD_IO ≤ 5.25 V, T_j-40 to 175 °C unless otherwise specified.

Table 16-1 DAC electrical parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
V00	Output voltage 00	-	0.7		1	V	AD_TEST	P_16.1
V01	Output voltage 01	-	1.5		1.8	V	AD_TEST	P_16.2
V10	Output voltage 10	-	2.2		2.8	V	AD_TEST	P_16.3
V11	Output voltage 11	-	3.6		4.7	V	AD_TEST	P_16.4
trise	Rising time	Transition from 00 to 11 10%-90% open	-	-	10	µs	AD_TEST	P_16.5
tfall	Falling time	Transition from 11 to 00 90%-10% open	-	-	10	µs	AD_TEST	P_16.6

17 ADC

A 10 bit ADC converter is integrated into device in order to give information about silicon temperature.

10 bit conversion can be read with *READ_COMMAND12* MSC frame. Conversion formula is $T = \text{MSCcode} * 290 / 1024 - 63$ (°C).

Conditions:

$5.5 \text{ V} \leq \text{VB_IN} \leq 18 \text{ V}$, $4.75 \text{ V} \leq \text{VDD_IO} \leq 5.25 \text{ V}$, $T_j -40$ to 175 °C unless otherwise specified.

Table 17-1 ADC electrical parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
ADC_bits	ADC effective number of bits	Guaranteed by design	-	10	-	-	-	P_17.1
Trange	Internal junction temperature range	Guaranteed by design	-50	-	200	°C	-	P_17.2
Tacc	Temperature accuracy	Guaranteed by design	-10	-	10	°C	-	P_17.3

18 Micro second channel (MSC) interface

18.1 Function description

The Micro Second Channel interface is an adapted high performance serial bus for power switch device.

HE9788N has one Micro Second Channel slave controller capable to receive up to 35 MHz downstream and to transmit up to 2.18 Mbaud upstream.

MSC Clock is supported with synchronous continuous mode. Master clock and data are implemented by LVDS. Slave data out signal is implemented by LVTTTL.

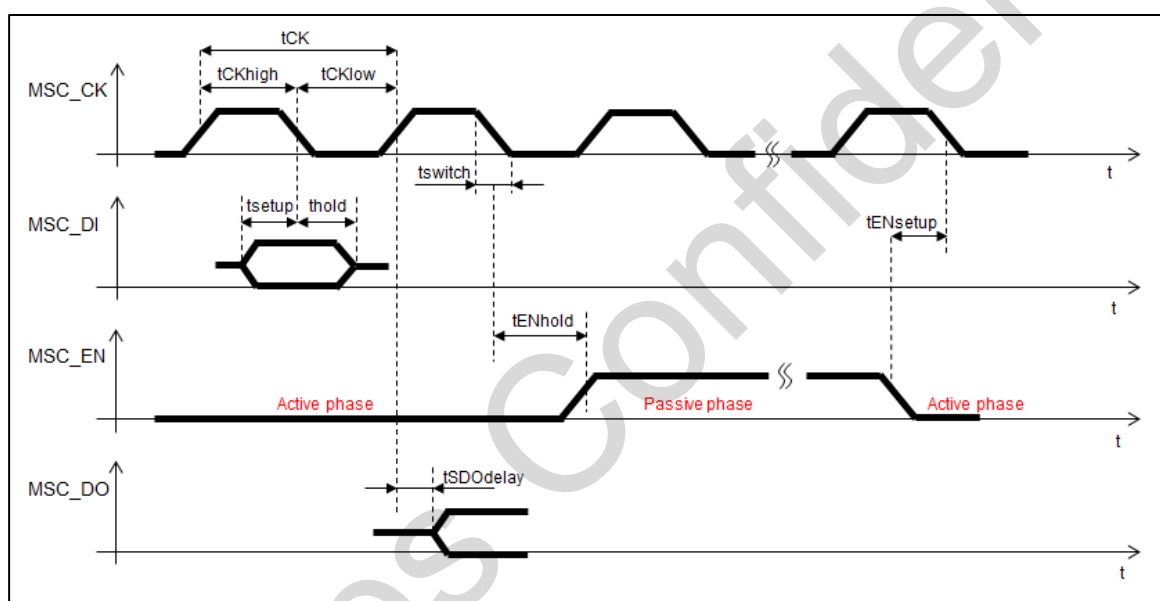


Figure 18-1 MSC communication timing diagram

Conditions:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, $4.75\text{ V} \leq V_{DD_IO} \leq 5.25\text{ V}$, T_j -40 to 175 °C unless otherwise specified.

Table 18-1 MSC communication timing parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
tCK	Cycle time	Tested by SCAN/	25	-	-	ns	-	P 18.1.1
tsetup	Data setup time	Tested by SCAN/	5	-	-	ns	-	P 18.1.2
thold	Data hold time	Tested by SCAN/	5	-	-	ns	-	P 18.1.3
Tswitch	Switching time for CL and SI measured between 0.1*V _{VDD3} and	Tested by SCAN/	-	-	3	ns	-	P 18.1.4

	0.9*VDD3							
tCKlow	CK low time	Tested by SCAN/	7.5	-	-	ns	-	P_18.1.5
tCKhigh	CK high time	Tested by SCAN/	7.5	-	-	ns	-	P_18.1.6
tENsetup	EN setup time (i.e. time between falling edge of EN and next falling edge of CK)	Tested by SCAN/	5	-	-	ns	-	P_18.1.7
tENhold	EN hold time (i.e. time between falling edge of CK and next rising edge of EN)	Tested by SCAN/	5	-	-	ns	-	P_18.1.8
tsDO /tck	Data out cycle time F_DO_SEL (CONFIG_REG2 [1:2]) = "11" F_DO_SEL="10" F_DO_SEL="01" F_DO_SEL="00"	Tested by SCAN/	-	128 64 32 16	-	-	-	P_18.1.9
fck	Clock range at CK as long as there is a clock at pins MSC_CK_P, MSC_CK_N. tCPP ≥ 2 * tCL	Tested by SCAN/	-	-	35	MHz	-	P_18.1.10
	Clock range at CK as long as there is a clock at pins MSC_CK_P, MSC_CK_N. tCPP ≥ 4 * tCL	Tested by SCAN/	-	-	40	MHz	-	P_18.1.11
-	tsDOdelay	Tested by SCAN/	-	-	160	ns	-	P_18.1.12

The MSC is used to receive the input command and data from CPU and to transmit an output data to CPU. Four signals are used according to the timing chart of Figure 0-2:

EN: Bus Enable

There is one input for chip select at pin [EN] This signal is LVTTTL Interface from Master to Slave. MSC uses inverted polarity for EN: a logic '1' is a 'passive level' and a logic '0' is a 'active level'. It is possible to drive multiple power devices with shared CL and DI lines and individual EN signal.

CL: Synchronous Serial Clock

The clock pins are [CLP] and [CLN], the differential clock. [CLP]-[CLN] is referred to as CL. The maximum downstream clock rate is CL= 35 MHz. There is an internal resistor between pins [CLP] and [CLN]. This signals are LVDS Interface from Master to Slave.

DI: Serial Input Data

Differential inputs for downstream data are pins [DIP] and [DIN]; the differential input signal [DIP]-[DIN] is referred to as DI. There is an internal resistor between pins [DIPP] and [DIN]. These signals are LVDS Interface from Master to Slave

DO: Serial Output Data

There is one push-pull output for upstream data at pin [DO]. Upstream is done with a lower clock rate f_{SDO} , selectable by the microcontroller; after a reset the upstream clock rate is $f_{SDO} = CL/64$. The upstream clock is synchronous with CL since it is derived from a clock divider. Therefore the CL signal must be always running independently whether a downstream transmission is running or not. This signal is LVTTTL Interface from Slave to Master.

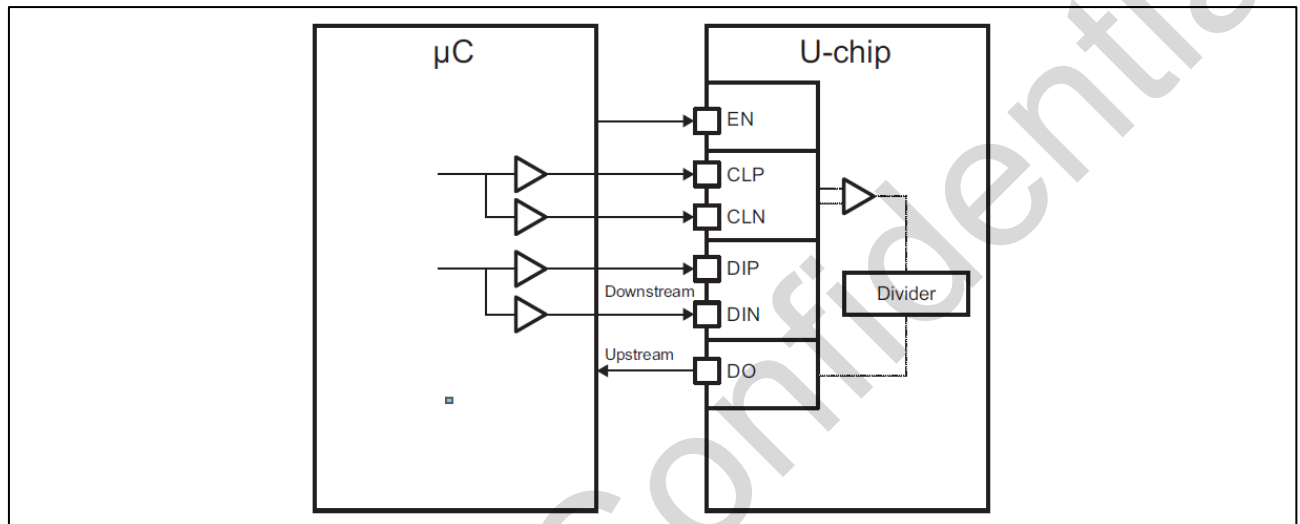


Figure 18-2 Communication diagram between μC and U-CHIP

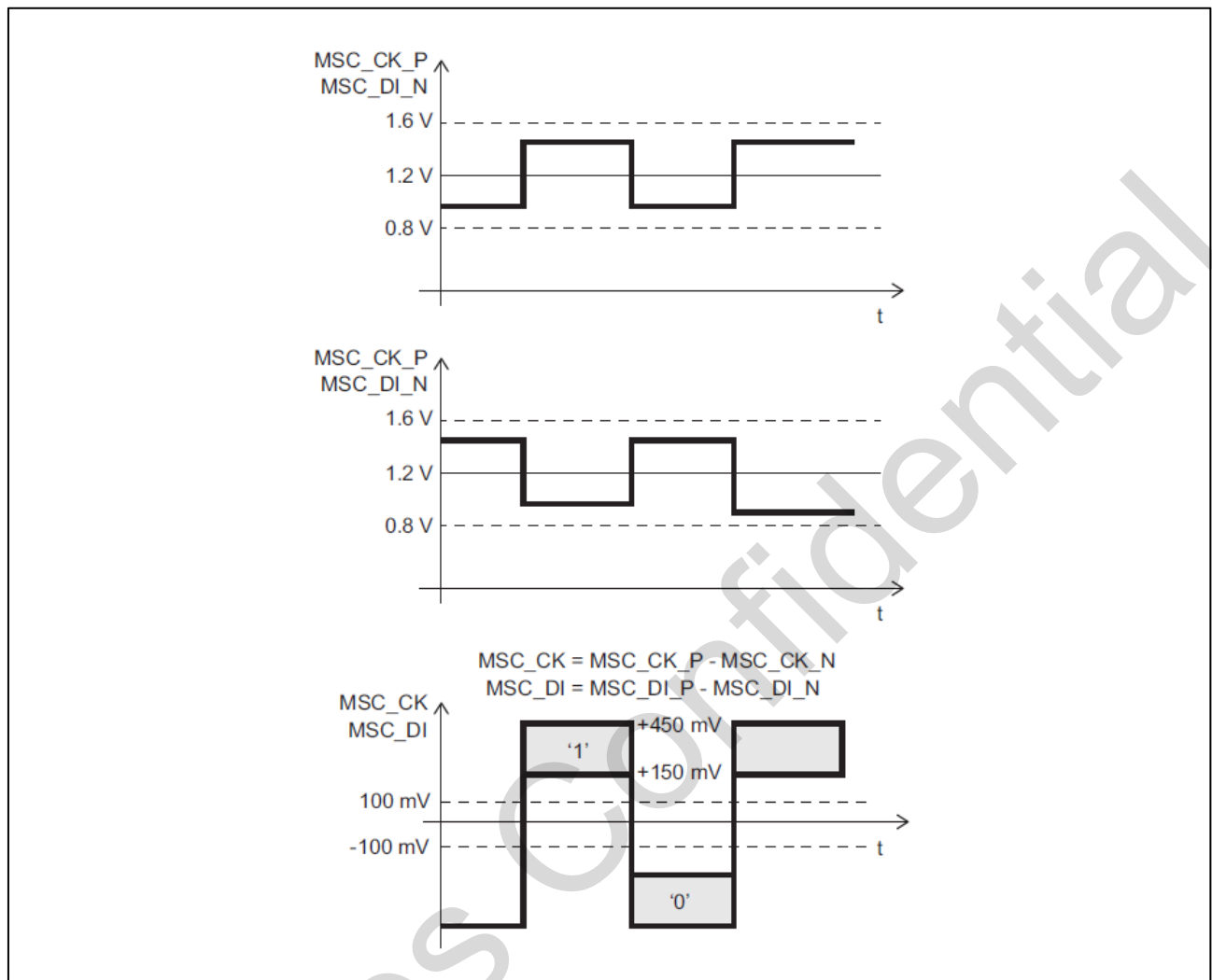


Figure 18-3 MSC voltage levels diagram

Conditions:

$5.5\text{ V} \leq V_{B_IN} \leq 18\text{ V}$, $4.75\text{ V} \leq V_{DD_IO} \leq 5.25\text{ V}$, $T_j -40$ to $175\text{ }^{\circ}\text{C}$ unless otherwise specified.

Table 18-2 MSC electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VCK_P, VCK_N	Input voltage range	Not to be tested. It is an application note	0.8	-	1.6	V	MSC_CK_P MSC_CK_N	P_18.1.13
VCKdiff	Differential input voltage $VCKdiff = VCK_P - VCK_N $	Not to be tested. It is an application note	150	-	450	mV	MSC_CK_P MSC_CK_N	P_18.1.14
VCKoff	Input voltage	Not to be tested.					MSC_CK_P	P_18.1.15

	offset $VCK_{off} = 0.5 \cdot (VCK_P + VCK_N)$	It is an application note	1	-	1.4	V	MSC_CK_N	
Rck	EXTERNAL Resistor between CK_P and CK_N	Not to be tested. It is an application note		100	-	Ω	MSC_CK_P MSC_CK_N	P_18.1.16
Rpu_N	Internal pull-up resistor	-	100	200	400	k Ω	MSC_CK_P MSC_CK_N	P_18.1.17
Rpd_P	Internal pulldown resistor	-	100	200	400	k Ω	MSC_CK_P MSC_CK_N	P_18.1.18
VCK_high	Differential input high detection level $VCK_{high} = VCK_{P_high} \wedge VCK_{N_high}$	-	-	-	100	mV	MSC_CK_P MSC_CK_N	P_18.1.19
VCK_low	Differential input low detection level $VCK_{low} = VCK_{P_low} \wedge VCK_{N_low}$	-	-100	-	-	mV	MSC_CK_P MSC_CK_N	P_18.1.20
VDI+,VDI-	Input voltage range	Not to be tested. It is an application note	0.8	-	1.6	V	MSC_DI_P MSC_DI_N	P_18.1.21
VDIdiff	Differential input voltage $VDIdiff = VDI_P - VDI_N $	Not to be tested. It is an application note	150	-	450	mV	MSC_DI_P MSC_DI_N	P_18.1.22
VDloff	Input voltage offset $VDloff = 0.5 \cdot (VDI_P + VDI_N)$	Not to be tested. It is an application note	1	-	1.4	V	MSC_DI_P MSC_DI_N	P_18.1.23
Rcl	Resistor between DI_P and DI_N	Not to be tested. It is an application note	-	100	-	Ω	MSC_DI_P MSC_DI_N	P_18.1.24
Rpu_N	Internal pull-up resistor	-	100	200	400	k Ω	MSC_DI_P MSC_DI_N	P_18.1.25
Rpd_P	Internal pulldown resistor	-	100	200	400	k Ω	MSC_DI_P MSC_DI_N	P_18.1.26
VDI_high	Differential input high detection level $VDI_{high} = VDI_{P_high} - VDI_{N_high}$	-	-	-	100	mV	MSC_DI_P MSC_DI_N	P_18.1.27

VDI_low	Differential input low detection level VDI_low= VDI_P_low- VDI_N_low	-	-100	-	-	mV	MSC_DI_P MSC_DI_N	P_18.1.28
VDO_L	DO output low level	VDDIO=5V or 3.3V I _{source} current=2mA	-	-	0.5	V	MSC_DO	P_18.1.29
VDO_H	DO output high level	VDDIO =5V or 3.3V I _{sink} current=2mA	VDDIO -0.5	-	-	V	MSC_DO	P_18.1.30
fDO	Maximum frequency	Tested by SCAN	fCK/128	fCK/64	fCK/16	MHz	MSC_DO	P_18.1.31
ENL	Low input level	-	-0.3	-	1.1	V	MSC_EN	P_18.1.32
ENH	High input level	-	2.3	-	VDD5 +0.3	V	MSC_EN	P_18.1.33
VHYST	Hysteresis	-	0.1	-	-	V	MSC_EN	P_18.1.34
IIN	Input current	-	-	-	32	μA	MSC_EN	P_18.1.35
RPU	Pull up resistor	-	50	-	250	kΩ	MSC_EN	P_18.1.36

18.2 Downstream communication

The enable input is active with inverted polarity - i.e. 'low level' during the active phases of command or data frames. An active enable signal validates the DI input signal. Outside the active phase (enable line is at high level) invalid data may occur at DI. The active phase of a downstream frame starts with the falling edge of the enable signal and ends with the rising edge of the enable signal. The enable signal changes its state with the rising edge of the clock CL. DI changes its state on rising edge and it is latched by HE9788N on the falling edge of CL. Downstream frames are synchronous serial frames. They support enable signal and command/data selection bit as part of the frame. Command/data selection bit allows distinguishing frames as command and data frames in the receiver circuit. Command frames and data frames may be sent in any sequence with a passive phase of at least 2 CL cycles after each frame.

18.2.1 Command frame

A command frame always starts with a high level bit (command selection bit). The number of the command bit of the active phase of a command frame NCB is fixed to 16. If the number of the command bit is not equal to NCB = 16 the frame will be ignored, the command will not be executed and the error flag (*TRANS_L*) will be set. If the MSC command frame has an invalid command the flag "*CMD_ERROR*" is set but no action on outputs is taken. The command frame is ignored.

The length of the command frame's passive phase tCPP must be a minimum of 2 * tCL.

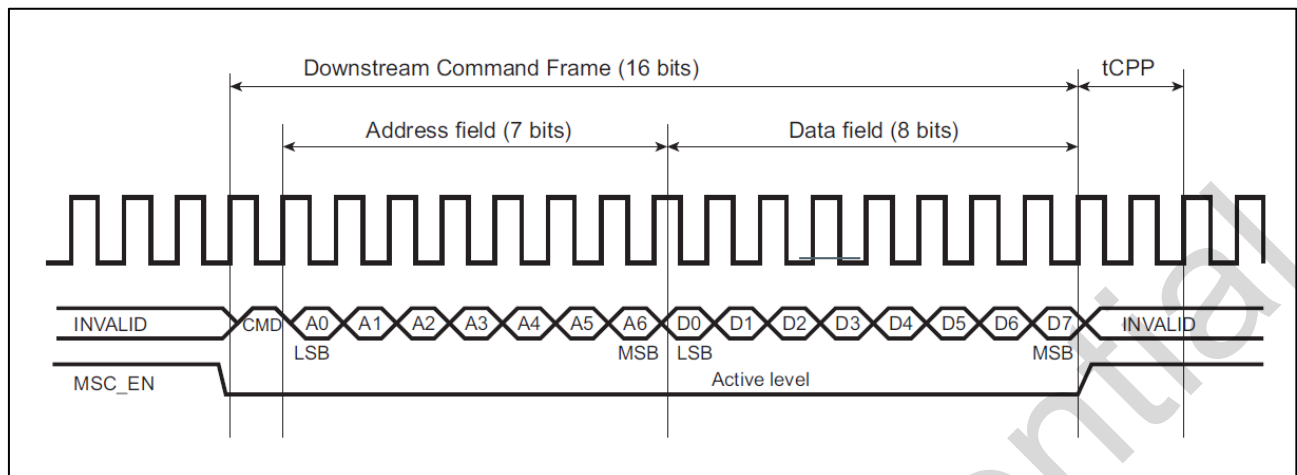


Figure 18-4 MSC Command Frame bit stream

Table 18-3 MSC Interface command frame

bit	Description
0	"1": command selection bit
1-7	Command. LSB first!
8-15	Data for the command. LSB first!

18.2.2 Data frame

A data frame always starts with a low level bit (data selection bit). The number of the data bit of the active phase of a data frame NDB is fixed to 31 bits:

If the number of the data bit is not equal to NDB = 31 the frame will be ignored and the error flag (*TRANS_L*) will be set, but no action on outputs is taken.

The length of the data frame's passive phase tCPP must be a minimum of $2 * tCL$.

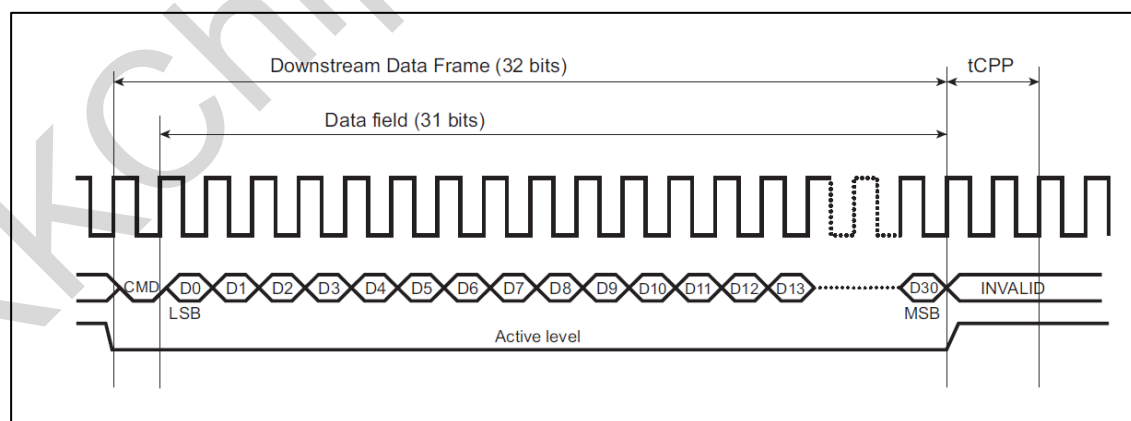


Figure 18-5 MSC Data Frame bit stream

Content of DATA frame (transmitted LSB first)

Table 18-4 MSC Interface Data frame

bit	Description
0	= "0": DATA selection bit
1-31	D0-D30 CONTROL REGISTER LSB first!

Table 18-5 Control register

Control Register							
D0	D1	D2	D3	D4	D5	D6	D7
LSD-INJ1	LSD-INJ2	LSD-INJ3	LSD-INJ4	LSD-O2H1	-	LSD-O2H2	-
D8	D9	D10	D11	D12	D13	D14	D15
LSD-SOL1	LSD-SOL2	LSD-LED1	LSD-LED2	LSD-RLY1	LSD-RLY2	LSD-RLY3	LSD-RLY4
D16	D17	D18	D19	D20	D21	D22	D23
LSD-RLY5	LSD-HSD-STR1	LSD-HSD-STR2	LSD-HSD-STR3	FET-PRD1	FET-PRD2	FET-PRD3	FET-PRD4
D24	D25	D26	D27	D28	D29	D30	-
FET-PRD5	IGN1	IGN2	IGN3	IGN4	IGN5	IGN6	-

Note: If RLY4, STR2, STR3 are not configured as starter their status is controlled by the data in control register.

If they are configured as starter their status is controlled by MSC dedicated command.

18.3 Upstream communication

The serial data output [DO] is the synchronous serial data signal of the upstream channel.

The polarity for [DO] is ,normal polarity'- i.e. a low level bit at [DO] is stored in the μ C as a logic ,0', and a high level bit at [DO] is stored in the μ C as a logic ,1'. The serial data output is single-ended.

The frequency is derived from CL by an internal divider to typ. $f_{SDO} = CL/64$. It can be adjusted via *MSC* to $f_{SDO} = CL/16 \dots CL/128$. The time for a bit is $T_{SDO} = 1/f_{SDO}$.

Each upstream frame consists of 16 bit:

- 1 start bit, always '0'
- 4-bit-upstream address field (A[0..3] with LSB first)
- 8 bit data upstream data field (D[0..7] with LSB first)
- 1 upstream parity bit (with even parity for the complete data frame)
- 2 f_{SDO} stop bit, always '1'.

There are 16 commands to perform read accesses.

All the read command READ-N Upstream Block are composed of 4 continued frames sent to μ C.

Within the execution of these read commands an upstream data frame is sent after the 2 stop bit of the prior upstream data frame and one additional interframe bit waiting time. If a new read command is received while the up-stream communication is active, the 16 bit upstream on-going is completed and the new read command is canceled. At the end of the upstream frame the latched flags contained in the register are cleared automatically, if the frame is interrupted by new read command, flags will not be cleared. The time from the read command to the first upstream frame of the answer is less than 100 μ s.

Outside the upstream frame the DO output is high impedance

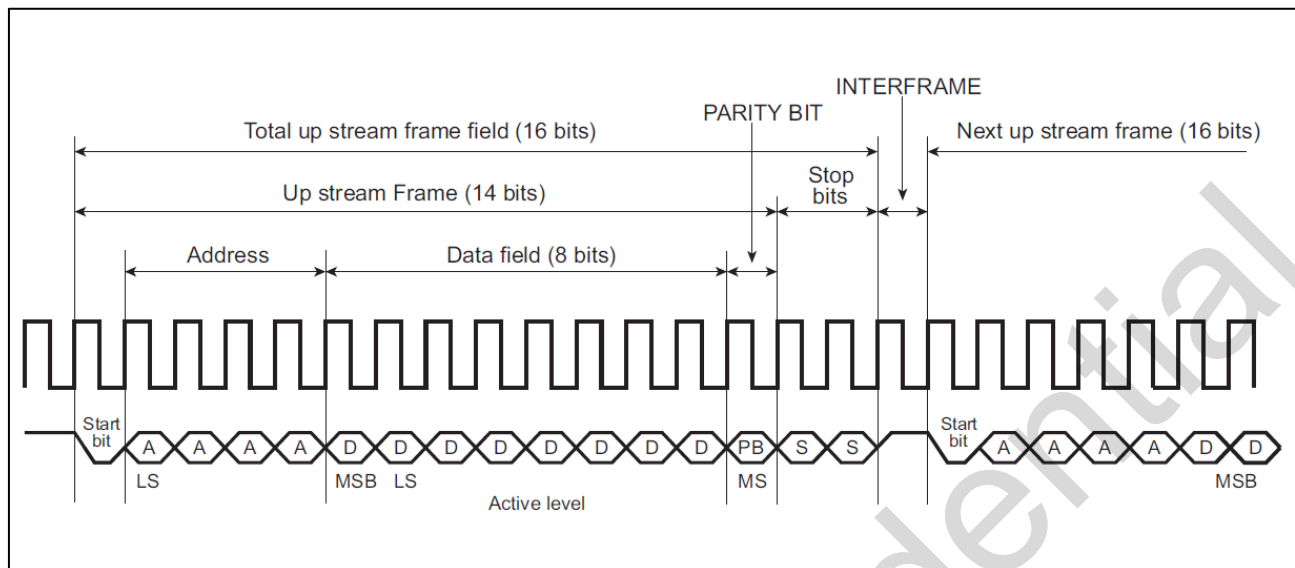


Figure 18-6 MSC upstream

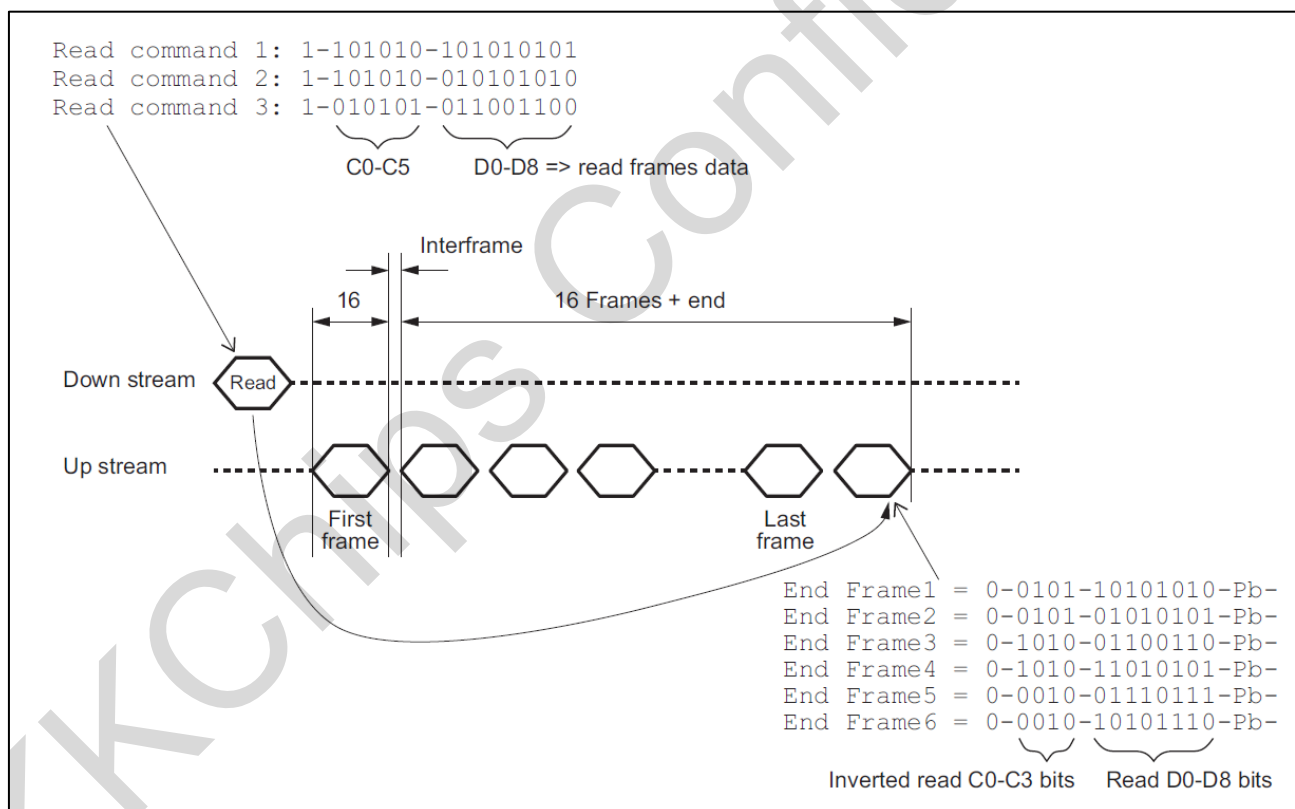


Figure 18-7 MSC upstream commands example

18.4 Micro Second Channel activity watchdog

MSC data frames are monitored to be sent in intervals shorter than tMSC_mon. If HE9788N

Micro second channel (MSC) interface

receives no valid data frame for longer than t_{MSC_mon} , it will switch off all the drivers, removing driver enable command, and the error flag ($TRANS_F$) is set to 1 and internal signal "OUT_DIS" will be set to "0". Note: $OUT_DIS = driver_en$ (Read11 frame 1); t_{MSC_mon} is the time needed to send OUT_DIS , it is different from time needed to turn off drivers (which can be different depending on driver type, the maximum time to turn off all drivers is about 300 μs).

To enable again the outputs, the μC has to read the $TRANS_F$ and then send the command Enable_Driver, and then outputs are reactivated with the first correct data frame. If the fault flag is not cleared the Enable_Driver command is ignored.

By default the Micro Second Channel activity watchdog is enabled and the monitoring time will start after writing of the internal signal OUT_DIS bit by Enable_Driver command. Each time HE9788N receives a valid data frame the t_{MSC_mon} timer is reset. This means that micro controller can drive the outputs only when the monitoring module is active.

It is also possible to enable /disable the Micro Second Channel activity watchdog through MSC_ACT_EN bit (CONFIG_REG2 D0), 0= disable and 1 = enable (reset value).

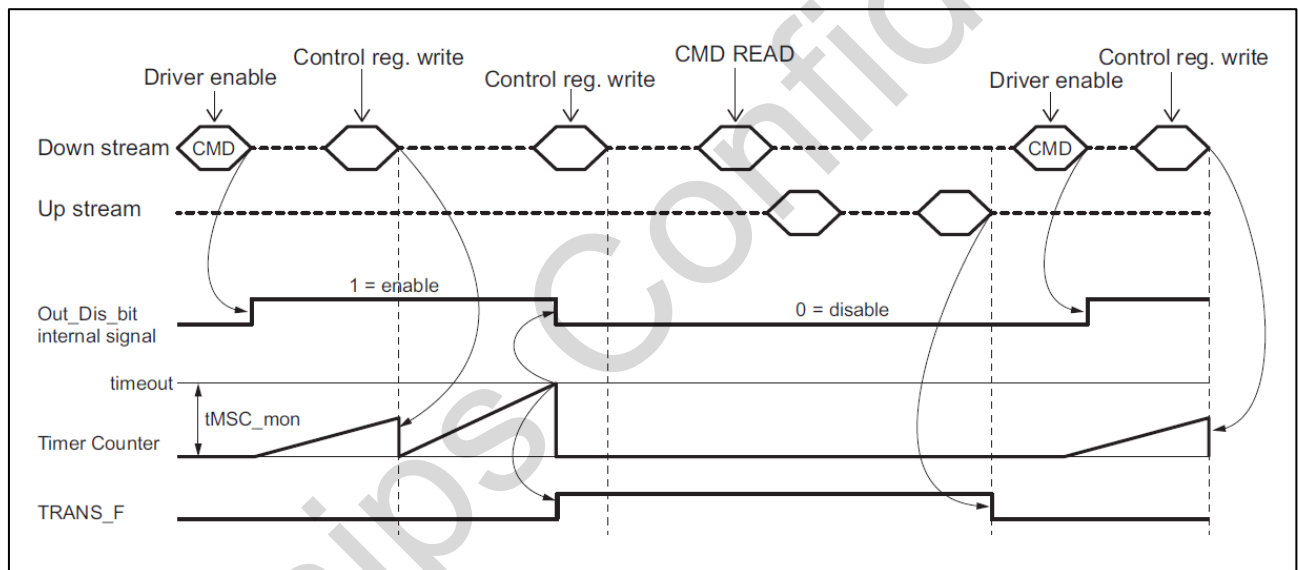


Figure 18-8 MSC activity watchdog time diagram

Table 18-6 MSC Interface Micro Second Channel activity watchdog

Parameter	min	typ	max	unit	Number
t_{MSC_mon}	100	142	185	μs	P_18.4.1

18.5 Downstream frame Bit Map

Table 18-7 MSC Interface DOWN STREAM FRAME Bit Map

ADDRESS							DATA								
	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
RD_COMMAND1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
RD_COMMAND2	1	0	1	0	1	0	1	1	0	1	0	1	0	1	0
RD_COMMAND3	0	1	0	1	0	1	0	0	1	0	1	0	1	0	1
RD_COMMAND4	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0
RD_COMMAND5	1	0	1	0	1	0	1	1	0	0	1	1	0	0	1
RD_COMMAND6	1	0	1	0	1	0	1	0	1	1	0	0	1	1	0
RD_COMMAND7	0	1	0	1	0	1	0	1	0	0	1	1	0	0	1
RD_COMMAND8	0	1	0	1	0	1	0	0	1	1	0	0	1	1	0
RD_COMMAND9	1	0	1	0	1	0	1	1	1	1	0	1	1	1	0
RD_COMMAND10	1	0	1	0	1	0	1	0	0	0	1	0	0	0	1
RD_COMMAND11	0	1	0	1	0	1	0	1	1	1	0	1	1	1	0
RD_COMMAND12	0	1	0	1	0	1	0	0	0	0	1	0	0	0	1
RD_COMMAND13	1	0	1	0	1	0	1	0	1	1	1	0	1	1	1
RD_COMMAND14	1	0	1	0	1	0	1	1	0	0	0	1	0	0	0
RD_COMMAND15	0	1	0	1	0	1	0	0	1	1	1	0	1	1	1
RD_COMMAND16	0	1	0	1	0	1	0	1	0	0	0	1	0	0	0
CONFIG-REG 1	0	0	1	1	1	1	1	PHOLD_EN	PHOLD_TIM[0:1]	LED1_PD_EN	LED2_PD_EN	BOOST_EN	WAKE_UP_TIMER		
CONFIG-REG 2	0	1	1	1	1	0	1	MSC_ACT_EN	F_DO_SEL[0:1]	VRS_MO_DE_SEL	IGN_CUR_RENT_CFG	IGN_DIAG	IPUP_PD_EN	IPUP_ID_MODE	

ADDRESS							DATA										
	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7		
RD_COMMAND3	1	1	1	1	0	0	1	RLY4_DL Y_OFF_EN	LSD/HSD ID_DRV1 -CFG	LSD/HSD DRV2-CFG	LSD/HSD DRV3- CFG	SRT2_EN	STR3_EN	T_SEQ_DELAY[0:1]			
RD_COMMAND4	1	1	0	0	0	0	1	PRD1_IDRV[0:1]		PRD2_IDRV[0:1]		PRD3_IDRV[0:1]		PRD4_IDRV[0:1]			
RD_COMMAND5	0	0	0	0	1	1	1	PRD5_IDRV[0:1]		PRD1_VDS[0:2]		PRD2_VDS[0:2]		PRD3_VDS[0:2]			
RD_COMMAND6	1	1	0	0	1	1	1	PRD3_VDS[0:2]		PRD4_VDS[0:2]		EOT_MODE		IGN_LSD _DIS			
RD_COMMAND7	0	0	1	1	0	0	1	PRD5_VDS[0:2]		TRK_EN[1]	TRK_EN[2]	TRK_EN[3]	DAC[0:1]				
RD_COMMAND8	1	0	1	1	0	1	1	VRS_DIAG	VRS_MODE[0:1]	VRS_HYS[2:0:2]		EN_FALLI NG_FILT		HYS_FB_ SEL			
RD_COMMAND9-0	0	1	0	0	1	0	1	0	1	WDA_RESPTIME[0:5]							
RD_COMMAND9-1	0	1	0	0	1	0	1	1	0	WDA_WI N_SEL	WDA_INI T						
RD_COMMAND10	0	0	0	1	1	0	1	WDA_RESP[0:7]									
RD_COMMAND11	0	1	1	0	1	1	1	LIN/KLIN ESEL	LIN_ER R_EN[0]	LIN_ERR_ EN[1]	LIN_ERR_ EN[2]	LIN_TX_ DIS_FOR_WDA_ ERR	VDD5_OF_F_SEL	O2H2_O C_FLT			
RD_COMMAND12	1	0	0	1	0	0	1	TNL_RST _EN	VB_IN_ O_V_RST_ _EN	VDD5_O V_RST_E N	O2H1_O C_FLT	FILTER_ MODE	WAKE_U P_CAN_RST	WAKE_U P_EOT_RST	WK_IN_RST		
RD_COMMAND13	1	1	0	1	1	0	1		WDA_P WR_CN	IPUPD_E N_HLS1	IPUPD_E N_HLS2	IPUPD_E N_HLS3	IDIAG_HI GH_INJ	IDIAG_HI GH_SOL	IDIAG_HI GH_PDRV		
RD_COMMAND14	1	1	1	0	0	1	1	RLY1_UC	RLY2_U C	RLY3_UC	RLY4_UC	RLY5_UC	STR1_UC	STR2_UC	STR3_UC		
RD_COMMAND15-0 (WUPT_0)	1	0	0	0	0	1	1	0	0	Wake_up_timer_SET_0[0:5]							

HE9788N

Multifunction IC for automotive engine management system



Micro second channel (MSC) interface

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7		
CONFIG-REG15-1 (WUPT_1)	1	0	0	0	0	1	0	0	1	Wake up timer_SET_1[6:11]							
CONFIG-REG15-2 (WUPT_2)	1	0	0	0	0	1	0	1	0	Wake up timer_SET_2[12:17]							
CONFIG-REG15-2 (WUPT_3)	1	0	0	0	0	1	1	1	1	Wake up timer_SET_3[18:23]							
CONFIG-REG16-0	0	0	1	0	0	1	1	0	1	PRD1_BLK[0:1]		PRD2_BLK[0:1]		PRD3_BLK[0:1]			
CONFIG-REG16-1	0	0	1	0	0	1	1	1	0	PRD4_BLK[0:1]		PRD5_BLK[0:1]		O2H2_SR	O2H2_OTC_TH		
CONFIG-REG17-0	0	0	1	0	0	1	0	0	0		O2H1_SR	O2H1_OTC_TH		O2H2_SR	O2H2_OTC_TH		
CONFIG-REG17-1	0	0	1	0	0	1	0	1	1	CAN_EN	CAN_TX_EN	CAN_WAKEUP_EN	CAN_PATERN_EN	CAN_RX_EN	CAN_2_5_MB		
STBY_NVM_ADD_REG	0	1	0	0	1	1	0	0	1	STBY_NVM_ADD[0:3]				MEM_VA_LID			
STBY_NVM_DATA_REG	1	0	1	1	0	0	0	STBY_NVM_DATA[0:7]									
CONFIG-REG20	1	1	1	1	1	1	0	BIST_EN	FIN_WA_P_KEY	WAKE_UP_TIMER_SEL	KEY_OC_RT_MAX_EN	CAN_LOP_EN	CAN_TDI	LIN_TX_DOM_ER_CFG	PDRV_OTC_2_H_DLY		
CONFIG-REG21	0	0	0	0	0	1	1	CAN_TX_DOM_ER_CFG	OL_RE_D	BUCK_SLOW_SR	CAN_TX_D_DOM_EN	CAN_PERM_REC_EN	CAN_PERM_DOM_EN	CAN_RX_D_REC_EN	CAN_ALU_TO_BIAS		
LOCK	1	0	0	1	1	1	1	0	1	0	1	0	1	0	1		
UNLOCK	1	0	0	1	1	1	1	1	0	1	0	1	0	1	0		
WDA_EN	1	0	0	1	1	1	1	0	1	1	0	0	1	1	0		
WDA_DIS	1	0	0	1	1	1	1	1	0	0	1	1	0	0	1		
SW-RESET	1	0	1	1	1	0	1	0	1	1	0	1	1	0	1		

	ADDRESS						DATA								
	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
EN-DRIVERS	0	1	1	0	0	0	1	1	0	0	1	1	0	0	1
DISABLE-DRIVERS	0	1	1	0	0	0	1	0	1	1	0	0	1	1	0
MRD_ON	0	1	1	0	0	0	1	1	1	0	0	0	1	1	1
MRD_OFF	0	1	1	0	0	0	1	0	0	1	1	1	0	0	0
MRD_UC	0	1	1	0	0	0	1	1	1	0	1	1	0	1	1
STR2_ON	0	1	0	0	1	0	0	0	0	1	0	1	0	1	0
STR2_OFF	0	1	0	0	1	0	0	1	1	0	1	0	1	0	1
STR3_ON	0	1	0	0	1	0	0	0	0	0	0	1	1	0	0
STR3_OFF	0	1	0	0	1	0	0	1	1	1	1	0	0	1	1
RLY4_ON	0	1	0	0	1	0	0	0	0	1	1	0	0	1	1
RLY4_OFF	0	1	0	0	1	0	0	1	1	0	0	1	0	0	0

Legenda:

Locked bit	Bit generated as pulse	Address	Default value	Bit in satellite logic, not affected by reset matrix clear conditions
------------	------------------------	---------	---------------	---

Locked bit: Bit generated as pulse Bit in satellite logic, not affected by reset matrix clear conditions.

18.5.1 CONFIG-REG 1

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONF1 REG 1	0	0	1	1	1	1	1	PHOLD D_EN	PHOLD_TIME[0: 1]		LED1_ PD_EN	LED2_ PD_EN	BOOS T_EN	WAKE_UP_TIM ER_START_ST OP[0:1]	
RW	DEFAULT							0	0	0	0	0	1	1	0
	RESET SOURCE							D0:D5 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN D6:D7 the read bits are from satellite logic on VB_STBY and are not affected by reset matrix							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							not locked bits							

[6:7] WAKE_UP_TIMER_START_STOP [0:1]

Start/stop Wake_up_timer counter

10: (stop) wake up timer stops (default)

01: (start) → 10 (stop) wake up timer counter value hold

10: (stop) → 01 (start) wake up timer starts from h'000001

01: (start) wake up timer counter is running

Application note: after a start/stop command 500us must be waited to have the command be effective in the EOT clock domain.

[5] BOOST_EN

Boost enable.

1: enabled (default)

0: disabled

[4] LED2_PD_EN

Enable the pull down current (open load diagnosis not available) for LED2 driver

0: pull down current disabled (LED mode enabled) (default)

1: enable

[3] LED1_PD_EN

Enable the pull down current (open load diagnosis not available) for LED1 driver

0: pull down current disabled (LED mode enabled) (default)

1: enable

[2:1] PHOLD_TIME[1:0]

Set the power hold timeout.

00: timeout disabled (default)

01: 500 ms

10: 5 minutes

11: 20 minutes

[0] PHOLD_EN

Enable the power hold function.

0: disabled (default)

1: enabled

18.5.2 CONFIG-REG 2

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFIG-REG 2	0	1	1	1	1	0	1	MSC_ACT_EN	F_DO_SEL[0:1]		VRS_MODE_SEL	IGN_CURR_RENT_CFG	IGN_DIAG	IPUPD_EN	IPUPD_MODE
RW	DEFAULT							1	1	0	0	0	0	0	1
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							not locked bits							

[7] IPUPD_MODE

Enable the fast charge current I_LS_PU1 required by driver fast off diagnosis.

0: fast charge current disabled

1: fast charge current enabled (default)

[6] IPUPD_EN

Enable the driver off diagnosis and the currents I_LS_PU2, I_LS_PD1 required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[5] IGN_DIAG.

0: IGN diagnosis off (default)

1: IGN diagnosis on

[4] IGN_CURRENT_CFG.

0: I1_HS_IGN current selected 15-30 mA (default)

1: I2_HS_IGN current selected 5-15 mA

[3] VRS_MODE_SEL: VRS mode

0: limited adaptive (default)

1: full adaptive

[2:1] F_DO_SEL [1:0] Upstream clock ratio selection.

00: fDO= fCL/ 16

01: fDO= fCL/ 32(default)

10: fDO= fCL/ 64

11: fDO= fCL/ 128

[0] MSC_ACT_EN: MSC activity monitoring enable

0: MSC activity monitoring function is disabled

1: MSC activity monitoring function is enabled (default)

18.5.3 CONFIG-REG 3

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFIG-REG 3	1	1	1	1	0	0	1	RLY4_DLY_OFF_EN	LSD/H SD_D RV1-CFG	LSD/H SD_D RV2-CFG	LSD/H SD_D RV3-CFG	SRT2_EN	STR3_EN	T_SEO_DELAY[0:1]	
RW	DEFAULT							1	1	1	1	1	1	0	0

	RESET SOURCE	D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN
	ACTIVATION	D0:D7 bits active as level
	CONTROL ACCESS	not locked bits

[7:6] SEO_DELAY_DELAY[1:0]

Delay from KEY_OFF for SEO function activation.

00: 100 ms (default)

01: 200 ms

10: 400 ms

11: 800 ms

[5] STR3_EN

0: disable

1: starter functions (DELAY OFF) enable for driver STR[3] (default)

[4] STR2_EN

0: disable

1: starter functions (DELAY OFF) enable for driver STR[2] (default)

[3] LSD/HSD_DRV3-CFG

1: LOW-SIDE configuration for STR3 driver (default)

0: HIGH-SIDE configuration for STR3 driver

[2] LSD/HSD_DRV2-CFG

1: LOW-SIDE configuration for STR2 driver (default)

0: HIGH-SIDE configuration for STR2 driver

[1] LSD/HSD_DRV1-CFG

1: LOW-SIDE configuration for STR1 driver (default)

0: HIGH-SIDE configuration for STR1 driver

[0] RLY4_DLY_OFF_EN

Enable delay off for RLY4 driver.

0: disabled

1: enabled (default)

18.5.4 CONFIG-REG 4

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFREG4	1	1	0	0	0	0	1	PRD1_IDRV[0:1]		PRD2_IDRV[0:1]		PRD3_IDRV[0:1]		PRD4_IDRV[0:1]	
RW	DEFAULT							0	0	0	0	0	0	0	0
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							not locked bits							

[7:6] PRD4_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR_V_00 20mA (default)

01: IDR_V_01 10mA

10: IDR_V_10 4mA

11: IDR_V_11 2mA

[5:4] PRD3_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR_V_00 20mA (default)

01: IDR_V_01 10mA

10: IDR_V_10 4mA

11: IDR_V_11 2mA

[3:2] PRD2_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR_V_00 20mA (default)

01: IDR_V_01 10mA

10: IDR_V_10 4mA

11: IDR_V_11 2mA

[1:0] PRD1_IDRV[1:0]

MOSFET pre-driver current driving strength
 00: IDR_V_00 20mA (default)
 01: IDR_V_01 10mA
 10: IDR_V_10 4mA
 11: IDR_V_11 2mA

18.5.5 CONFIG-REG 5

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFREG5	0	0	0	0	1	1	1	PRD5_IDRV[0:1]		PRD1_VDS[0:2]			PRD2_VDS[0:2]		
RW	DEFAULT							0	0	0	0	0	0	0	0
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							not locked bits							

[7:5] PRD2_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH_DS_000 150mV (default)

001: VTH_DS_001 245mV

010: VTH_DS_010 325mV

011: VTH_DS_011 405mV

100: VTH_DS_100 525mV

101: VTH_DS_101 660mV

110: VTH_DS_110 950mV

[4:2] PRD1_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH_DS_000 150mV (default)

001: VTH_DS_001 245mV

010: VTH_DS_010 325mV

011: VTH_DS_011 405mV

100: VTH_DS_100 525mV

101: VTH_DS_101 660mV

110: VTH_DS_110 950mV

[1:0] PRD5_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR_V_00 20mA (default)

01: IDR_V_01 10mA

10: IDR_V_10 4mA

11: IDR_V_11 2mA

18.5.6 CONFIG-REG 6

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFREG6	1	1	0	0	1	1	1	PRD3_VDS[0:2]			PRD4_VDS[0:2]			EOT_MODE	IGN_L SD_DISS
RW	DEFAULT							0	0	0	0	0	0	1	1
	RESET SOURCE							D0:D5, D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN D6 the read bit is from satellite logic on VB_STBY and is not affected by reset matrix							

	ACTIVATION	D0:D7 bits active as level
	CONTROL ACCESS	not locked bits

[7] IGN_LSD_DIS

Disable for LSD stage in igniter driver to be used with IGBT load

0: LSD enable

1: LSD disabled (default)

[6] EOT_MODE

Bit going to Wake Up Timer Logic

Selection for EOT function

0: counter without wake up function EOT function 1

1: counter with wake up function EOT function 2 (default)

Application note: EOT_MODE bit can be written only when WAKE_UP_TIMER_EN_SEL=0

(CONFIG_REG20 D2, MSC Start/Stop) and WAKE_UP_TIMER_START_STOP[0:1]=10

(CONFIG_REG1 D7-D6, Stop condition) when WAKE_UP_TIMER_START_STOP[0:1] is

effective in EOT clock domain

[5:3] PRD4_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH_DS_000 150mV (default)

001: VTH_DS_001 245mV

010: VTH_DS_010 325mV

011: VTH_DS_011 405mV

100: VTH_DS_100 525mV

101: VTH_DS_101 660mV

110: VTH_DS_110 950mV

[2:0] PRD3_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH_DS_000 150mV (default)

001: VTH_DS_001 245mV

010: VTH_DS_010 325mV

011: VTH_DS_011 405mV

100: VTH_DS_100 525mV

101: VTH_DS_101 660mV

110: VTH_DS_110 950mV

18.5.7 CONFIG-REG 7

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CO NFI GR EG 7	0	0	1	1	0	0	1	PRD5_VDS[0:2]			TRK_EN[1]	TRK_EN[2]	TRK_EN[3]	DAC[0:1]	
RW	DEFAULT							0	0	0	1	1	1	1	1
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							not locked bits							

[7:6] DAC

A/D test output voltage selection

00: 0.25*Vref

01: 0.50*Vref

10: 0.75*Vref

11: 1.0*Vref (default)

[5] TRK_EN[3]

Enable for tracking regulator 3

0: disabled

1: enabled (default)

[4] TRK_EN[2]

Enable for tracking regulator 2

0: disabled

1: enabled (default)

[3] TRK_EN[1]

Enable for tracking regulator 1

0: disabled

1: enabled (default)
 [2:0] PRD5_VDS[2:0]
 MOSFET pre-driver VDS threshold
 000: VTH_DS_000 150mV (default)
 001: VTH_DS_001 245mV
 010: VTH_DS_010 325mV
 011: VTH_DS_011 405mV
 100: VTH_DS_100 525mV
 101: VTH_DS_101 660mV
 110: VTH_DS_110 950mV

18.5.8 CONFIG-REG 8

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFGR EG 8	1	0	1	1	0	1	1	VRS_D IAG	VRS_MODE[0:1]		VRS_HYS2[0:2]			EN_FALLING_FILTER	HYS_FB_SEL
RW	DEFAULT							0	0	1	0	0	0	0	0
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							not locked bits							

[7] HYS_FB_SEL:
 0: VRS hyst. Feedback connected before adaptive filter (default)
 1: VRS hyst. Feedback connected after adaptive filter
 [6] EN_FALLING_FILTER:
 0: Falling edge filter disabled (default)
 1: Falling edge filter enabled
 [5:3] VRS_HYST[2:0]
 000: Hys current = 17 μ A (Hys VRS = 347 mV with 10 k Ω ext resistors) (default)
 001: Hys current = 5 μ A (Hys VRS=100mV with 10 k Ω ext resistors)
 010: Hys current = 10 μ A (Hys VRS=200mV with 10 k Ω ext resistors)
 011: Hys current = 17 μ A (Hys VRS=347mV with 10 k Ω ext resistors)
 100: Hys current = 32 μ A (Hys VRS=644mV with 10 k Ω ext resistors)
 101: Hys current = 51 μ A (Hys VRS=967mV with 10 k Ω ext resistors)
 110: Hys current = 17 μ A (Hys VRS=347mV with 10 k Ω ext resistors)
 111: Hys current = 0 μ A (used only for test purpose)
 [2:1] VRS_MODE[1:0]
 Internal auto-adaptive hysteresis OFF allows to configure hysteresis by MSC. Internal autoadaptive hysteresis ON selects higher hysteresis between the one configured by MSC and the one internally computed by peak voltage.
 Internal auto-adaptive filter time ON works properly only in full adaptive mode, in limited adaptive mode filter time ON is fixed to 4 us. Internal auto-adaptive filter time OFF allows to bypass the filter in both modalities.
 00: Internal auto-adaptive hysteresis OFF, internal auto-adaptive filter time OFF
 01: Internal auto-adaptive hysteresis ON, internal auto-adaptive filter time OFF
 10: Internal auto-adaptive hysteresis OFF, internal auto-adaptive filter time ON (default)
 11: Internal auto-adaptive hysteresis ON, internal auto-adaptive filter time ON
 [0] VRS_DIAG: VRS diagnosis enable, only for full adaptive mode
 0: diagnosis function is disabled (default)
 1: diagnosis function is enabled

18.5.9 CONFIG-REG 9

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
--	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Micro second channel (MSC) interface

CO NFI GR EG 9_0	0	1	0	0	1	0	1	0	1	WDA_RESPTIME[0:5]						
RW	DEFAULT									1	1	1	1	1	1	
	RESET SOURCE							D2:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_ov, vdd5_uv, vddio_uv, err_cnt>7, sw_reset, after_run_reset, RSTN_IN								
	ACTIVATION							D2:D7 bits active as level								
	CONTROL ACCESS							D2:D7 not locked bits								

[7:2] WDA_RESPTIME[5:0]

Response-time = WDA_RESPTIME[5:0] * 1.6 ms

The error counter is incremented by one on a controller write access to this register

	C	C	C	C	C	C	C	D0	D1	D2	D3	D4	D5	D6	D7
	0	1	0	0	1	0	1								
CO NFI GR EG 9_1	0	1	0	0	1	0	1	1	0	WDA_ WIN _SEL	WDA_I NIT				
RW	DEFAULT									1	0				
	RESET SOURCE							D2:D3 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_ov, vdd5_uv, vddio_uv, err_cnt>7, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D2:D3 bits active as level							
	CONTROL ACCESS							D2:D3 locked bits							

[3] WDA_INIT

Monitoring module reset

0: disabled (default)

1: enabled

[2] WDA_WIN_SEL

Select the time base for window watchdog generation

0: 39 kHz

1: 64 kHz (default)

18.5.10 CONFIG-REG 10

	C	C	C	C	C	C	C	D0	D1	D2	D3	D4	D5	D6	D7
	0	1	2	3	4	5	6								
CO NFI GR EG 10	0	0	0	1	1	0	1	WDA_RESP[0:7]							
RW	DEFAULT							0	0	0	0	0	0	0	0
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_ov, vdd5_uv, vddio_uv, err_cnt>7, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							D0:D7 not locked bits							

WDA_RESP[0:7]

Q&A WD response according to Table 4-3.

18.5.11 CONFIG-REG 11

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONF11	0	1	1	0	1	1	1	LIN/KLINE SEL	LIN_ERR_N[0]	LIN_ERR_N[1]	LIN_ERR_N[2]	LIN_TX_EN	LIN_TX_DIS_FOR_WDA_ERR	VDD5_OFF_SEL	O2H2_OC_FLT
RW	DEFAULT							0	0	0	0	0	1	0	0
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							D0:D7 not locked bits							

[7] O2H2_OC_FLT

Select the OVC filter time for O2H2.

0: select I_LS_ocvflt (default)

1: select I_LS_ocvflt x 2

[6] VDD5_OFF_SEL

VDD5 disabled by VB_IN UV:

0 VDD5 depends on VB_IN (default)

1: VDD5 does not depend on VB_IN

[5] LIN_TX_DIS_FOR_WDA_ERR

Disable the LIN transmitting when Watchdog ERR_CNT>4

0: LIN transmitting stays enabled when Watchdog ERR_CNT>4

1: LIN transmitting disabled when Watchdog ERR_CNT>4 (default)

[4] LIN_TX_EN

Enable the LIN transmission function

0: disabled (default)

1: enabled

[3] LIN_ERR_EN[2]

LIN error handling permanent dominant

0: LIN error handling disabled (default)

1: LIN error handling enabled

[2] LIN_ERR_EN[1]

LIN error handling permanent recessive

0: LIN error handling disabled (default)

1: LIN error handling enabled

[1] LIN_ERR_EN[0]

LIN error handling permanent TX timeout

0: LIN error handling disabled (default)

1: LIN error handling enabled

[0] LIN/KLINE SEL

Select between LIN/K-LINE sel

0: LIN Mode (default)

1: K-LINE Mode

18.5.12 CONFIG-REG 12

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONF12	1	0	0	1	0	0	1	TNL_RST_EN	VB_IN_OVR_ST_EN	VDD5_OV_RST_EN	O2H1_OC_FLT	FILTER_MODE	WAKE_UP_CAN_RST	WAKE_UP_EOT_RST	WK_IN_RST
RW	DEFAULT							1	1	1	0	0		0	0
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							

	CONTROL ACCESS	D0:D7 not locked bits
--	----------------	-----------------------

[7] WK_IN_RST

Activation of WK_IN_RST, active on level

1: clearing of WK_IN_DET

0: no functionality (default)

[6] WAKE_UP_EOT_RST

Activation of WAKE_UP_EOT_RST, active on level

1: clearing of WAKE_UP_EOT_DET

0: no functionality (default)

[5] WAKE_UP_CAN_RST

Activation of WAKE_UP_CAN_RST, active on level

1: clearing of WAKE_UP_CAN_DET

0: no functionality (default)

[4] FILTER_MODE

Select the filter time (Tflt_diagoff) of all the drivers.

0: selected Tflt_diagoff1 (100us) (default)

1: selected Tflt_diagoff2 (600us)

[3] O2H1_OC_FLT

Select the OVC filter time for O2H1.

0: select I_LS_ovcflt (default)

1: select I_LS_ovcflt x 2

[2] VDD5_OV_RST_EN

Enable the contribution of VDD5_OV in the reset matrix

0: disabled

1: enabled (default)

[1] VB_IN_OV_RST_EN

Enable the contribution of VB_IN_OV on BUCK /VDD5/ VTRK regulator in the reset matrix

0: disabled

1: enabled (default)

[0] TNL_RST_EN

enable the RSTN activation for TNL time on the positive edge of KEY_IN

1: enable (default)

0: disable

18.5.13 CONFIG-REG 13

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CO NFI GR EG 13	1	1	0	1	1	0	1		WDA_ PWR_ CNT_O FF_DI S	IPUPD_ EN_S TR 1	IPUPD_ EN_S TR 2	IPUPD_ EN_S TR 3	IDIAG_ HIGH_ INJ	IDIAG_ HIGH_ SOL	IDIAG_ HIGH_ PDRV
RW	DEFAULT							0	0	0	0	0	1	1	1
	RESET SOURCE							D1:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D1:D7 bits active as level							
	CONTROL ACCESS							D1 locked bit D2:D7 not locked bits							

[7] IDIAG_HIGH_PDRV

Increase the currents required for pre-drivers off diagnosis.

0: I_LS_PU2 (40 µA (min)) I_LS_PD1(60 µA (min))

1: I_LS_PU3 (100 µA (min)) I_LS_PD2(350 µA (min)) (default)

[6] IDIAG_HIGH_SOL

Increase the currents required for Solenoids off diagnosis.

0: I_LS_PU2 (40 µA (min)) I_LS_PD1(60 µA (min))

1: I_LS_PU3 (100 µA (min)) I_LS_PD2(350 µA (min)) (default)

[5] IDIAG_HIGH_INJ

Increase the currents required for Solenoids off diagnosis.

0: I_LS_PU2 (40 µA (min)) I_LS_PD1(60 µA (min))

1: I_LS_PU3 (100 µA (min)) I_LS_PD2(350 µA (min)) (default)

[4] IPUPD_EN_STR3

Enable the driver off diagnosis and the currents required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[3] IPUPD_EN_STR2

Enable the driver off diagnosis and the currents required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[2] IPUPD_EN_STR1

Enable the driver off diagnosis and the currents required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[1] WDA_PWR_CNT_OFF_DIS

Disable the power down due to watchdog pwr_cnt overflow.

0: enabled (default)

1: disabled

[0] Not used

18.5.14 CONFIG-REG 14

	C	C	C	C	C	C	C	D0	D1	D2	D3	D4	D5	D6	D7
	0	1	2	3	4	5	6								
CO NFI GR EG 14	1	1	1	0	0	1	1	RLY1_ UC	RLY2_ UC	RLY3_ UC	RLY4_ UC	RLY5_ UC	STR1_ UC	STR2_ UC	STR3_ UC
RW	DEFAULT							0	0	0	0	0	0	0	0
	RESET SOURCE							D0:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							D0:D7 not locked bits							

[7] STR3_UC

0→1: micro command for fast diagnostic activation for relay driver STR3

0: no functionality (default)

1: no functionality

[6] STR2_UC

0→1: micro command for fast diagnostic activation for relay driver STR2

0: no functionality (default)

1: no functionality

[5] STR1_UC

0→1: micro command for fast diagnostic activation for relay driver STR1

0: no functionality (default)

1: no functionality

[4] RLY5_UC

0→1: micro command for fast diagnostic activation for relay driver RLY5

0: no functionality (default)

1: no functionality

[3] RLY4_UC

0→1: micro command for fast diagnostic activation for relay driver RLY4

0: no functionality (default)

1: no functionality

[2] RLY3_UC

0→1: micro command for fast diagnostic activation for relay driver RLY3

0: no functionality (default)

1: no functionality

[1] RLY2_UC

0→1: micro command for fast diagnostic activation for relay driver RLY2

0: no functionality (default)

1: no functionality

[0] RLY1_UC

0→1: micro command for fast diagnostic activation for relay driver RLY1

0: no functionality (default)

1: no functionality

18.5.15 CONFIG-REG 15

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFIGREG15-0(WUPT_0)	1	0	0	0	0	1	1	0	0	Wake up timer_SET_0[0:5]					
RW	DEFAULT									0	0	0	0	0	0
CONFIGREG15-1(WUPT_1)	1	0	0	0	0	1	0	0	1	Wake up timer_SET_1[6:11]					
RW	DEFAULT									0	0	0	0	0	0
CONFIGREG15-2(WUPT_2)	1	0	0	0	0	0	1	1	0	Wake up timer_SET_2[12:17]					
RW	DEFAULT									0	0	0	0	0	0
CONFIGREG15-3(WUPT_3)	1	0	0	0	0	1	1	1	1	Wake up timer_SET_3[23:28]					
RW	DEFAULT									0	0	0	0	0	0
	RESET SOURCE							D2:D7 The read bits are from satellite logic on VB_STBY and are not affected by reset matrix							
	ACTIVATION							D2:D7 bits active as level							
	CONTROL ACCESS							D2:D7 not locked bits							

Application note: the minimum time distance between two consecutive access to the Wake up timer SET registers is 2.5 us.

Wake up timer_SET_x Bits go to and are stored in Wake Up Timer Logic.

18.5.16 CONFIG-REG 16

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFIGREG 16-0	0	0	1	0	0	1	1	0	1	PRD1_BLK[0:1]		PRD2_BLK[0:1]		PRD3_BLK[0:1]	
RW	DEFAULT									1	0	1	0	1	0
CONFIGREG 16-1	0	0	1	0	0	1	1	1	0	PRD4_BLK[0:1]		PRD5_BLK[0:1]		O2H_P DRV_1	O2H_P DRV_3
RW	DEFAULT									1	0	1	0	0	0
	RESET SOURCE							D2:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D2:D7 bits active as level							
	CONTROL ACCESS							D2:D7 not locked bits							

[7] O2H_PDRV_3

VDS comparator selection for ON diagnosis

0: PDRV3 takes VDS3 comparator and PDRV4 works as independent driver

1: PDRV3 takes VDS4 comparator and PDRV4 driver cannot be used

[6] O2H_PDRV_1

VDS comparator selection for ON diagnosis

0: PDRV1 takes VDS1 comparator and PDRV2 works as independent driver

1: PDRV1 takes VDS2 comparator and PDRV2 driver cannot be used

PRDx_BLK[1:0]

Blanking time on VDS detection in ON state for Predriver

00: 6 μ s01: 12 μ s (default)10: 18 μ s11: 24 μ s**18.5.17 CONFIG-REG 17**

	C	C	C	C	C	C	C	D	D	D	D3	D4	D	D6	D7
	0	1	2	3	4	5	6	0	1	2			5		
CONFIGREG 17_0	0	0	1	0	0	1	0	0	0		O2H1_SR	O2H1_OC_TH		O2H2_SR	O2H2_OC_TH
RW	DEFAULT									0	0	1	0	0	1
	RESET SOURCE							D2:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN							
	ACTIVATION							D2:D7 bits active as level							
	CONTROL ACCESS							D2:D7 not locked bits							

[7] O2H2_OC_TH

Select OVC threshold

0: threshold1 I_ovc1_o2h 4.5A typ

1: threshold2 I_ovc2_o2h 10A typ (default)

[6] O2H2_SR

Select SR threshold

0: typ 0,5 V/ μ s (default)1: typ 4 V/ μ s

[4] O2H1_OC_TH

Select OVC threshold

0: threshold1 I_ovc1_o2h 4.5A typ

1: threshold2 I_ovc2_o2h 10A typ (default)

[3] O2H1_SR

Select SR threshold

0: typ 0,5 V/ μ s (default)1: typ 4 V/ μ s

	C	C	C	C	C	C	C	D	D	D2	D3	D4	D5	D6	D7
	0	1	2	3	4	5	6	0	1						
CONFIGREG 17_1	0	0	1	0	0	1	0	1	1	CAN_EN	CAN_TX_EN	CAN_WAK_EUP_EN	CAN_PATTERN_EN	CAN_RX_EN	CAN_2_5_MB
RW	DEFAULT									1	1	1	1	1	0
	RESET SOURCE							D2:D7 The read bits are from satellite logic on VB_STBY and are not affected by reset matrix							
	ACTIVATION							D2:D7 bits active as level							
	CONTROL ACCESS							D2:D6 not locked bits D7 locked bit							

[7] CAN_2_5_MB

CAN mode selection:

0: CAN with 2Mbaud (default)

1: CAN FD with 5Mbaud with reduced EMC performance.

[6] CAN_RX_EN:

CAN Receiver enable:

0: disable

1: enable (default)

[5] CAN_PATTERN_EN

Normal pattern wake up enable:

0: disable

1: enable (default)
 [4] CAN_WAKEUP_EN
 Enable wake up by CAN:
 0: disable
 1: enable (default)
 [3] CAN_TX_EN
 CAN Transmitter enable:
 0: disable
 1: enable (default)
 [2] CAN_EN
 CAN enable
 0: disable
 1: enable (default)
 All Bits go to and are stored in CAN Logic

18.5.18 CONFIG-REG STBY_NVM

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
STBY_NVM_ADDR	0	1	0	0	1	1	0	0	1	STBY_NVM_ADD[0:3]				MEM_VALID	
RW	DEFAULT									0	0	0	0	0	
	RESET SOURCE							D2:D6 The read bits are from satellite logic on VB_STBY and are not affected by reset matrix							
	ACTIVATION							D2:D6 bits active as level							
	CONTROL ACCESS							D2:D6 not locked bits							

[6] MEM_VALID
 STBY NVM memory valid bit:
 0: not valid (default)
 1: valid
 [5:2] STBY_NVM_ADD
 STBY NVM write address
 0000: address of the byte to write (default)
 All Bits go to and are stored in Wake Up Timer Logic

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
STBY_NVM_DATA	1	0	1	1	0	0	0	STBY_NVM_DATA[0:7]							
RW	DEFAULT									0	0	0	0	0	
	RESET SOURCE							D0:D7 The read bits are from satellite logic on VB_STBY and are not affected by reset matrix							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							D0:D7 not locked bits							

[7:0] STBY_NVM_ADD
 Byte to write in STBY NVM
 All Bits go to and are stored in Wake Up Timer Logic

18.5.19 CONFIG-REG 20

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CON FIGR EG 20	1	1	1	1	1	1	0	BIST_EN	FIN_WAKE	WAKE_UP_TIMER_EN_SEL	KEY_OC_RETRY_MAX_EN	CAN_LOOP_EN	CAN_TDI	LIN_TX_DOM_ERR_CFG	PDRV_O2H_DLY
RW	DEFAULT							0	0	0	1	0	0	0	0
	RESET SOURCE							D0:D1; D4:D7 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN D2 the read bits are from satellite logic on VB_STBY and are not affected by reset matrix							
	ACTIVATION							D0 bit active as pulse D1:D7 bits active as level							
	CONTROL ACCESS							D0:D3 not locked bits D4:D7 locked bits							

[7] PDRV_O2H_DLY

Selection of delay time for switching from low to high current in predriver on/off transitions in o2h configuration.

0: 150 us selected (default)

1: 300 us selected

[6] LIN_TX_DOM_ERR_CFG

LIN Tx enable after dominant error timeout:

0: re-enable as soon as Tx become recessive (default)

1: wait for error flag read

[5] CAN_TDI

CAN transmission depends on WDA:

0: WDA does not affect CAN (default)

1: disable transmission if WDA_INT active

[4] CAN_LOOP_EN

CAN looping mode enable:

0: disable (default)

1: enable

[3] KEY_OC_RETRY_MAX_EN

Enable a maximum number of retry after OVC, when wake up by key:

0: disable

1: enable(default)

Bits going to Wake Up Timer Logic

[2] WAKE_UP_TIMER_EN_SEL

Select the source of activation for the wake_up_timer

0: from MSC start/stop commands (default)

1: from KEY_IN 1->0

Bits going to Wake Up Timer Logic

[1] FIN_WAKE

0: satellite logics execute auto clear 1 second after power up (default)

1: micro confirmed power up, no auto clear after 1 second

[0] BIST_EN

Bist enable for regulators

0: disable (default)

1: enable

18.5.20 CONFIG-REG 21

	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
CONFIG-REG 21	0	0	0	0	0	0	1	CAN_TX_DOM_ERR_CFG	OL_RED	BUCK_SLOW_SR	CAN_TXD_DOM_EN	CAN_PERM_REC_EN	CAN_PERM_DOM_EN	CAN_RXD_REC_EN	CAN_AUTO_BIAS
RW	DEFAULT							0	0	0	1	1	1	1	1
	RESET SOURCE							D0:D2 V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN D3:D7 the read bits are from satellite logic on VB_STBY and are not affected by reset matrix							
	ACTIVATION							D0:D7 bits active as level							
	CONTROL ACCESS							D0 locked bit D1:D7 not locked bits							

[7] CAN_AUTO_BIAS

0: auto biasing disabled

1: auto biasing enabled (default)

[6] CAN_RXD_REC_EN

0: error handling disable

1: error handling enable (default)

[5] CAN_PERM_DOM_EN

0: error handling disable

1: error handling enable (default)

[4] CAN_PERM_REC_EN

0: error handling disable

1: error handling enable (default)

[3] CAN_TXD_DOM_EN

0: error handling disable

1: error handling enable (default)

All Bits going to CAN Logic

[2] BUCK_SLOW_SR

0: fast slew rate selected (default)

1: slow slew rate selected

[1] OL_RED

Select open load current:

0: 0.8 mA max (default)

1: 0.2 mA max

[0] CAN_TX_DOM_ERR_CFG

CAN Tx enable after dominant error timeout:

0: re-enable as soon as Tx become recessive (default)

1: wait for error flag read

18.5.21 Commands

RD_COMMAND1.. RD_COMMAND16

These commands allow the reading of information that will be output on Upstream.

Table 18-8 Commands

Name	ADDRESS							DATA							
	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
RD_COMMAND1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
RD_COMMAND2	1	0	1	0	1	0	1	1	0	1	0	1	0	1	0
RD_COMMAND3	0	1	0	1	0	1	0	0	1	0	1	0	1	0	1
RD_COMMAND4	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0
RD_COMMAND5	1	0	1	0	1	0	1	1	0	0	1	1	0	0	1
RD_COMMAND6	1	0	1	0	1	0	1	0	1	1	0	0	1	1	0
RD_COMMAND7	0	1	0	1	0	1	0	1	0	0	1	1	0	0	1
RD_COMMAND8	0	1	0	1	0	1	0	0	1	1	0	0	1	1	0
RD_COMMAND9	1	0	1	0	1	0	1	1	1	1	0	1	1	1	0
RD_COMMAND10	1	0	1	0	1	0	1	0	0	0	1	0	0	0	1
RD_COMMAND11	0	1	0	1	0	1	0	1	1	1	1	0	1	1	0
RD_COMMAND12	0	1	0	1	0	1	0	0	0	0	1	0	0	0	1
RD_COMMAND13	1	0	1	0	1	0	1	0	1	1	1	0	1	1	1
RD_COMMAND14	1	0	1	0	1	0	1	1	0	0	0	1	0	0	0
RD_COMMAND15	0	1	0	1	0	1	0	0	1	1	1	0	1	1	1
RD_COMMAND16	0	1	0	1	0	1	0	1	0	0	0	1	0	0	0

Lock Command

Registers that have this function are locked as default.

Registers are unlocked by the UNLOCK command and locked again by the LOCK command. Before the LOCK command the registers stay unlocked.

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
LOCK	1	0	0	1	1	1	1	0	1	0	1	0	1	0	1

Unlock Command

Enables the write right to registers under lock.

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
UNLOCK	1	0	0	1	1	1	1	1	0	1	0	1	0	1	0

Question and Answer Watch Dog Enable command

WD is enabled as default.

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
WDA_EN	1	0	0	1	1	1	1	0	1	1	0	0	1	1	0

Question and Answer Watch Dog Disable command

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
WDA_DIS	1	0	0	1	1	1	1	1	0	0	1	1	0	0	1

SW-Reset command

This command generates internal reset initiated by the CPU's software that clears all the configuration and diagnostic registers and switches off all the drivers.

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
SW-RESET	1	0	1	1	1	0	1	0	1	1	0	1	1	0	1

EN_Drivers command

The command Enable Drivers sets the bit Drivers_EN to “1” (UPS1 Frame 10 bit D1).

With Drivers_EN =1 all drivers controlled through control register can be activated using the Data Frame.

After a reset or command Disable Drivers_EN = “0” and all drivers are disabled.

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
EN-DRIVERS	0	1	1	0	0	0	1	1	0	0	1	1	0	0	1

Disable Driver

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
DISABLE-DRIVERS	0	1	1	0	0	0	1	0	1	1	0	0	1	1	0

Main relay Driver ON

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
MRD_ON	0	1	1	0	0	0	1	1	1	0	0	0	1	1	1

Main relay Driver OFF

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
MRD_OFF	0	1	1	0	0	0	1	0	0	1	1	1	0	0	0

MRD_OFF_UC command

The Main Relay Driver is switched off for the time required to perform the off diagnosis and then switched on again. The command can be activated if firstly is unlocked.

W	C0	C1	C2	C3	C4	C5	C6	D0	D1	D2	D3	D4	D5	D6	D7
MRD_UC	0	1	1	0	0	0	1	1	1	0	1	1	0	1	1

STR2 driver ON/OFF

It has effect only if delay-off function for this driver is active (Config3[4] = 1).

STR2_ON	0	1	0	0	1	0	0	0	0	1	0	1	0	1	0
STR2_OFF	0	1	0	0	1	0	0	1	1	0	1	0	1	0	1

STR3 driver ON/OFF

It has effect only if delay-off function for this driver is active (Config3[5] = 1).

STR3_ON	0	1	0	0	1	0	0	0	0	0	0	1	1	0	0
STR3_OFF	0	1	0	0	1	0	0	1	1	1	1	0	0	1	1

RLY4 driver ON/OFF

It has effect only if delay-off function for this driver is active (Config3[0] = 1).

RLY4_ON	0	1	0	0	1	0	0	0	0	1	1	0	0	1	1
RLY4_OFF	0	1	0	0	1	0	0	1	1	0	0	1	1	0	0

18.6 Upstream frame

See downstream paragraph for configuration register bits description. Bits highlighted in orange are clear on read. Bits highlighted in green come from satellite logics, they need VB_STBY in order to be correctly read.

18.6.1 Upstream Bit Map Read1 (CONFIG-REG 1-4)**Table 18-9** MSC Interface Upstream Bit Map Read1

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	CONFIG-REG 1							
FRAME2	0	0	1	0	CONFIG-REG 2							
FRAME3	0	0	0	1	CONFIG-REG 3							
FRAME4	0	0	1	1	CONFIG-REG 4							
source					logic on VB_IN (main logic)							
access					read							

Frame[1]

[7:6] WAKE_UP_TIMER_START_STOP:

Read start "10" or stop "01" status, (never read "00" or "11").

These bits are valid only if WUT is driven by MSC.

18.6.2 Upstream Bit Map Read2 (CONFIG-REG 5-8)**Table 18-10** MSC Interface Upstream Bit Map Read2

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	CONFIG-REG 5							
FRAME2	0	0	1	0	CONFIG-REG 6							
											EOT _MO DE(1)	
FRAME3	0	0	0	1	CONFIG-REG 7							
FRAME4	0	0	1	1	CONFIG-REG 8							

Micro second channel (MSC) interface

source	FRAME[1][3][4] FRAME[2]D[0:5] FRAME[2]D[7] logic on VB_IN (main logic) FRAME[2]D[6] logic on VB_STBY(Wake Up Timer logic)
access	read

1. Bit in satellite logic.

18.6.3 Upstream Bit Map Read3 (CONFIG-REG 11-13, 16-0)

Table 18-11 MSC Interface Upstream Bit Map Read3

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	CONFIG-REG 11							
FRAME2	0	0	1	0	CONFIG-REG 12							
FRAME3	0	0	0	1	CONFIG-REG 13							
FRAME4	0	0	1	1	STR_DELAY _OFF_STAT US		CONFIG-REG 16-0					
					STR 2	STR 3						
source				logic on VB_IN (main logic)								
access				read								

FRAME[4]:

[1] 1:STR3 delay-off function active

0: STR3 delay-off function inactive

[0] 1: STR2 delay-off function active

0: STR2 delay-off function inactive

18.6.4 Upstream Bit Map Read4 (CONFIG-REG 16-1 + Wake up timer_SET)

Table 18-12 MSC Interface Upstream Bit Map Read4

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	RLY 4_D ELA Y_O FF_ STA TUS	MRD _ST _ATU S	CONFIG-REG 16_1					
					RLY 4	MRD						
					Wake up timer_SET [23:0]							
FRAME2	0	0	1	0	Wake up timer_SET_0[0:7](1)							
FRAME3	0	0	0	1	Wake up timer_SET_1[8:15](1)							
FRAME4	0	0	1	1	Wake up timer_SET_2[16:23](1)							
source					FRAME[1][0:7]							

	logic on VB_IN (main logic) FRAME[2:4]D[0:7] logic on VB_STBY (Wake Up Timer logic)
access	read

1. Bit in satellite logic.

FRAME[1]:

[1] 1: Main Relay Driver On

0: Main Relay Driver Off

[0] 1: RLY4 delay-off function active

0: RLY4 delay-off function inactive

FRAME[2:4]: Wake Up Timer setting

Bits coming from WUT Logic

18.6.5 Upstream Bit Map Read5 (CONFIG-REG 17-0 + WAKE_UP_TIMER VALUE)

Table 18-13 MSC Interface Upstream Bit Map Read5

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0		TNL	CONFIG-REG 17_0					
						TNL _RE _SET (1)						
					TIMER CONT VALUE[23:0]							
FRAME2	0	0	1	0	Wake up timer _CNT_0[0:7](2)							
FRAME3	0	0	0	1	Wake up timer _CNT_1[8:15](2)							
FRAME4	0	0	1	1	Wake up timer _CNT_2[16:23](2)							
source					FRAME[1]D[0:7] logic on VB_IN (main logic) FRAME[2:3]D[0:7] logic on VB_STBY (Wake Up Timer logic)							
access					FRAME[1]D[1] clear-on-read FRAME[1]D[0] FRAME[1]D[2:7] FRAME[2:4]D[0:7] read							

1. Clear on Read.

2. Bit in satellite logic.

FRAME[1]:

[1] 1: After run reset (TNL) happened

0: After run reset (TNL) not happened, clear on read

FRAME[2:4]: Wake Up Timer counting

18.6.6 Upstream Bit Map Read6 (DRIVER DIAGNOSIS)

Table 18-14 MSC Interface Upstream Bit Map Read6

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	DIAG INJ1				DIAG INJ2			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME2	0	0	1	0	DIAG INJ3				DIAG INJ4			

Micro second channel (MSC) interface

					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME3	0	0	0	1	DIAG O2H1							
					STA	STG	OPL	OVC				
FRAME4	0	0	1	1	DIAG O2H2							
					STA	STG	OPL	OVC				
source					logic on VB_IN (main logic)							
access					clear-on-read							

Clear on Read

FRAME[1:4]: Driver diagnosis for driver

STA: driver diagnosis: error in driver status

The driver status is detected by reading the STG comparator in on and off, it should be aligned with command (msc command and enabling).

0: aligned, no error

1: not aligned, error

STG/STB: driver diagnosis: short to ground/battery

0: no fault (default)

1: for low-side and igniter drivers: short to ground fault

for high-side driver: short to battery fault

OPL: driver diagnosis: open load

0: no fault (default)

1: open load fault

OVC: driver diagnosis: over current

0: no fault (default)

1: over current fault

18.6.7 Upstream Bit Map Read7 (DRIVER DIAGNOSIS)

Table 18-15 MSC Interface Upstream Bit Map Read7

	ADDRESS				READ3 DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	DIAG LED1				DIAG LED2			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME2	0	0	1	0	MRD DIAG				DIAG RLY1			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME3	0	0	0	1	DIAG RLY2				DIAG RLY3			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME4	0	0	1	1	DIAG RLY4				DIAG RLY5			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
source					logic on VB_IN (main logic)							
access					clear-on-read							

Clear on Read

18.6.8 Upstream Bit Map Read8 (DRIVER DIAGNOSIS)

Table 18-16 MSC Interface Upstream Bit Map Read8

	ADDRESS				READ4 DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	DIAG STR1				DIAG STR2			
					STA	STG/STB	OPL	OVC	STA	STG/STB	OPL	OVC
FRAME2	0	0	1	0	DIAG STR3				DIAG PRD1			
					STA	STG/STB	OPL	OVC	STA	STG	OPL	OVC
FRAME3	0	0	0	1	DIAG PRD2				DIAG PRD3			

Micro second channel (MSC) interface

					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME4	0	0	1	1	DIAG PRD4				DIAG PRD5			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
source					logic on VB_IN (main logic)							
access					clear-on-read							

Clear on Read

18.6.9 Upstream Bit Map Read9 (DRIVER DIAGNOSIS)

Table 18-17 MSC Interface Upstream Bit Map Read9

	ADDRESS				READ4 DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	DIAG IGN1				DIAG IGN2			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME2	0	0	1	0	DIAG IGN3				DIAG IGN4			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME3	0	0	0	1	DIAG IGN5				DIAG IGN6			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
FRAME4	0	0	1	1	DIAG SOL1				DIAG SOL2			
					STA	STG	OPL	OVC	STA	STG	OPL	OVC
source					logic on VB_IN (main logic)							
access					clear-on-read							

Clear on Read

18.6.10 Upstream Bit Map Fast Read10 (SAFETY WDA)

Table 18-18 MSC Interface Upstream Bit Map Read10

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	WDA REQULO							
					REQU[0:3]				ERR_CNT[0:2]			WDA_INT
FRAME2	0	0	1	0	WDA REQUHI							
					RESP_TOO_EARLY	NO_RESPONSE	W_R_E_S_P	CHRT	RESP_Z0	RESP_ERR	RESP_CNT[0:1]	
FRAME3	0	0	0	1	WDA PWR_RST_CNT							
					RST_CNT[0:2]			PWR_CNT[0:2]				WDA_RST
FRAME4	0	0	1	1	WDA RESPTIME							
					RESPTIME[0:5]						WDA_WINS_EL	WDA_INIT
source					logic on VB_IN (main logic)							
access					FRAME[1]D[0:7] FRAME[2]D[0] FRAME[2]D[2] FRAME[2]D[4:7] FRAME[3]D[0:7] FRAME[4]D[0:7] read FRAME[2]D[1] FRAME[2]D[3]							

clear-on-read

Clear on Read

FRAME[1]: WATCHDOG REQULO

[7] WDA_INT

1: ERR_CNT[2:0]>4

Default: 1

[6:4] ERR_CNT[2:0]

Value of ERR_CNT[2:0]

Default: 0x6

[3:0] REQU[3:0]

4-bit question

FRAME[2]: WATCHDOG REQUIHI

[7:6] RESP_CNT[1:0]

Counter for receiving the 4 response bytes

Default: 0xC

[5] RESP_ERR

1: 1 byte of 32-bit response is incorrect: one of the response bytes in the current sequencer run is wrong, reading WDA status register after the 4th byte write implies that this flag will always be read as 0.

reset to zero at each sequencer-run

[4] RESP_V0

1: Controller set response time to 0ms

a correct response within the time window nevertheless increments the error counter by one

0: Response-time is greater than 0ms

[3] CHRT

1: Controller has changed response time

reset to zero after a read access and after the next sequencer run

[2] W_RESP

1: if one of the RESP_BYTE_x was incorrect during the previous sequencer run;

0: otherwise

[1] NO_RESP

1: in case of no response at all timer is restarted automatically

reset to zero after a read access

[0] RESP_TOO_EARLY

1: in case the 4 response bytes arrive before time window starts during the previous sequencer run;

Reset to zero at each sequencer run

FRAME[3]: WATCHDOG PWR_RST_CNT

[7] WDA_RST

See reset matrix

[5:3] PWR_CNT[2:0]

Current value of PWR_CNT register

[2:0] RST_CNT[2:0]

Current value of RST_CNT register

FRAME[4]: WATCHDOG RESPTIME

[7:6] CONFIG-REG9_1

[5:0] RESPTIME[5:0]

CONFIG-REG9_0

18.6.11 Upstream Bit Map Read11 (SAFETY+BIST+VRS+LIN+ASIC_REV)**Table 18-19 MSC Interface Upstream Bit Map Read11**

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	SAFETY/VRS							
					TRANS_L	TRANS_F	CMD_ERR	FREQ_ERR	CLK_MON_FAULT	OSC_STUCK	OL_RED	VRS_DIAG
FRAME2	0	0	1	0	BIST							
					V5V	VDD5	IGN1	IGN2	IGN3	IGN4	IGN5	IGN6
FRAME3	0	0	0	1	BIST/LIN							
					BIST_END	INJ1	INJ2	INJ3	INJ4	DRIVER_EN	LIN/KLINES TATUS	LOCK
FRAME4	0	0	1	1	ASIC REVISION[0:7]							
					1	0	0	0	0	0	1	0
source					logic on VB_IN (main logic)							
access					FRAME[1]D[6] FRAME[3][5:7] FRAME[4]D0:7 read FRAME[1]D[0:5] FRAME[1]D[7] FRAME[2]D[0:7] FRAME[3]D[0:4] clear-on-read							

Clear on Read

FRAME[1]: SAFETY/VRS

[0] TRANS_L

Wrong command frame or data frame down stream length (longer then 16 bits)

0: no fault (default)

1: down stream frame length incorrect

[1] TRANS_F

No valid the data frame for longer than tMSC_mon

0: no fault (default)

1: no data stream within tMSC_mon time out

[2] CMD_ERR

This bit is address error(C0-C5) of previous command Frame

0: no fault (default)

1: command error

[3] FREQ_ERR

Main OSC and checker OSC running with more than +/- 20% freq difference

0: no fault (default)

1: clocks out of frequency or stucked

[4] CLK_MON_FAULT

Main OSC and checker OSC running with more than +/- 30% freq difference

0: no fault (default)

1: clocks out of frequency or stucked (drivers disabled)

[5] OSC_STUCK

Main OSC or checker OSC stuck

0: no fault (default)

1: clocks stucked (drivers disabled)

[6] OL_RED

CONFIG_REG21 BIT 1

[7] VRS_DIAG

VRS diagnosis result

0: no fault detected (default)

1: generic fault detected

FRAME[2]: BIST

[0] BIST_V5V

Bist result for V5V regulator

0: bist pass (default)

1: bist fail

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[1] BIST_VDD5
Bist result for VDD5 regulator
0: bist pass (default)
1: bist fail
[2:7] BIST_IGN
Bist result for Igniter drivers
0: bist pass (default)
1: bist fail
FRAME[3]: BIST/LIN
[0] BIST_END
End of Bist operation
0: bist not end
1: bist end
[1:4] BIST_INJ
Bist result for INJECTOR drivers
0: bist pass (default)
1: bist fail
[5] DRIVER_EN
Drivers enabled or disabled by MSC command
0: drivers disabled (default)
1: drivers enabled
[6] LIN_KLINE_STATUS
LIN / K-LINE Mode (CONFIG-REG11 [0])
0: LIN mode (default)
1: KLINE mode
[7] MSC_LOCK
MSC command and bit lock
0: unlocked
1: locked (default)
FRAME [4]: ASIC REVISION FOR AA
[0:7] ASIC_REVISION

18.6.12 Upstream Bit Map Read12 (PHOLD+POWER_UP+ADC)

Table 18-20 MSC Interface Upstream Bit Map Read12

1. Bit in satellite logic.

	ADDRESS				DATA								
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7	
FRAME1	0	0	0	0	PHOLD								
					PHOLD_TIMER_STATUS	PHOLD_TIMER [0]	PHOLD_TIMER [1]	PHOLD_TIMER [2]	PHOLD_TIMER [3]	PHOLD_TIMER [4]	PHOLD_TIMER [5]	PHOLD_TIMER [6]	
FRAME2	0	0	1	0	PHOLD								
					PHOLD_TIMER [7]	PHOLD_TIMER [8]	PHOLD_TIMER [9]	PHOLD_TIMER [10]	PHOLD_TIMER [11]	PHOLD_TIMER [12]	PHOLD_TIMER [13]	PHOLD_TIMER [14]	
FRAME3	0	0	0	1	POWER UP					EOT_STATUS		ADC	
					KEY_INTERRUPT	WK_IN_DET	WAKE_UP_EVENT_DET	WAKE_UP_CANCEL_DET	FAULT_WARN	EOT ⁽¹⁾	ADC[0]	ADC[1]	
FRAME4	0	0	1	1	ADC								
					ADC[2]	ADC[3]	ADC[4]	ADC[5]	ADC[6]	ADC[7]	ADC[8]	ADC[9]	
source					FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0:4] FRAME[3]D[6:7] FRAME[4]D[0:7] logic on VB_IN (main logic) FRAME[3]D[5] logic on VB_STBY (Wake Up Timer logic)								
access					read								

FRAME[1:2]: POWER_HOLD TIMER

It is the counter used to reach the timeout selected by PHOLD_TIME[0:1] configuration bits.

The run time is derived from the PHOLD_TIMER[14:0] according to the formula:

run time = PHOLD_TIMER[14:0] * 279 ms

FRAME[3]: POWER_UP

[0] KEY_IN_DET

Latch status of activation of KEY_IN signal

0: low level detected on KEY_IN signal

1: high level on KEY_IN signal

[1] WK_IN_DET

Latch status of activation of WK_IN signal.

0: no positive edge detected on WK_IN signal

1: positive edge on WK_IN signal

[2] WAKE_UP_EOT_DET

Latch status of activation of Wake_Up_Timer counter

0: Wake_Up_Timer end counting not detected

1: Wake_Up_Timer end counting detected

[3] WAKE_UP_CAN_DET

Latch status of activation of CAN signal

0: no wake up activity detected on CAN

1: wake up activity detected on CAN

[4] FAULT_WARN

0: no fault warning active

1: at least one of fault warning contributors (oscillator fault, V5Vpre_mon_ov,

V5Vpre_mon_uv, V5Vpre_ov, V5Vpre_uv, V5Vana_ov, V5Vana_uv, V5Vdig_ov, V5Vdig_uv, gnd_dig_loss) is active.

[5] EOT_STATUS:

1: EOT active (WAKE_UP_TIMER_START_STOP[0:1] = 01, start)

0: EOT inactive (WAKE_UP_TIMER_START_STOP[0:1] = 10, stop)

[6:7] ADC[0:1]

FRAME[4]

[0:7] ADC[2:9]

Temperature= (ADC[9:0] * 290 / 1024) - 63 [°C]

18.6.13 Upstream Bit Map Read13 (SUPPLY UV/OV)

Table 18-21 MSC Interface Upstream Bit Map Read13

ADDRESS					DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	SUPPLY OVERVOLTAGE/UNDERVOLTAGE							
						V5VA_UV	V5VA_OV	V5VD_UV	V5VD_OV	VB_IN_UV	VB_IN_OV1	VB_IN_OV2
FRAME2	0	0	1	0	SUPPLY OVERVOLTAGE/UNDERVOLTAGE							
					VB_STBY_UV ⁽¹⁾	GND_DIG_LOSS	VDDIO_UV	VDD5_UV	VDD5_OV	VPRE_UV	VPRE_OV	
FRAME3	0	0	0	1	SUPPLY OVERVOLTAGE/UNDERVOLTAGE							
					WAKE_UP_CAN_DET_AUTO	TRIM_VALID	TRK_STB[1]	TRK_STB[2]	TRK_STB[3]	TRK_OVC[1]	TRK_OVC[2]	TRK_OVC[3]
FRAME4	0	0	1	1	SUPPLY OVERVOLTAGE/UNDERVOLTAGE							
					EXT_TRK_UV	INT_TRK_UV[1]	INT_TRK_UV[2]	INT_TRK_UV[3]	EXT_TRK_OV	INT_TRK_OV[1]	INT_TRK_OV[2]	INT_TRK_OV[3]
source					FRAME[1]D[0:7] FRAME[2]D[1:7] FRAME[3]D[0:7] FRAME[4]D[0:7] logic on VB_IN (main logic) FRAME[2]D[0] logic on VB_STBY (Wake Up Timer logic)							
access					FRAME[3]D[1] read FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0] FRAME[3]D[2:7] FRAME[4]D[0:7] clear-on-read							

1. Bit in satellite logic.

Clear on Read

FRAME[1:4]: SUPPLY OVERVOLTAGE/UNDERVOLTAGE

0: no fault (default)

1: overvoltage/undervoltage detected on the corresponding pin

VB_STBY_UV is functional only if VB_STBY is present, if VB_STBY is totally disconnected no diagnosis related to VB_STBY will be available (bit latched in wut logic).

GND_DIG_LOSS comparator for digital ground level

0: the digital ground supply is connected to ground.

1: the digital ground supply is disconnected from ground.

TRIM_VALID

0: trimming content not written or not valid (parity check fail) (default)

(in this case the trimming content is anyway used)

1: trimming content written and valid

WAKE_UP_CAN_DET_AUTO

0: No Wakeup pattern detected in sleep mode with UCHIP-ON and CAN_AUTO_BIAS =

1(CONFIG-REG 21 D7) (default)

1: Wakeup pattern detected

18.6.14 Upstream Bit Map Read14 (THERMAL WARNING AND CAN)**Table 18-22 MSC Interface Upstream Bit Map Read14**

ADDRESS					DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	OVERTEMPERATURE							
					OVT [1]	OVT [2]	OVT [3]	OVT [4]	OVT [5]	OVT [6]	OVT [7]	OVT [8]
FRAME2	0	0	1	0	OVERTEMPERATURE							
					OVT [9]	OVT [10]	OVT [11]	OVT [12]	OVT [13]	OVT [14]	OVT [15]	OVT [16]
FRAME3	0	0	0	1	OVERTEMPERATURE				CONFIG-REG 17_1			
					OVT [17]	OVT [18]	OVT [19]	VDD_CAN_OV (1)	CAN_EN (1)	CAN_TX_EN (1)	CAN_RX_EN (1)	CAN_WAKEUP_EN (1)
FRAME4	0	0	1	1	CONFIG-REG 17_1			CAN DIAG				
					CAN_PATTERN_EN (1)	CAN_25_MB (1)	CAN_TXD_DOM (1)	CAN_PERRM_REC (1)	CAN_PERRM_DOM (1)	CAN_RXD_REC (1)	CAN_SILENT (1)	CAN_SUP_LOW (1)
source					FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0:2] logic on VB_IN (main logic) FRAME[3]D[4:7] FRAME[4]D[0:7] logic on VB_STBY (Can logic)							
access					FRAME[3]D[4:7] FRAME[4]D[0:1] read FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0:3] FRAME[4]D[2:7] clear on read							

1. Bit in satellite logic.

Clear on Read

FRAME[1:3]: OVERTEMPERATURE

0: no fault (default)

1: overtemperature detected on the corresponding block

FRAME[3:4]:

[3]: VDD_CAN_OV

0: no fault

1: VDD_CAN_OV overvoltage detected

CONFIG_REG 17-1

FRAME[4]: CAN DIAG, see CAN section for detailed description.

18.6.15 Upstream Bit Map Read15 (MEM_STBY_CONFIG+CONFIG_REG20+UCs)**Table 18-23 MSC Interface Upstream Bit Map Read15**

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	LIN_ERROR				STBY_NVM_ADD_REG			
					LIN_PERM	LIN_PERM	LIN_TXD	STBY_NVM_ADD[0:3] (1)				MEM_VA_LID

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					REC	DOM	DOM						(1)	
FR AM E2	0	0	1	0	CONFIG_REG_21									
					BUCK_SL OW_SR	CAN_T XD_DO M _EN (1)	CAN_P ERM_ REC_E N (1)	CAN_PER M_ DOM_EN (1)	CAN_RX D_REC_ EN (1)	CAN_ AUTO _BIAS(1)	KEY_IN_P IN _FLT	WAKE_I N_PIN_F LT		
FR AM E3	0	0	0	1		CONFIG-REG 20								
					CAN_TX_ DOM_ER R_ CFG(1)	FIN_W AKE	WAKE_ UP_TI MER_E N_SEL(1)	KEY_OC_ RERTY_M A X_EN(1)	CAN_LO OP_EN	CAN_ TDI	LIN_TX_D OM_ERR_ C FG	PDRV_O 2H_DLY		
FR AM E4	0	0	1	1	CONFIG-REG 14 (UCOMMANDS STATUS)									
					RLY1_UC	RLY2_ UC	RLY3_ UC	RLY4_U C	RLY5_U C	STR1_ UC	STR2_UC	STR3_U C		
source					FRAME[1]D[0:2] FRAME[2]D[0] FRAME[2]D[6:7] FRAME[3]D[1] FRAME[3][4:7] FRAME[4]D[0:7] logic on VB_IN (main logic)									
access					FRAME[1]D[3:7] FRAME[2]D[0:7] FRAME[3]D[0:7] FRAME[4]D[0:7] read FRAME[1]D[0:2] clear-on-read									

1. Bit in satellite logic.

Clear on Read

FRAME [1]

[7] MEM_VALID (see 4.19)

1: memory has been written and validated by micro by reading back data.

0: memory was not validated or standby-power-supply is removed

[6:3] STBY_NVM_ADD

[2] 1: LIN TxD dominant timeout error occurs

0: no LIN TxD dominant timeout error occurs

Clear on read

[1] 1: LIN permanent dominant error occurs

0: no LIN permanent dominant error occurs

Clear on read

[0] 1: LIN permanent recessive error occurs

0: no LIN permanent recessive error occurs

Clear on read

FRAME[2]

[7] 1: The WK_IN pin (filtering on low to high transition) is at high state.

0: The WK_IN pin (filtering on high to low transition) is at low state.

[6] 1: The KEY_IN pin (filtering on low to high transition) is at high state.

0: The KEY_IN pin (filtering on high to low transition) is at low state.

FRAME[3]

FRAME[4]

[7:0] µcommands:

1 if driver µcommand is still active and driver is off, 0 otherwise

0 if driver µcommand is still active and driver is on, 1 otherwise

See driver section for detailed description of µcommands.

18.6.16 Upstream Bit Map Read16 (MEM_STBY_DATA)

Table 18-24 MSC Interface Upstream Bit Map Read16

	ADDRESS				DATA							
	A0	A1	A2	A3	D0	D1	D2	D3	D4	D5	D6	D7
FRAME1	0	0	0	0	MEM_STBY_DATA(Byte0) ⁽¹⁾							
FRAME2	0	0	1	0	MEM_STBY_DATA(Byte1) ⁽¹⁾							
FRAME3	0	0	0	1	MEM_STBY_DATA(Byte2) ⁽¹⁾							
FRAME4	0	0	1	1	MEM_STBY_DATA(Byte3) ⁽¹⁾							
source					logic on VB_STBY (Wake Up Timer logic) ⁽¹⁾							
access					read							

1. Bit in satellite logic.

FRAME[1:4]: BYTES READ FROM MEM+STBY

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Package information

19 Package information.

ETQFP package information

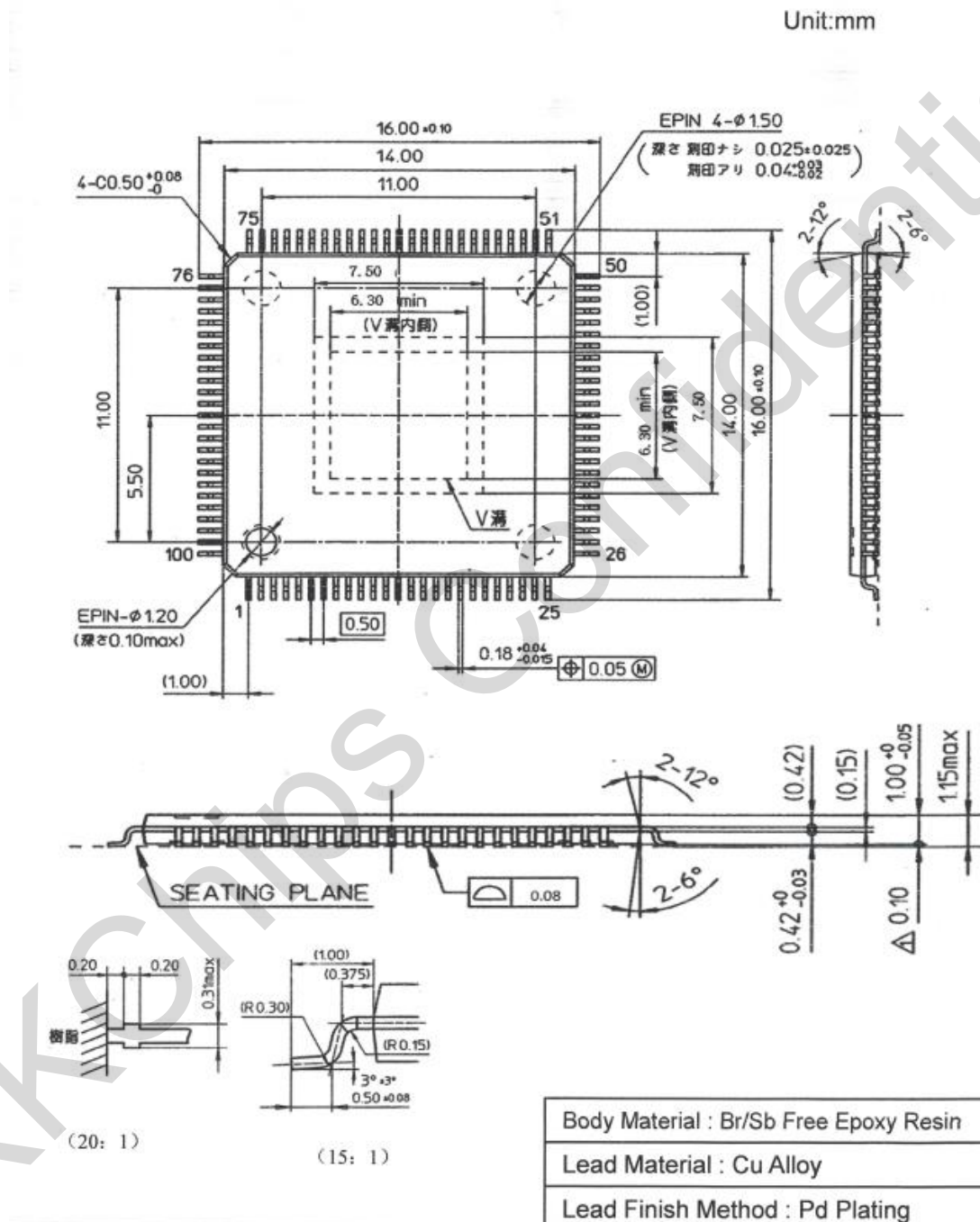
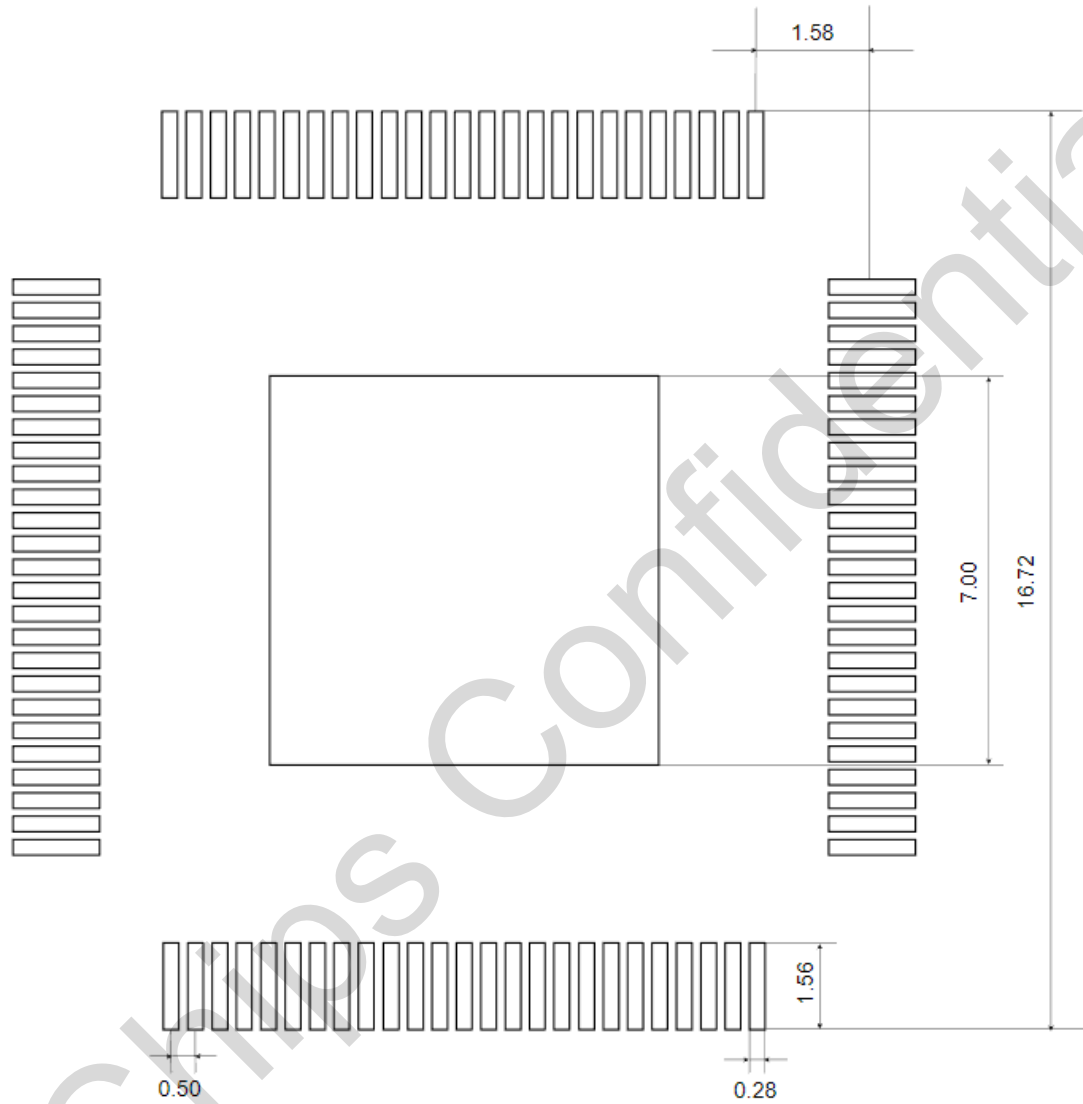


Figure 19-1 ETQFP package outline

20 Recommended pad size



Unit: mm

Figure 20-1 Recommended pad size

21 Appendix A Device behavior

21.1 Summary of all drivers block

Table 21-1Summary of all drivers block

Drivers		Min OVC value(A)	Rdson (Ω)	Vclamp Typ(V)	STB	STG	OPL	OVC
INJ		3	0.6	55	√	√	√	√
SOL		3	0.47	55	√	√	√	√
O2H		3/7.8	0.2	50	√	√	√	√
RLY		1	1.5	50	√	√	√	√
HSLs	HS	1	1.5	-3.5	√	√	√	√
	LS	1	1.5	45	√	√	√	√
LED		0.07	20	45	√	√	√	√
MRD		1	-	50	√	√	√	√
IGN		0.1	7	-	√	√	√	
PreMOS		-	-	-	√	√	√	√

21.2 Slew rate and on/off delay time

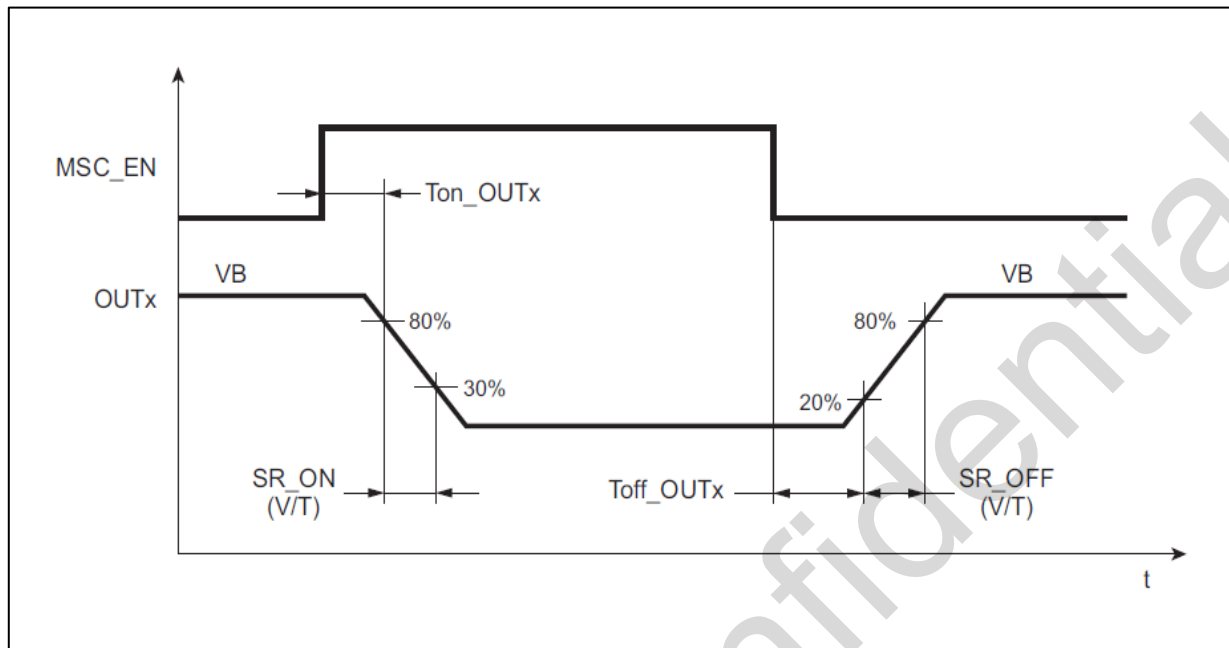


Figure 0-1 Slew rate and on/off delay time

Ton_OUTX is the on delay time of OUTX;
 Toff_OUTX is the off delay time of OUTX;
 SR_ON is the on slew rate of OUTX;
 SR_OFF is the off slew rate of OUTX.

21.3 Power up/down scenarios

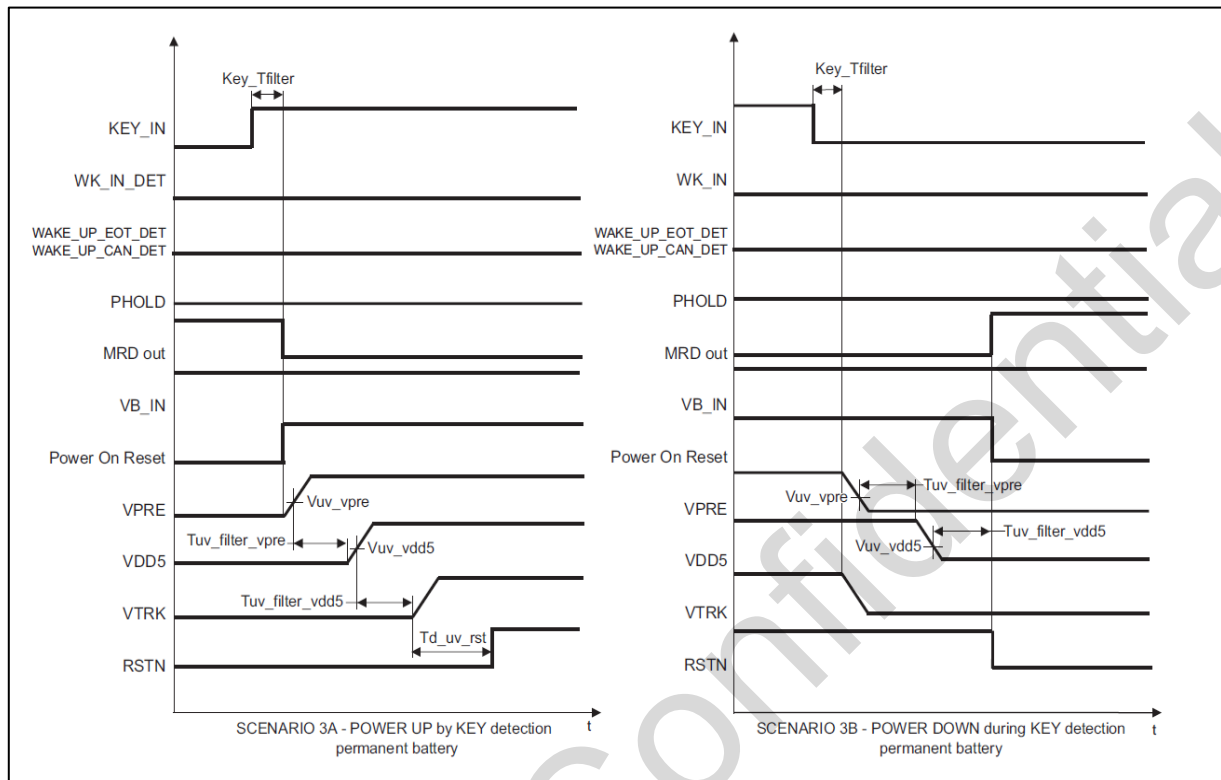


Figure 0-2 Slew rate and on/off delay time

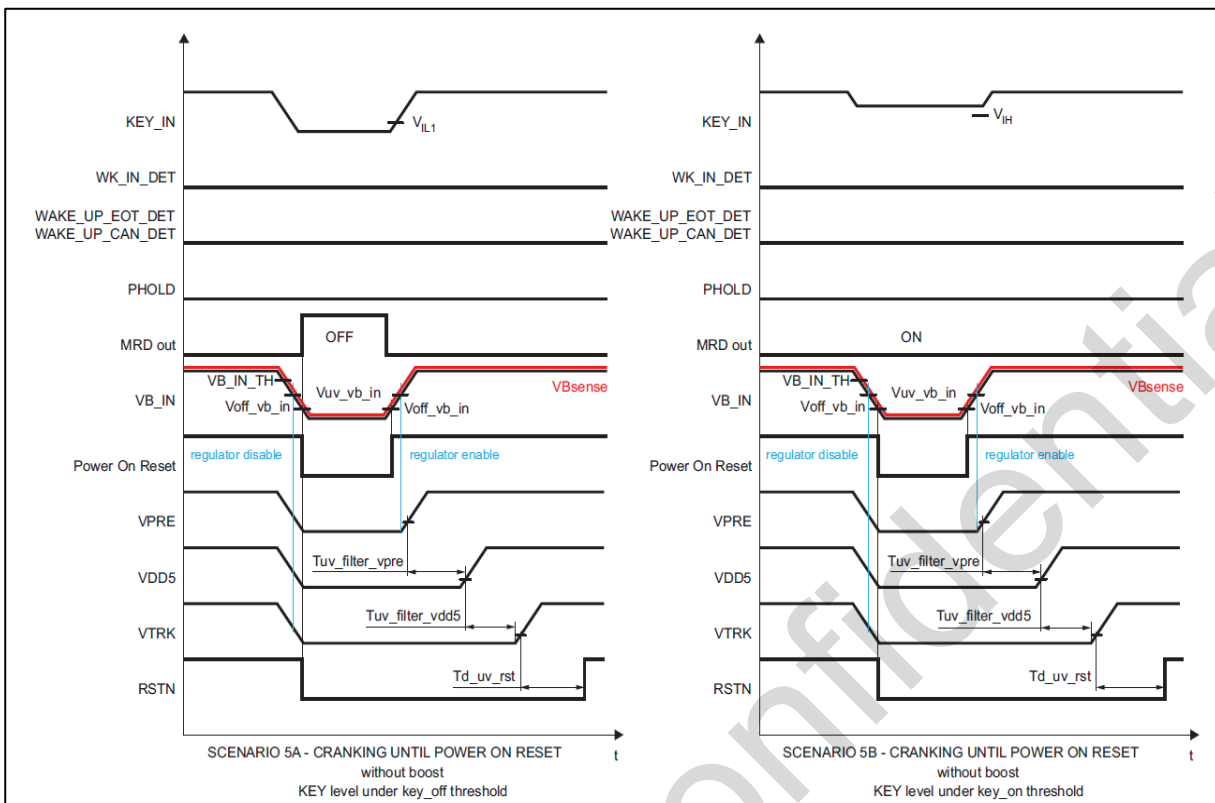


Figure 0-3 Power up permanent battery without boost and cranking

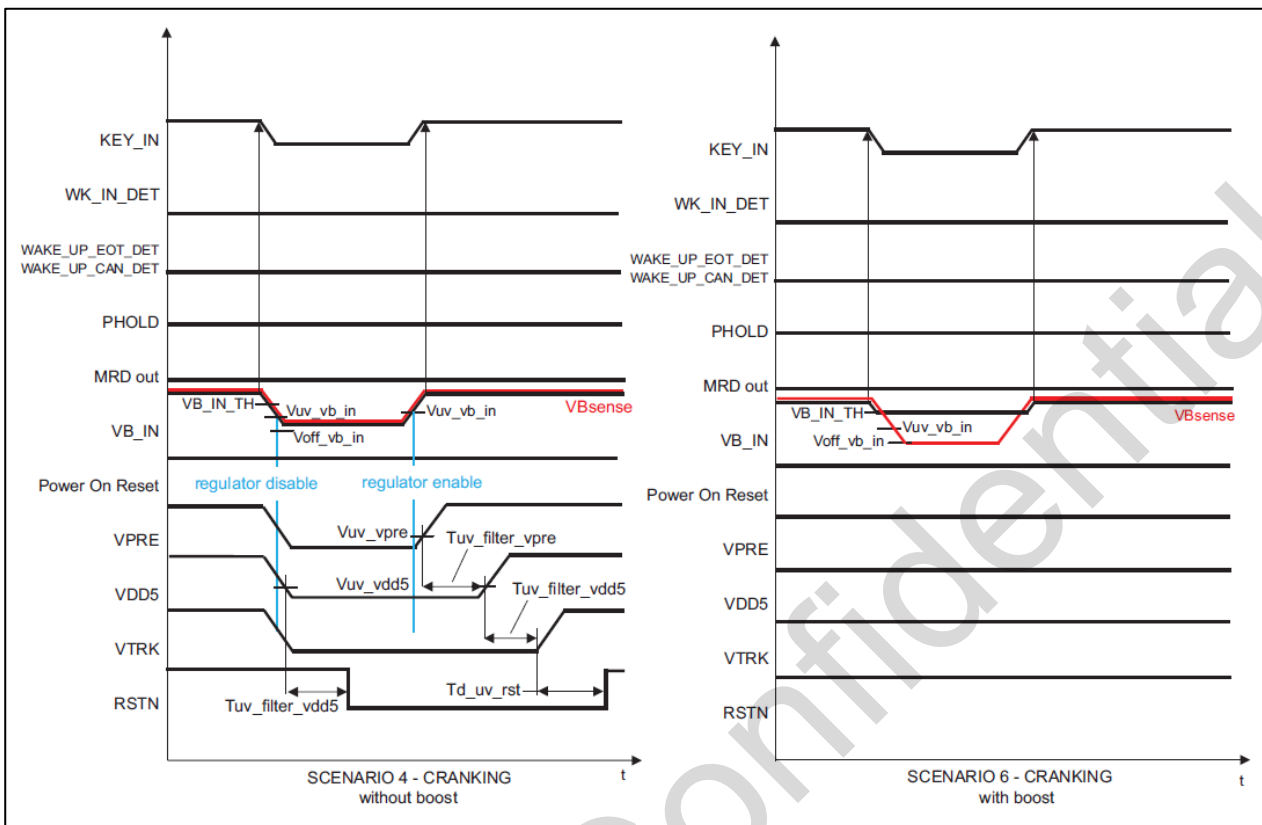


Figure 0-4 Power up permanent battery with boost and cranking

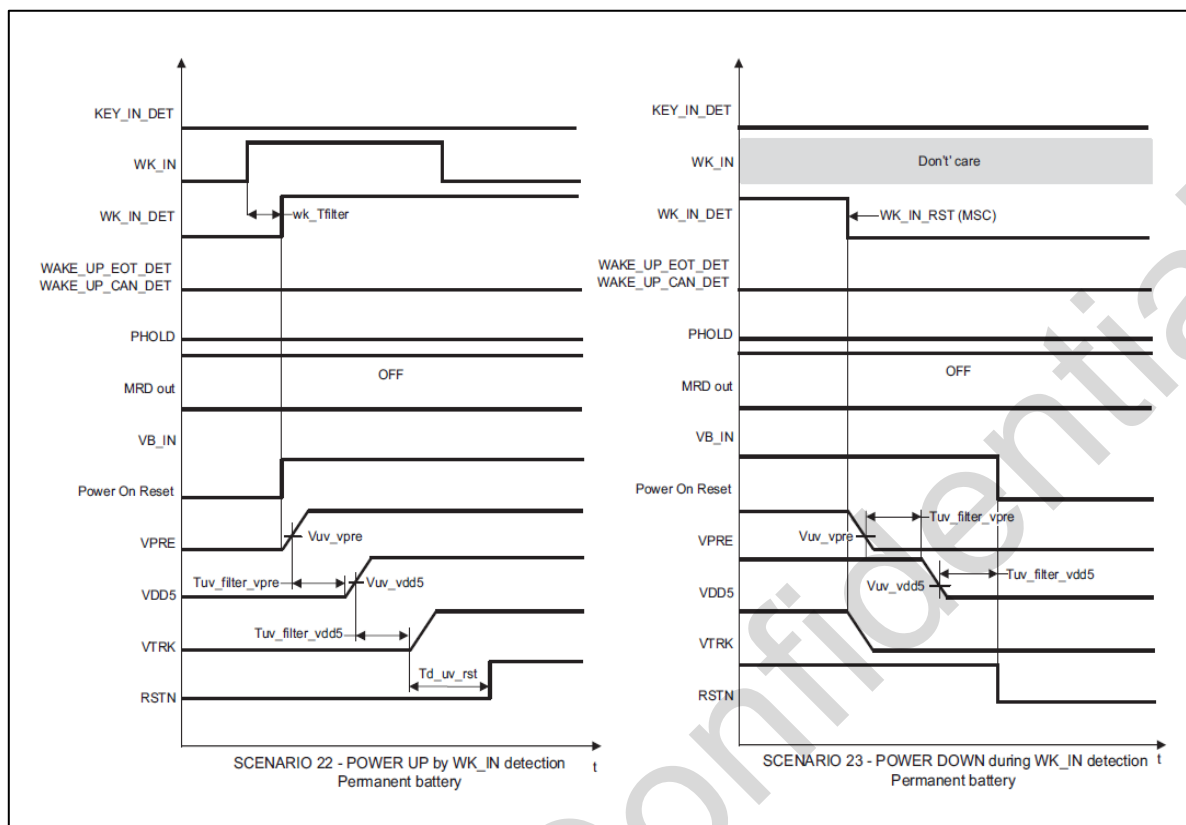


Figure 0-5 Power up with WK_IN and power down with WK_IN in permanent battery condition

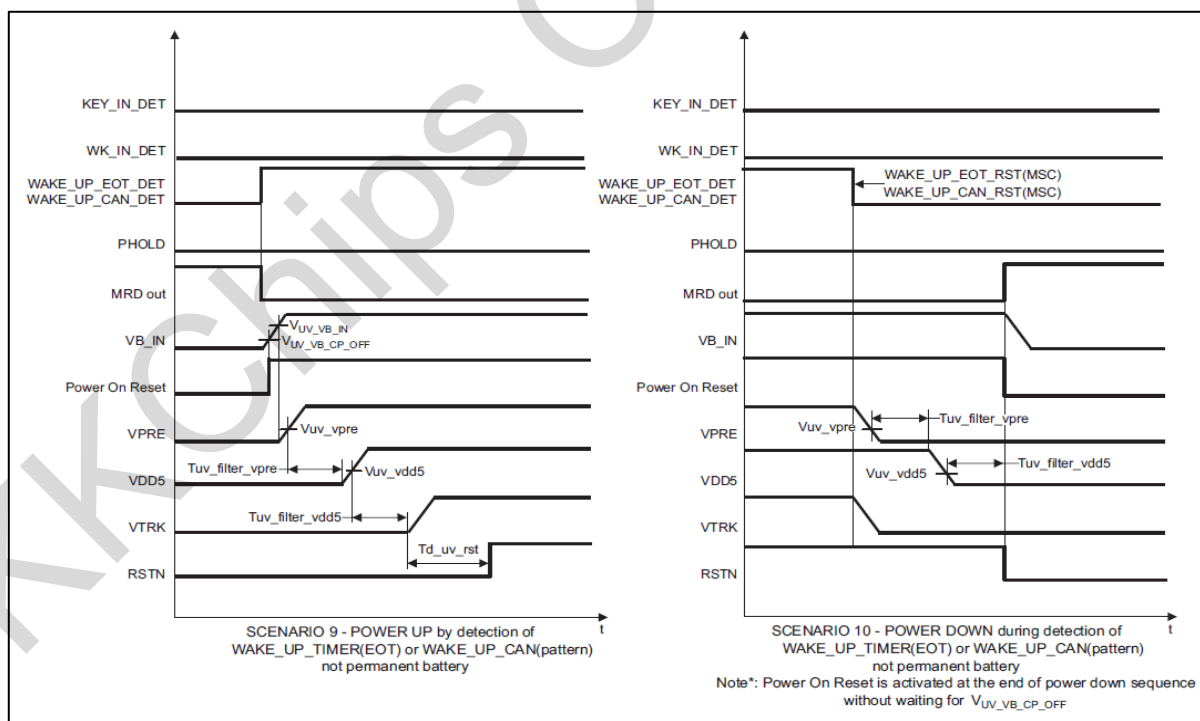


Figure 0-6 Power up with WAKE_UP_EOT/CAN and power down with WAKE_UP_EOT/CAN

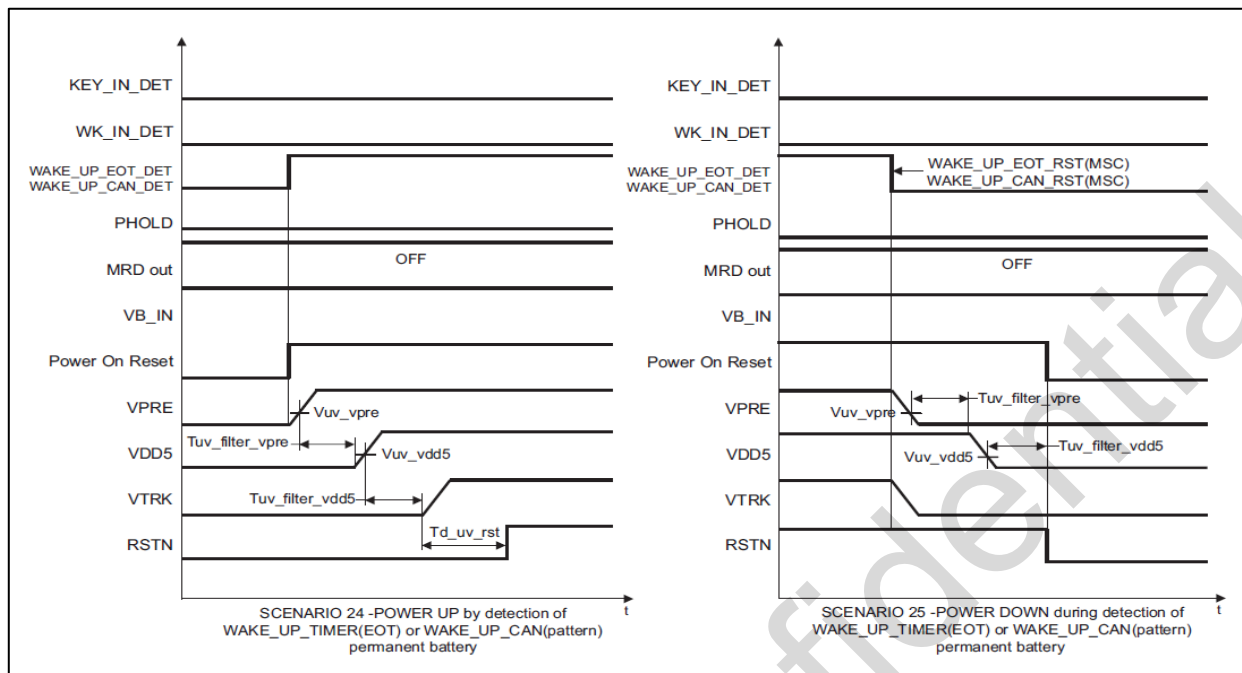


Figure 0-7 Power up with WAKE_UP_EOT/CAN and power down with WAKE_UP_EOT/CAN

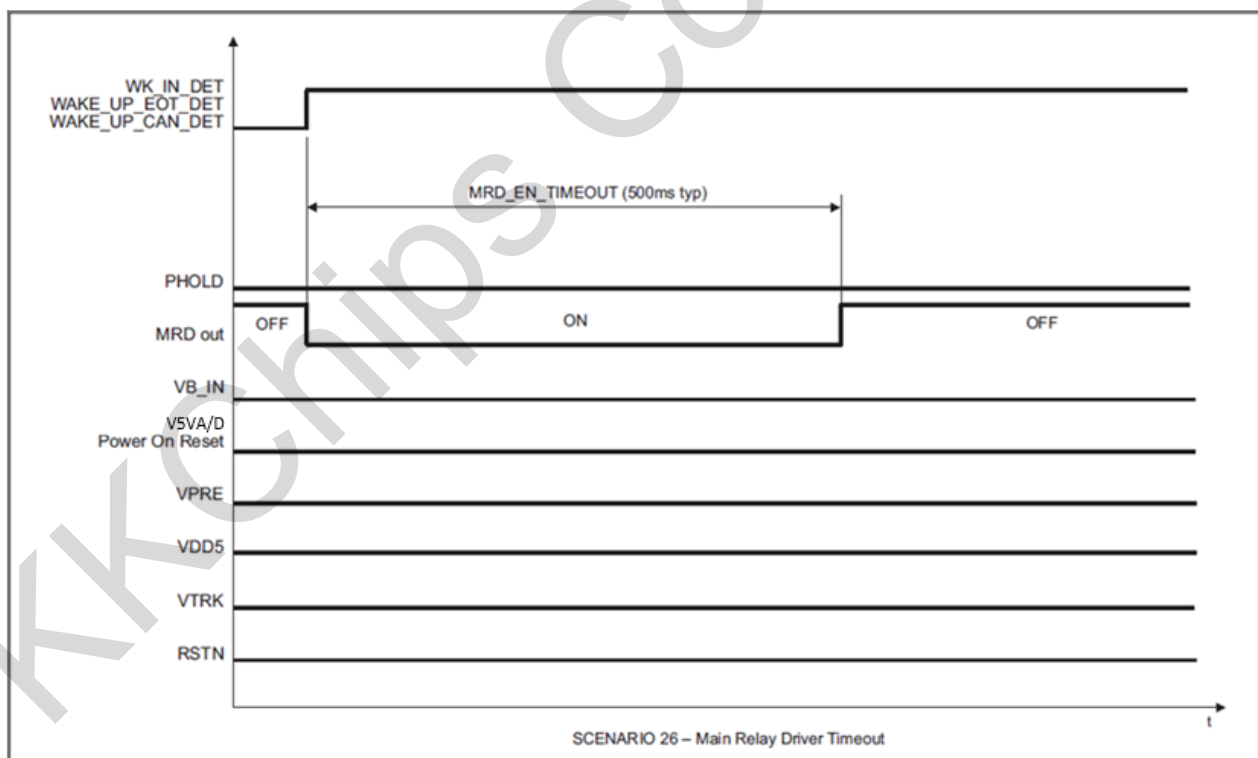


Figure 0-8 Main relay driver timeout

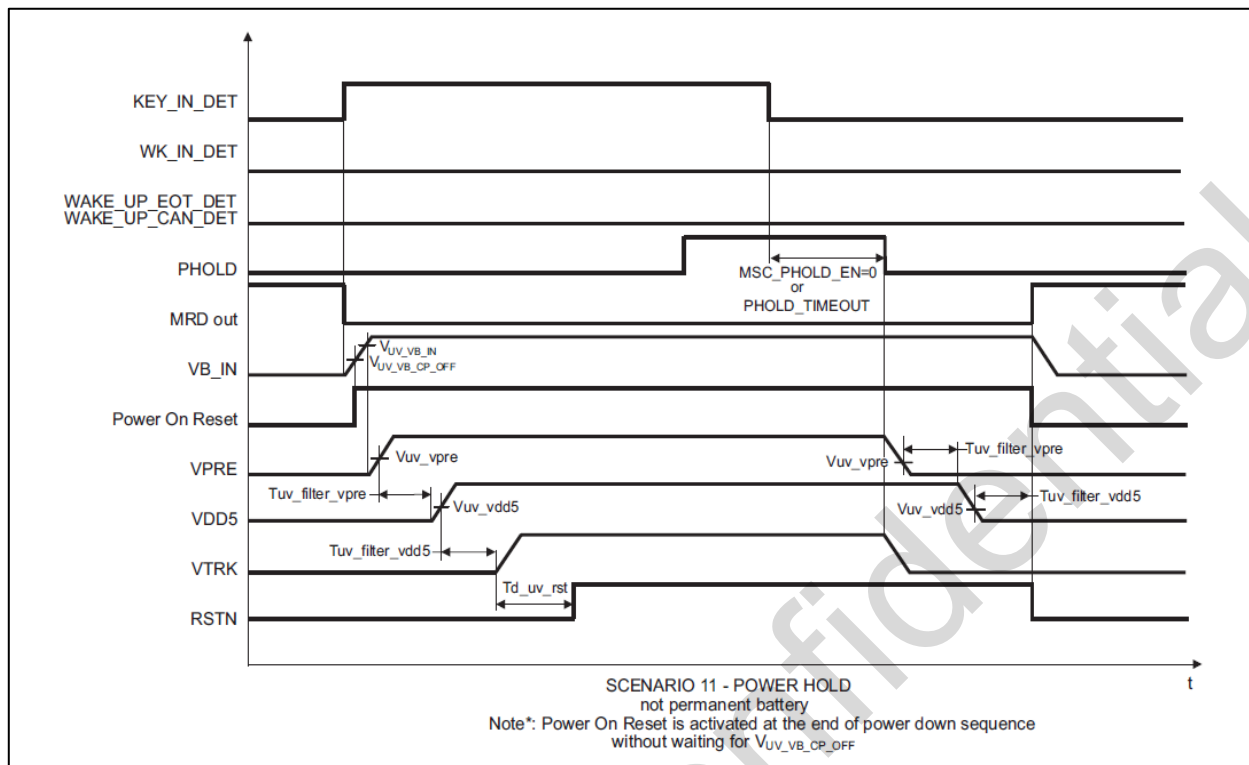


Figure 0-9 Power HOLD

According to reset matrix power down is also produced by Watchdog PWR_CNT counter overflow during KEY_IN low. In these cases the power down sequence is actuated without considering VDD5 undervoltage

A.4 Main relay scenario

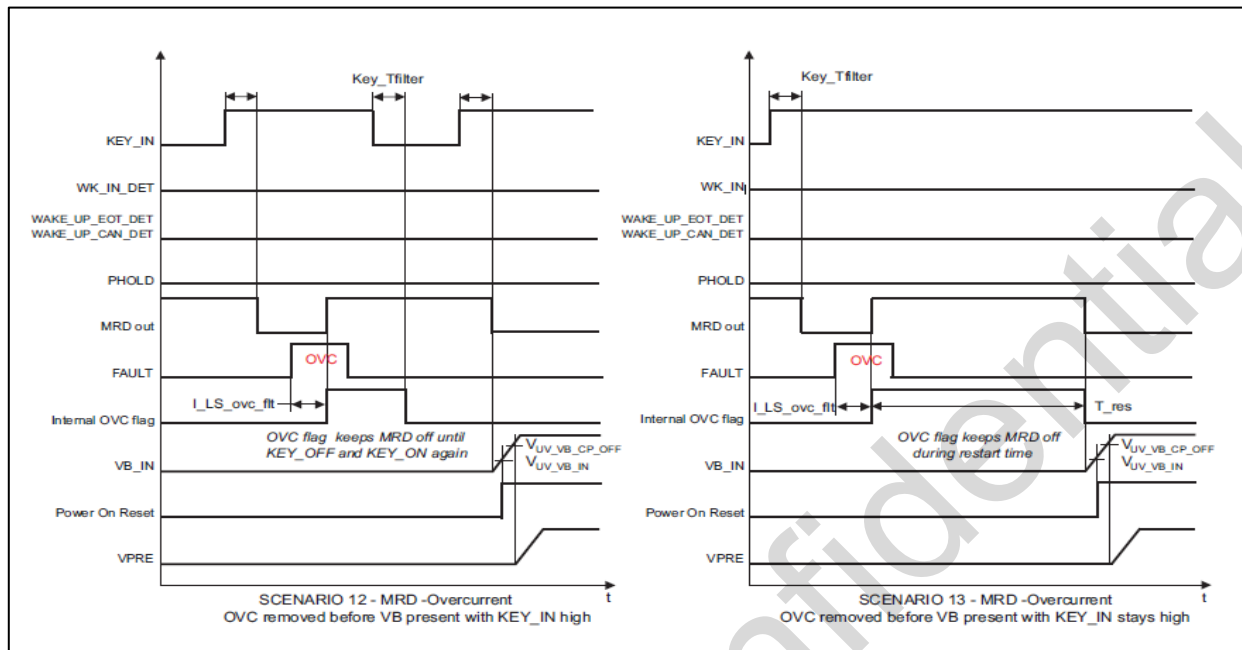


Figure 0-10 Overcurrent OVC removed before VB present with KEY_IN high and Overcurrent OVC

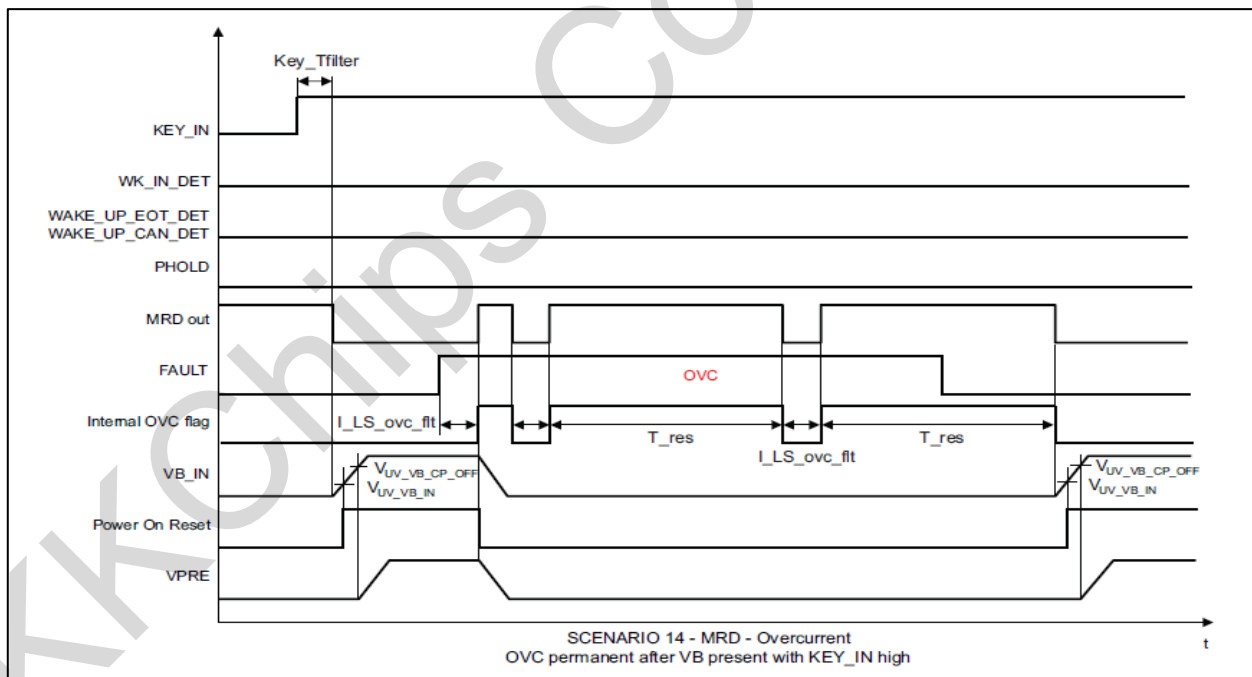


Figure 0-11 Overcurrent OVC permanent after VB present with KEY_IN high

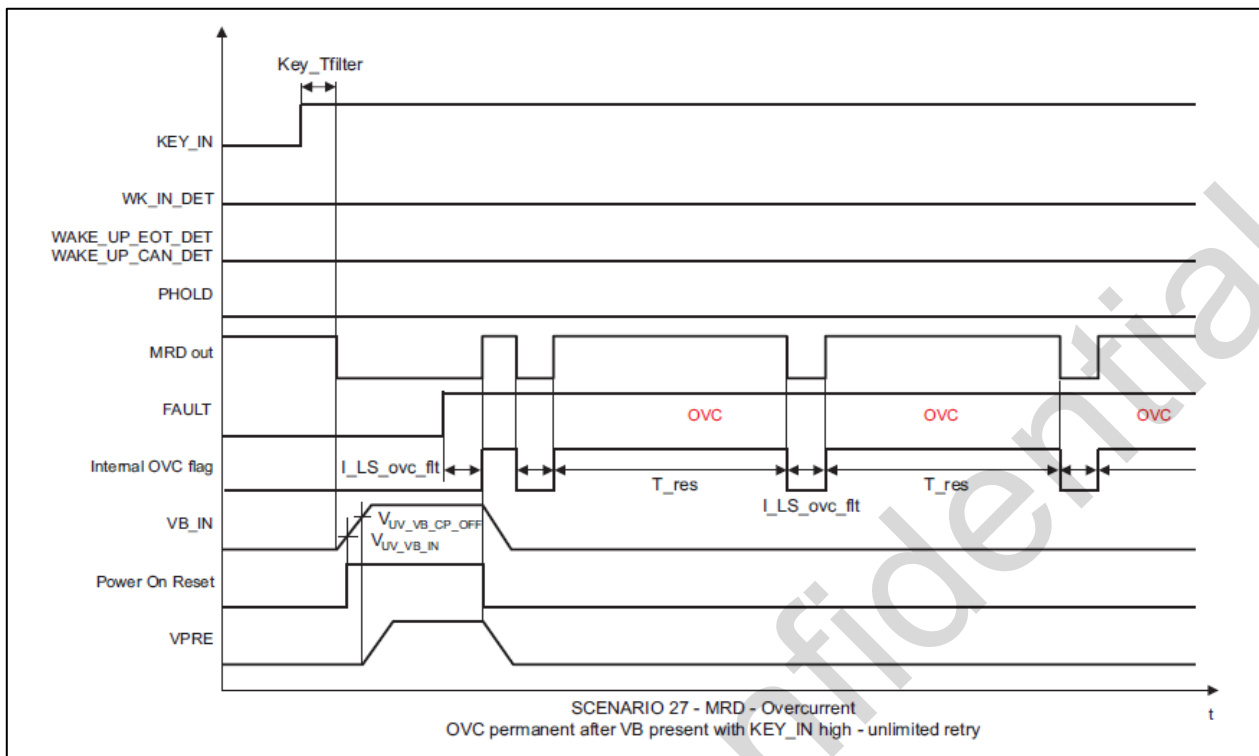


Figure 0-12 Overcurrent OVC permanent after VB present with KEY_IN high - unlimited retry

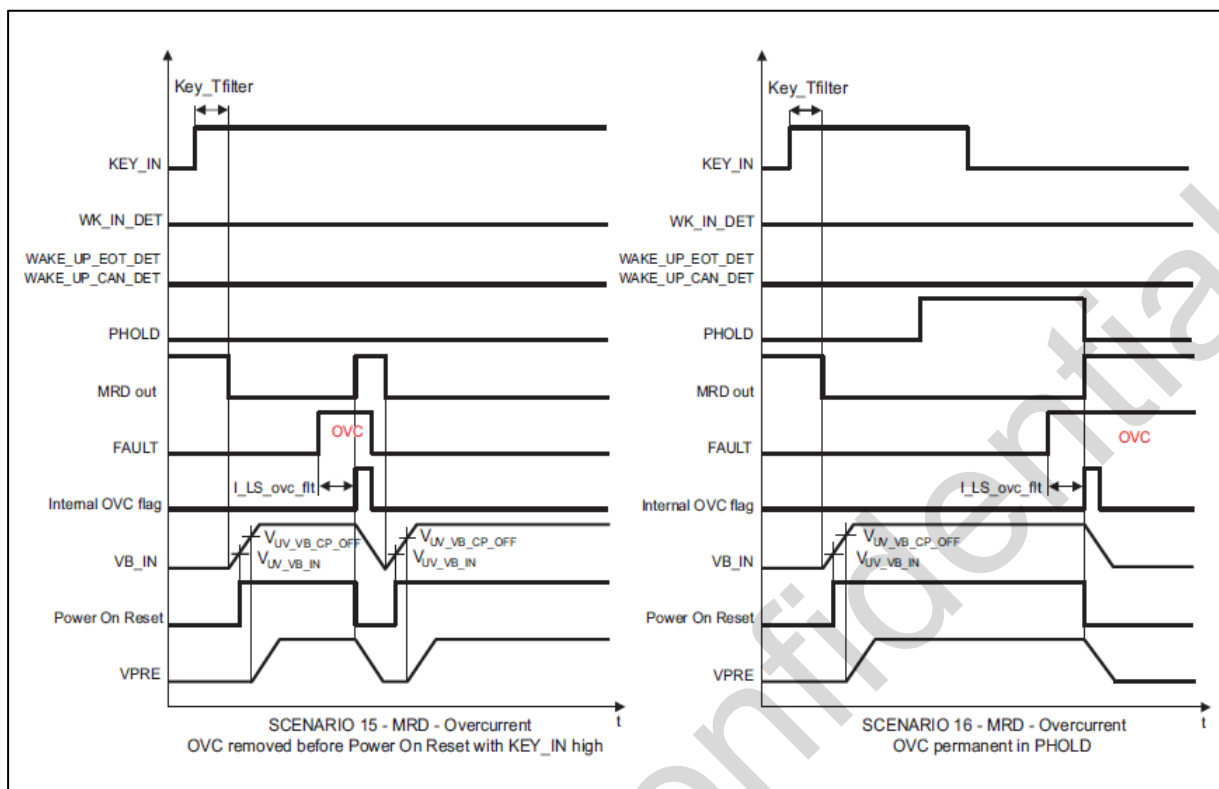


Figure 0-13 Overcurrent OVC removed before Tres activation with KEY_IN high and Overcurrent

Appendix B Application diagrams

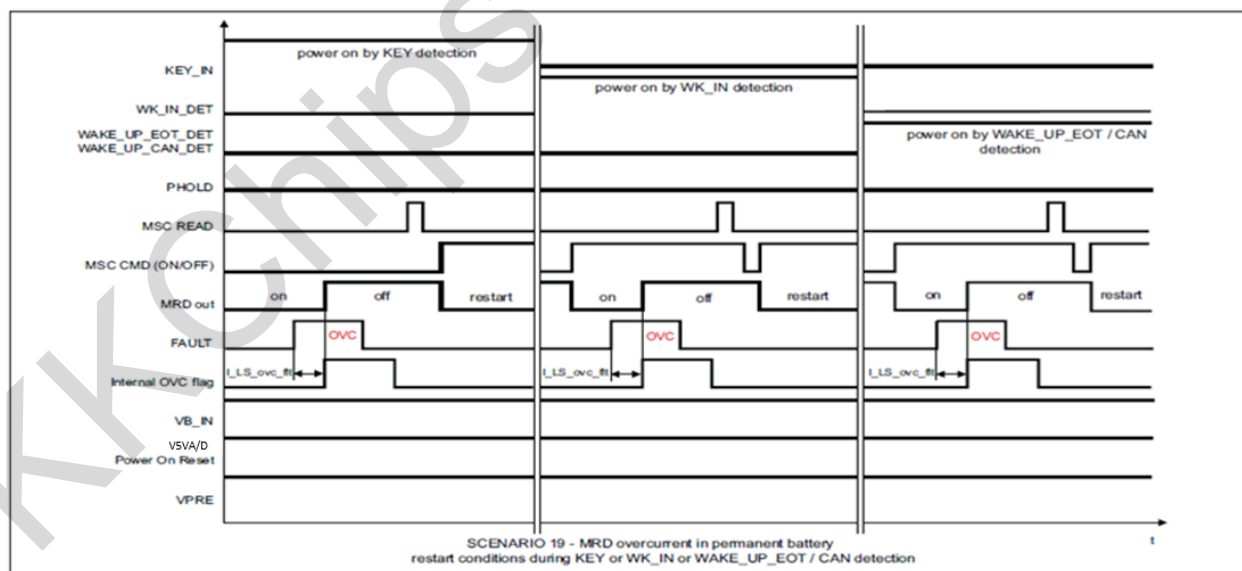


Figure 0-14 Overcurrent in permanent battery restart conditions during KEY or WK_IN or

In permanent battery when power on by KEY the MSC CMD can only restart driver after OVC. The MSC CMD OFF has no effect

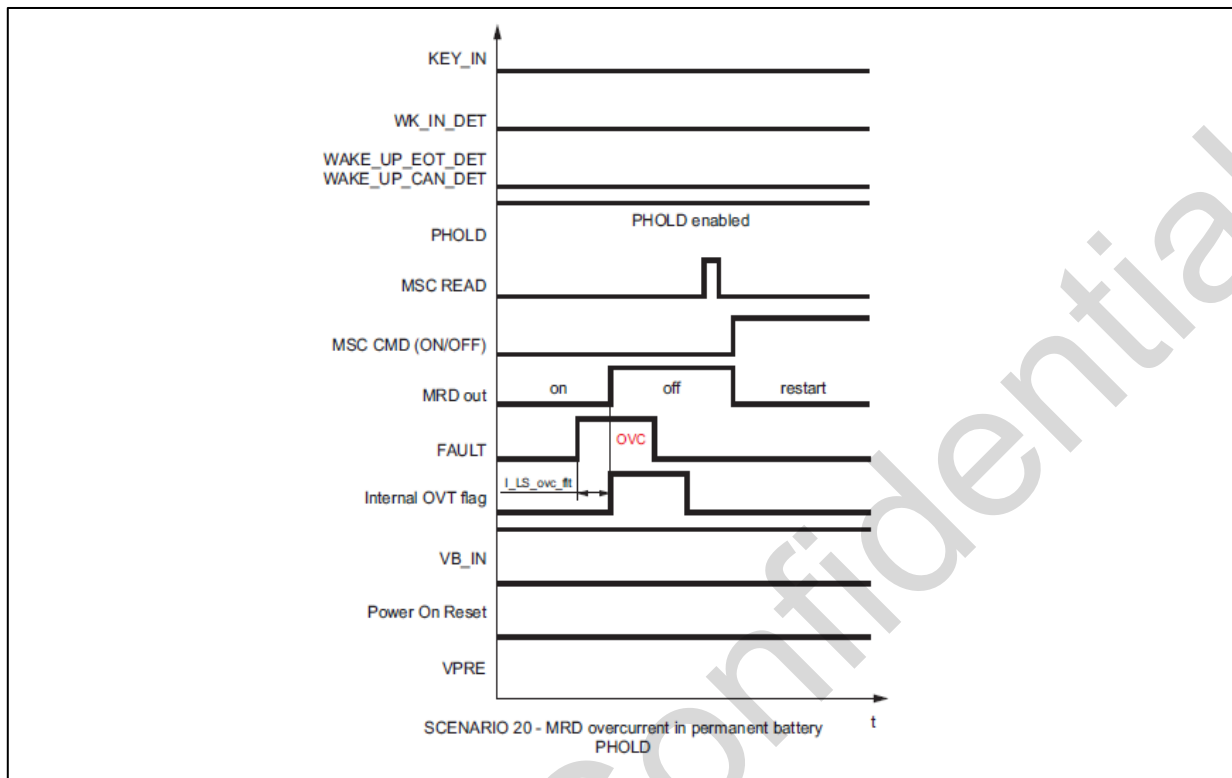


Figure 0-15 Overcurrent in permanent battery PHOLD

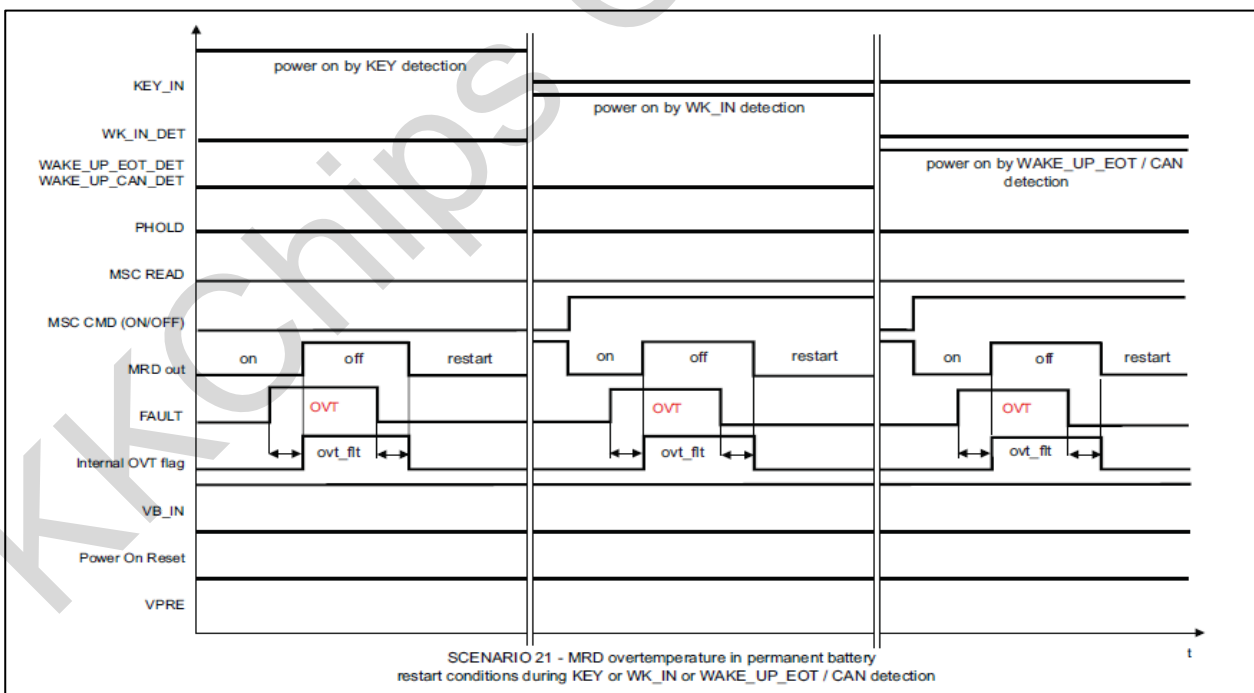
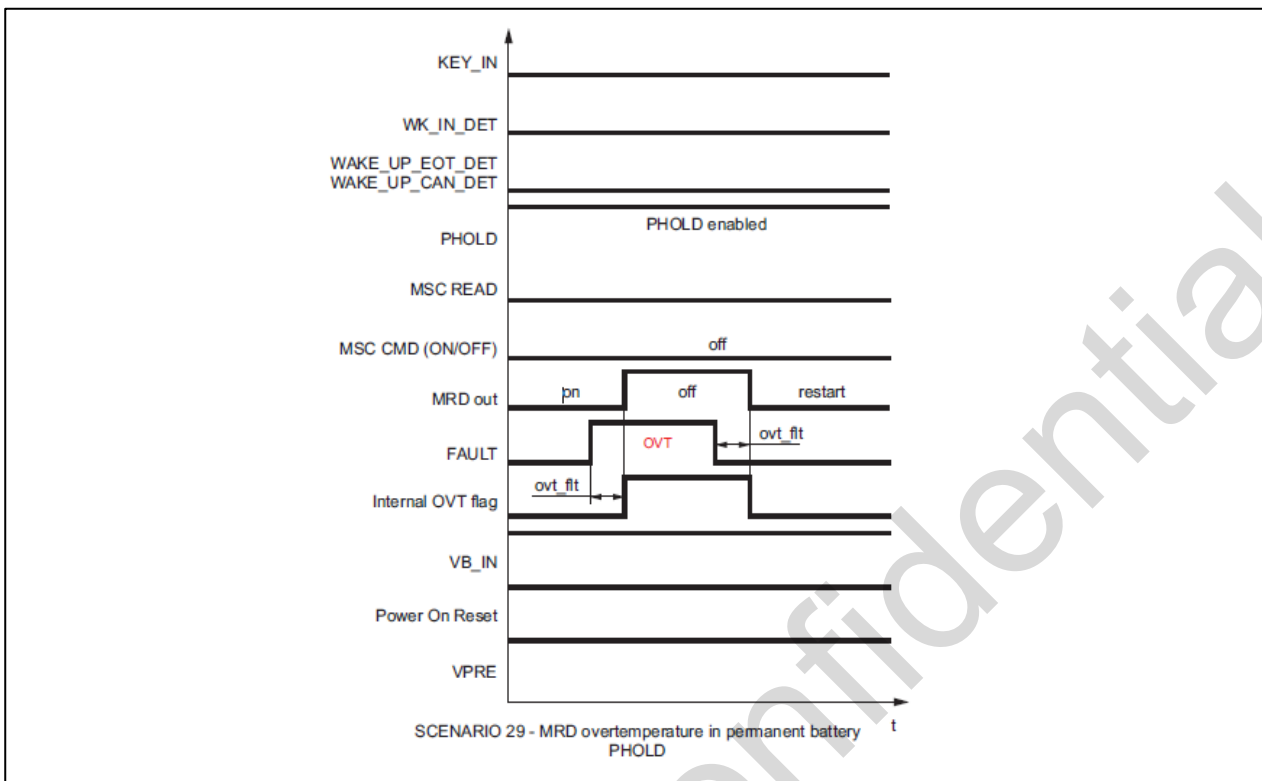


Figure 0-16 Overtemperature in permanent battery restart conditions during KEY or WK_IN or

**Figure 0-17** Overtemperature in permanent battery PHOLD

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Appendix A Device behavior

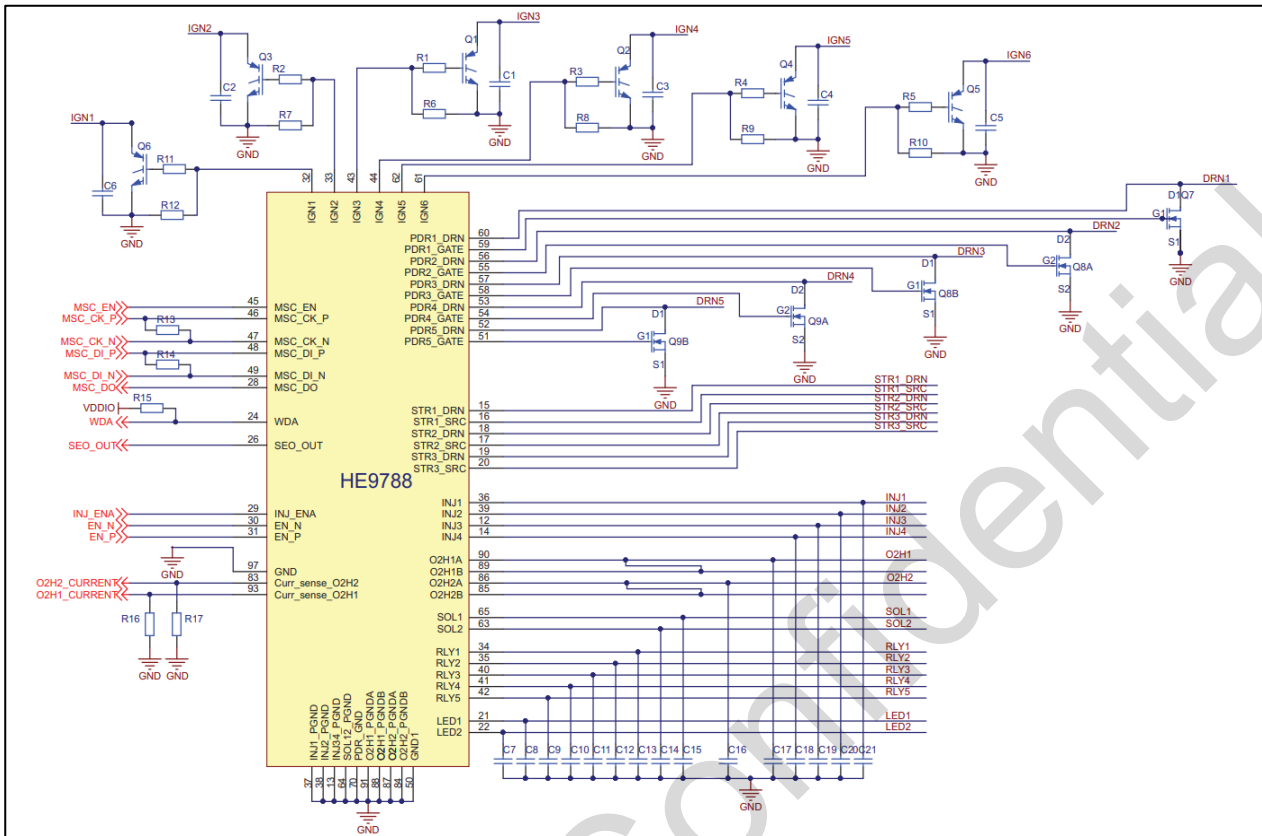


Figure 0-18 Driver/predriver section application diagram

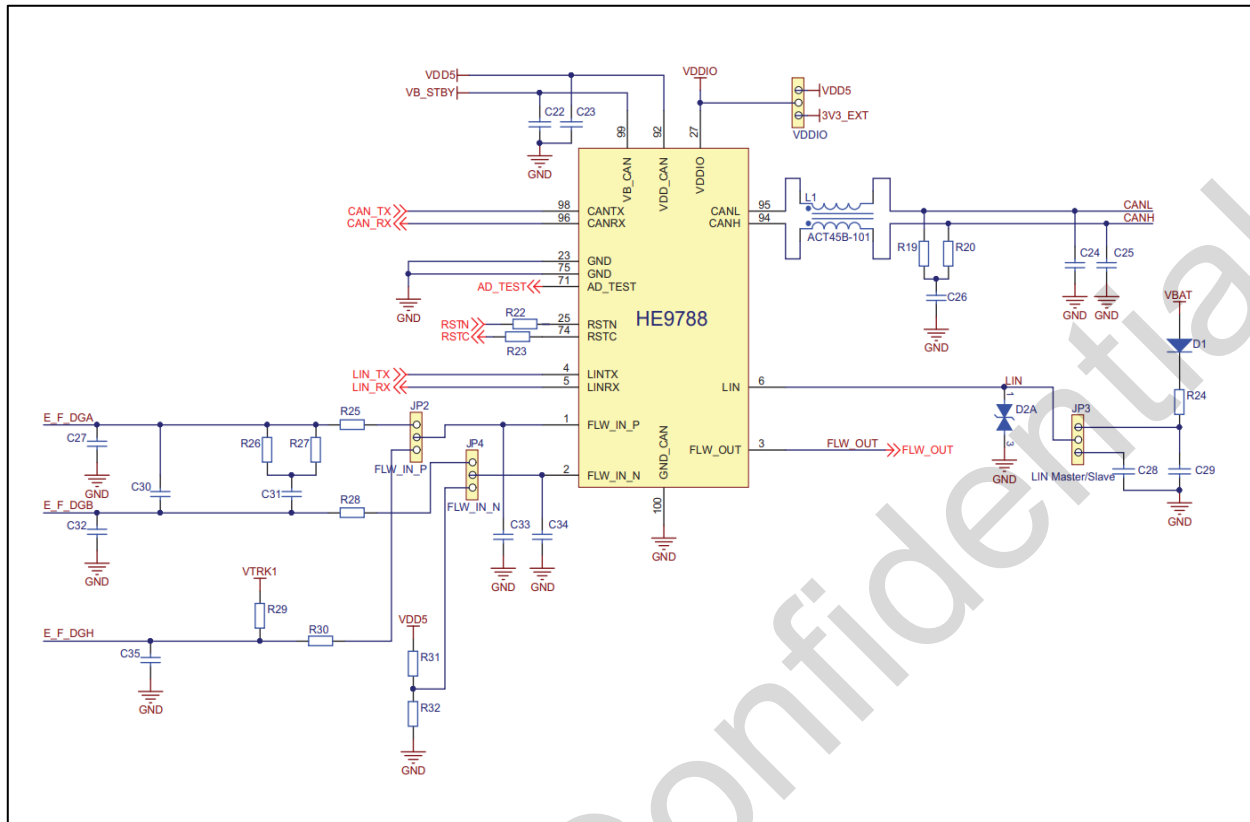


Figure 0-19 Interface section application diagram

21.4 Bill of material

Table 21-2 Bill of material

Symbol	Parameter	Min	Typ	Max	Unit	Note	Number
C39	Battery filter capacitor	-	4.7	-	nF	-	P_21.4.1
D3	Reverse battery protection diode	-	-	-	-	Choice: STTH4R02	P_21.4.2
D5,D7	TVS diode	-	-	-	-	Choice: TPSMB36A	P_21.4.3
C42	battery decoupling capacitor	-	100		nF	-	P_21.4.4
C41	battery decoupling capacitor	-	100		μF	-	P_21.4.5
L2	Boost inductance	-	-	-	-	see Table 6-2	P_21.4.6
Q10	Boost MOSFET	-	-	-	-	choice: STD20NF06L see Table 6-1 and Table 6-2	P_21.4.7
D4	Boost diode	-	-	-	-	choice: STPS10H100C see Table 6-2	P_21.4.8
C36	Cout_boost	36	40	150	μF	see Table 6-2	P_21.4.9
C53	Cout_boost_ALU	0	330	470	μF	see Table 6-2	P_21.4.10
R33	Rg	-	2.2	-	Ω	see Table 6-2	P_21.4.11
R34	Rg_pd	-	150	-	kΩ	see Table 6-2	P_21.4.12
C37	External charge pump capacitor	80	100	120	nF	see Table 6-13	P_21.4.13

R35	Rsense	10	22	30	kΩ	see Table 6-2	P_21.4.14
C43	filter capacitor	-	100	-	nF	Optional	P_21.4.15
R36	Y_IN protection resistor	-	1	-	kΩ	-	P_21.4.16
R37	WK_IN protection resistor	-	1	-	kΩ	-	P_21.4.17
C46,C47,C48,C49,C44,C54	filter capacitor	-	100	-	nF	-	P_21.4.18
C38	CBST	-	47	-	nF	see Table 6-4	P_21.4.19
L3	Lvpre	15	22	29	μH	see Table 6-4	P_21.4.20
D6	Dvbuck_CTRL	-	-	-	-	Choice: STPS5L60-Y see Table 6-4	P_21.4.21
C40	Cvpre	15	40	150	μF	see Table 6-4	P_21.4.22
Q11	-	-	-	-	-	choice: STD20NF06L see Table 6-8 and Table 6-9	P_21.4.23
C45	CVDD5	5	20	60	μF	see Table 6-9	P_21.4.24
C50,C51,C52	CVS1/2/3	0.38	0.47	20	μF	see Table 6-9	P_21.4.25
R1,R2,R3,R4,R5, R11	IGBT gate resistor	-	1	-	kΩ	-	P_21.4.26
R6,R7,R8,R9,R10, R12	IGBT Gate discharge resistor	-	4.7	0	kΩ	-	P_21.4.27
C1,C2,C3,C4,C5,C6	filter capacitor	-	1.5	-	nF	-	P_21.4.28
R13,R14	MSC differential line adapting resistor	-	100	-	Ω	-	P_21.4.29
R15	WDA pull-up external resistor	50	-	250	kΩ	See Table 4-6	P_21.4.30
Q7, Q8A, Q8B, Q9A, Q9B	-	-	-	-	-	choice: STL15DN4F	P_21.4.31
C7,C8,C9, C10,C11, C12,C13,C14,C15,C16, C17,C18,C19,C20,C21	filter capacitor	-	4.7	-	nF	-	P_21.4.32
R16, R17	-	-	5.1	-	kΩ	See Table 7-6	P_21.4.33
C22,C23	decoupling capacitor	-	100	-	nF	-	P_21.4.34
R22,R23	protection resistor	-	1	-	kΩ	-	P_21.4.35
C27, C32	filter capacitor (VRS sensor configuration)	-	100	-	pF	-	P_21.4.36
C30	filter capacitor (VRS sensor configuration)	-	470	-	pF	-	P_21.4.37
R26,R27	filter resistor (VRS sensor configuration)	-	33	-	kΩ	-	P_21.4.38
C31	filter capacitor (VRS sensor configuration)	-	100	-	nF	-	P_21.4.39
R25,R28	Protection and reference resistors (VRS sensor configuration)	-	10 1%	-	kΩ	see Section 11.2	P_21.4.40
C35	filter capacitor (Hall sensor configuration)	-	1	-	nF	-	P_21.4.41
R29	pull-up resistor (Hall sensor configuration)	-	27	-	kΩ	-	P_21.4.42
R30	Protection and reference resistor (Hall sensor configuration)	-	27	-	kΩ	-	P_21.4.43
R31,R32	Protection and reference resistor (Hall sensor configuration)	-	33	-	kΩ	-	P_21.4.44
C33,C34	filter capacitor	-	470	-	pF	place close to pin	P_21.4.45
L1	CAN Choke	-	-	-	-	choice: ACT45B-101	P_21.4.46
R19, R20	CAN termination resistor	-	60	-	Ω	-	P_21.4.47
C26	CAN termination capacitor	-	4.7	-	nF	-	P_21.4.48

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Appendix A Device behavior

C24,C25	-	-	47	-	pF	-	P_21.4.49
D2A	-	-	-	-	-	27V TVS	P_21.4.50
D1	Vser Diode	-	-	-	-	Choice: BAS21	P_21.4.51
R24	Lin pull-up resistor	-	1	-	kΩ		P_21.4.52
C28	-	-	-	-	-	Choice: 220pF	P_21.4.53
C29	-	-	-	-	-	Choice: 1 nF (K-Line), 680 pF (LIN)	P_21.4.54

21 Customer Service

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22 Revision history

Table 22-1 Document revision history

Date	Revision	Changes
2025/06/10	0.0.1	1. Based on the modifications of the DA version. 2. Increase the SEO_OUT_L parameter. 3. Modify the test conditions for VTHdomLP and VTHrecLP. 4. Revise Figure 3-2 illustration. 5. Delete the description of the "read 1" register section. 6. The upper limit of the IMAX_VS is changed to 300 mA.